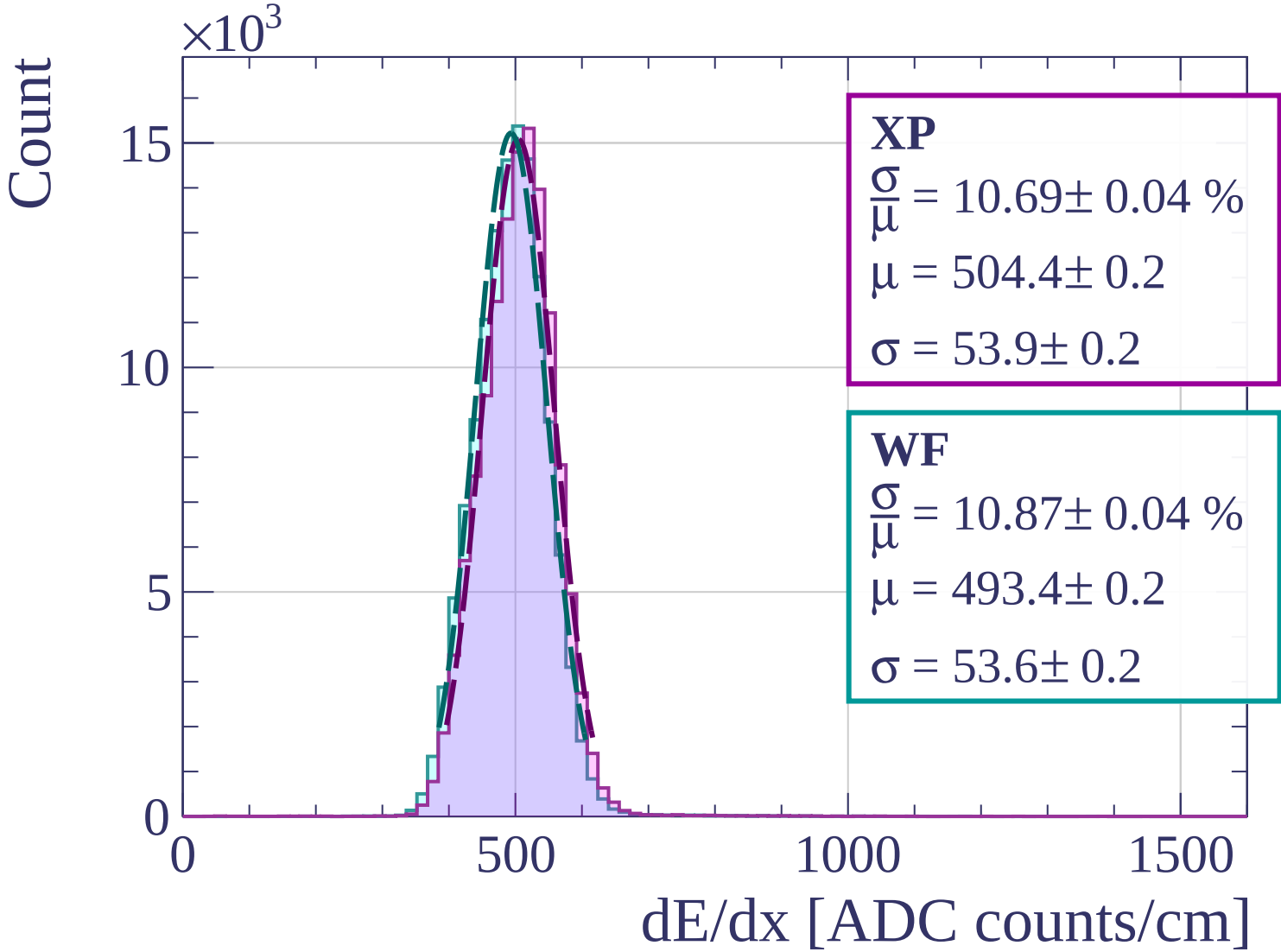
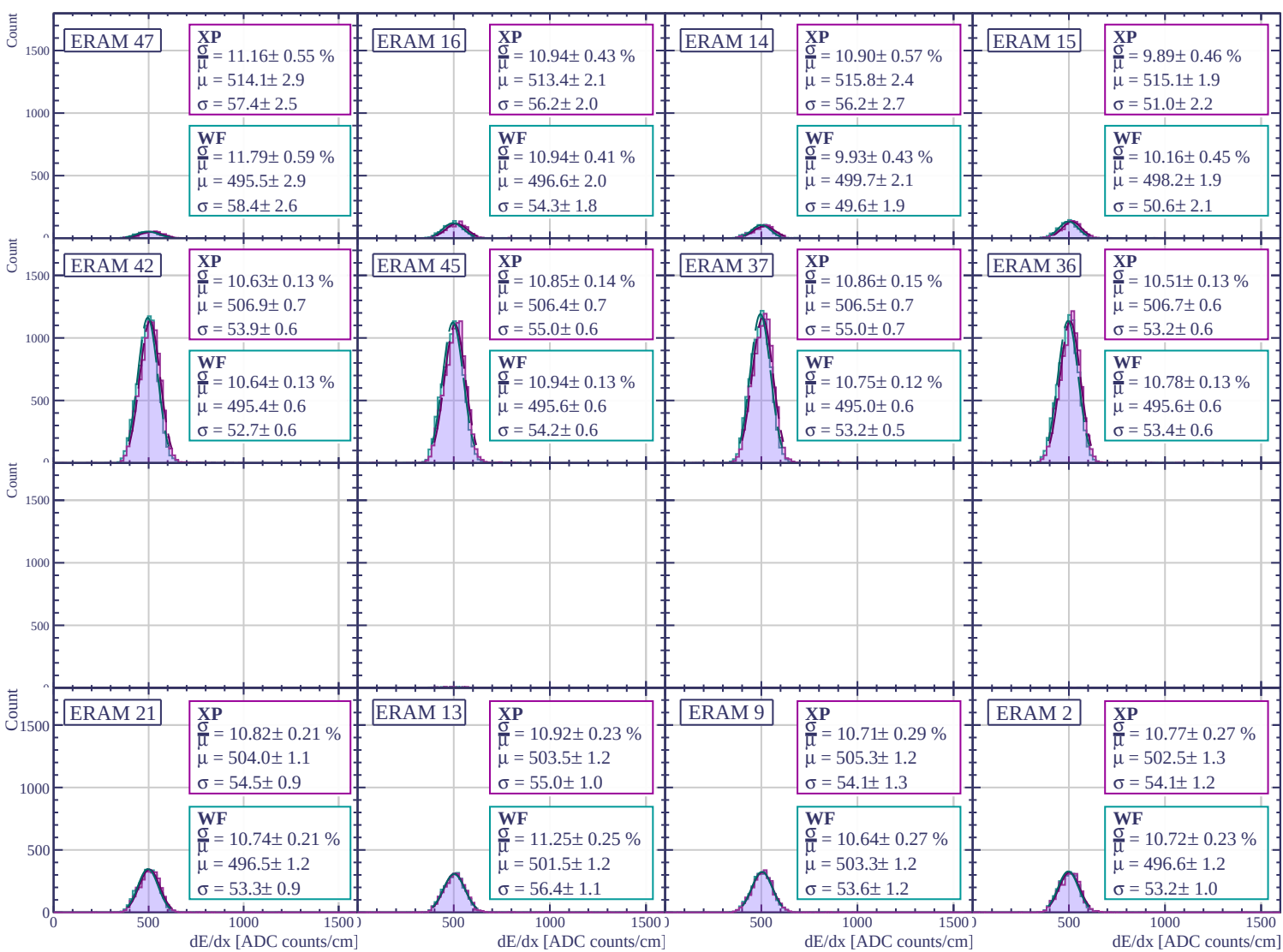
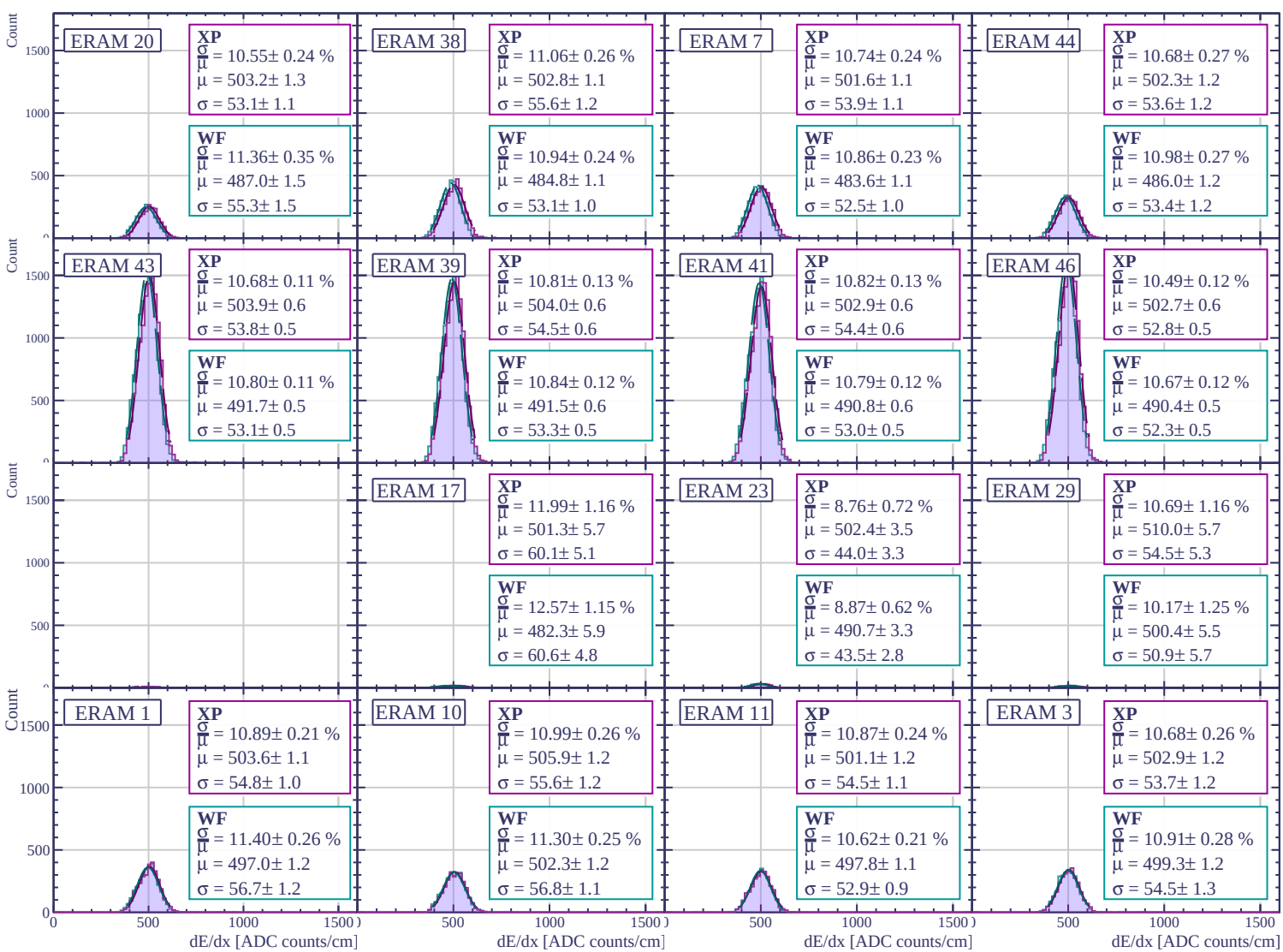
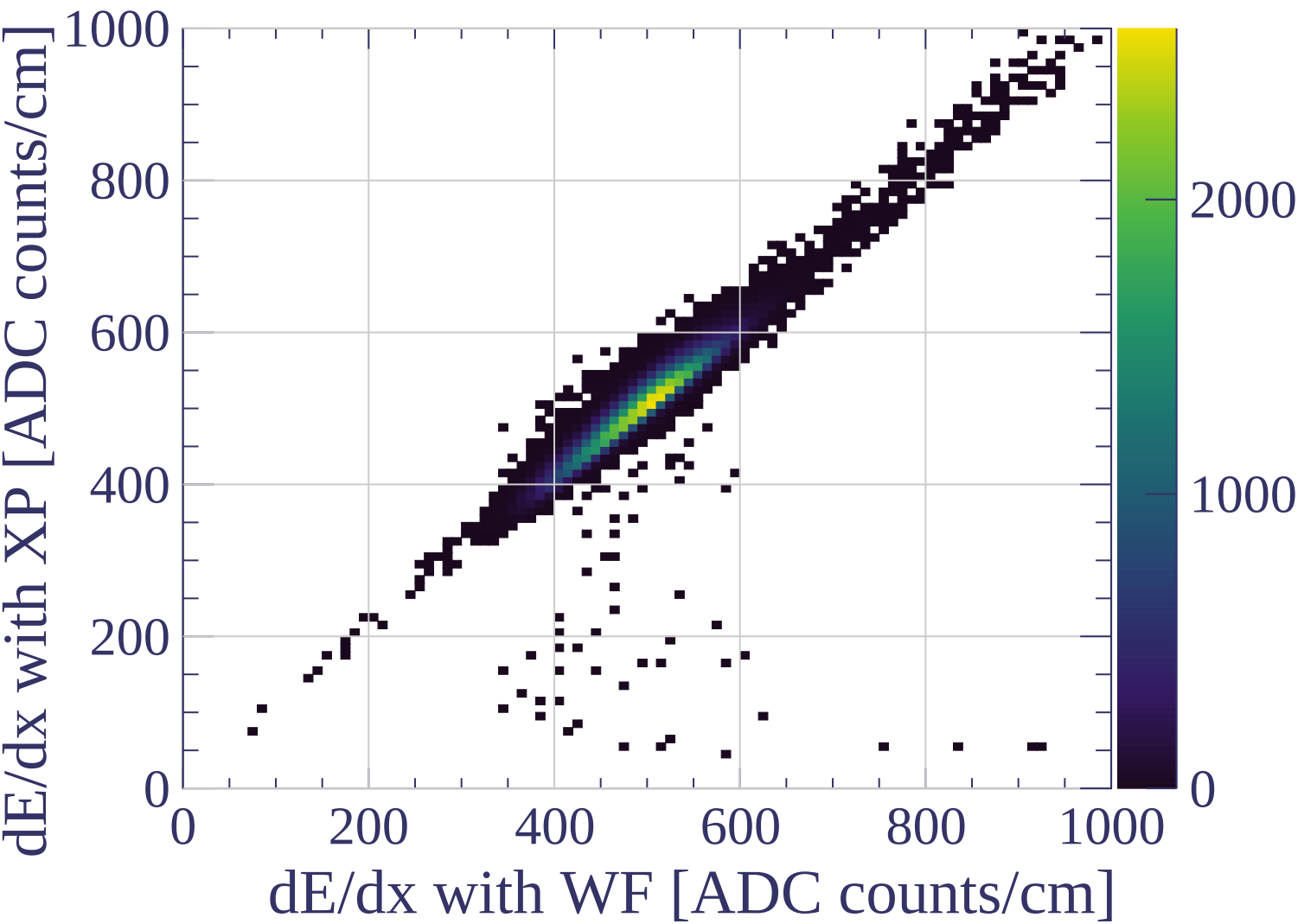
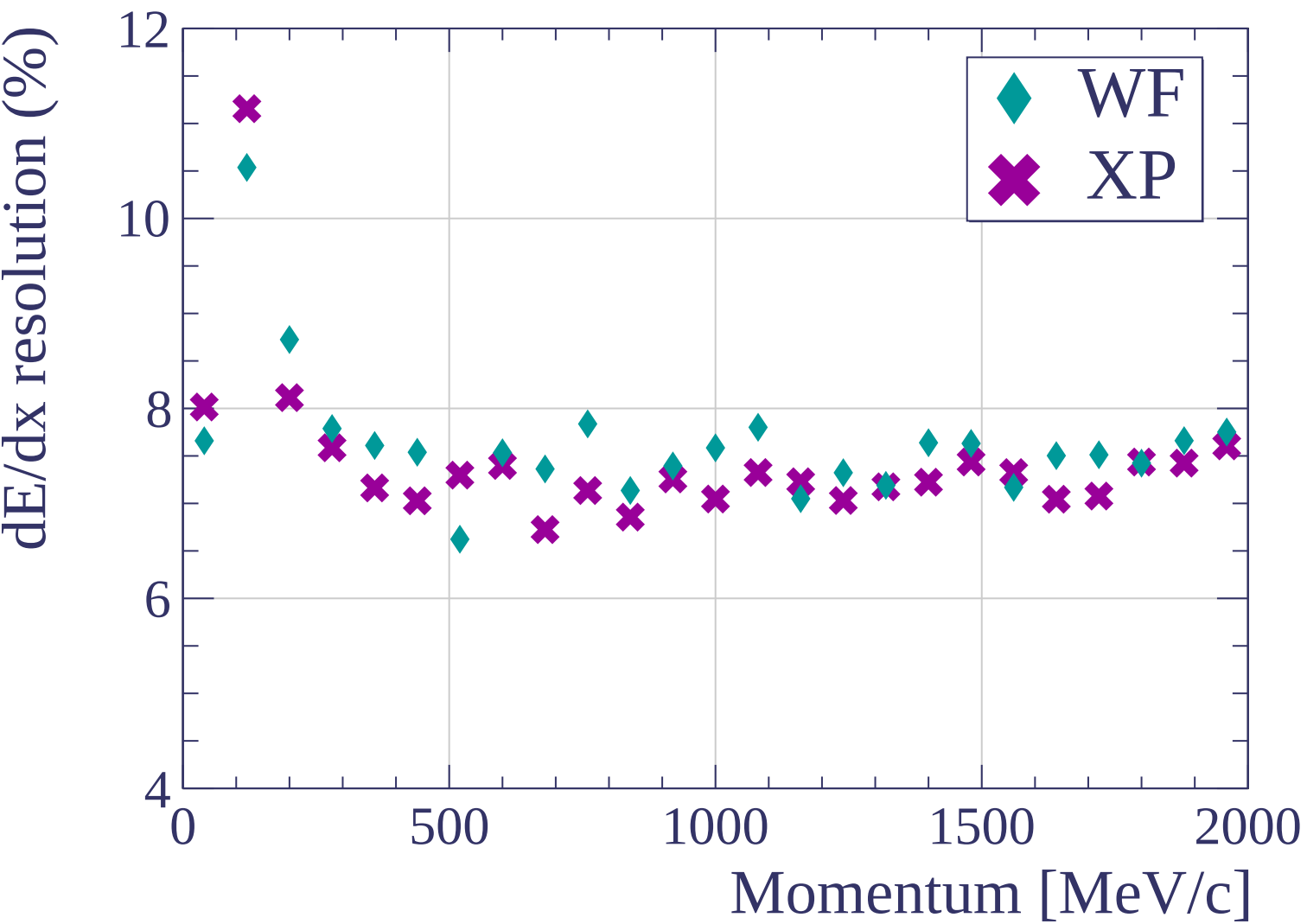


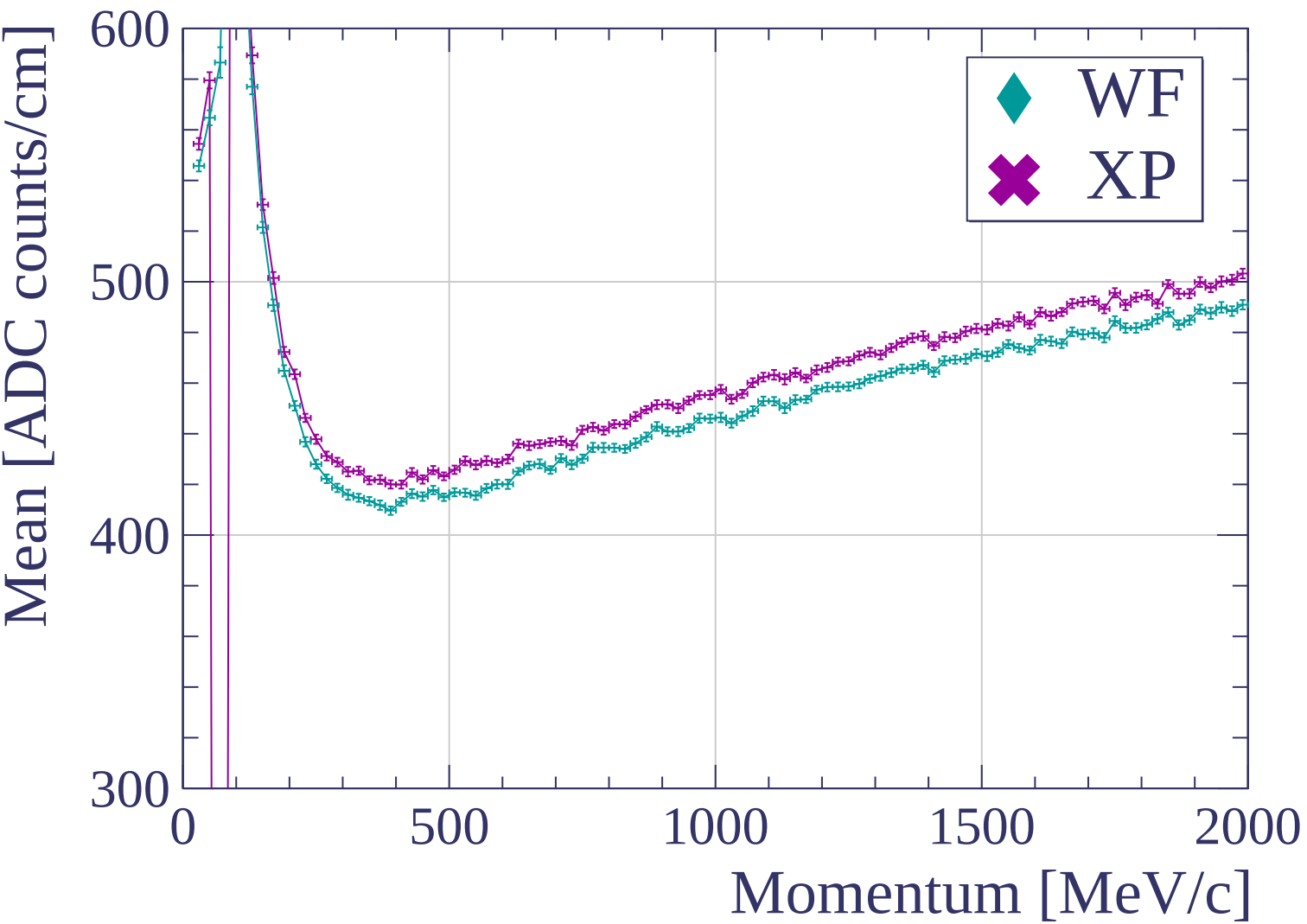


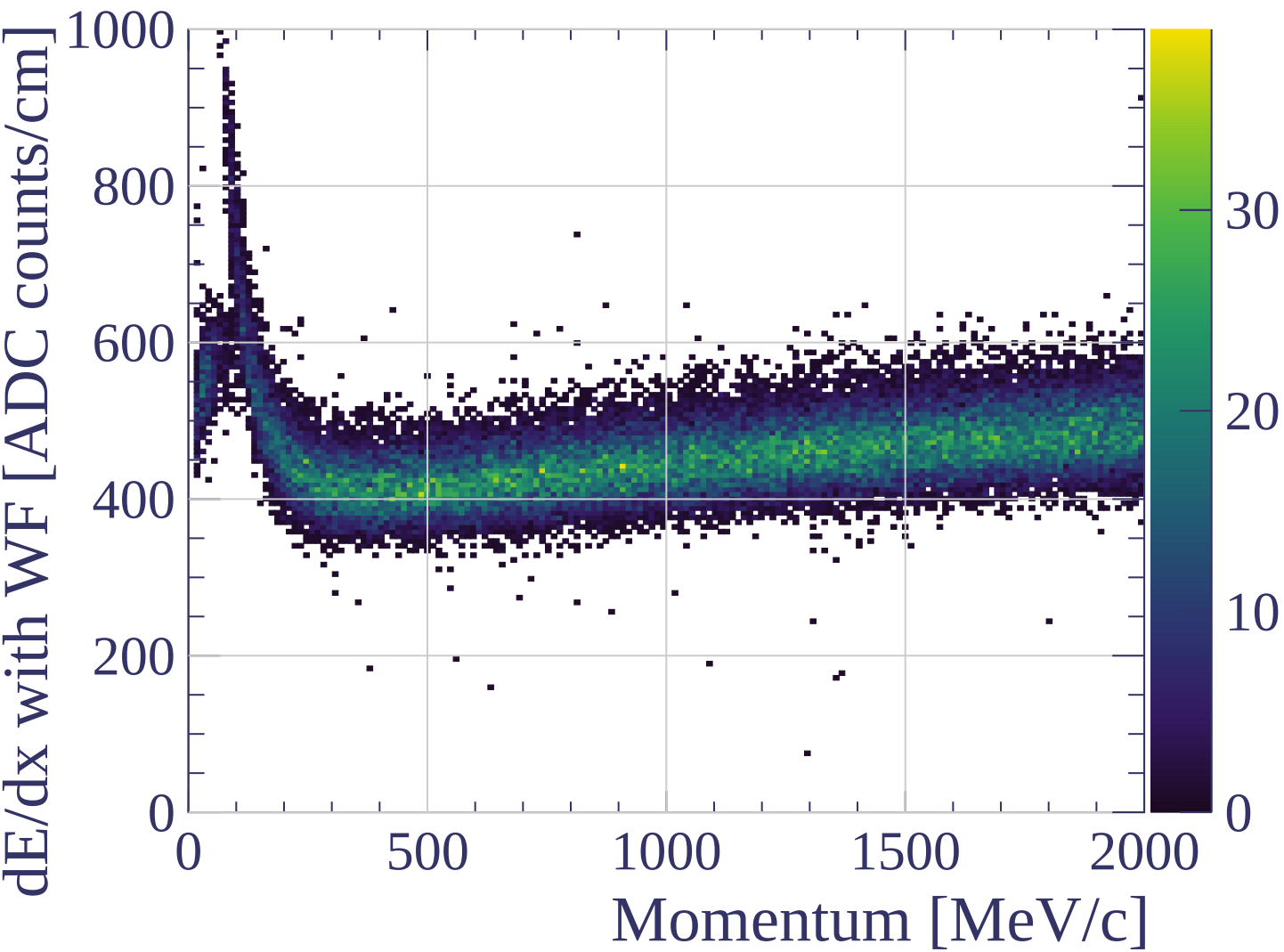


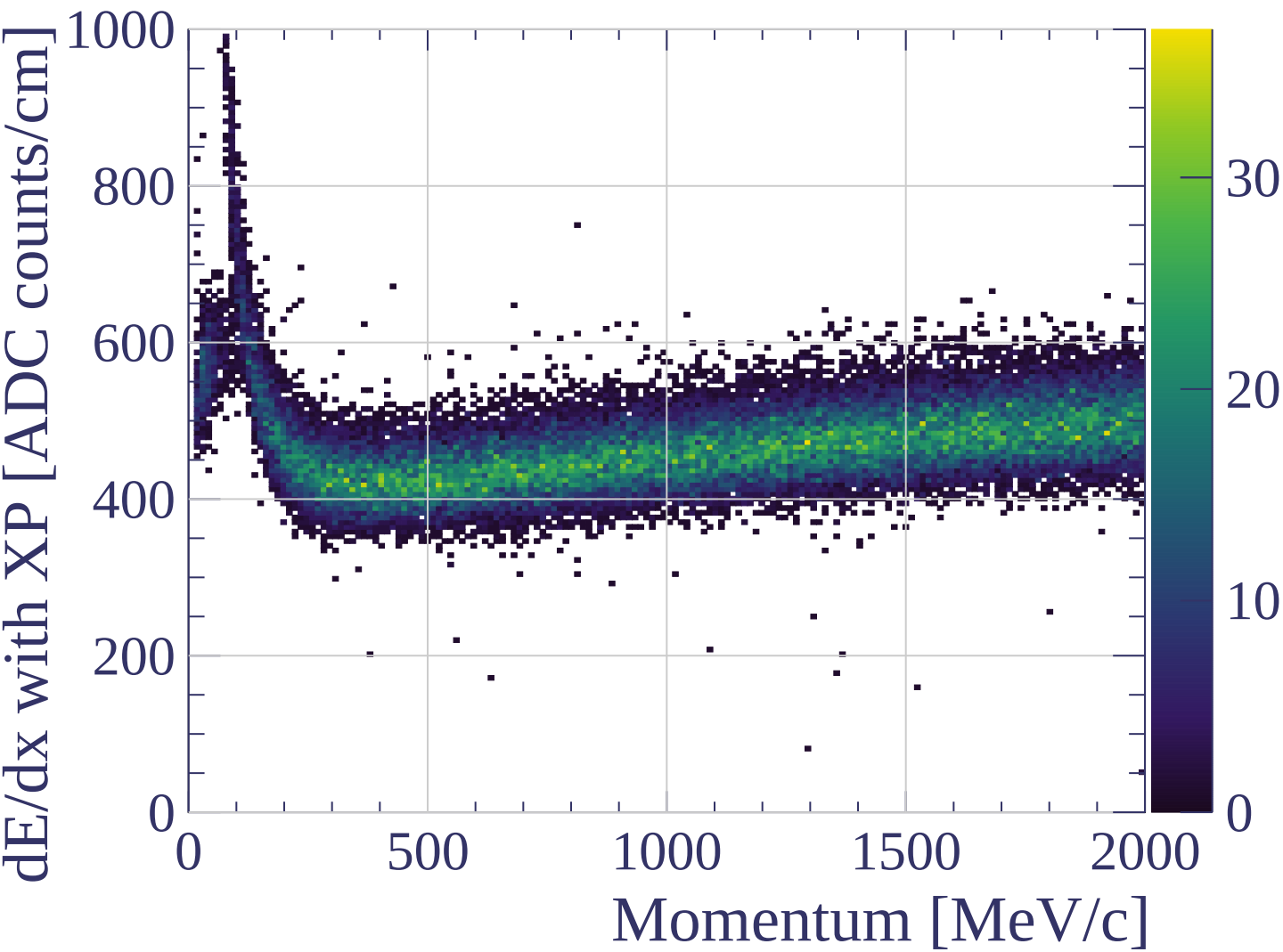


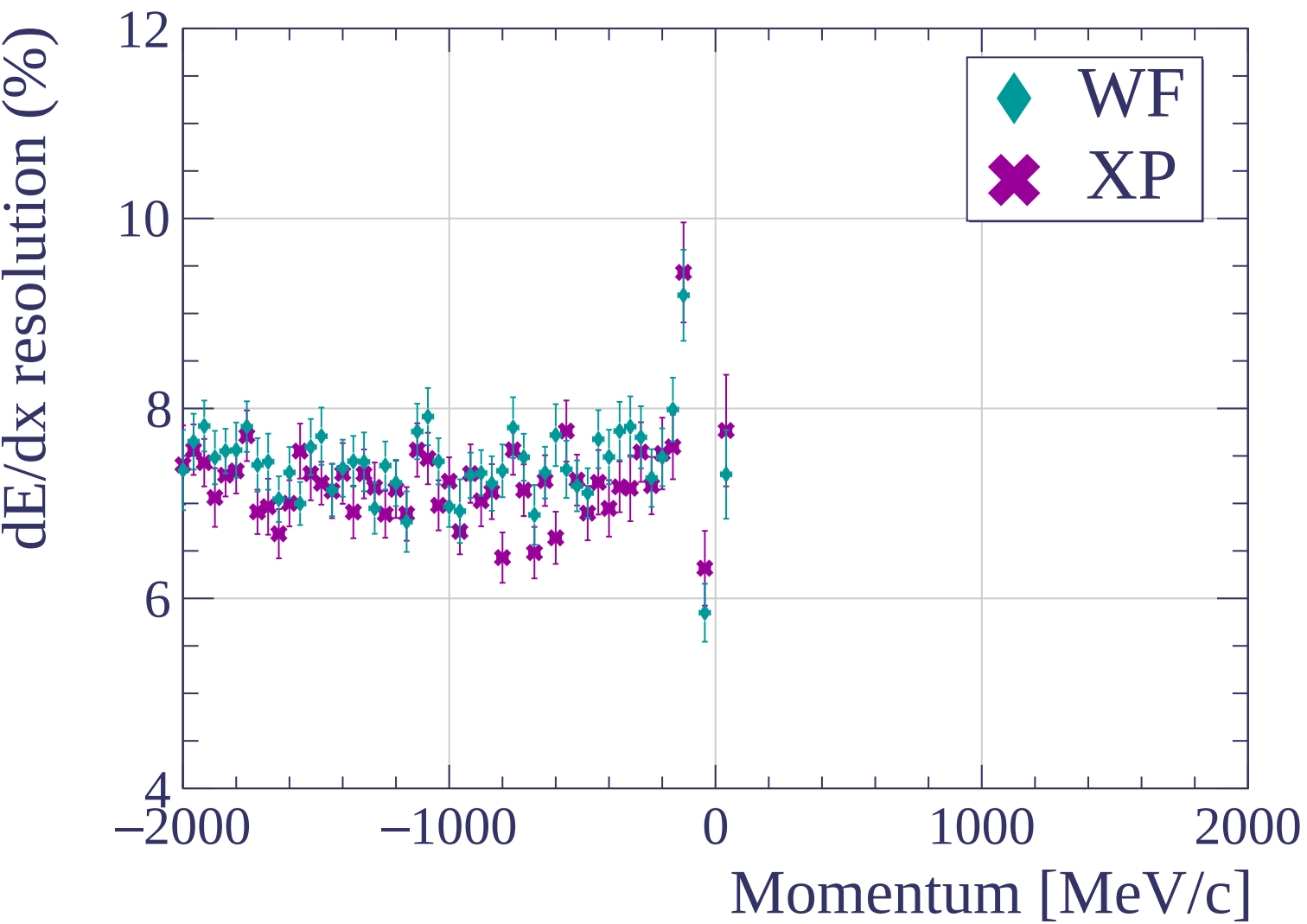


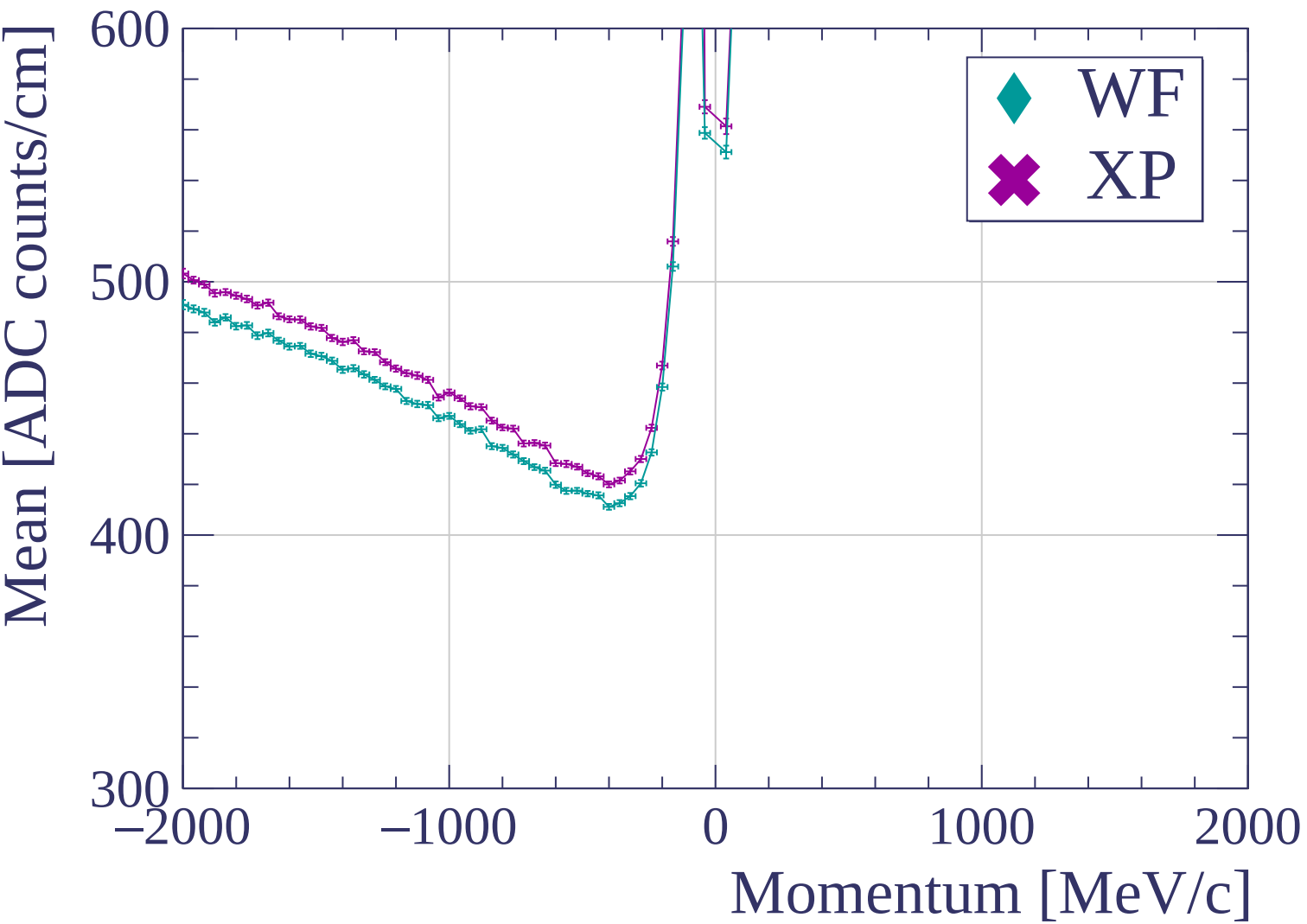


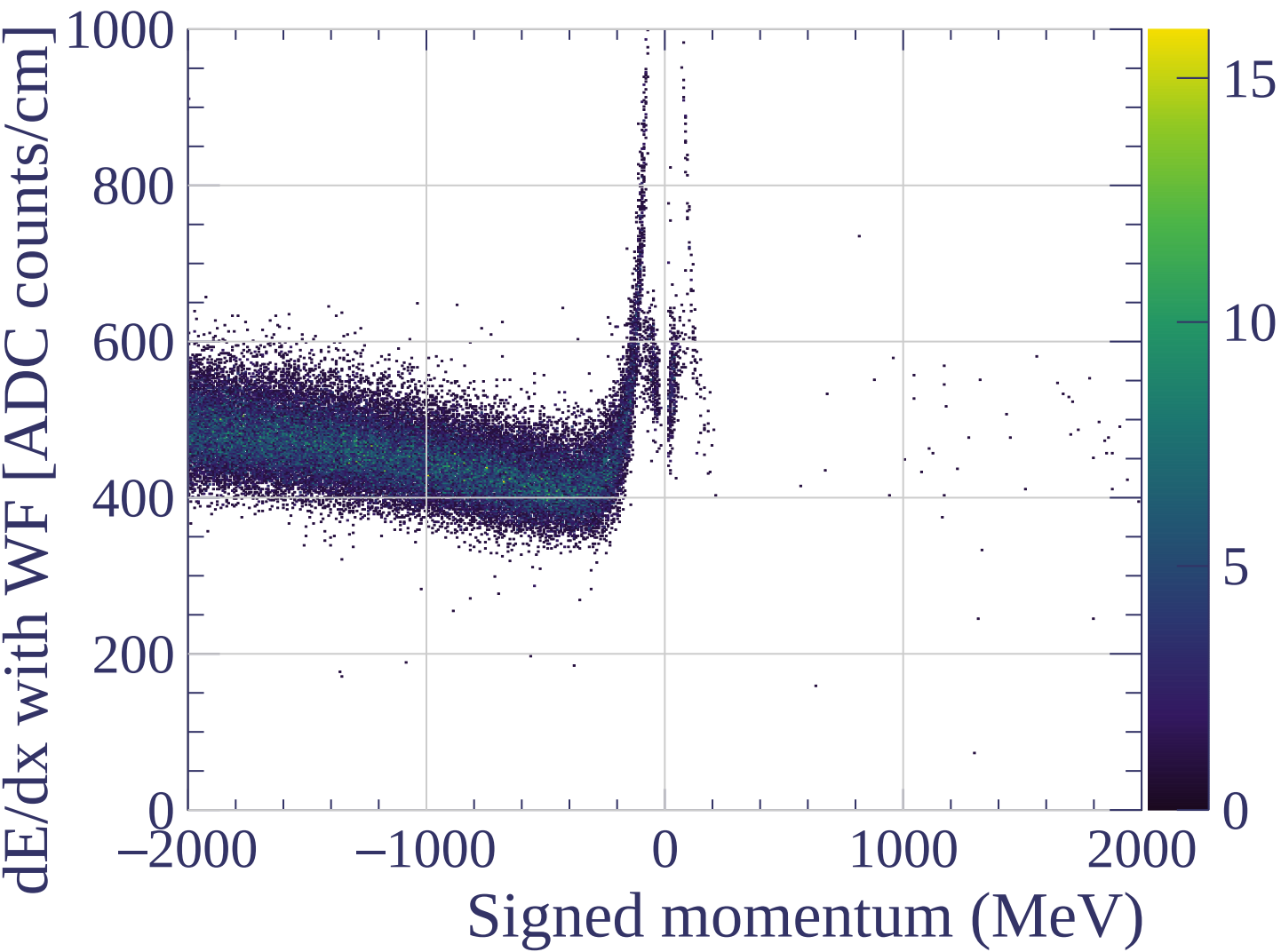


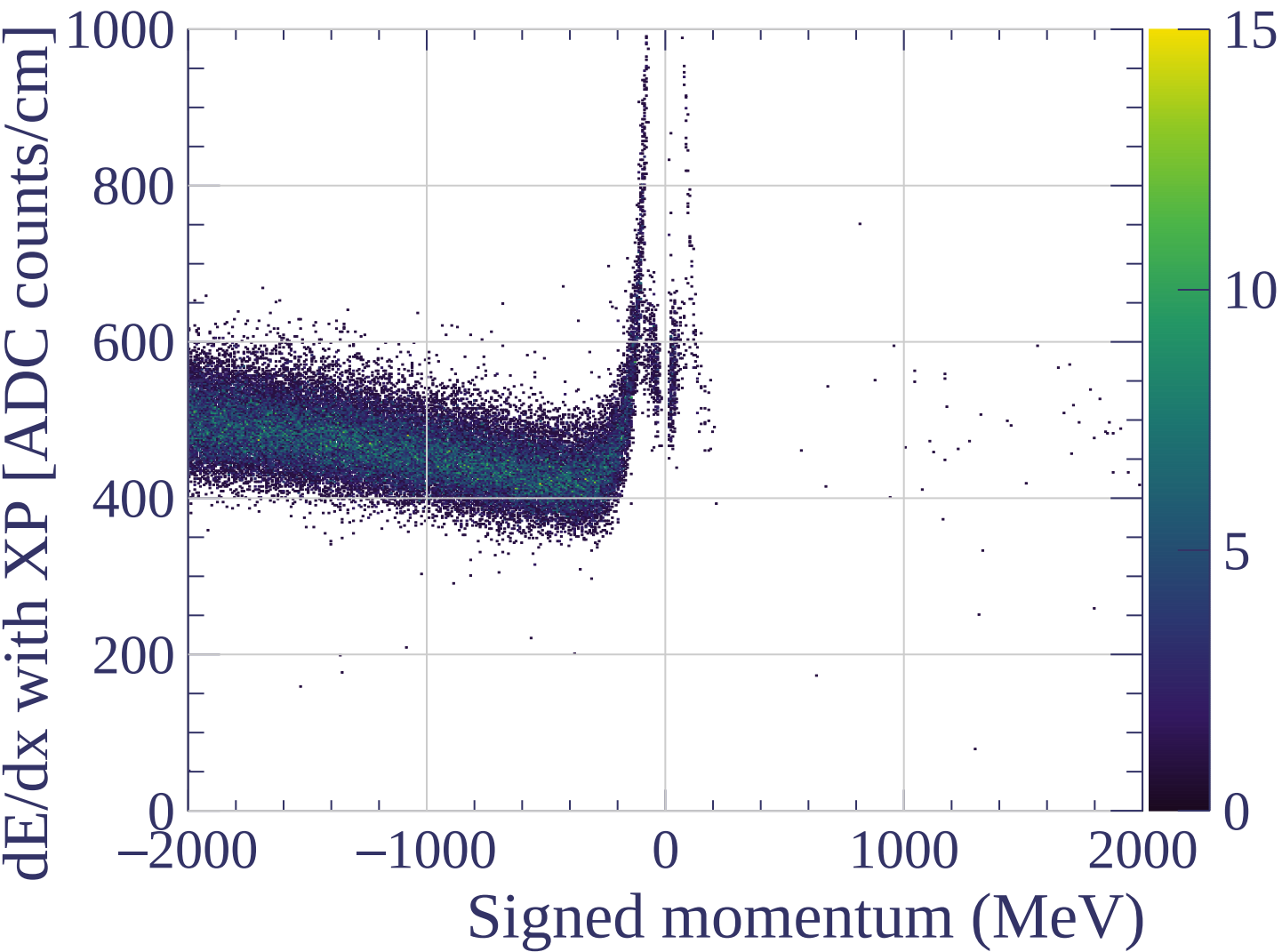




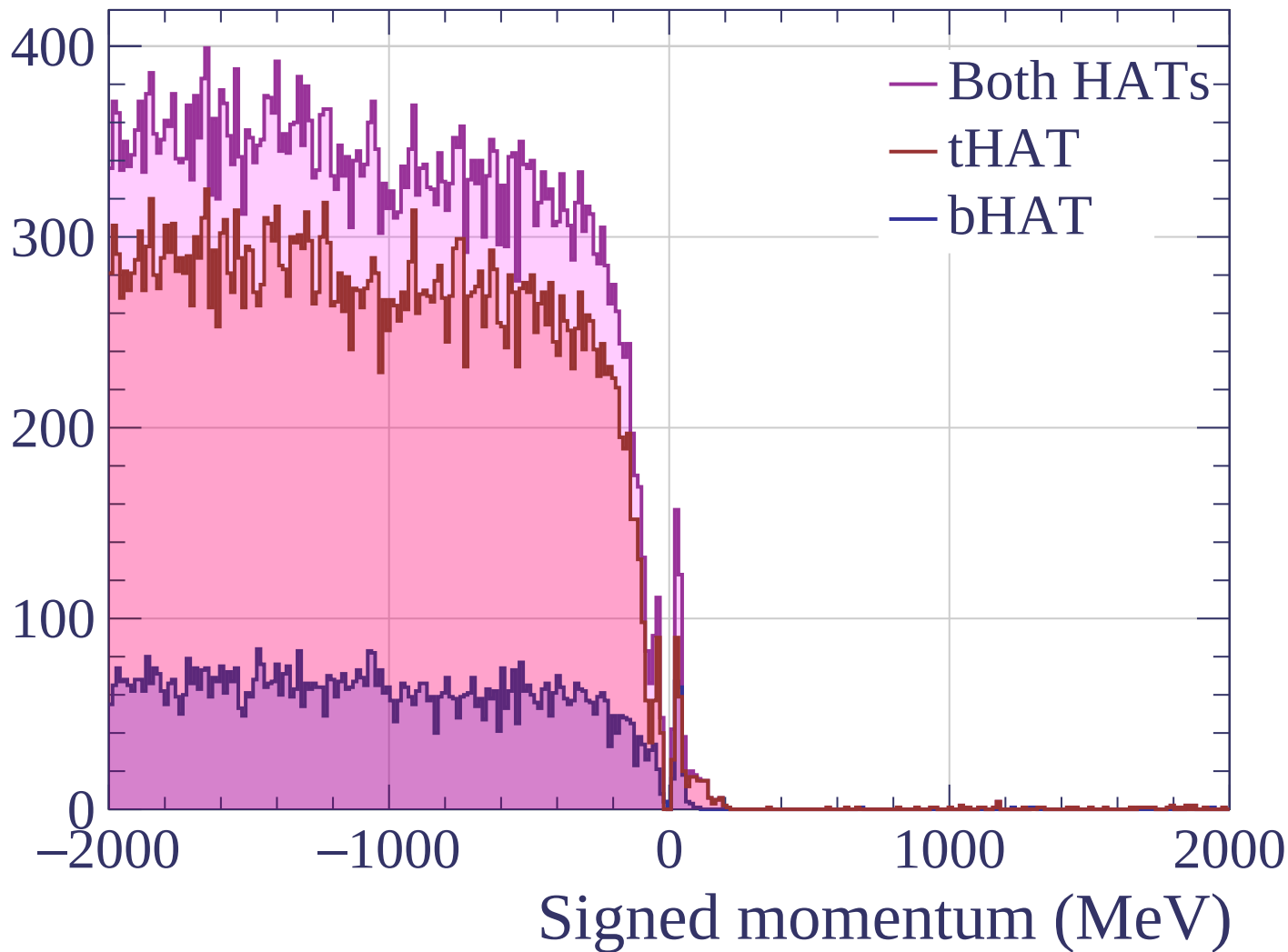


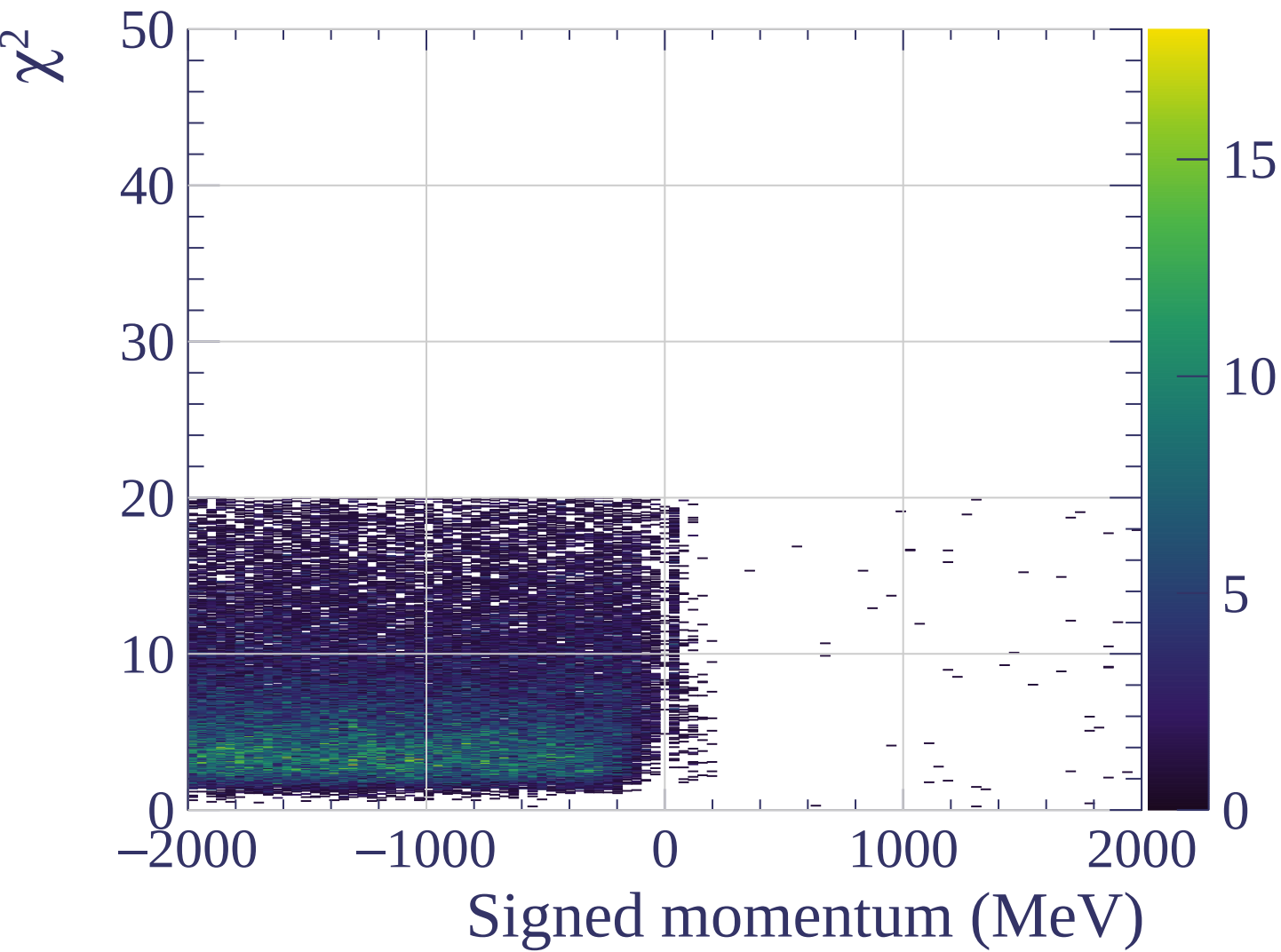


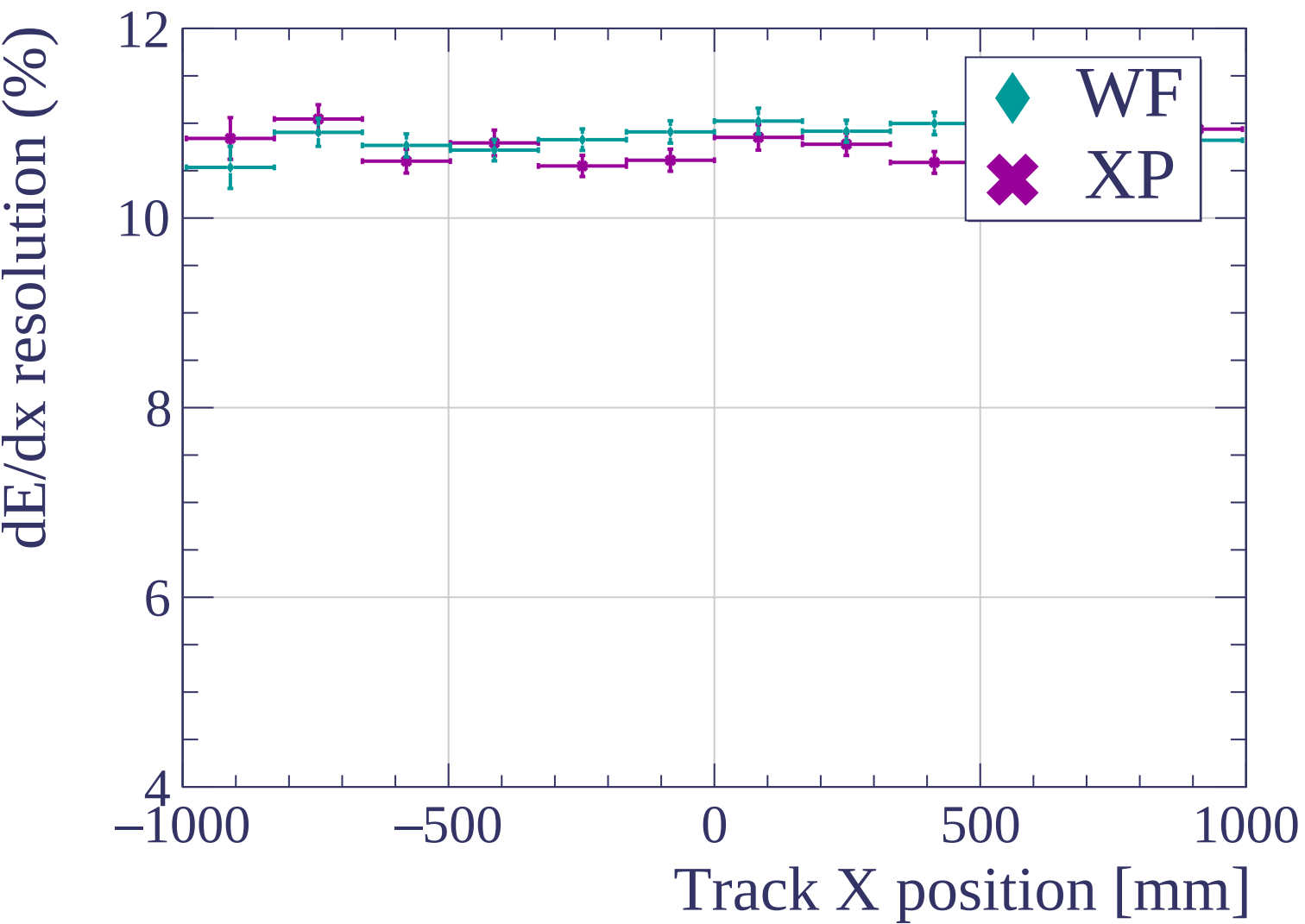


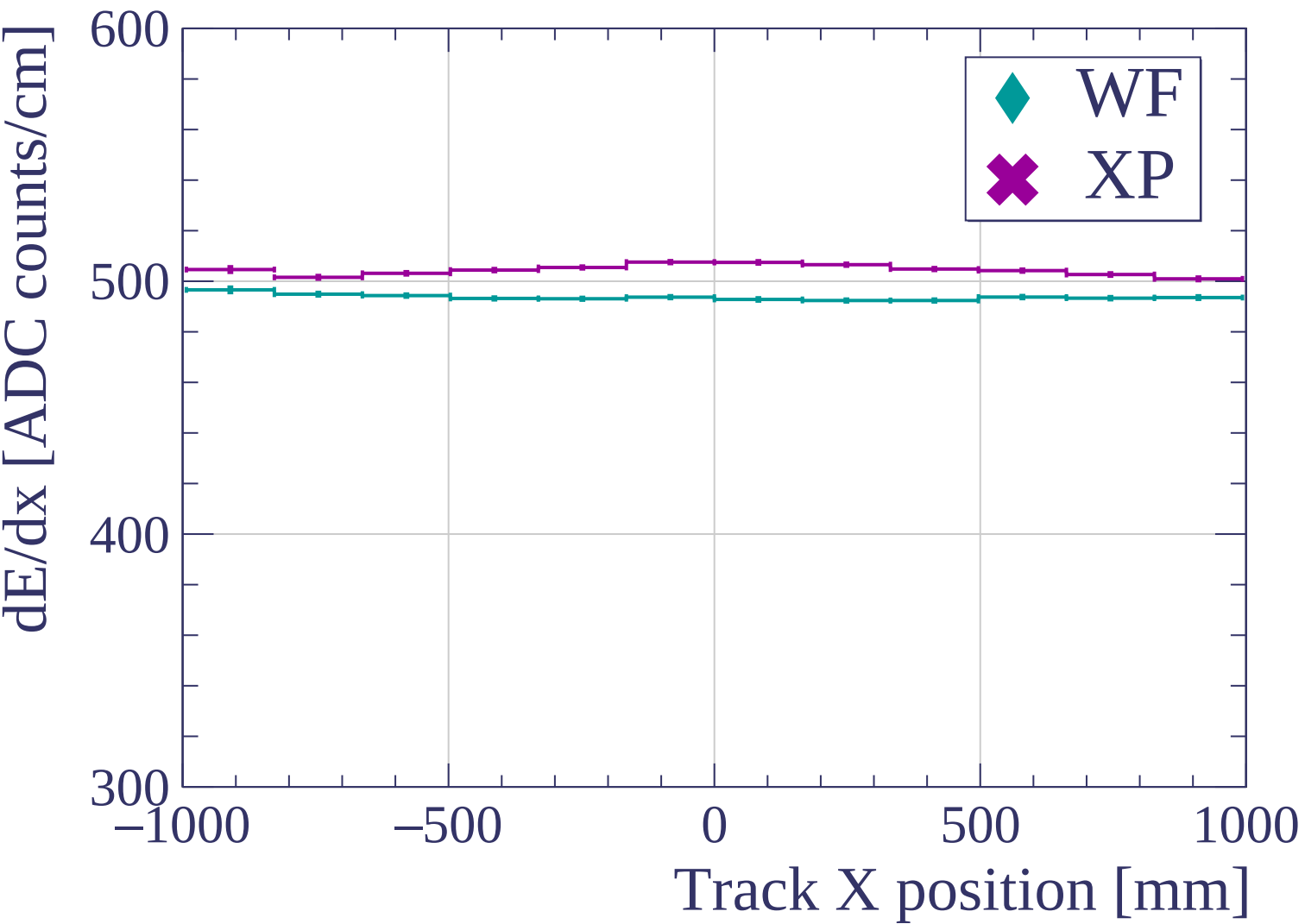


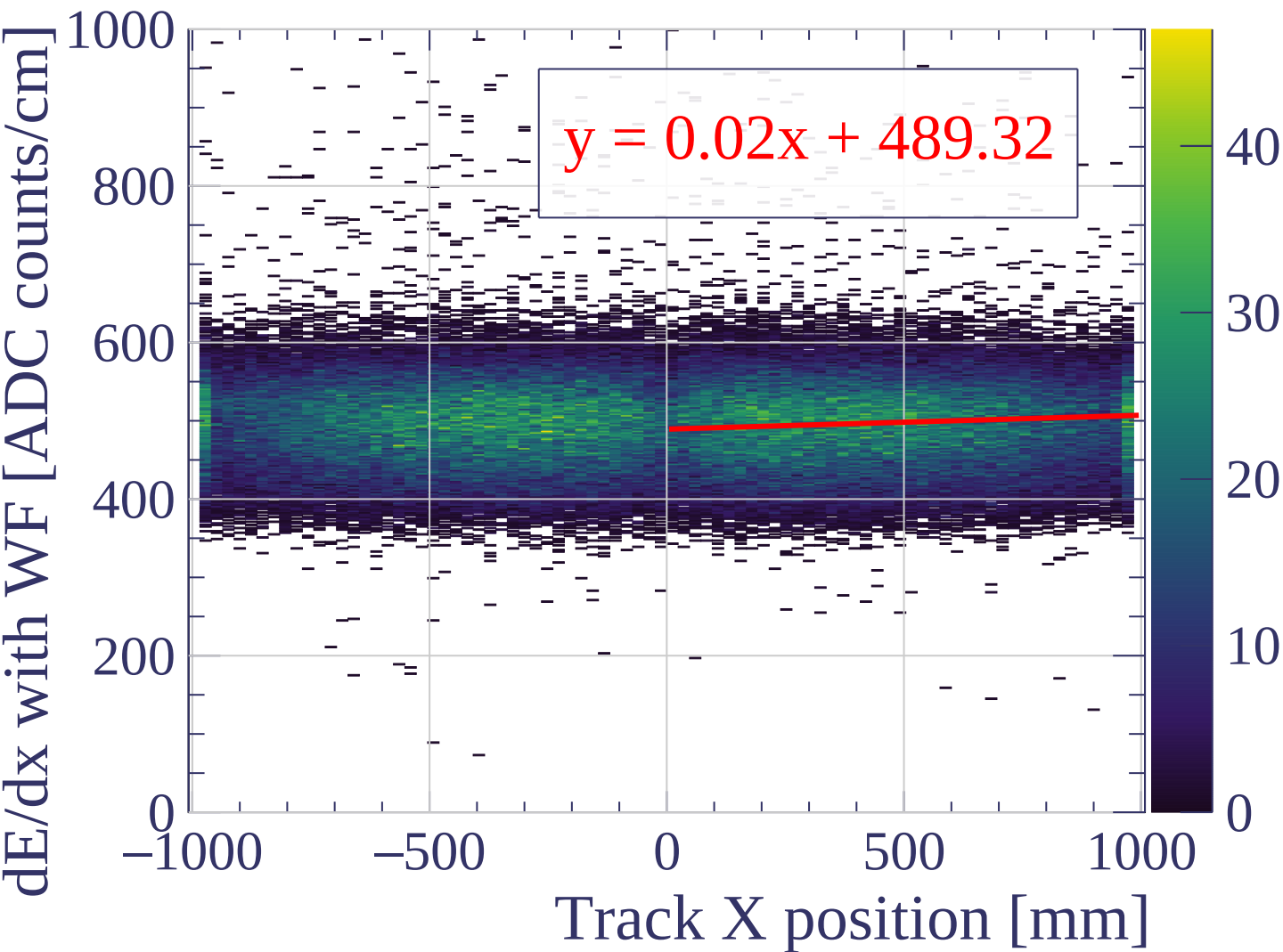
Count

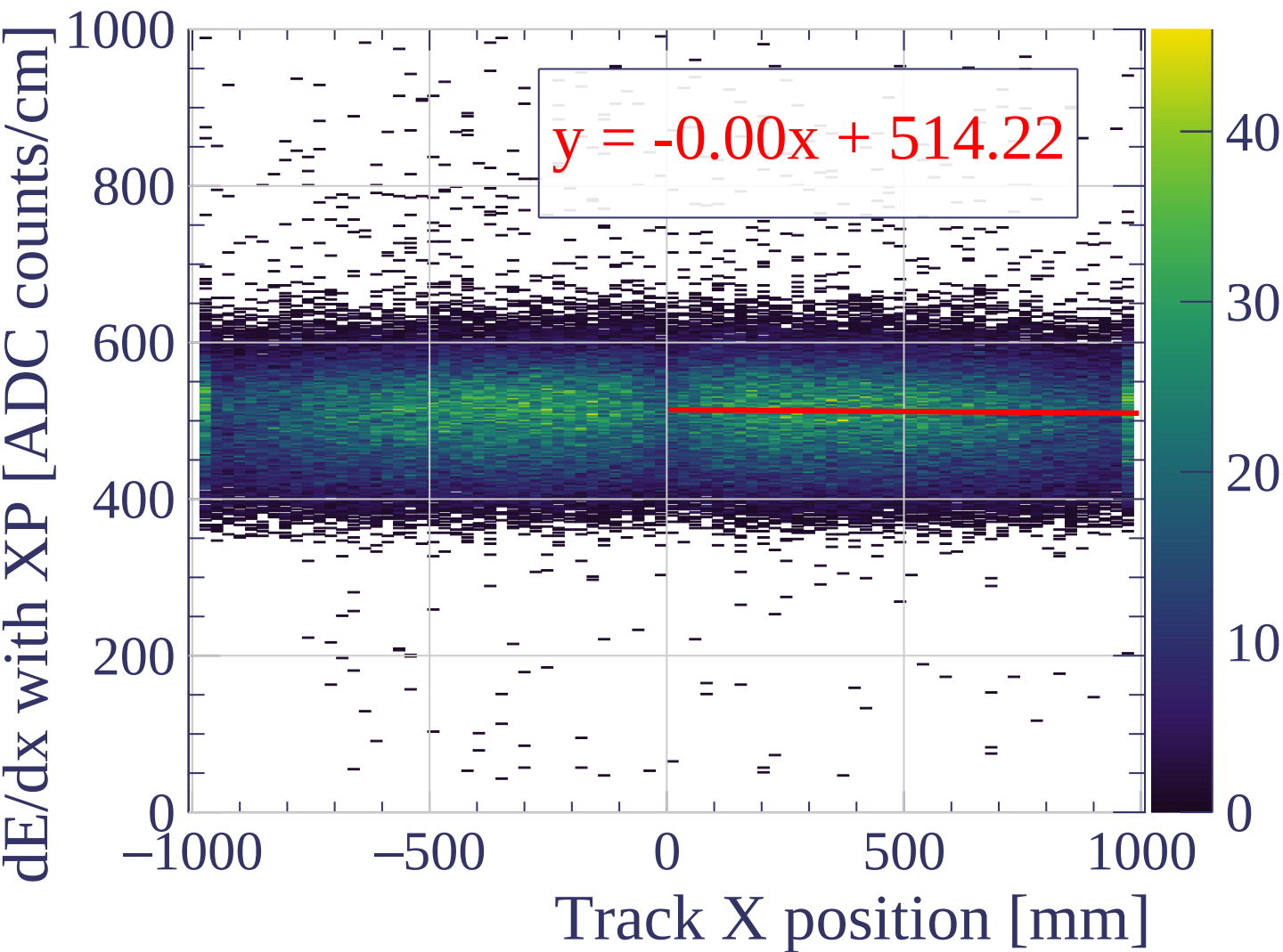


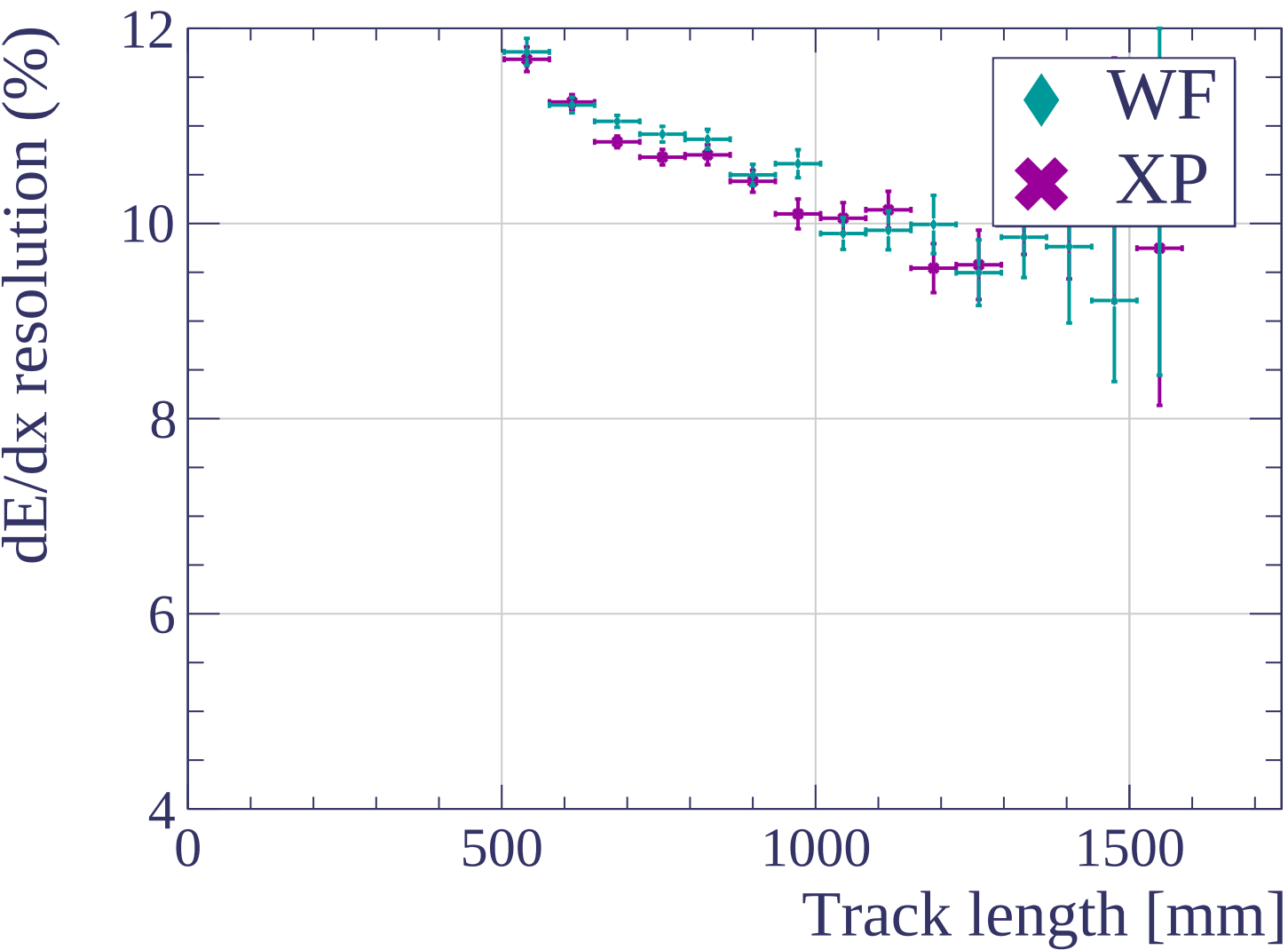


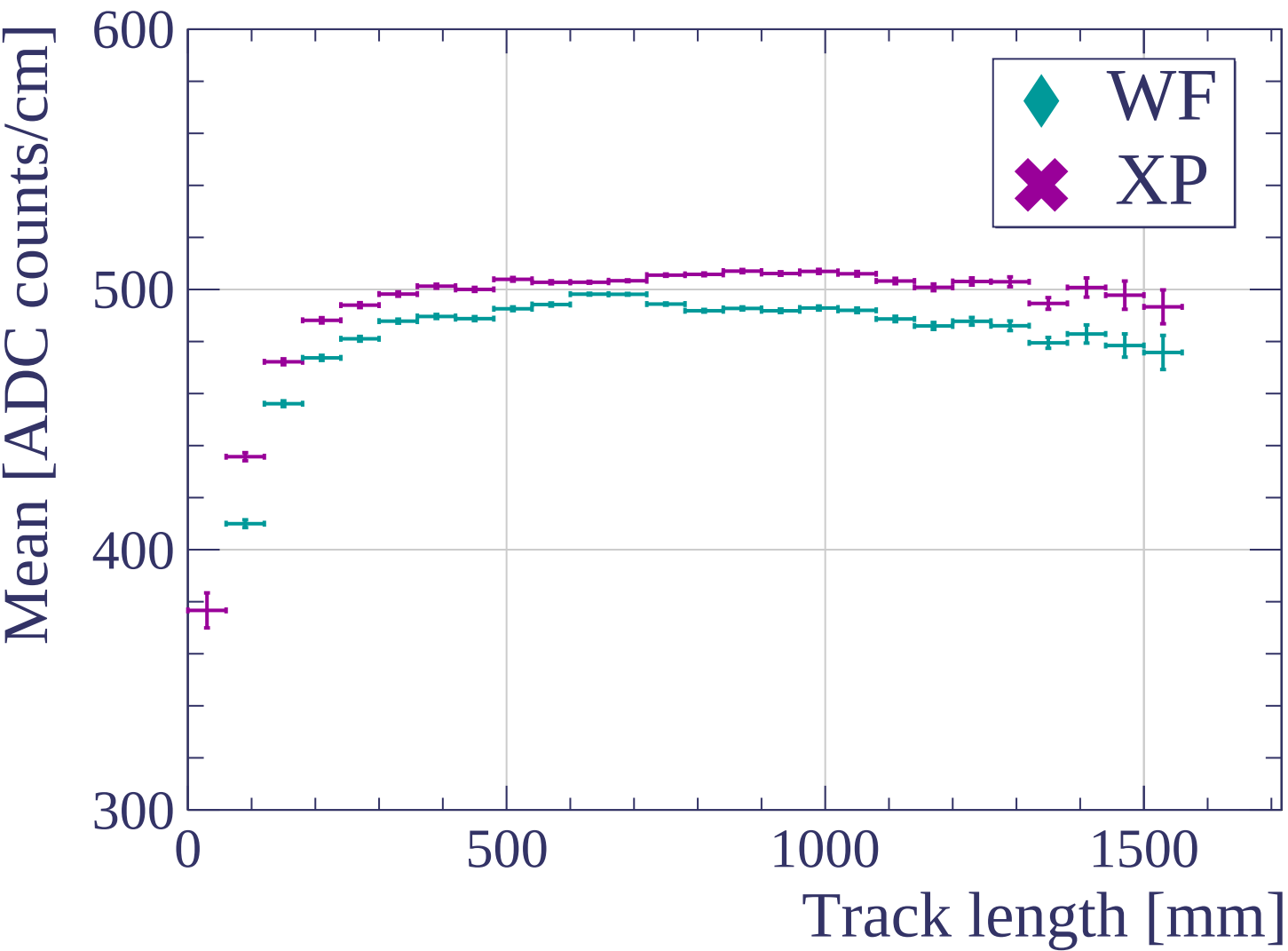


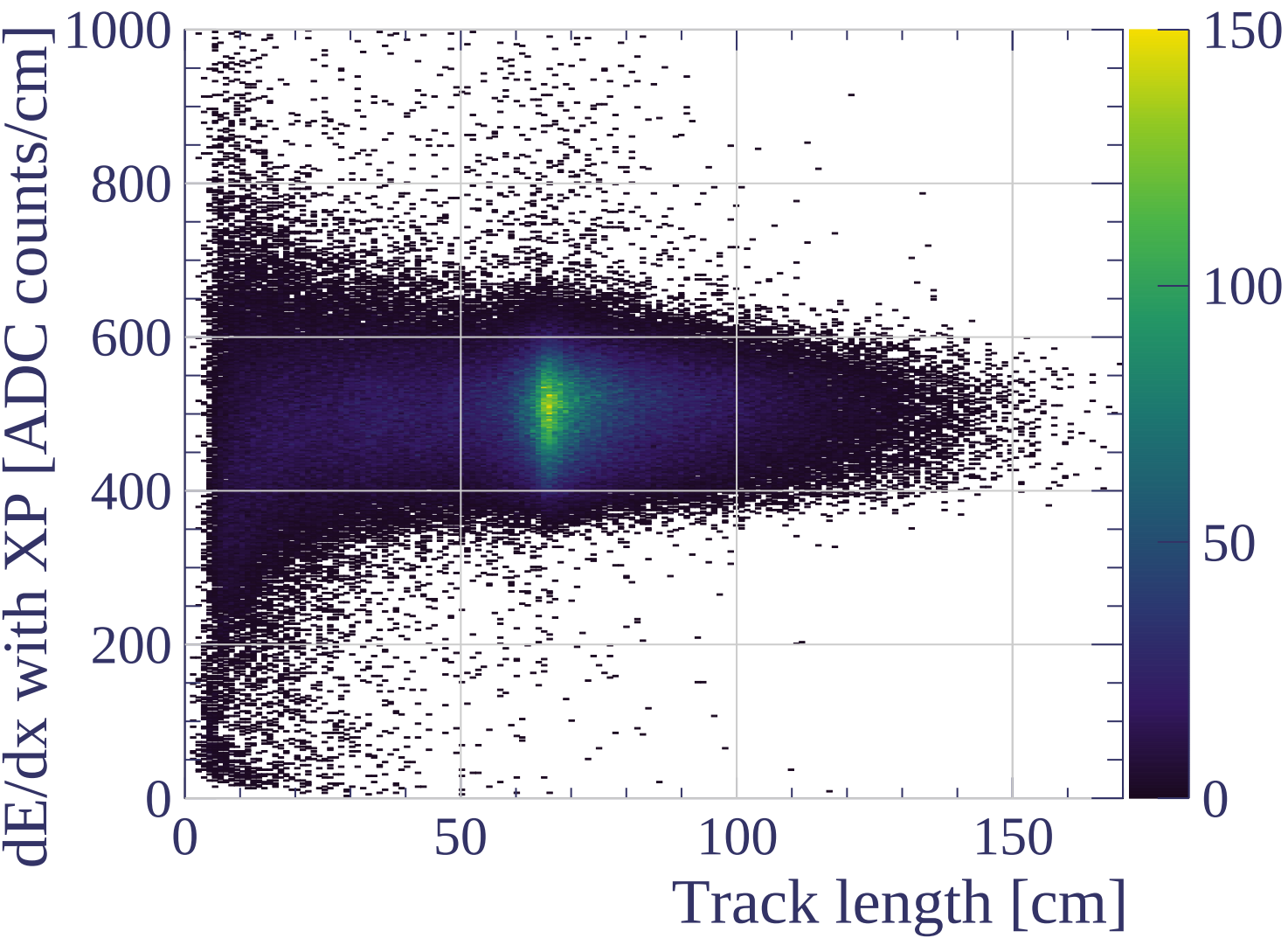


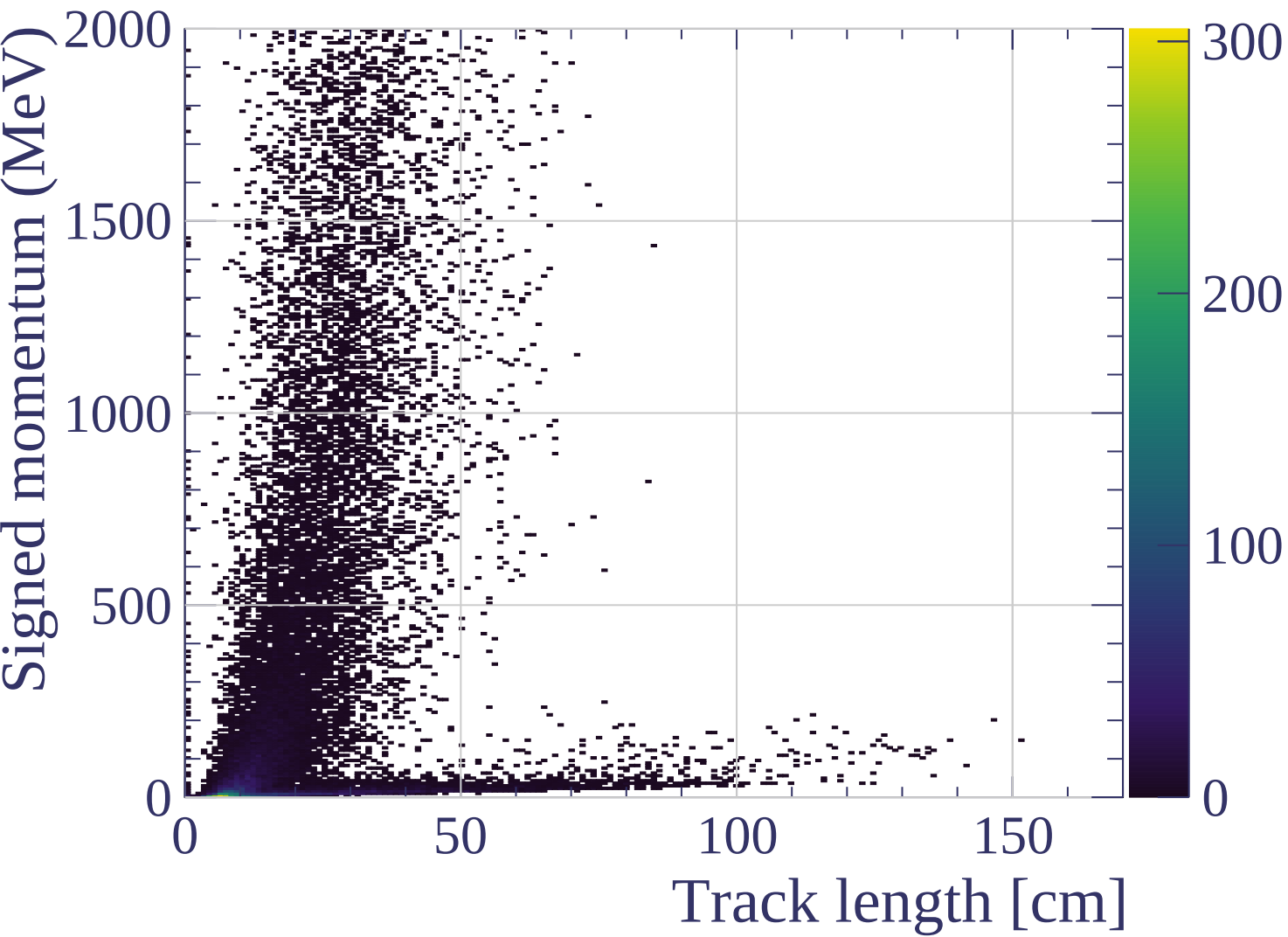


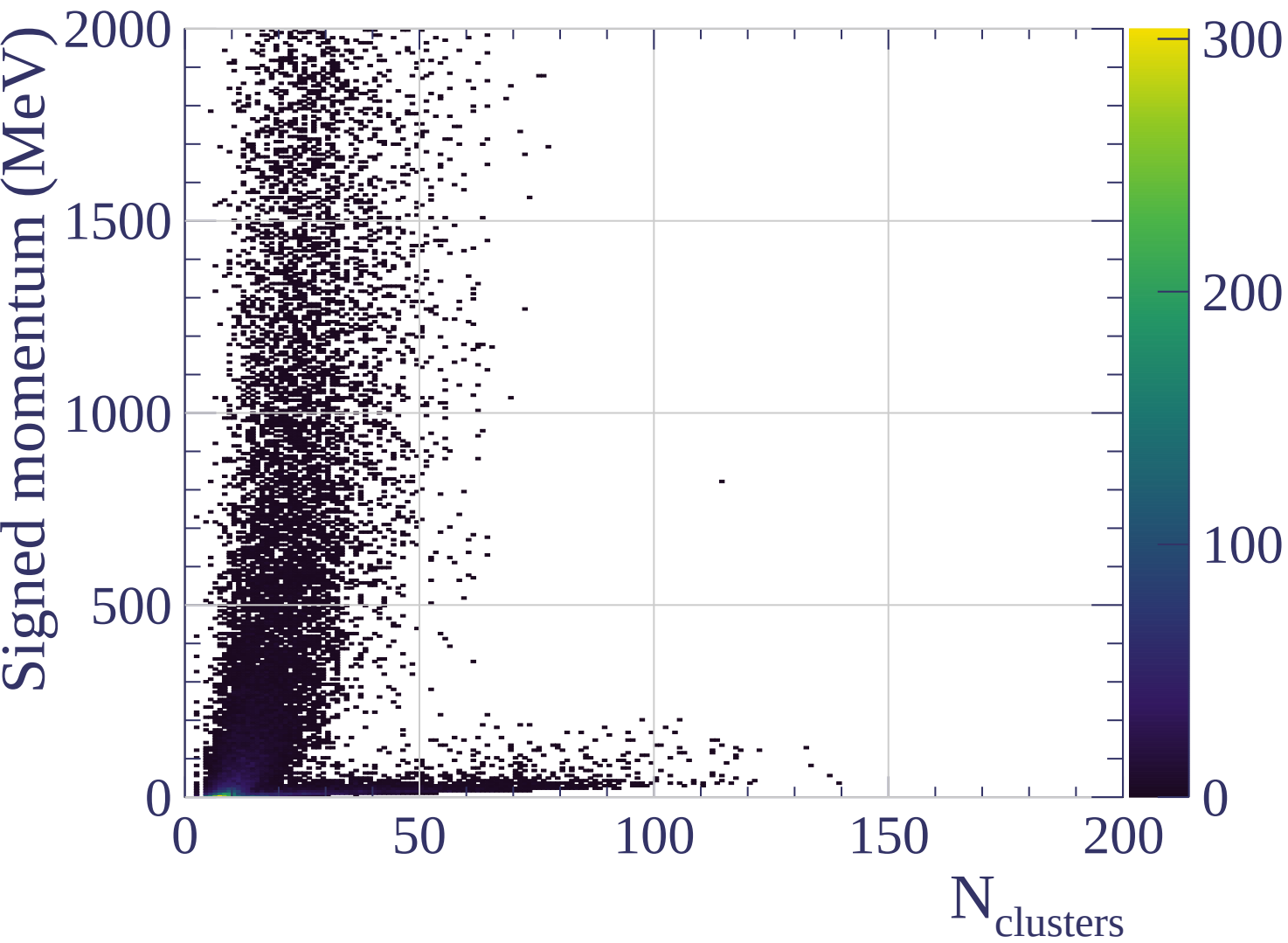




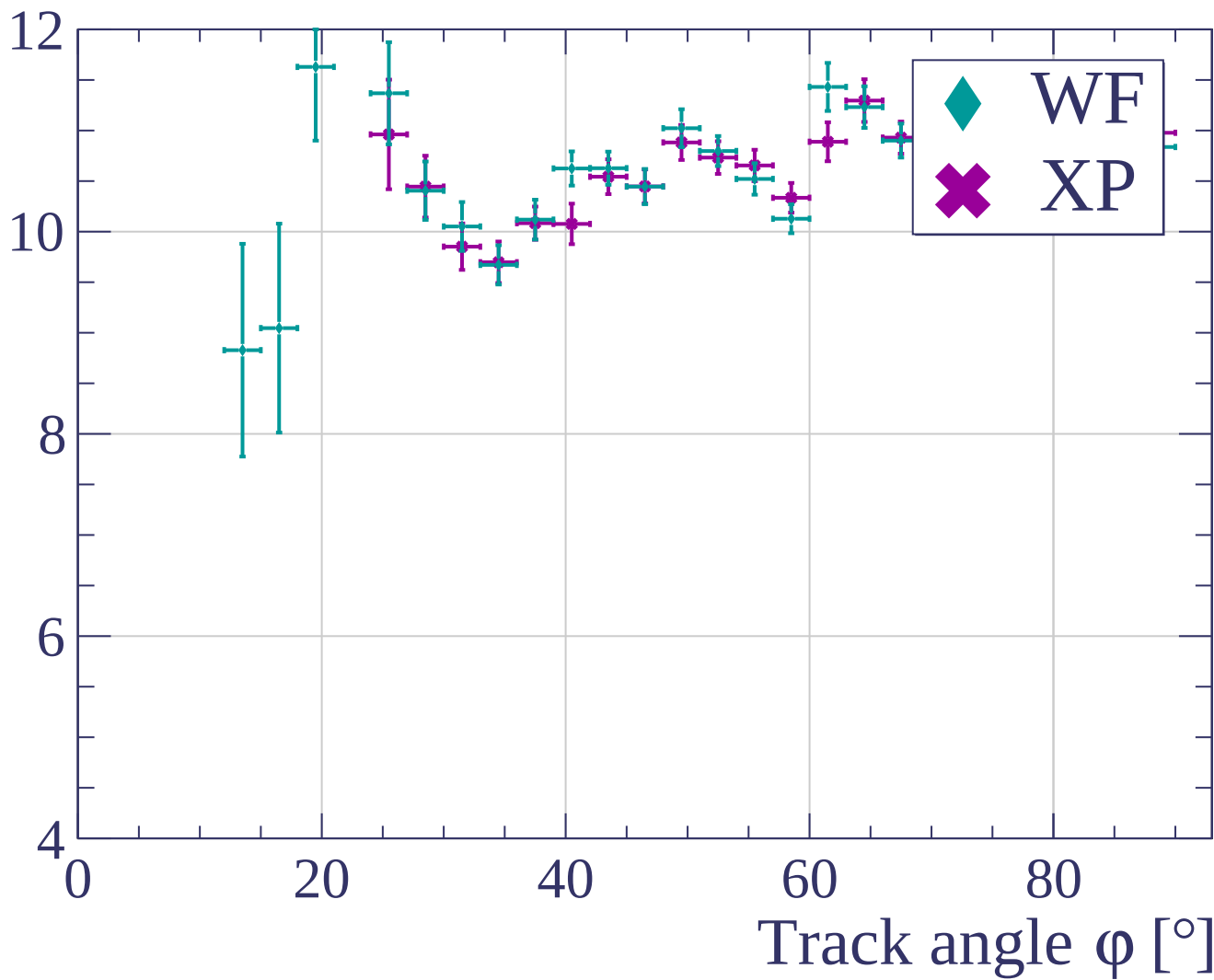


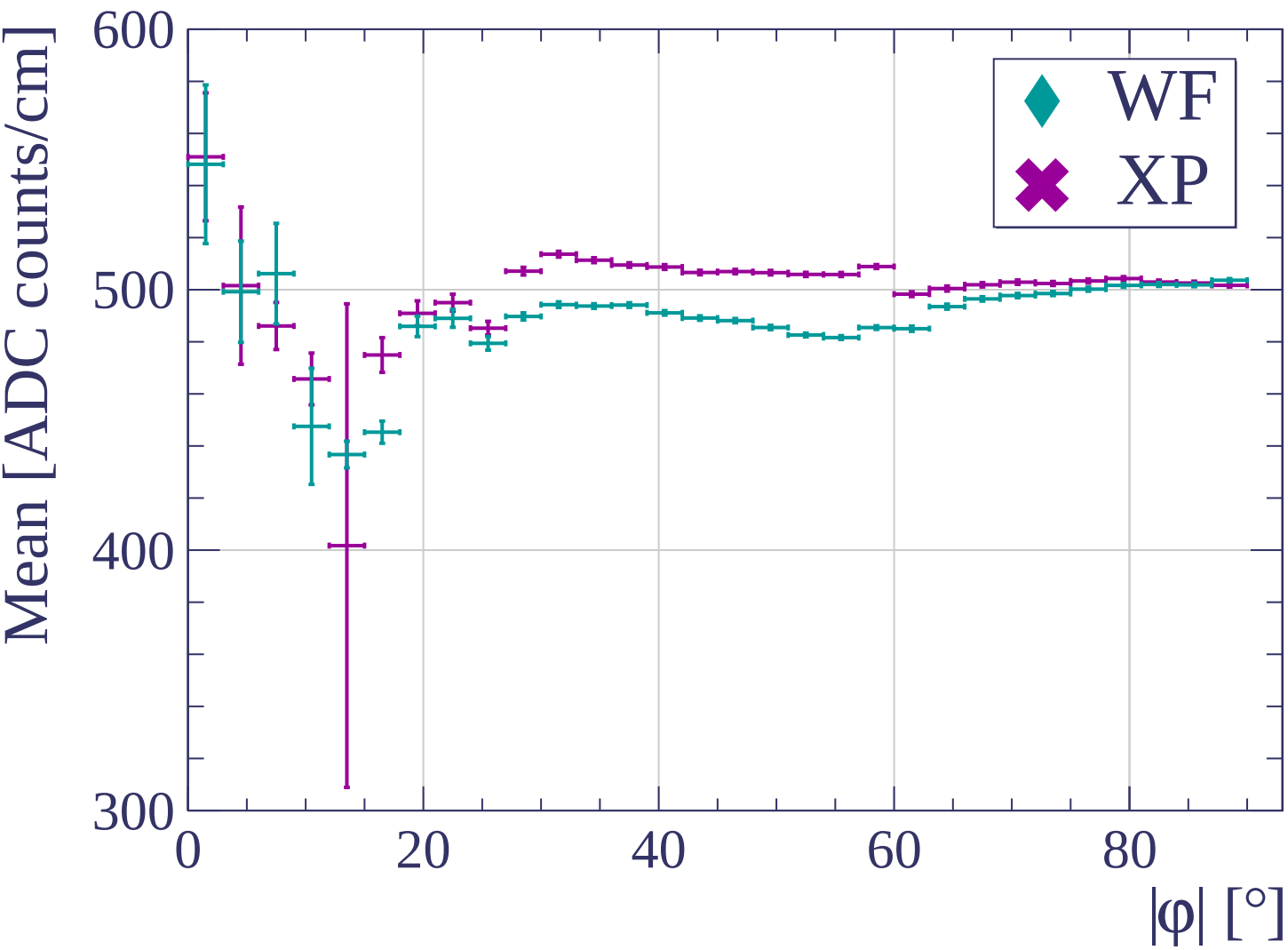




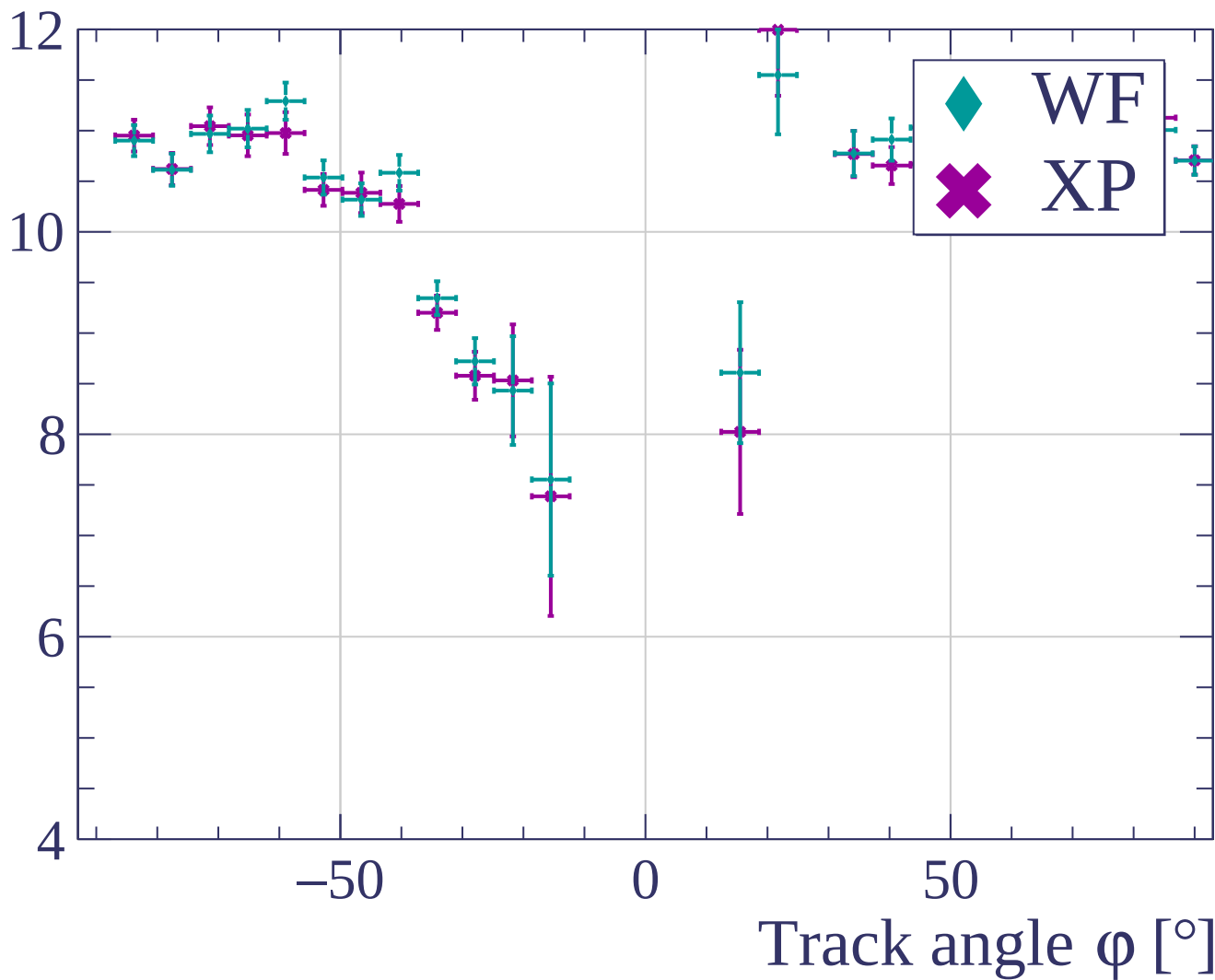


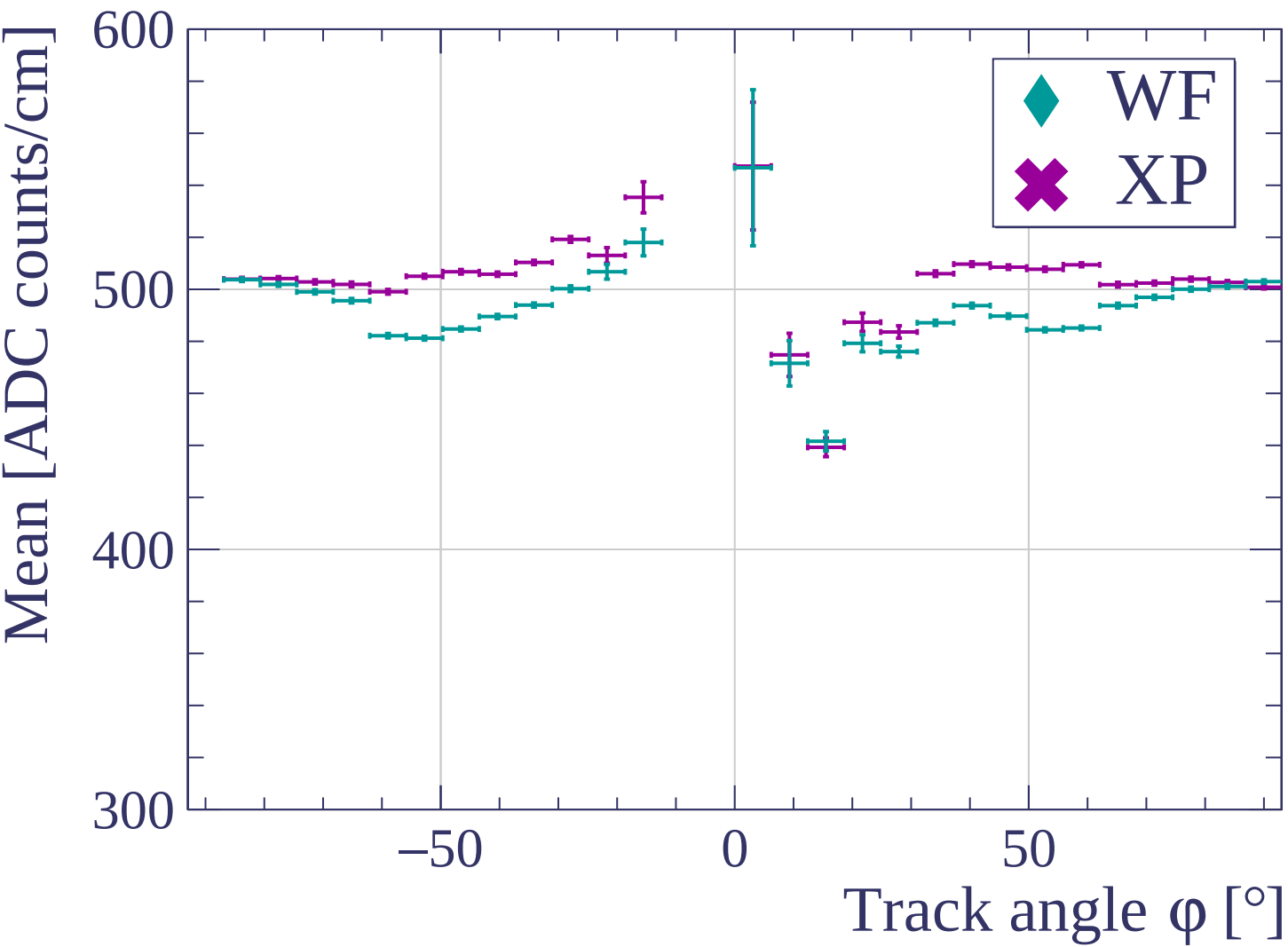
dE/dx resolution (%)

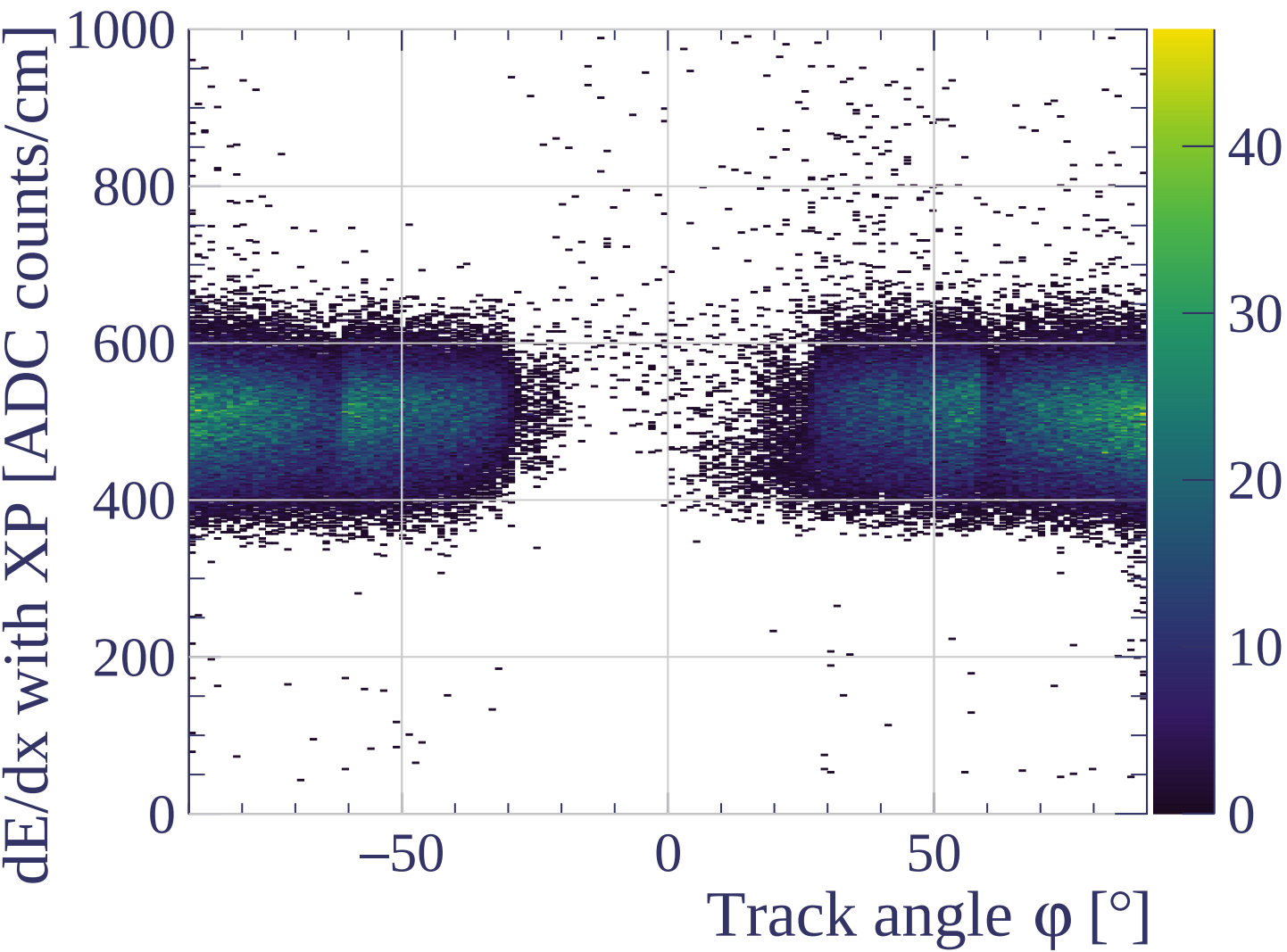




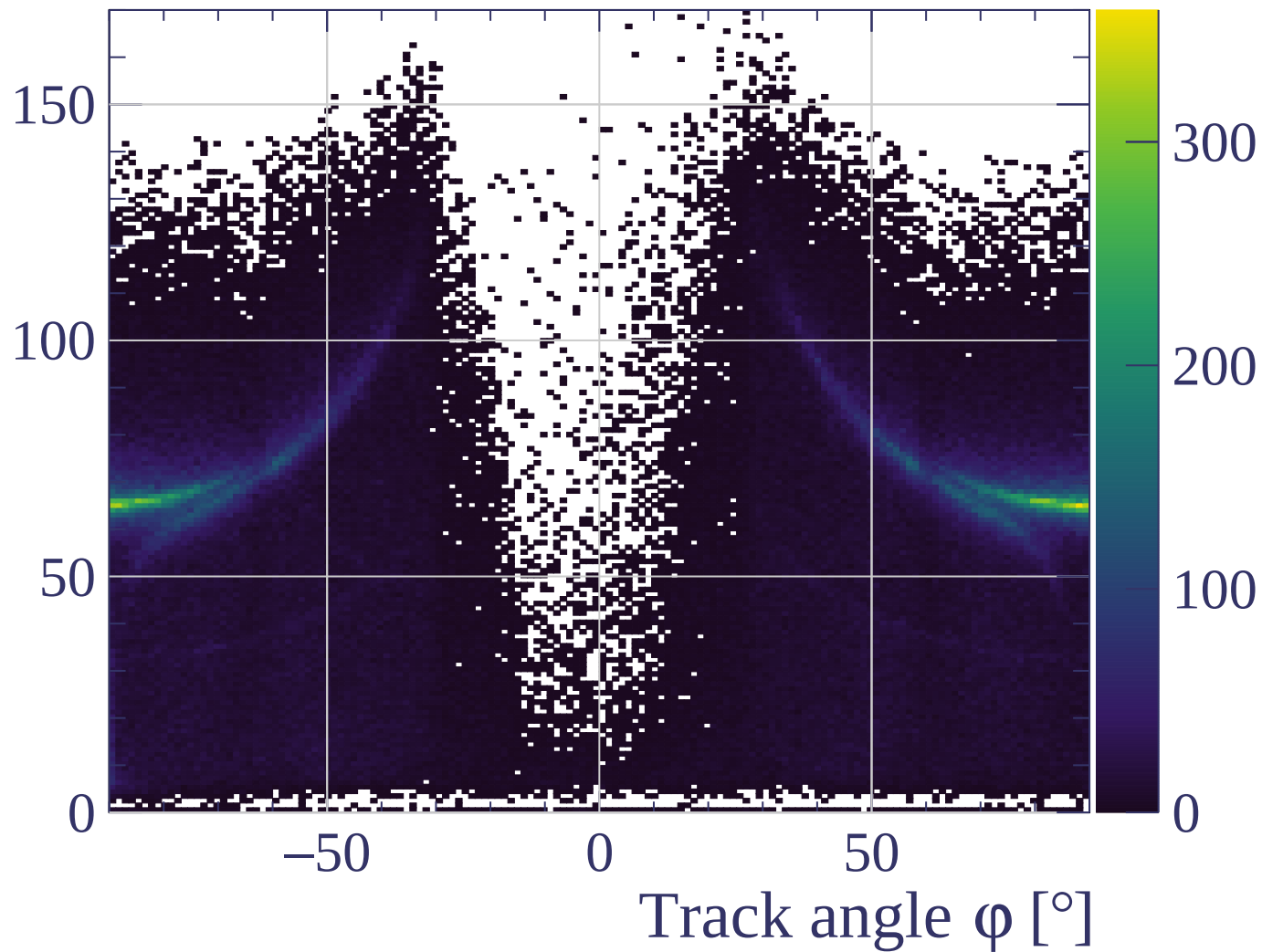
dE/dx resolution (%)

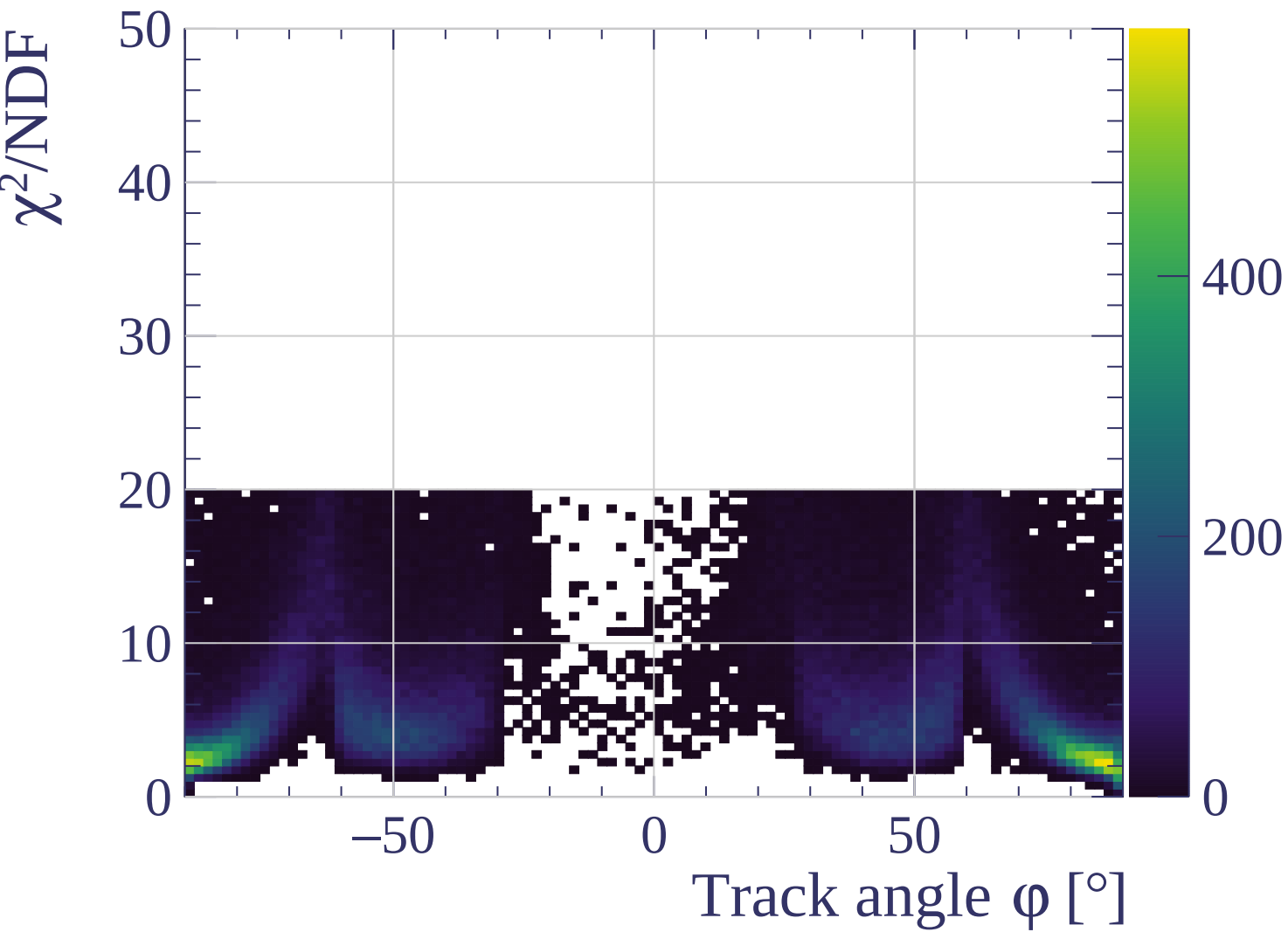


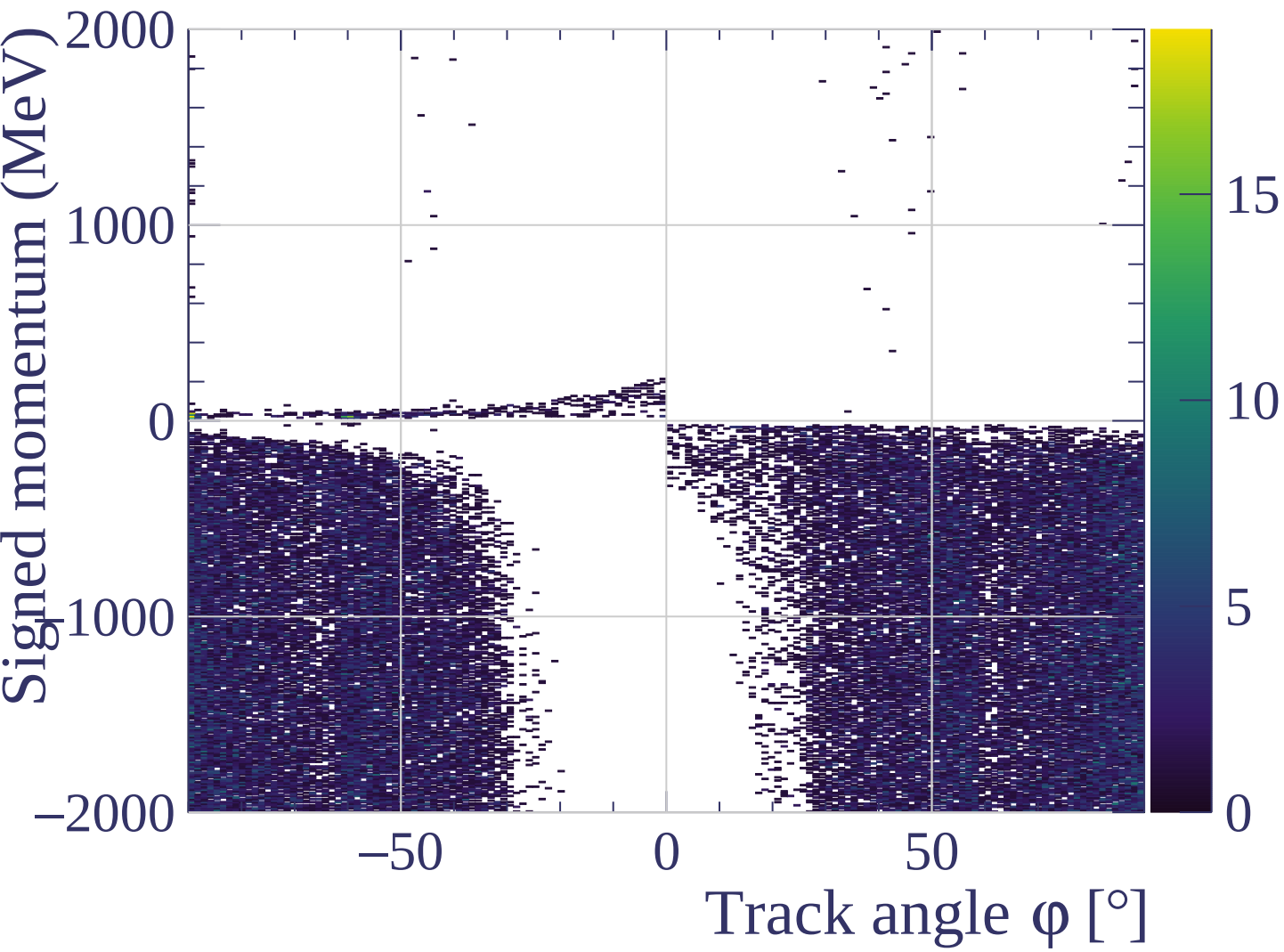


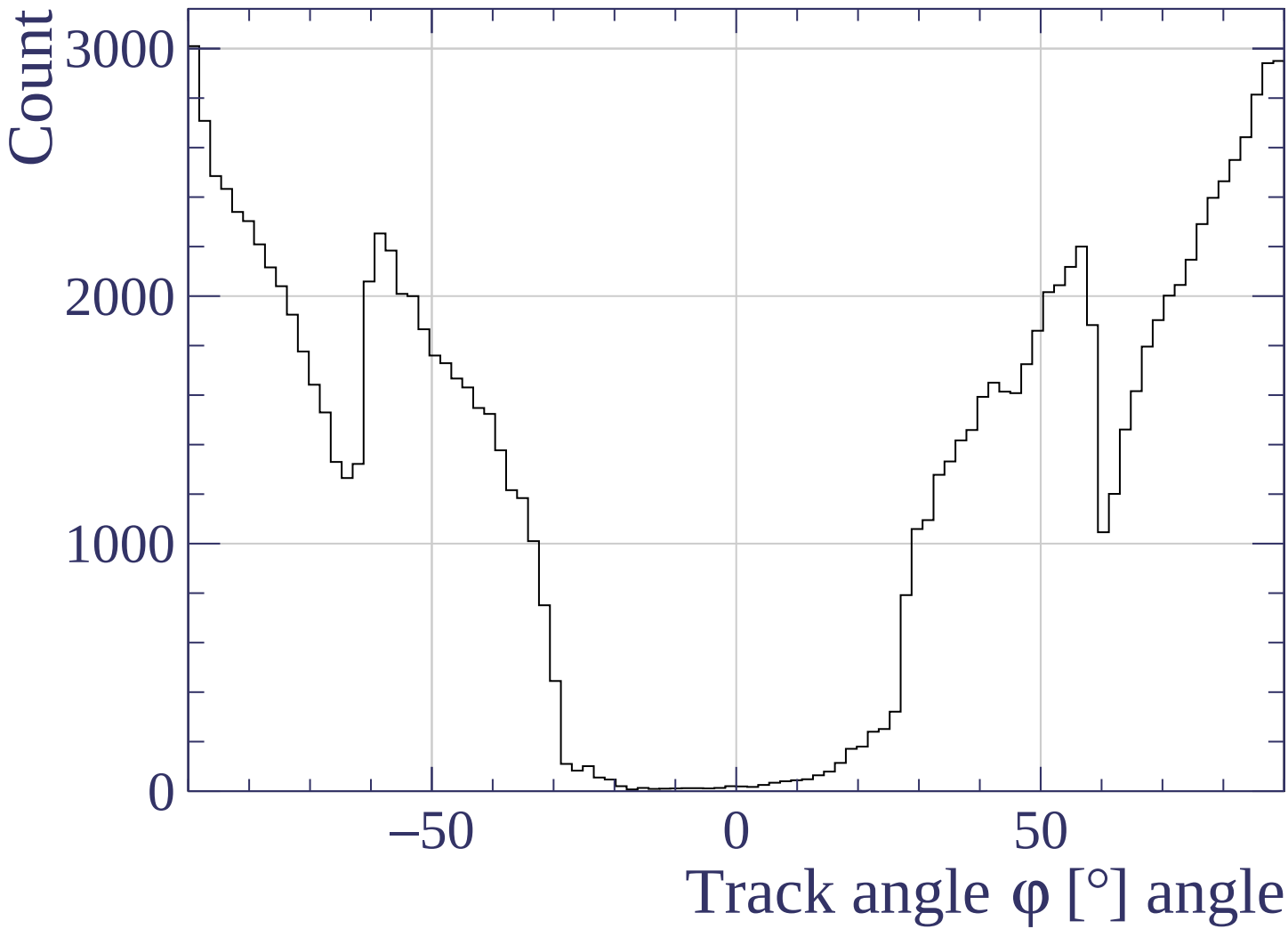


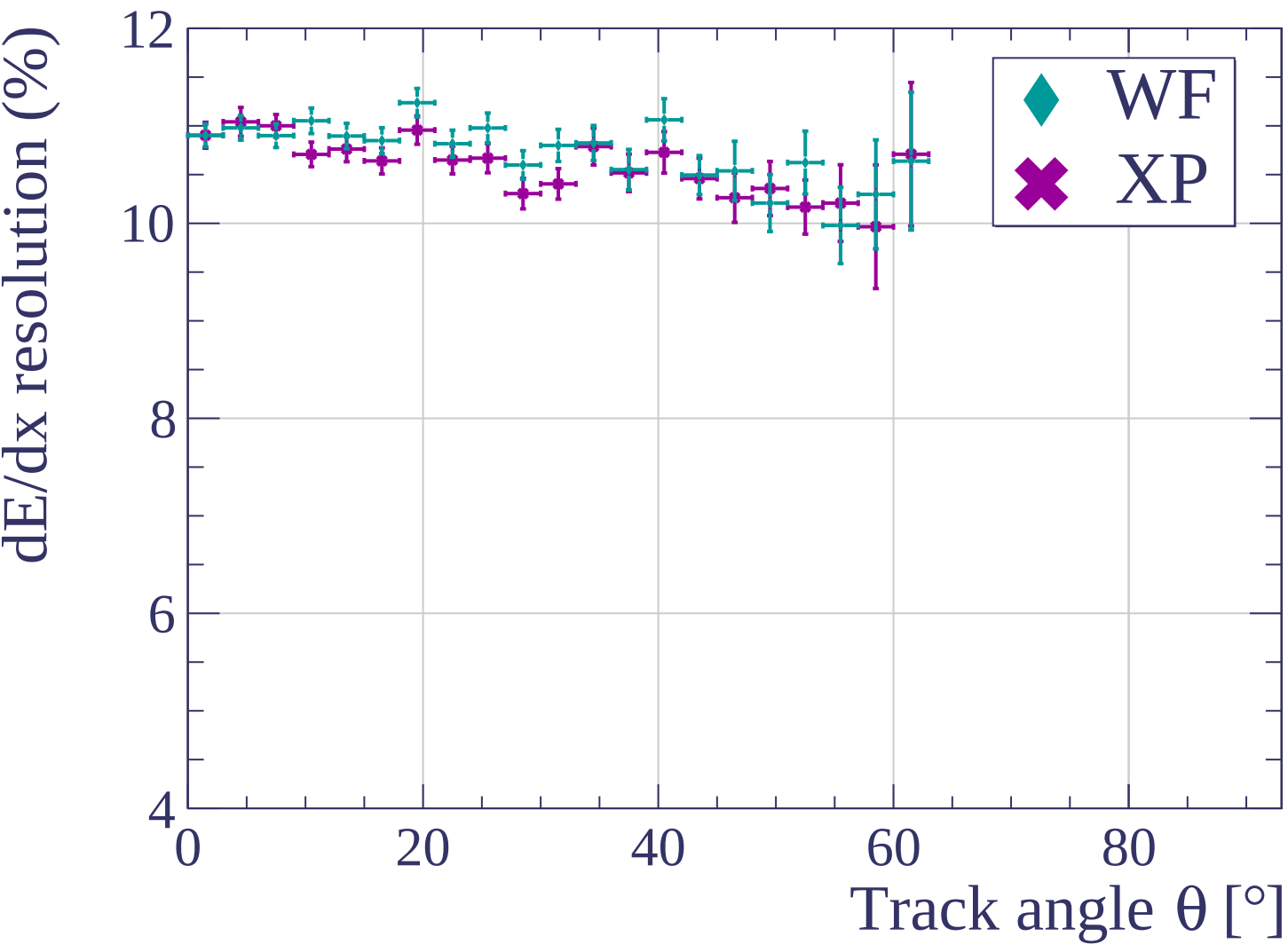
Track length [cm]

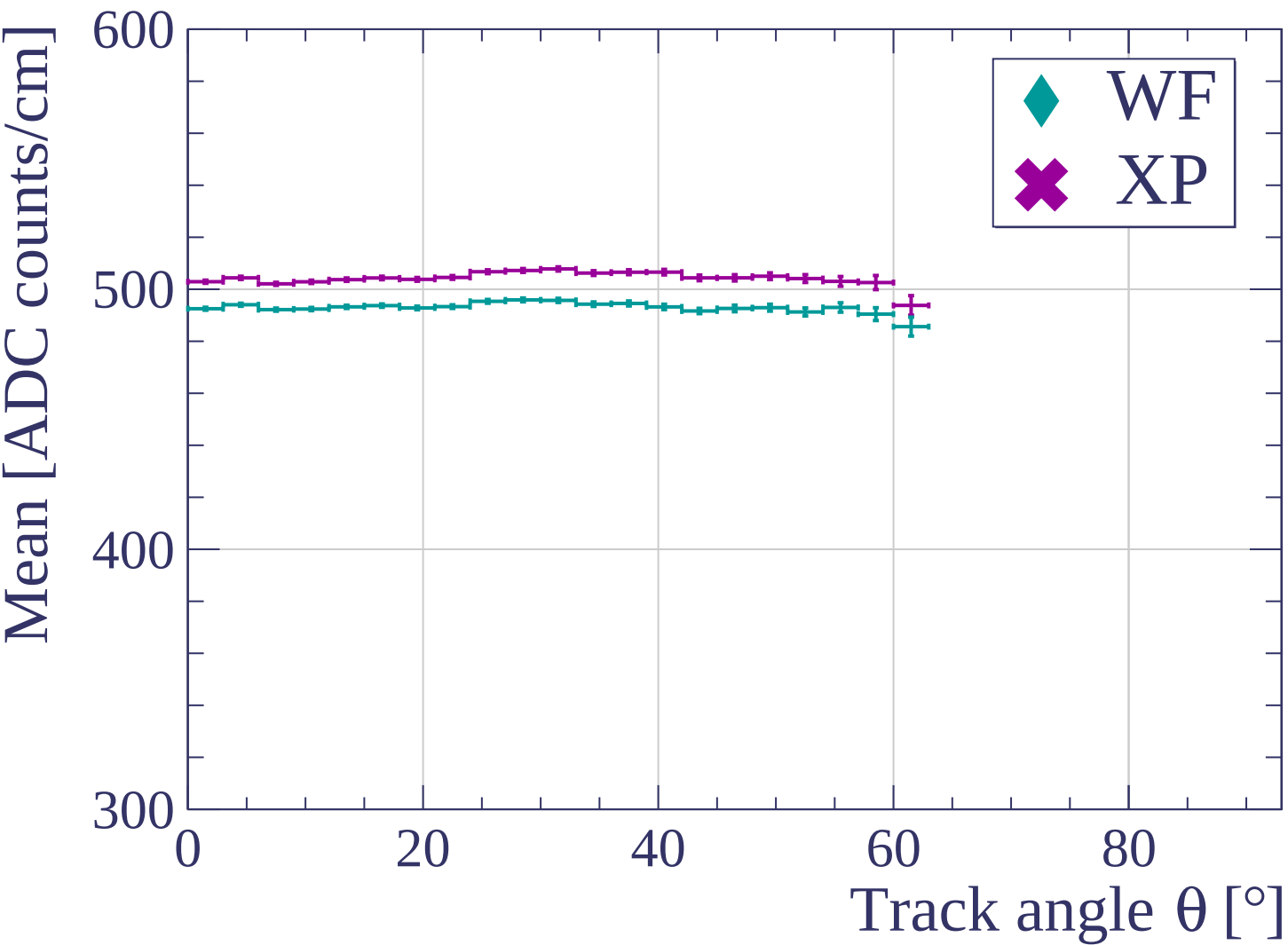




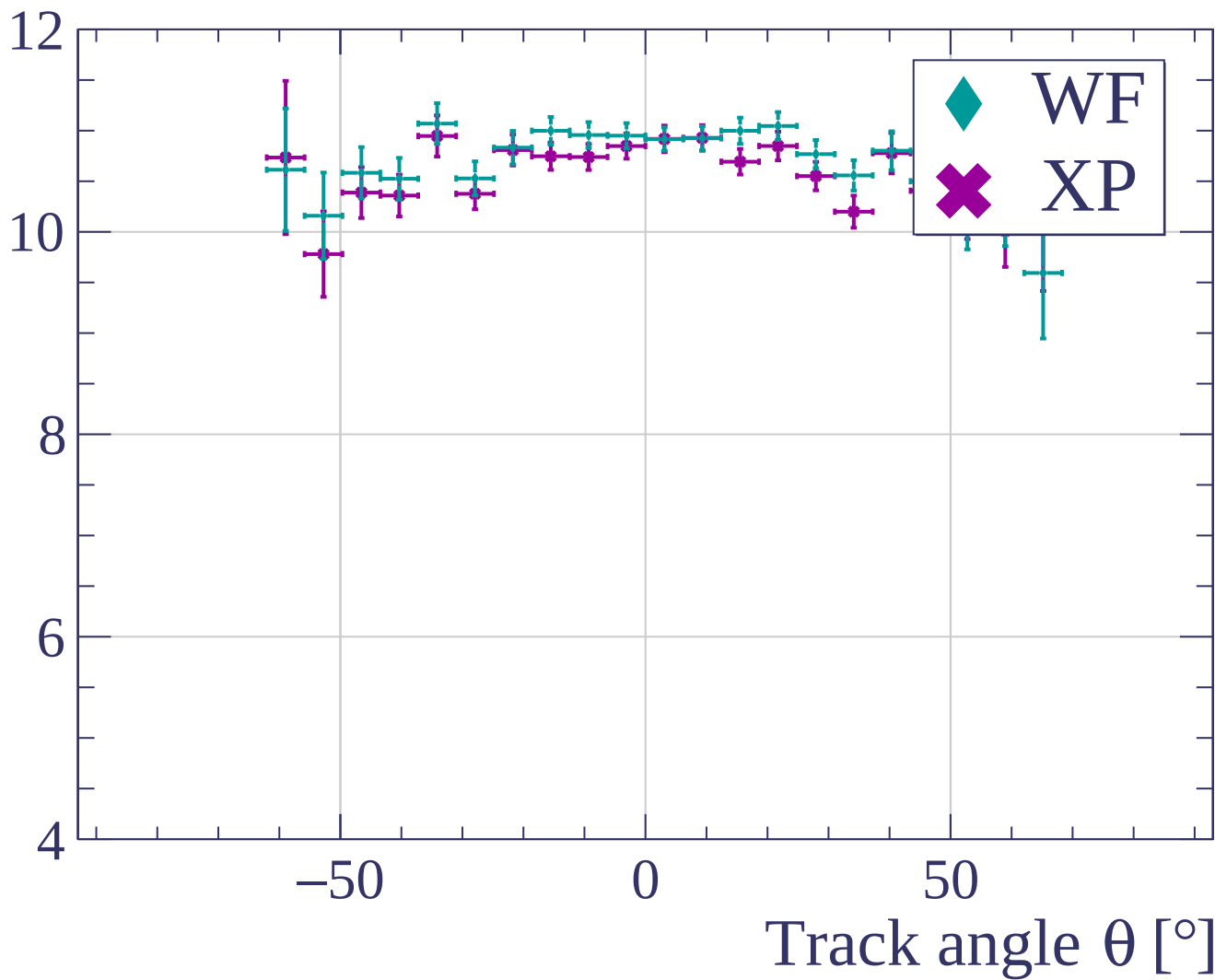


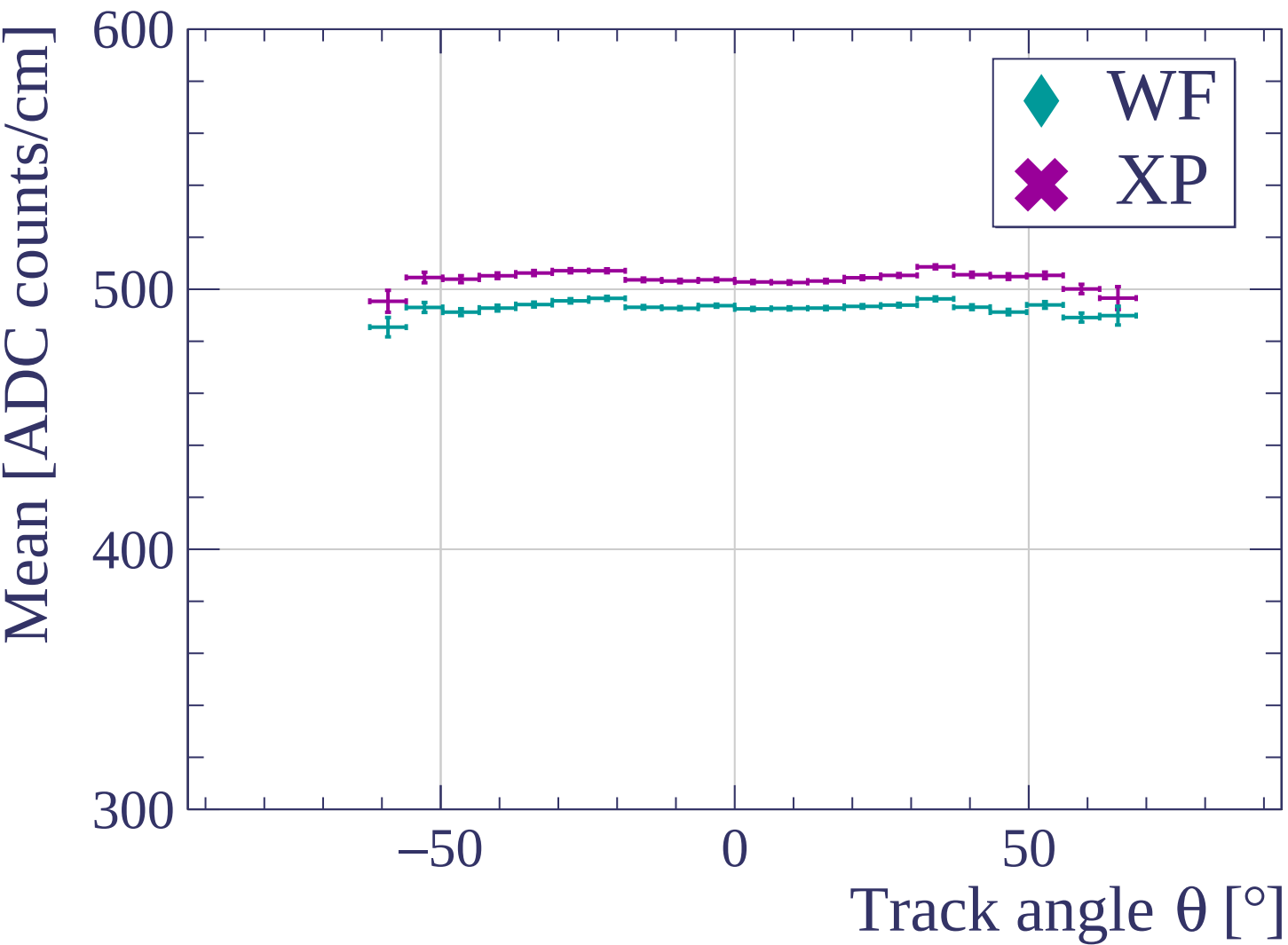


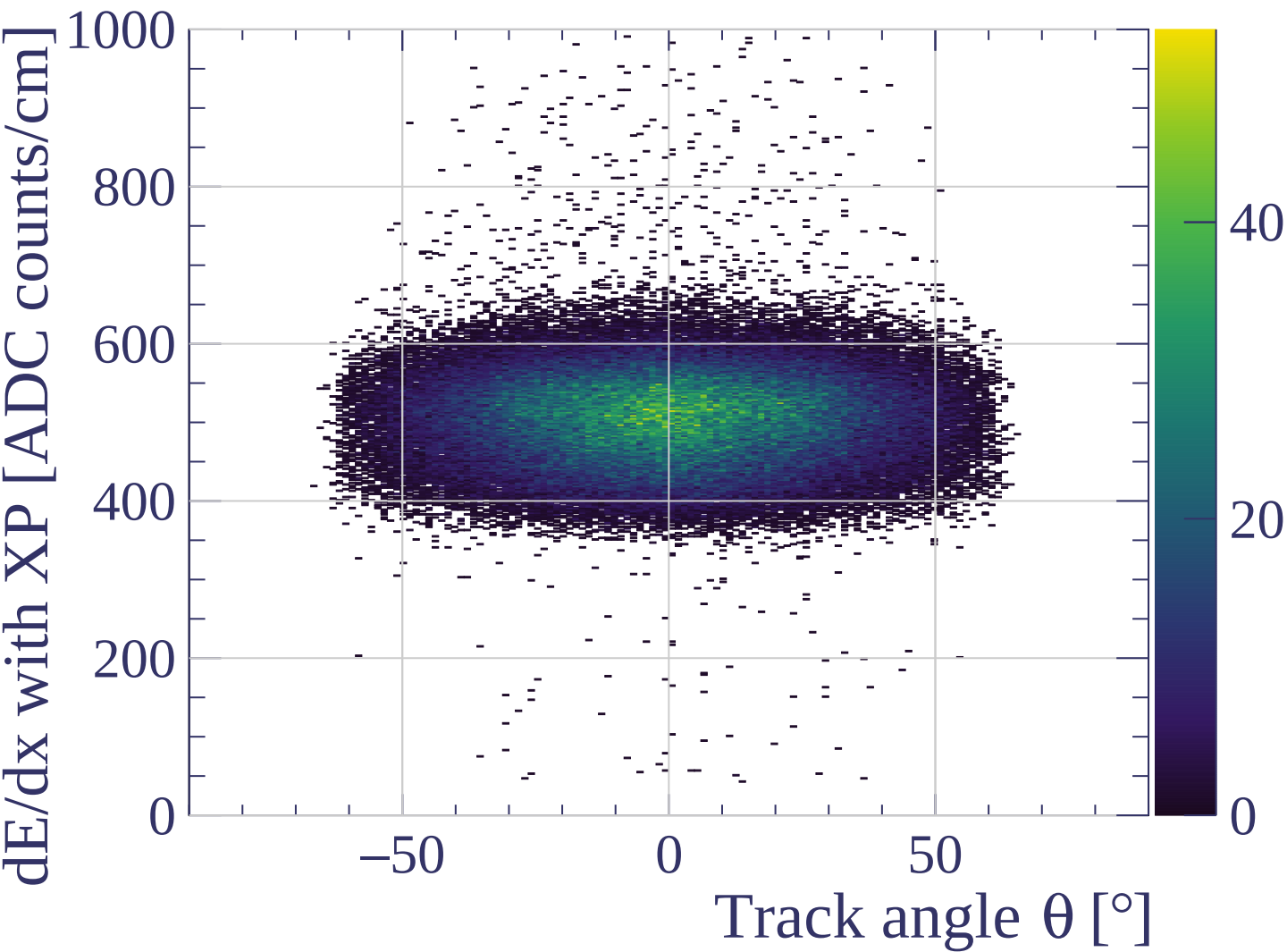




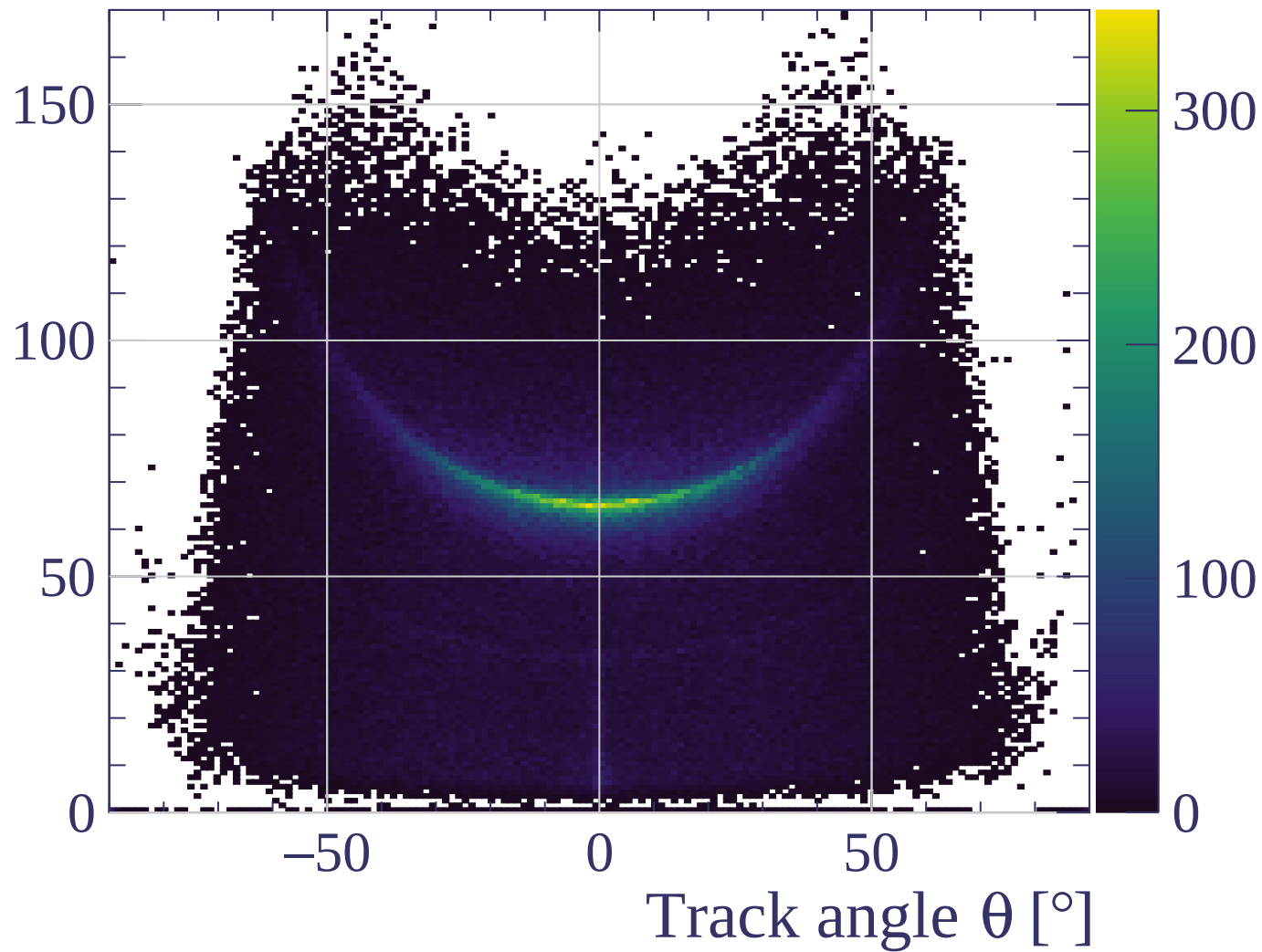
dE/dx resolution (%)

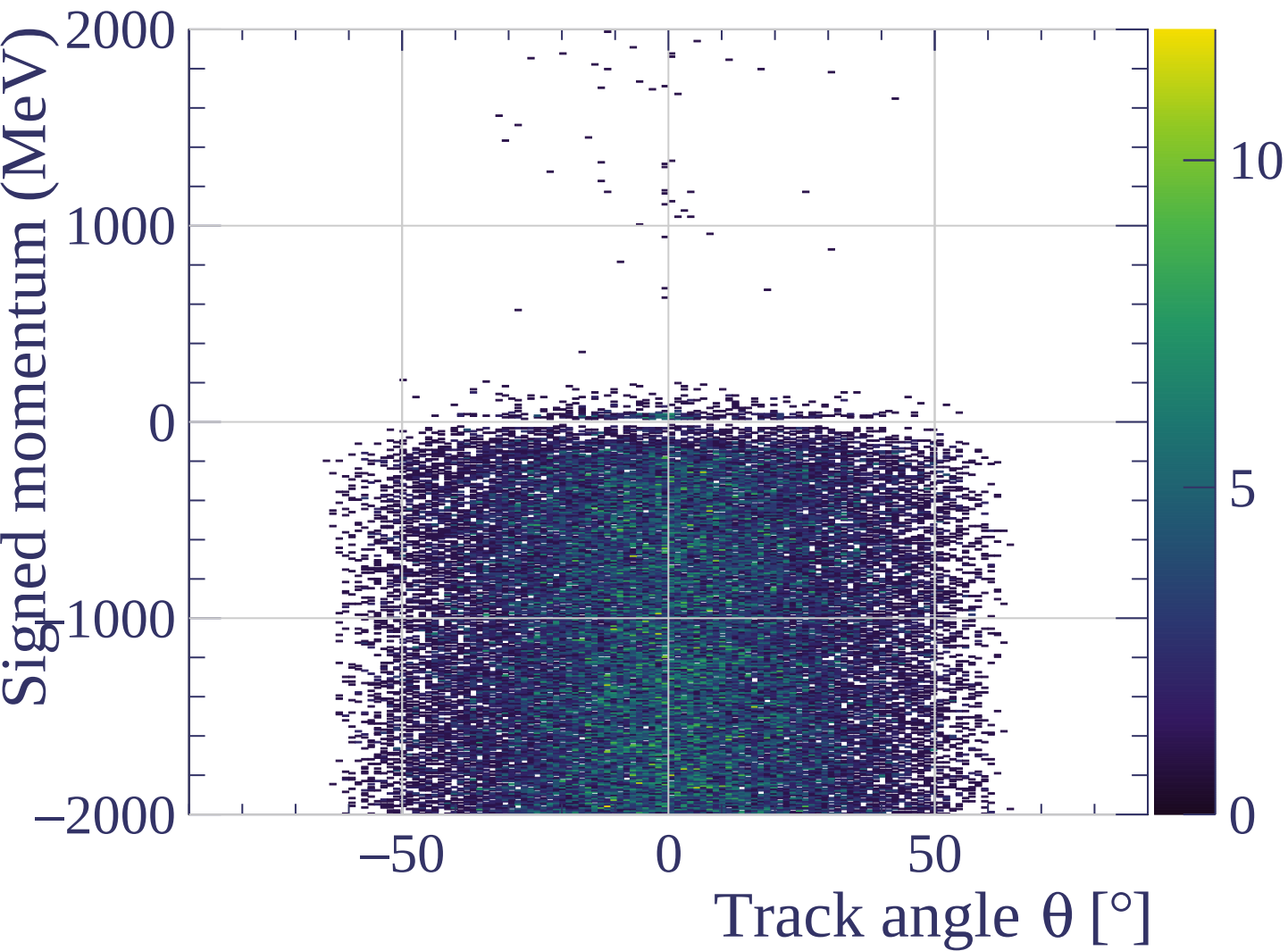


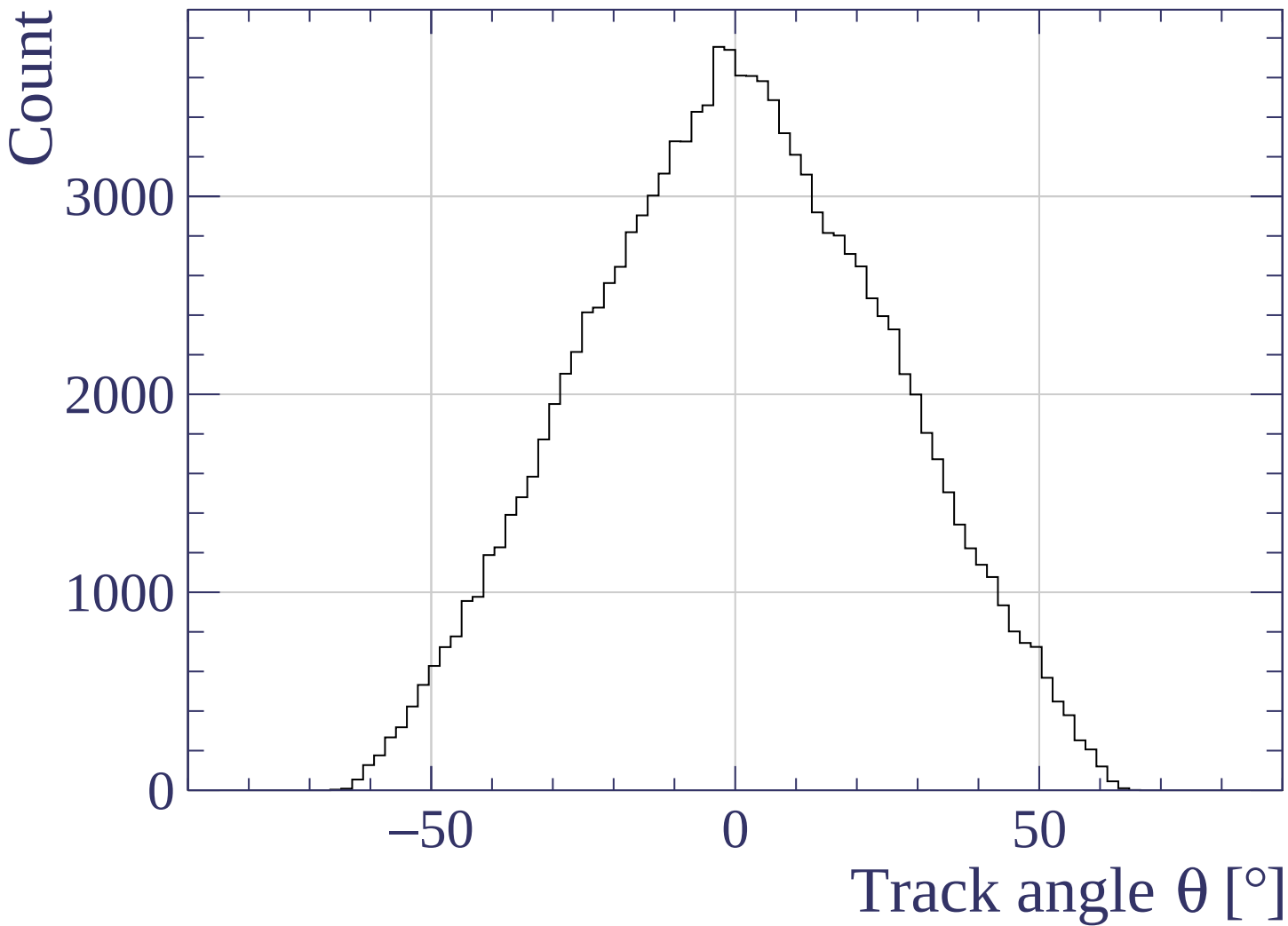




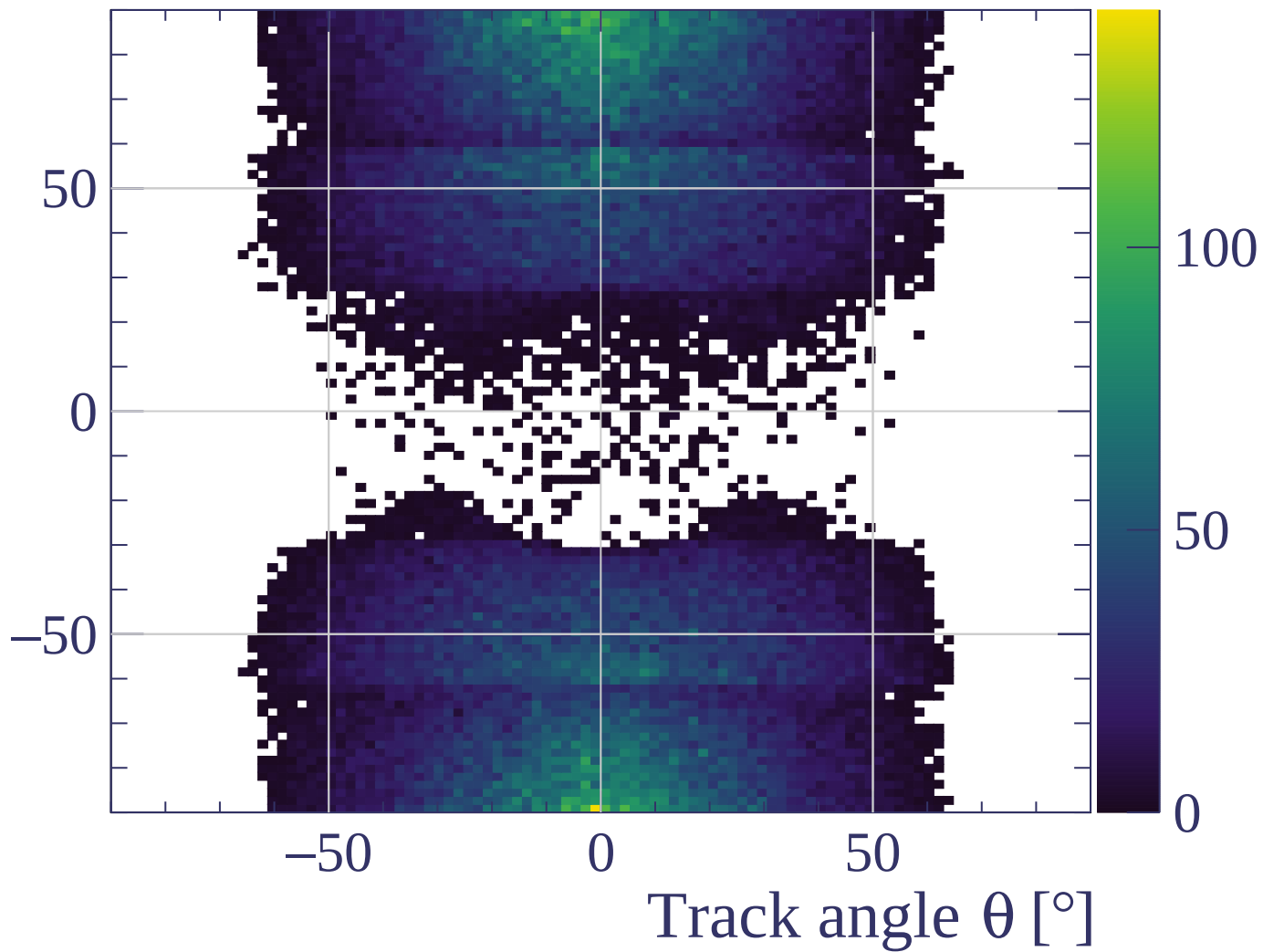
Track length [cm]



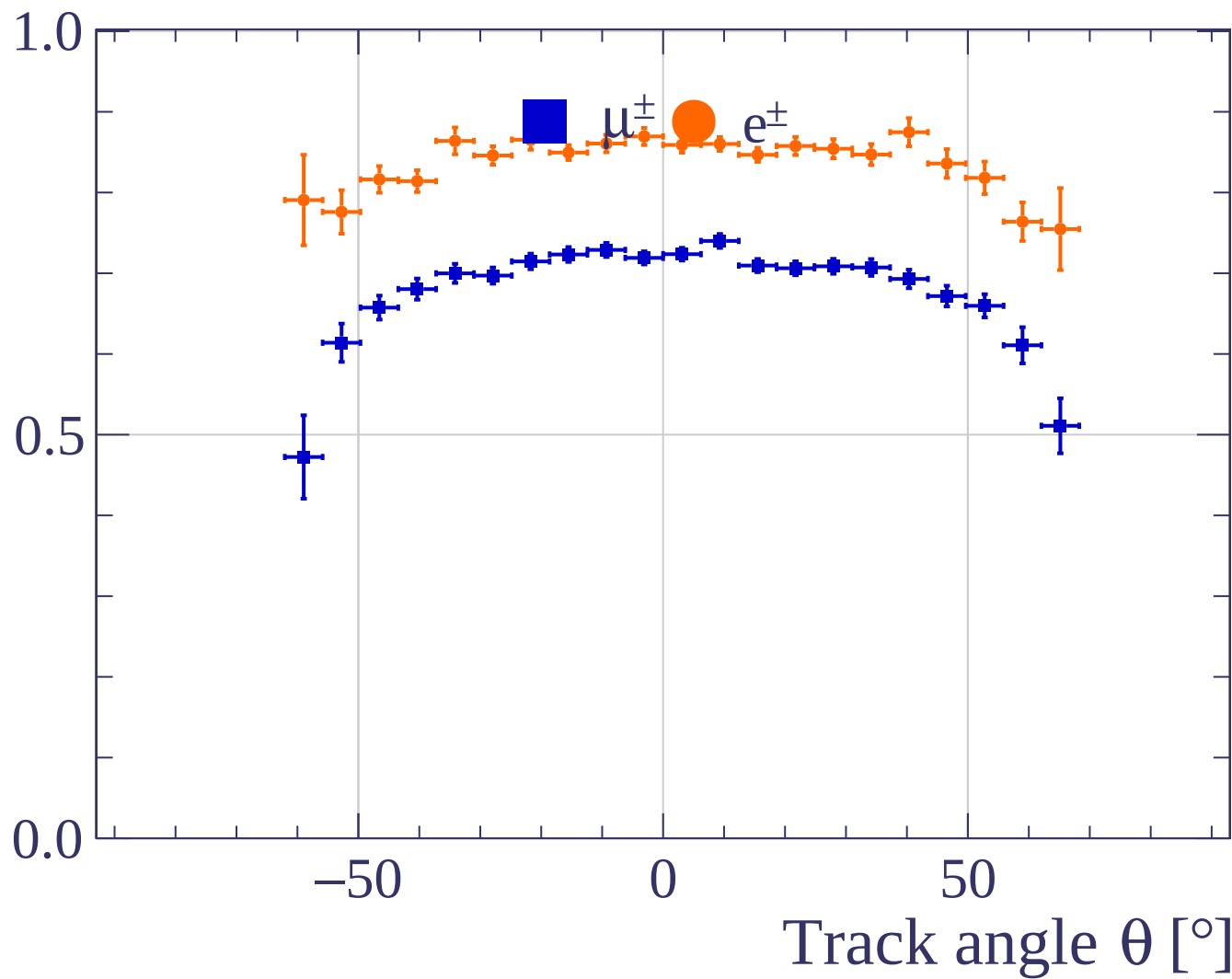




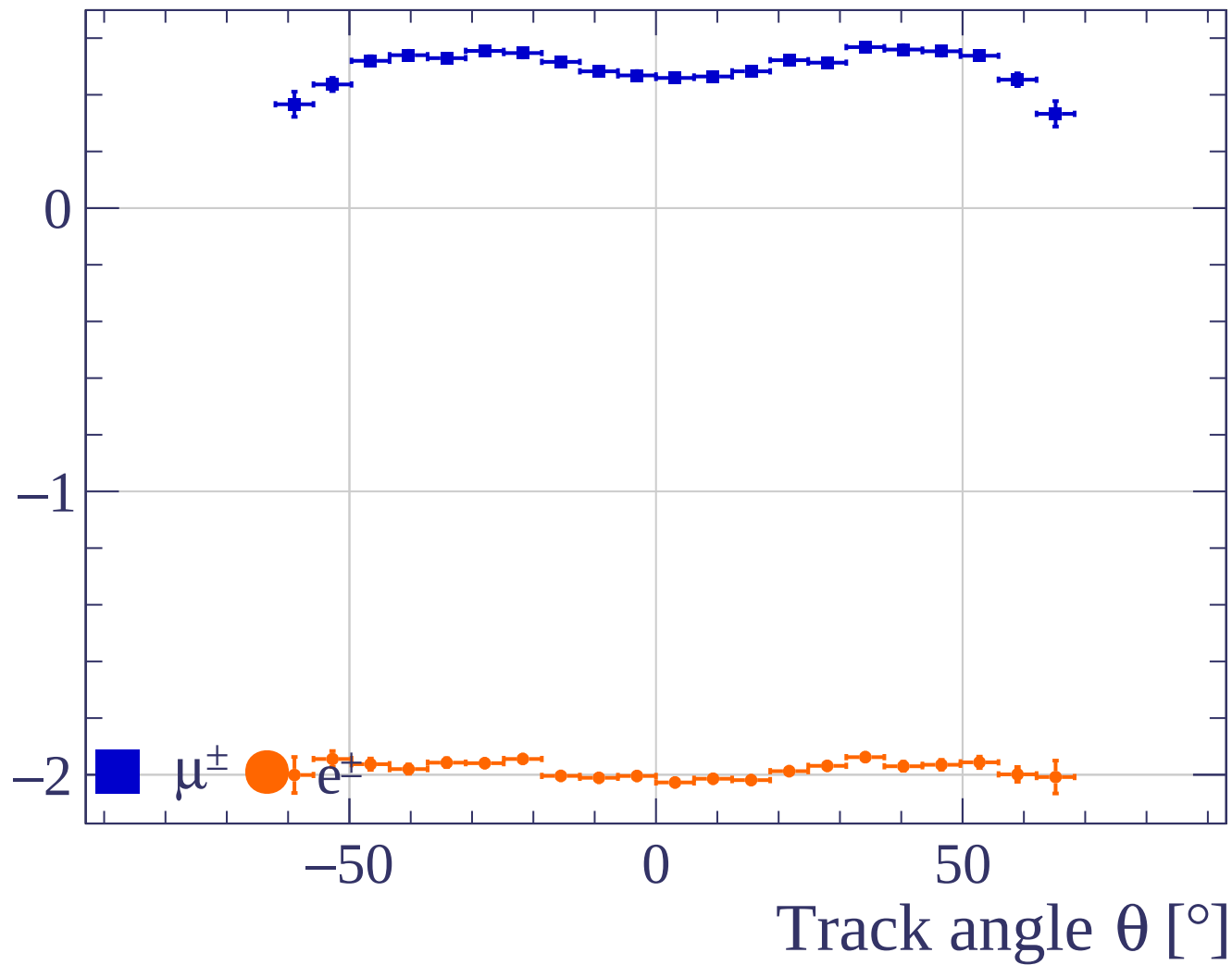
Track angle ϕ [°]



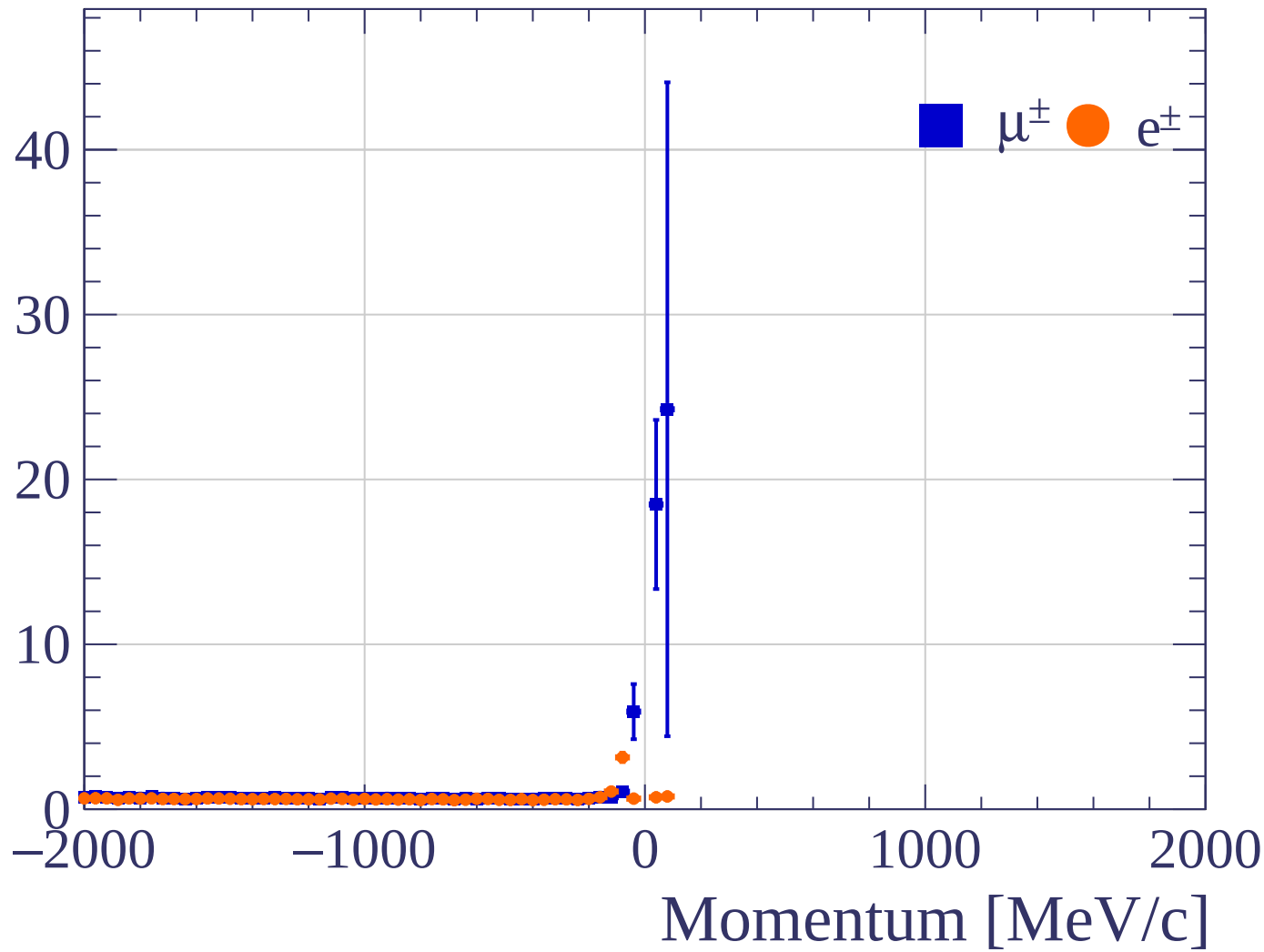
Pulls standard deviation



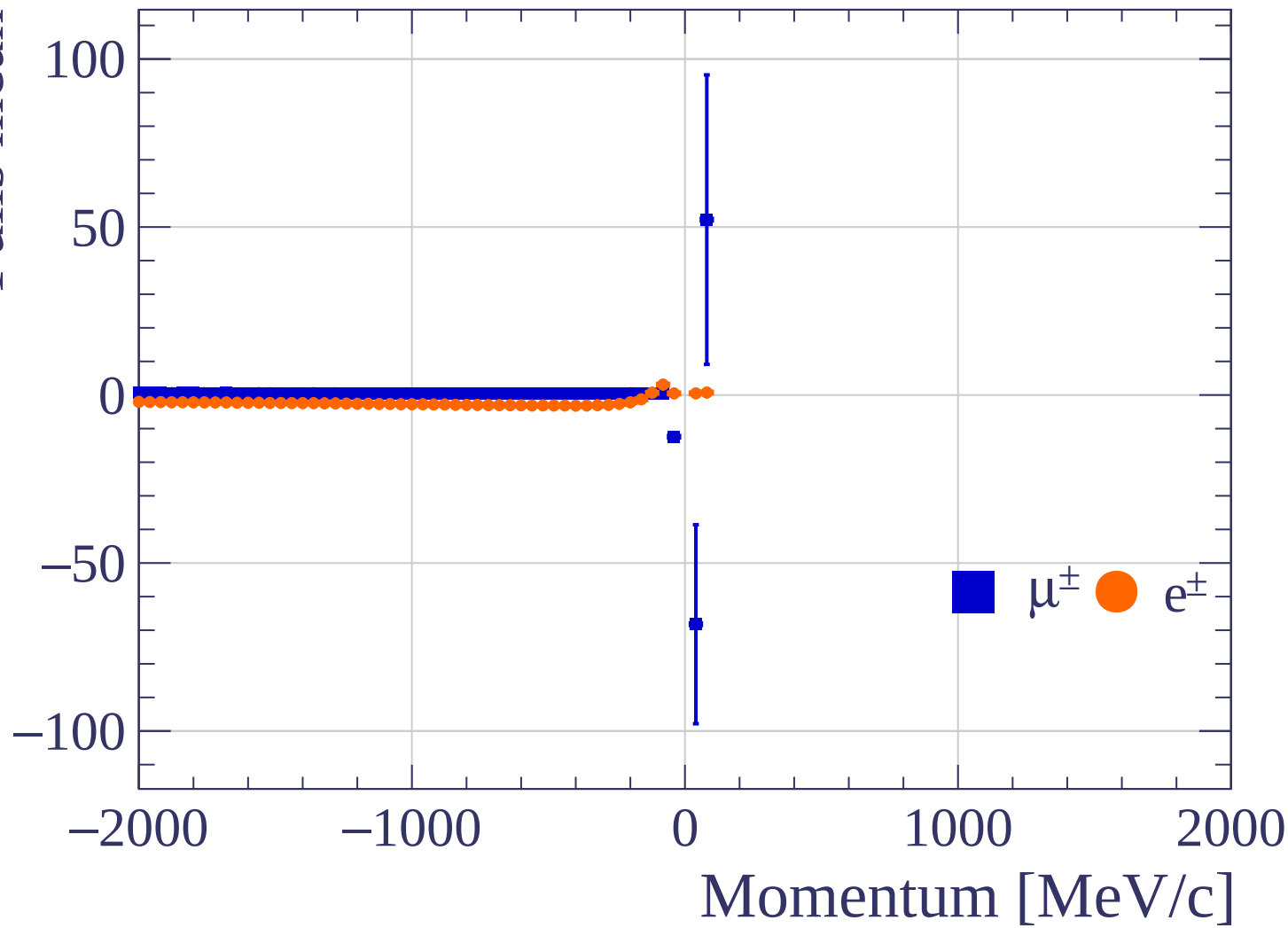
Pulls mean



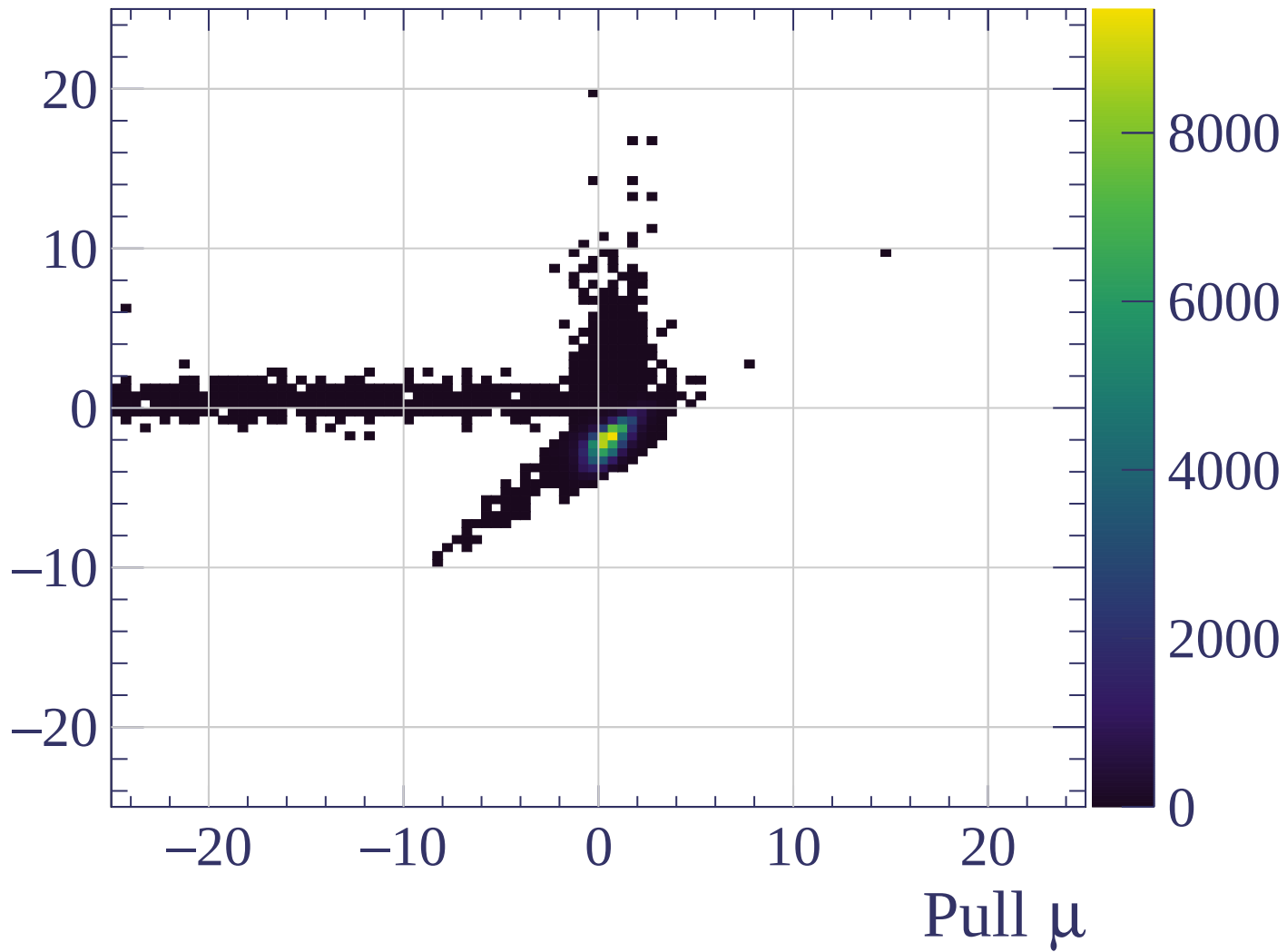
Pulls standard deviation

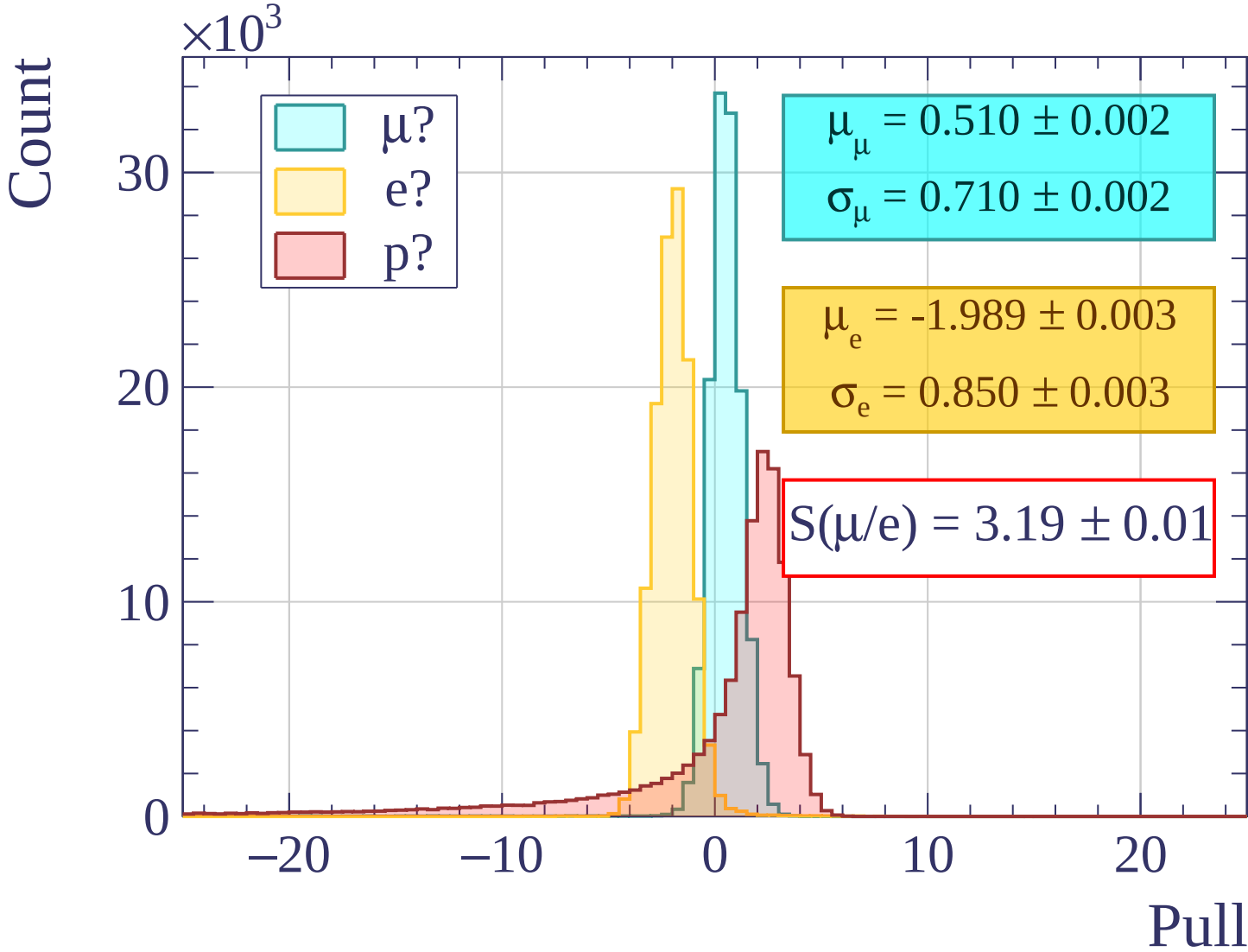


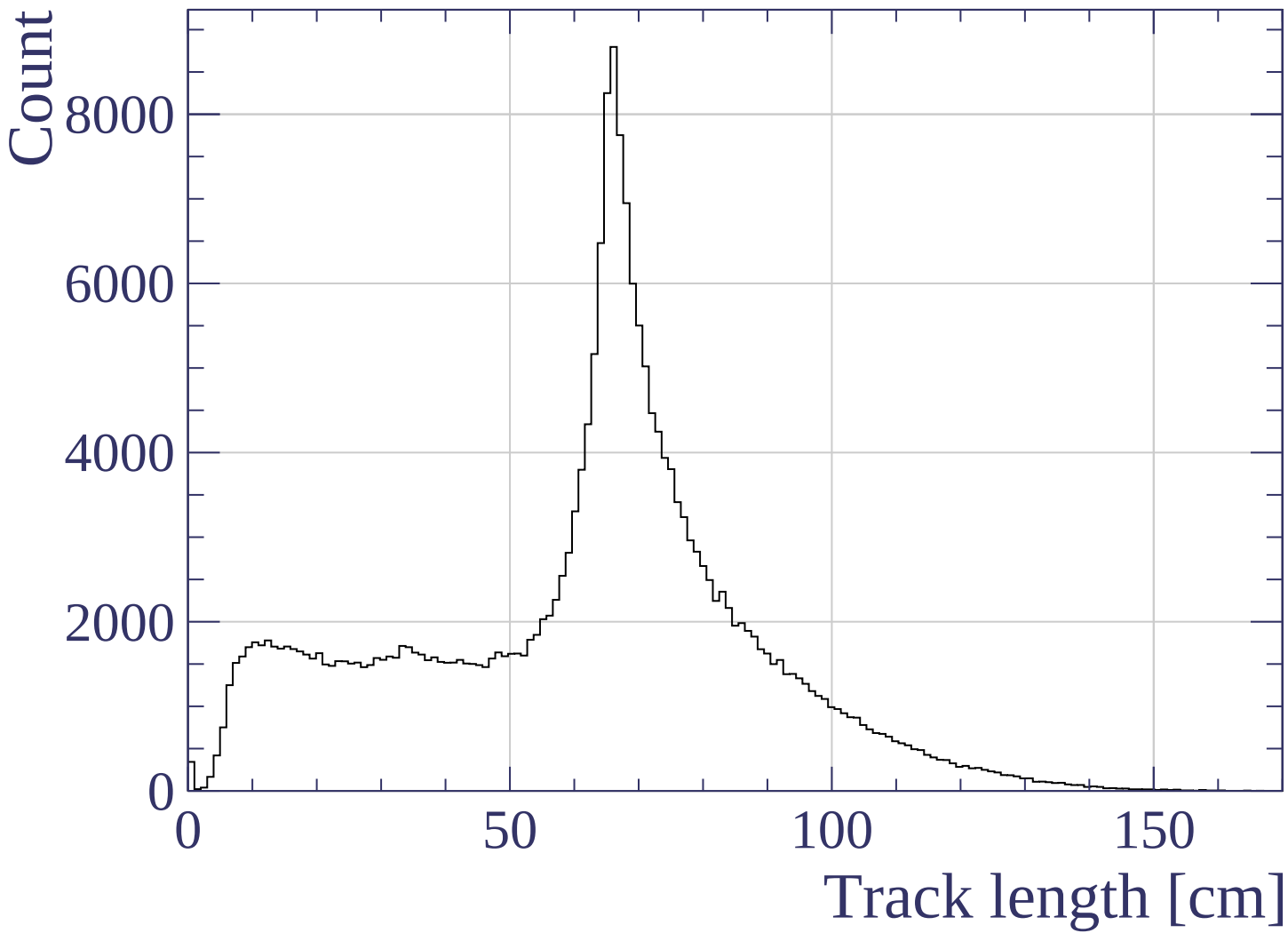
Pulls mean

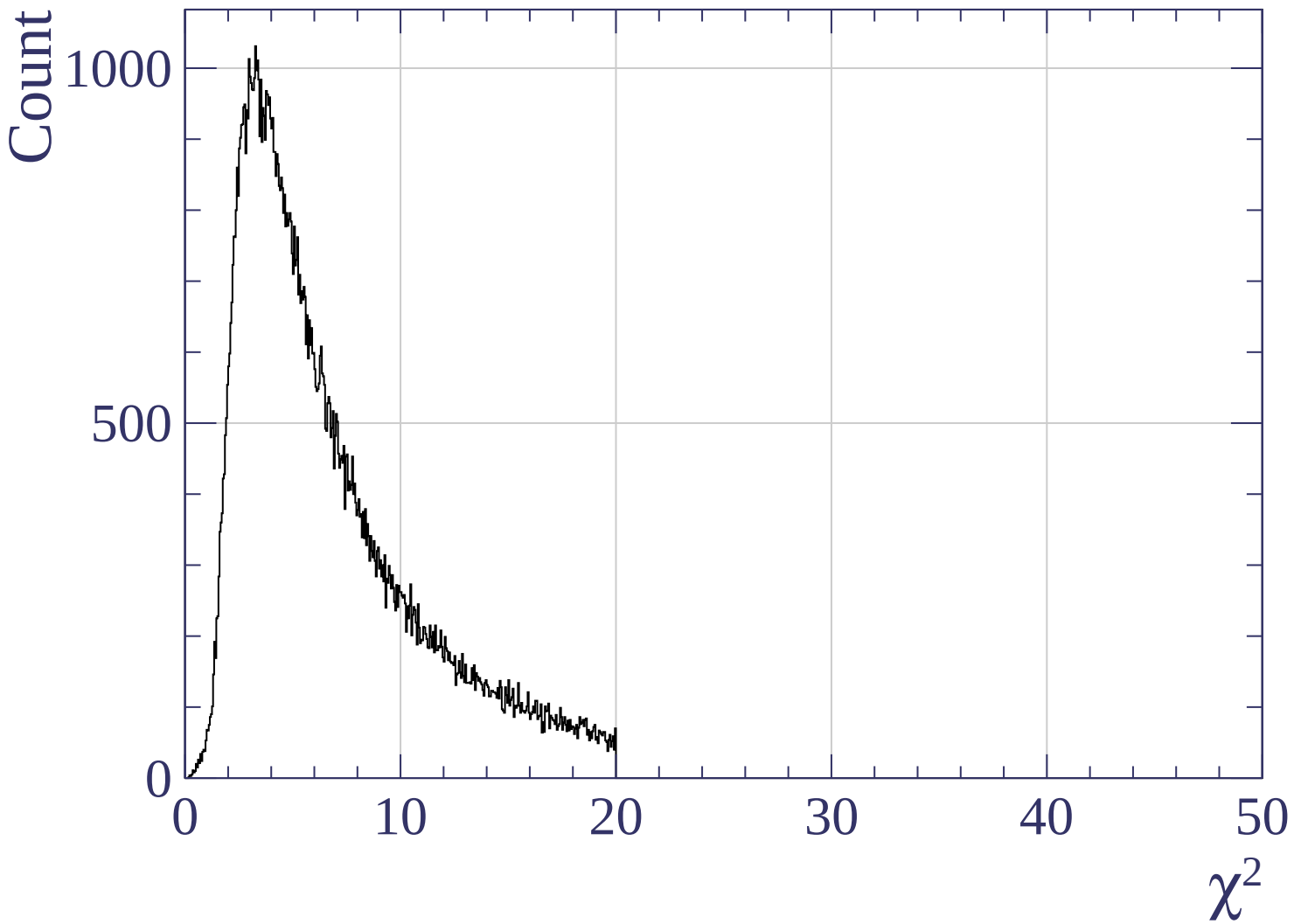


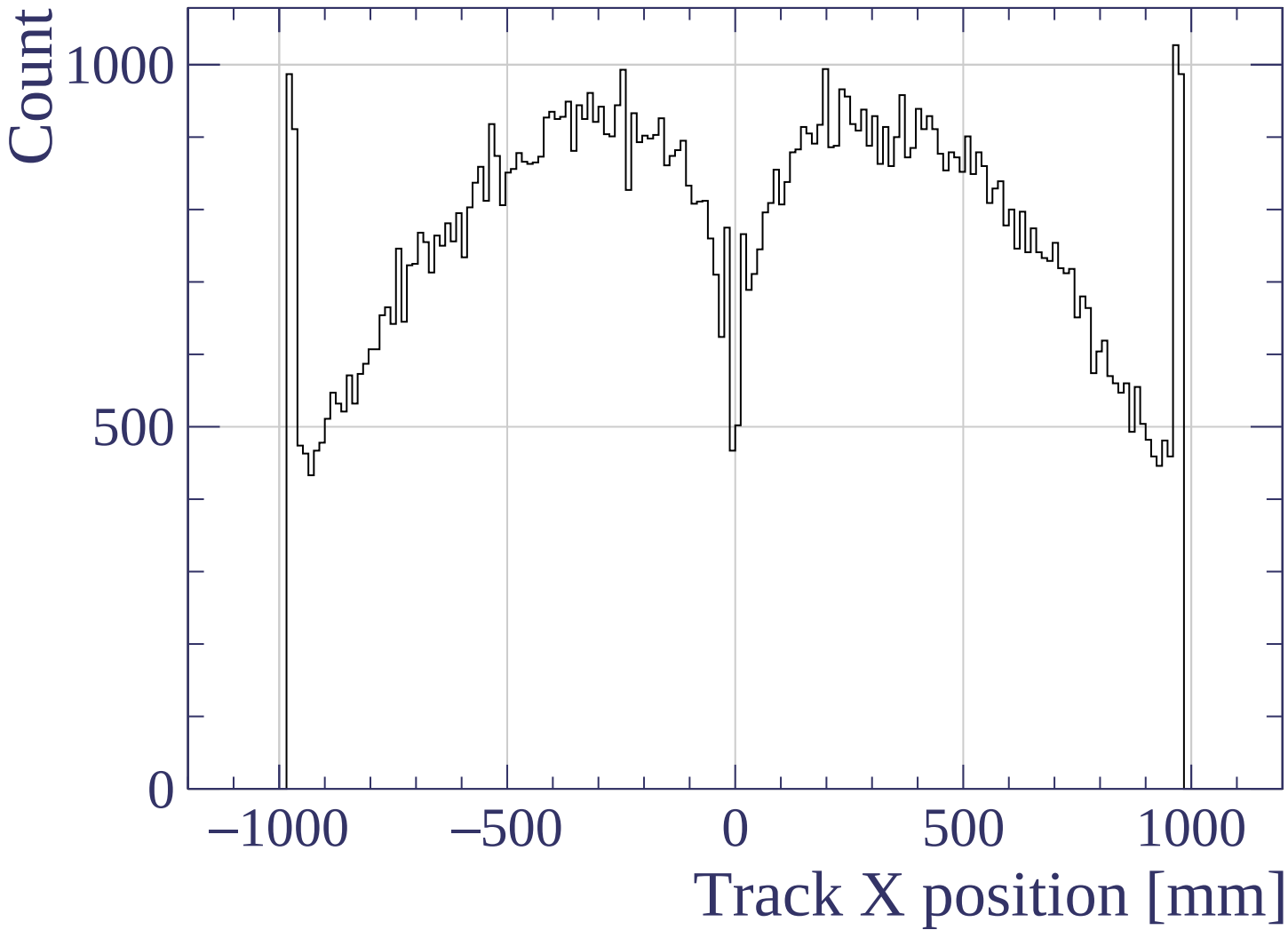
Pull e

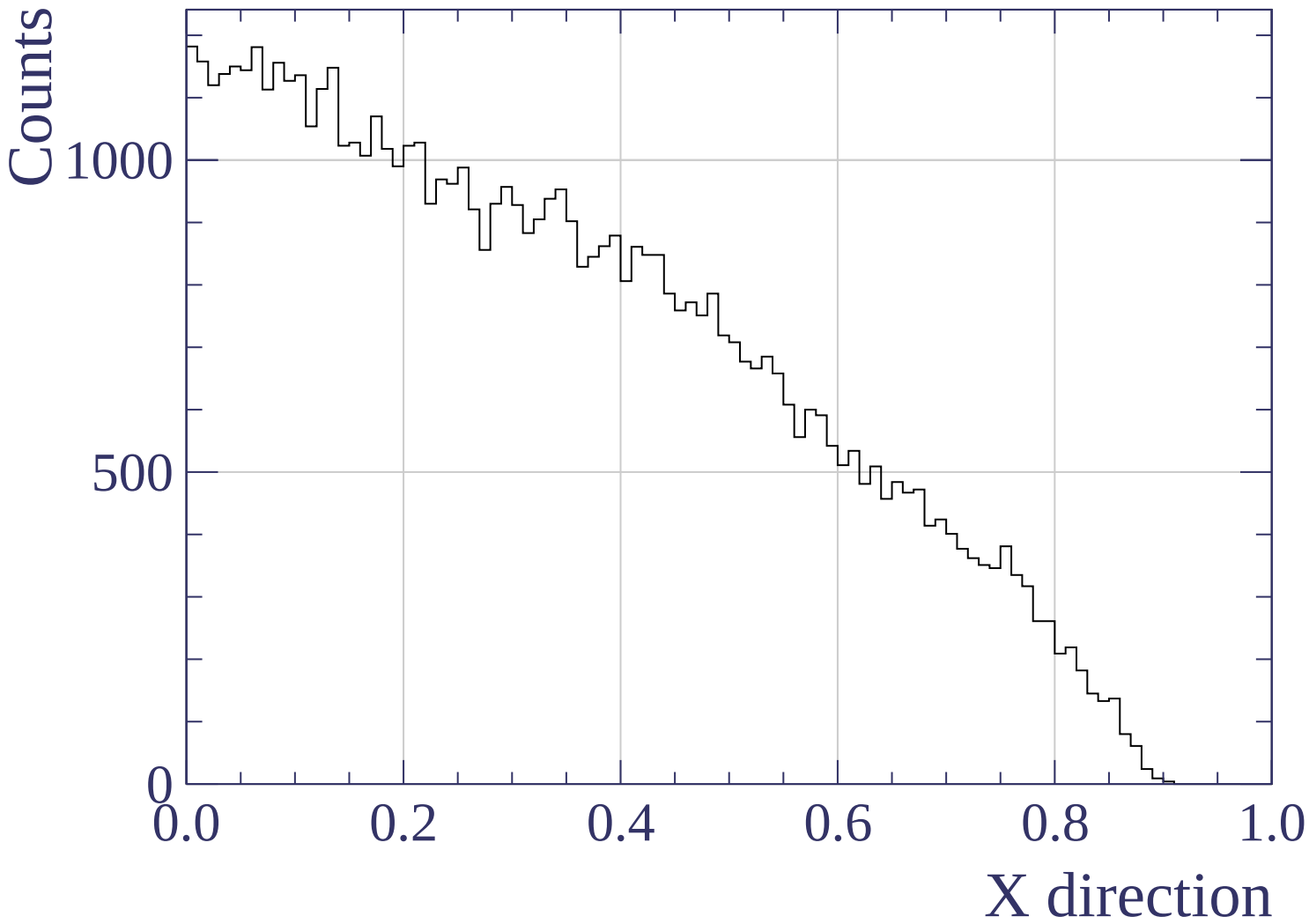


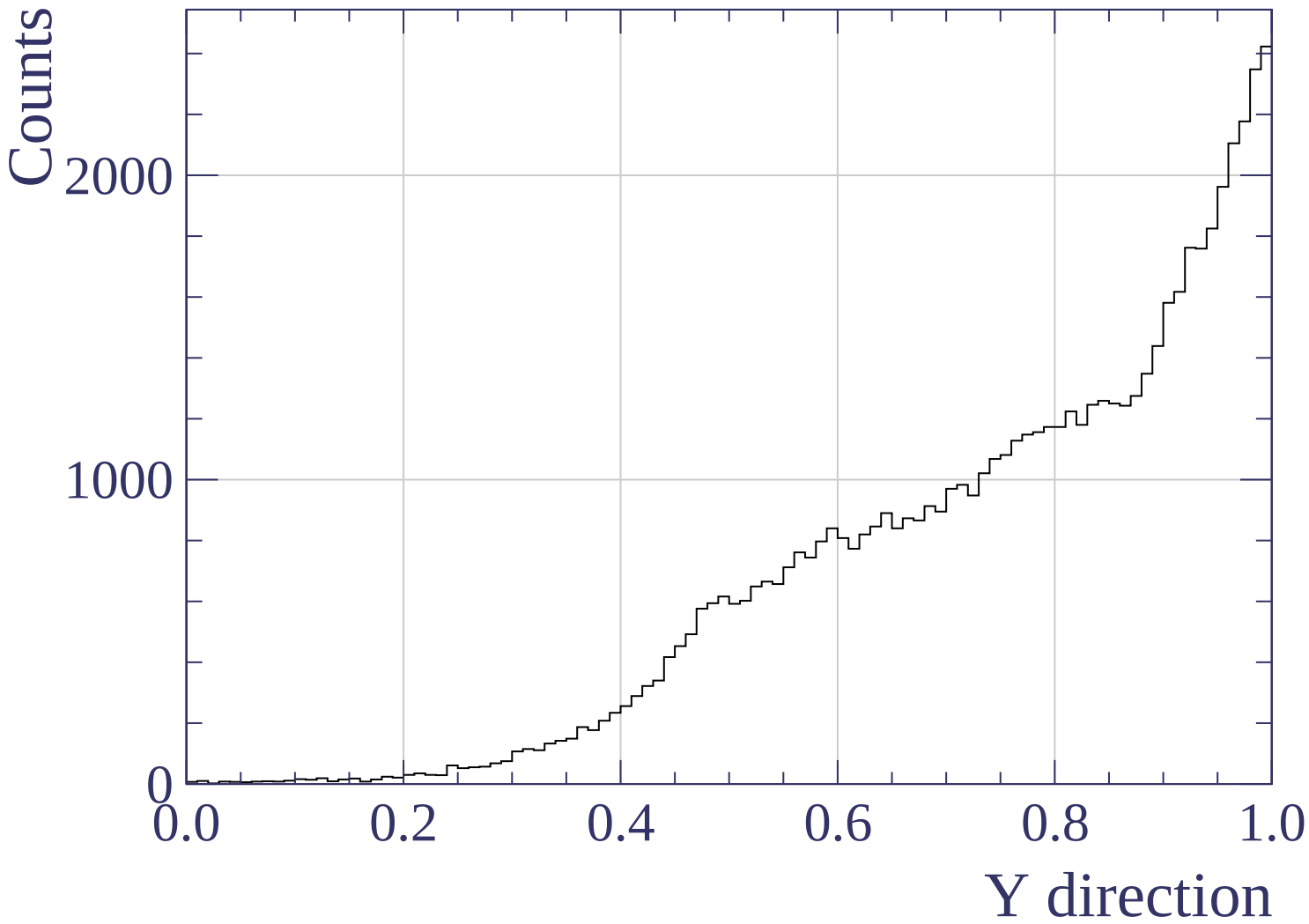


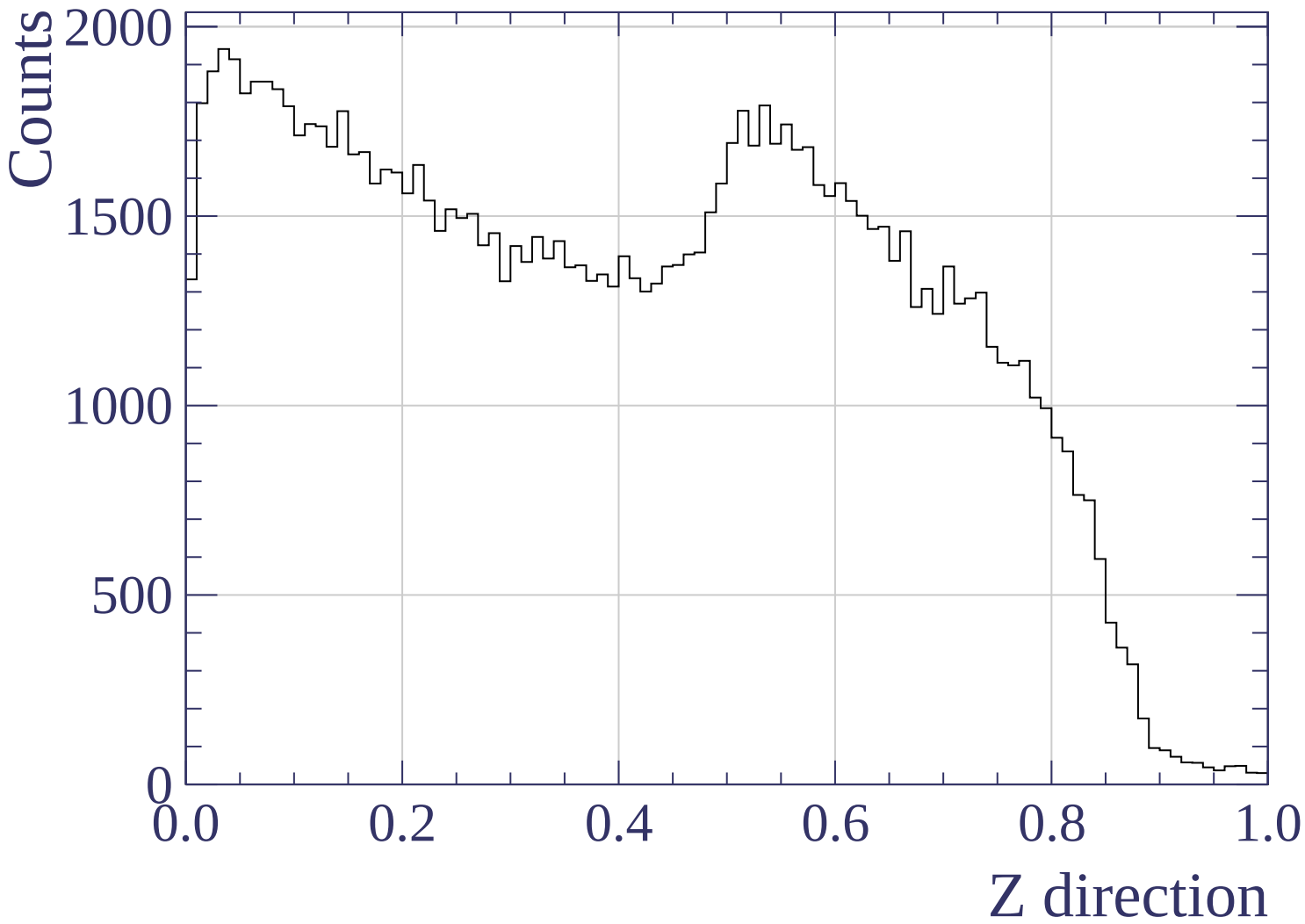


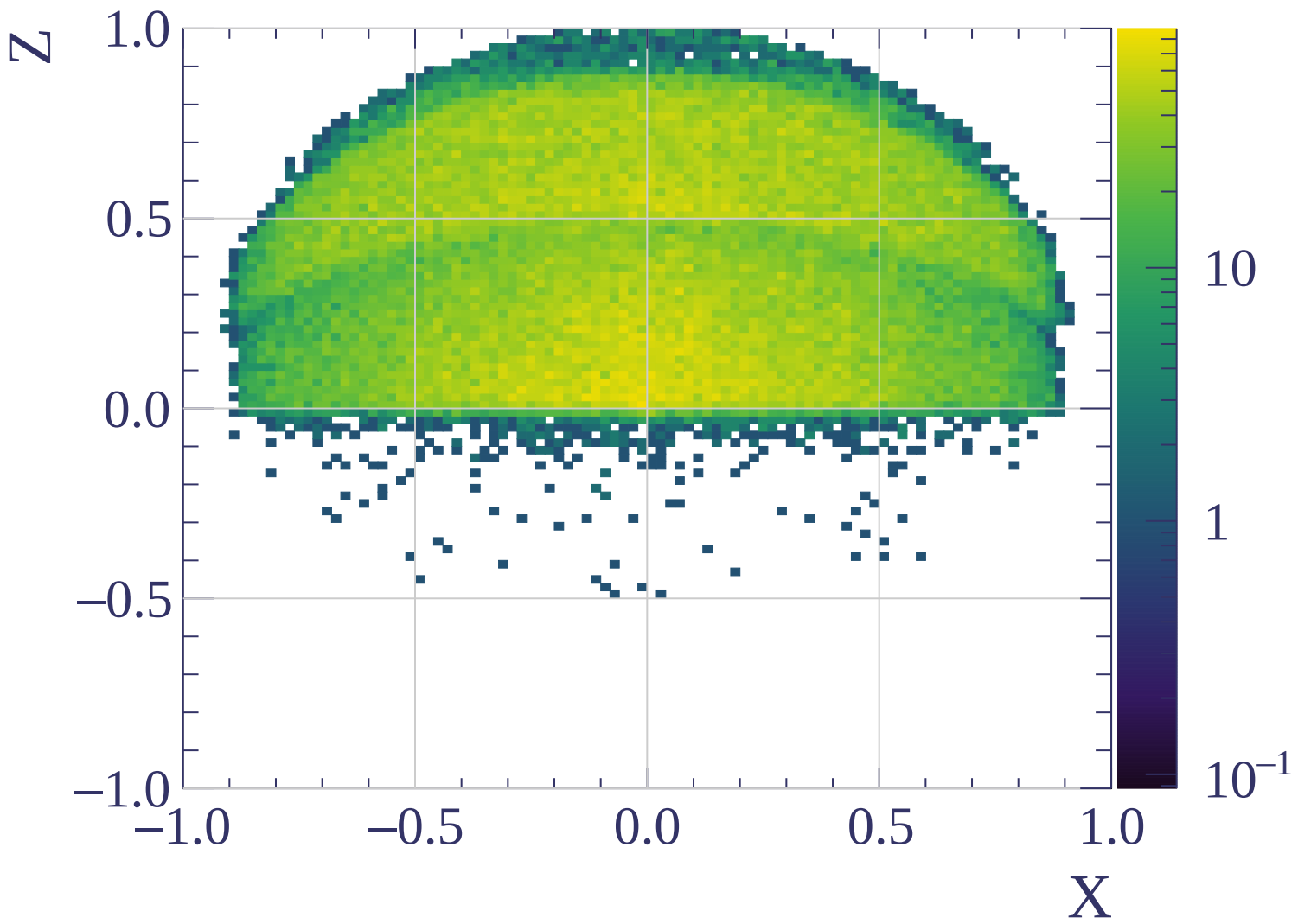


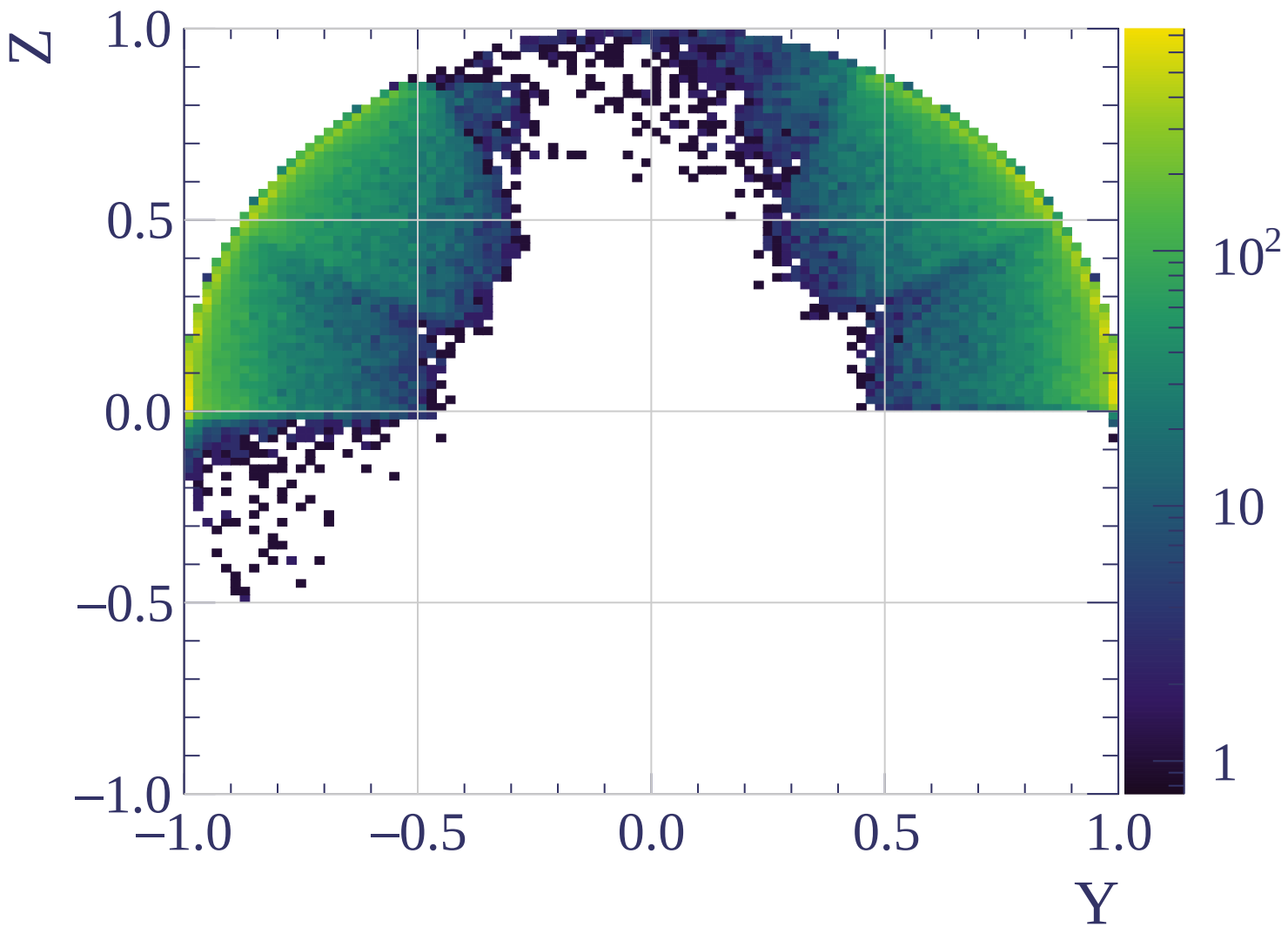


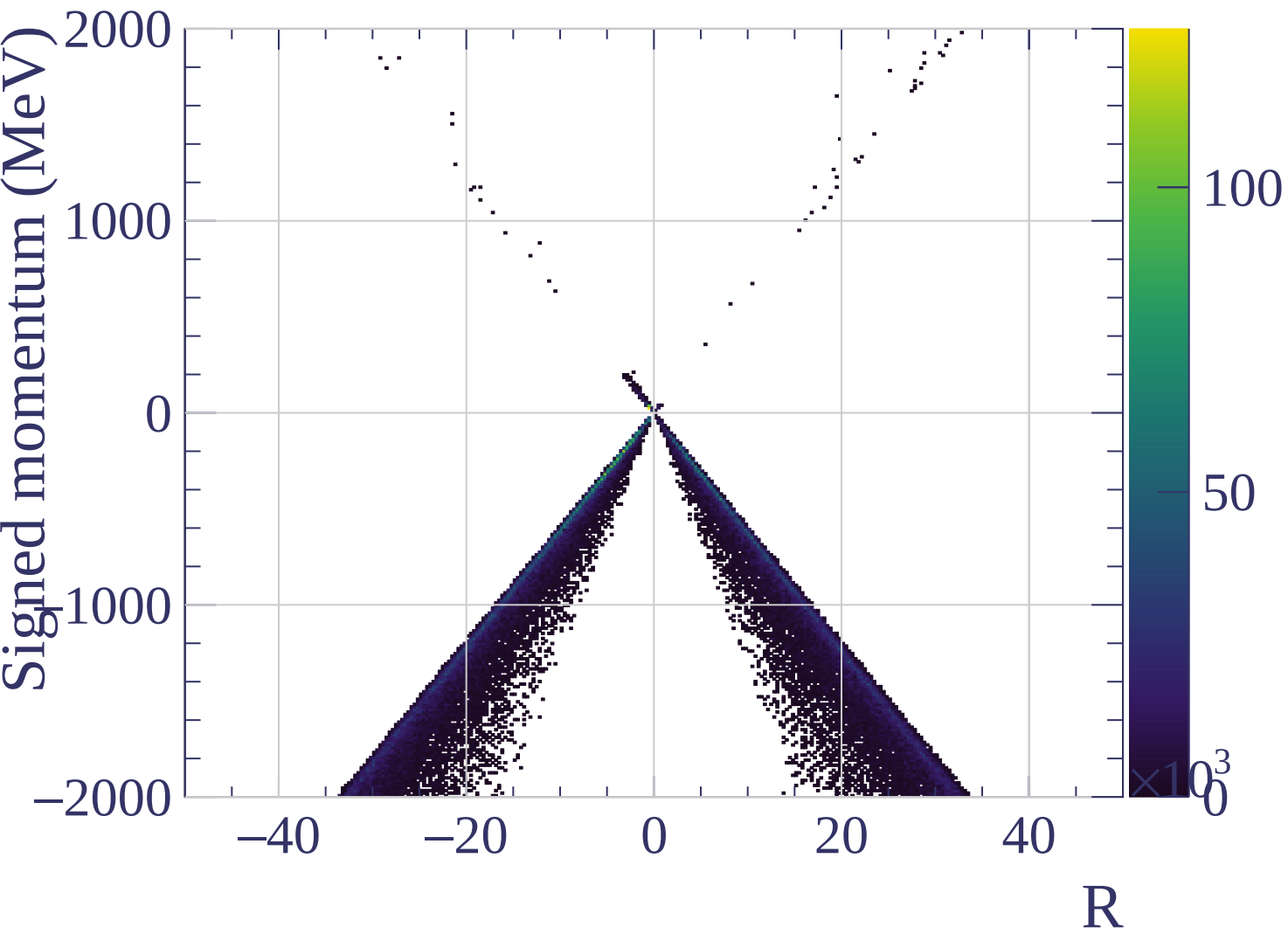


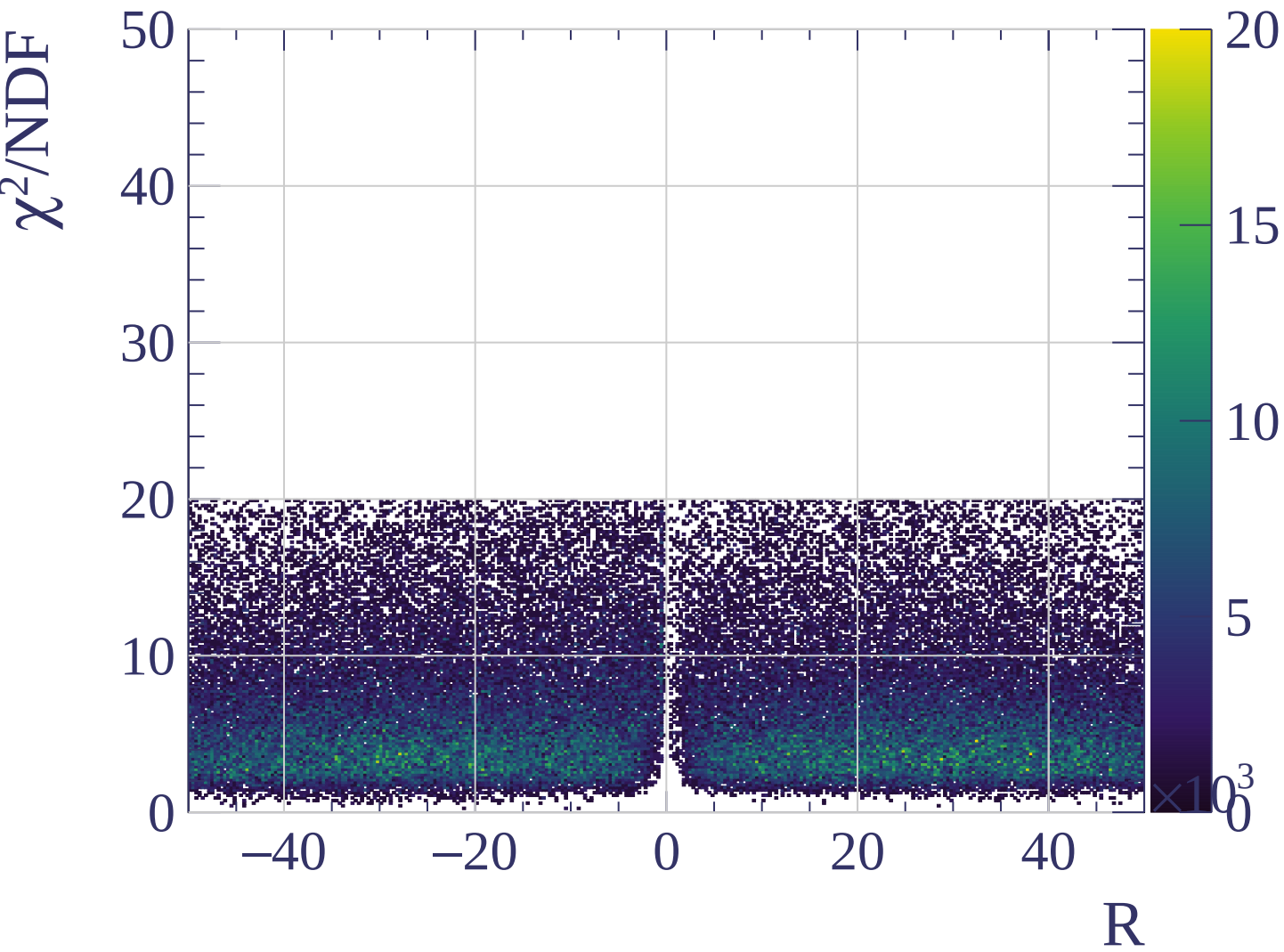












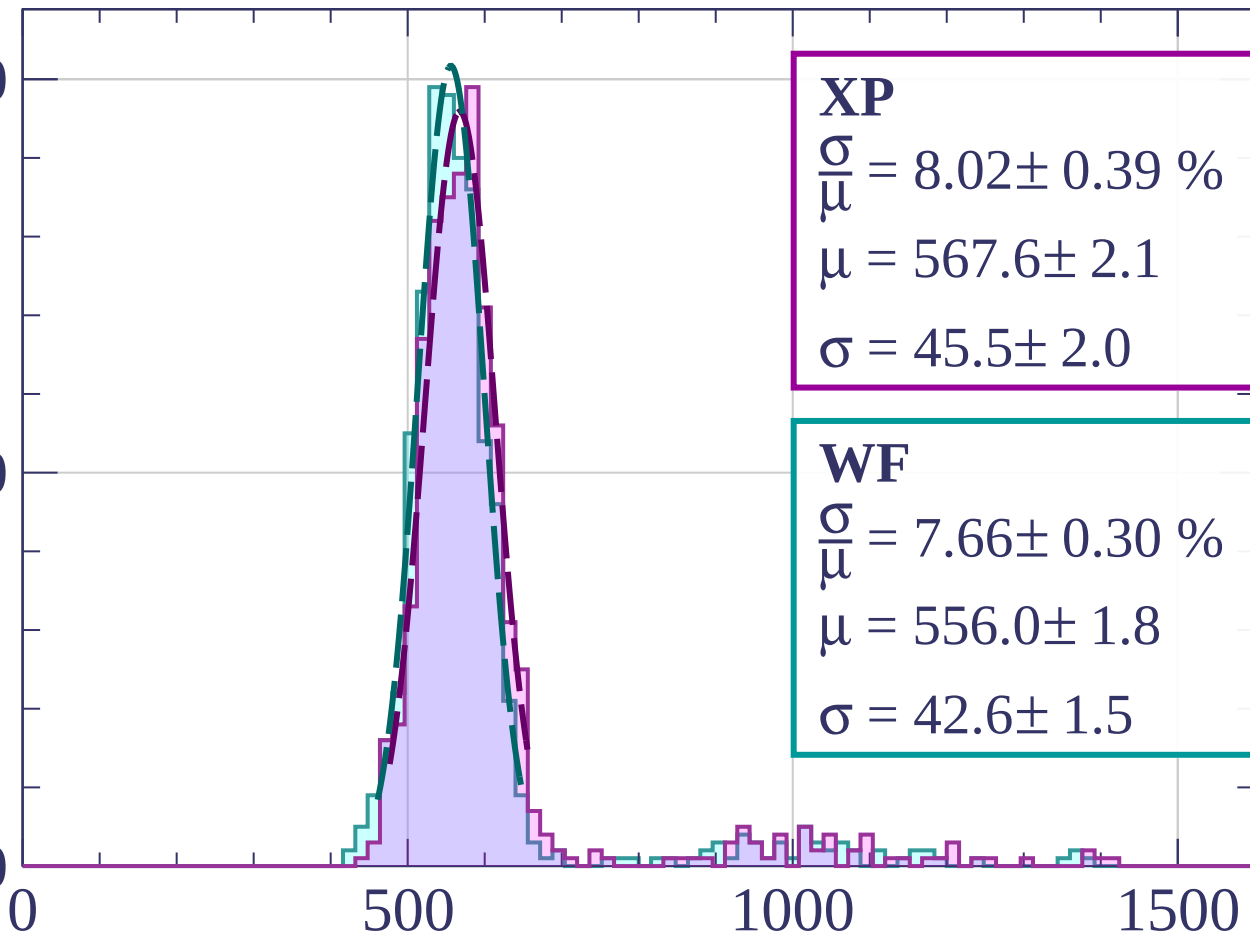
Energy loss | $0 < p < 80$

Count

100

50

0



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.39 \%$$

$$\mu = 567.6 \pm 2.1$$

$$\sigma = 45.5 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.30 \%$$

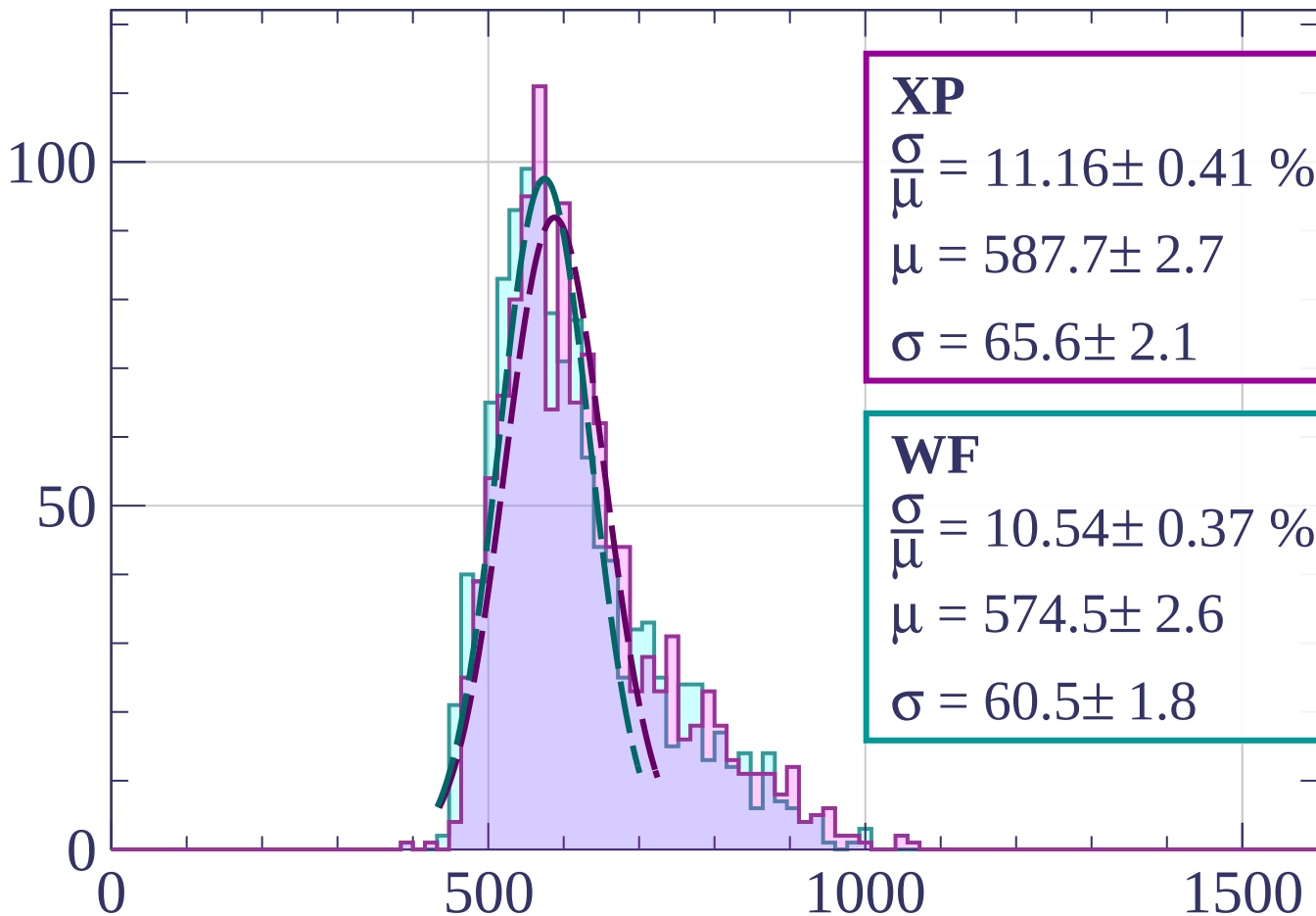
$$\mu = 556.0 \pm 1.8$$

$$\sigma = 42.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 11.16 \pm 0.41 \%$$

$$\mu = 587.7 \pm 2.7$$

$$\sigma = 65.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.37 \%$$

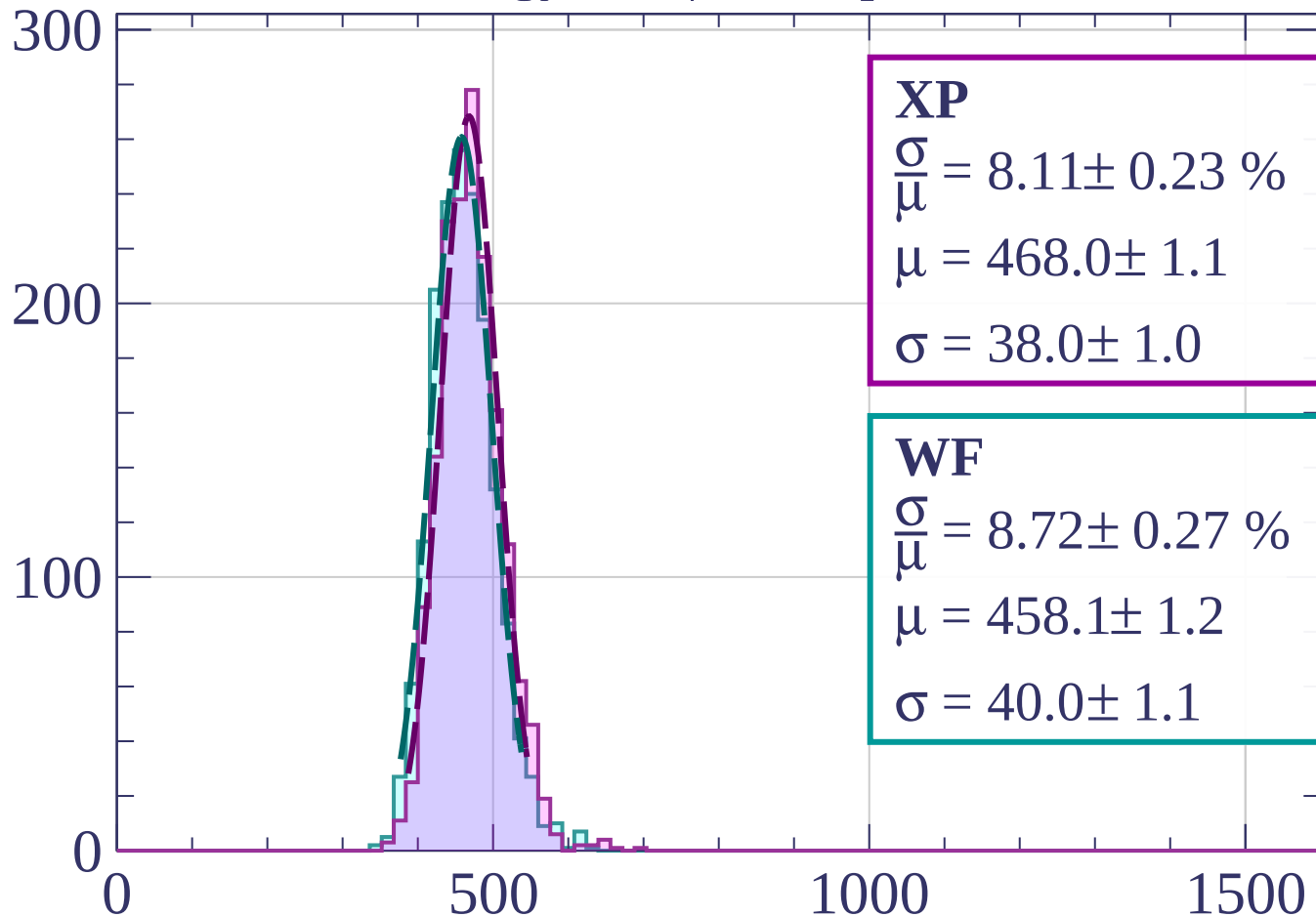
$$\mu = 574.5 \pm 2.6$$

$$\sigma = 60.5 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 8.11 \pm 0.23 \%$$

$$\mu = 468.0 \pm 1.1$$

$$\sigma = 38.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.72 \pm 0.27 \%$$

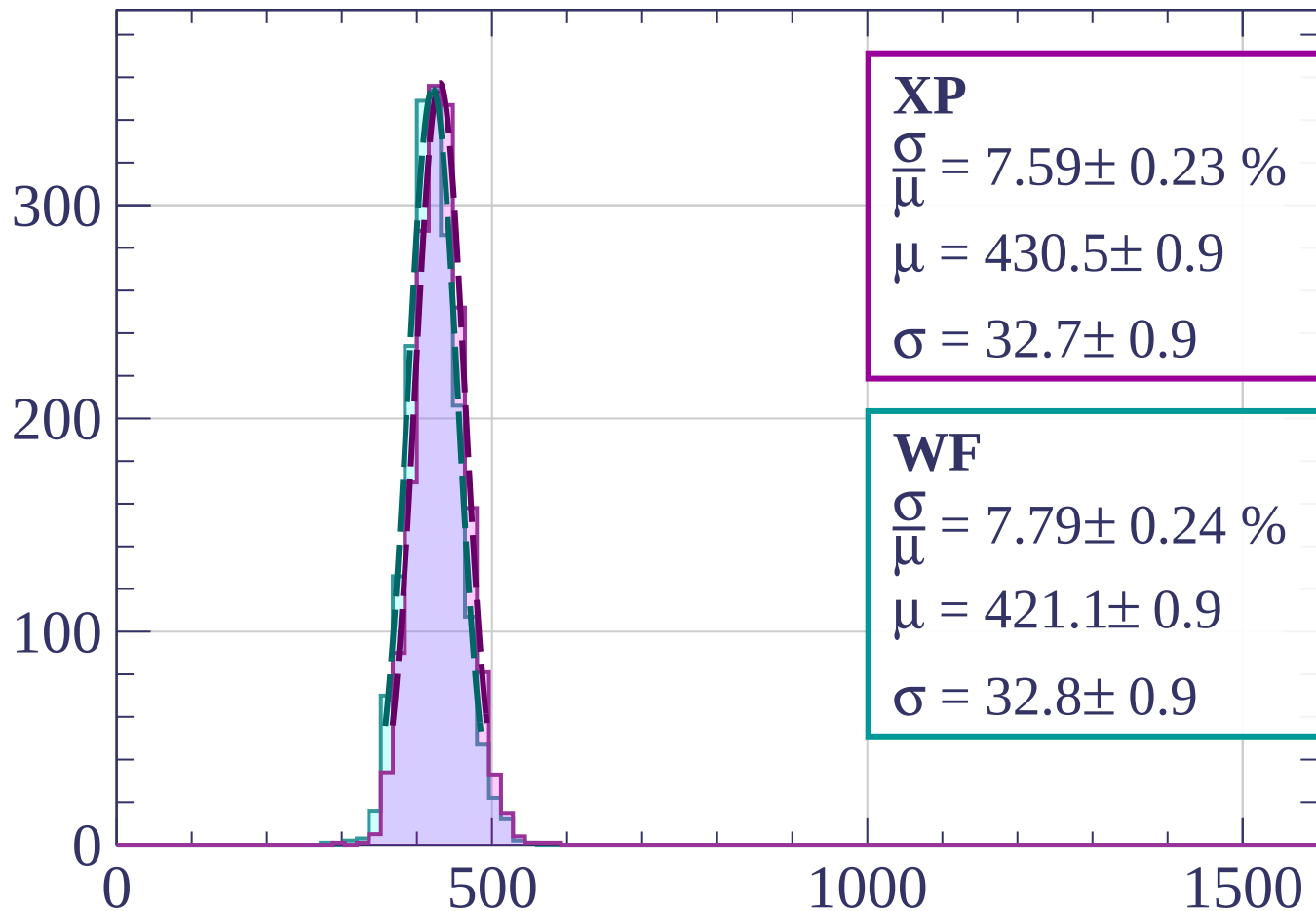
$$\mu = 458.1 \pm 1.2$$

$$\sigma = 40.0 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 7.59 \pm 0.23 \%$$

$$\mu = 430.5 \pm 0.9$$

$$\sigma = 32.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.24 \%$$

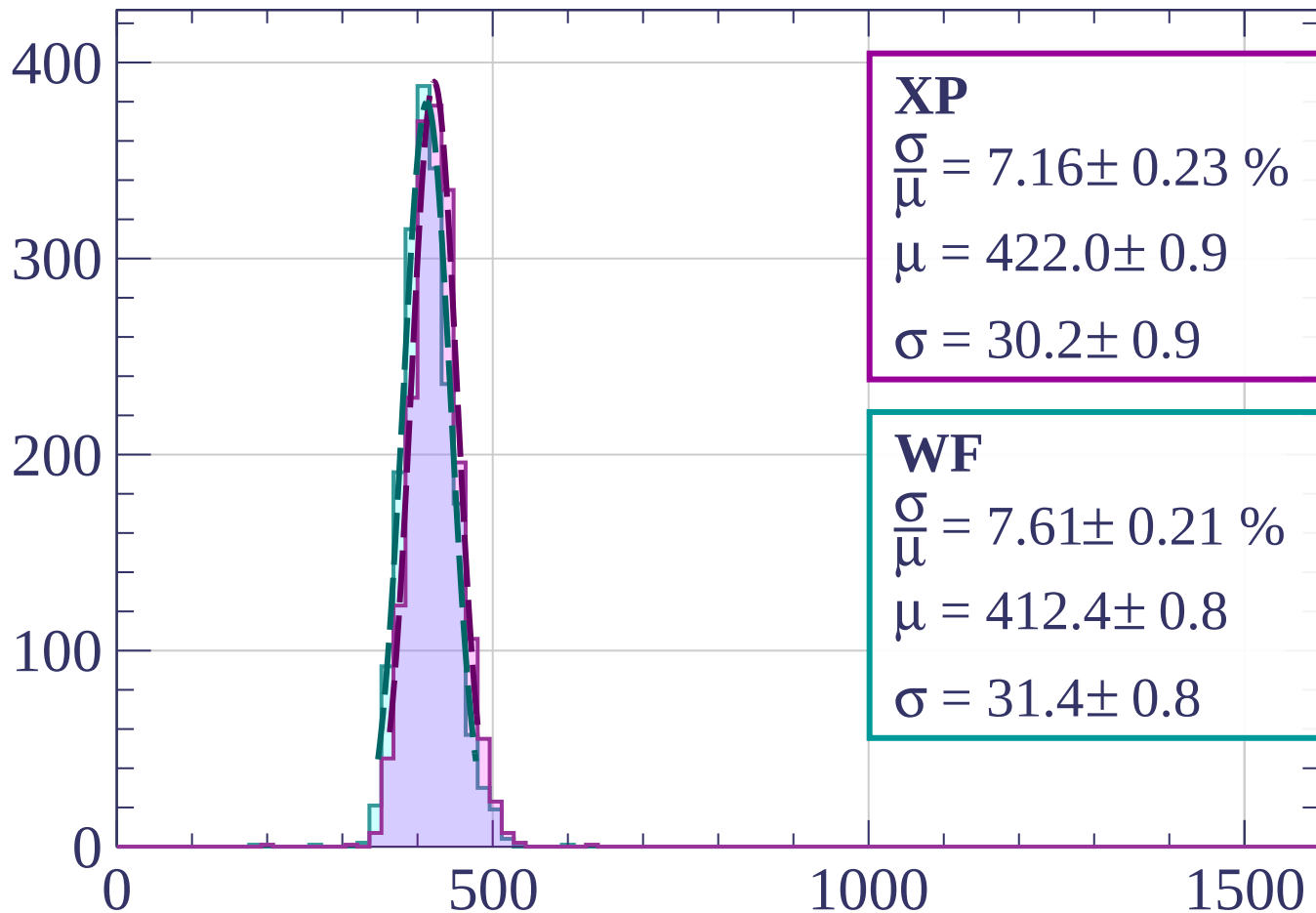
$$\mu = 421.1 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 7.16 \pm 0.23 \%$$

$$\mu = 422.0 \pm 0.9$$

$$\sigma = 30.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.21 \%$$

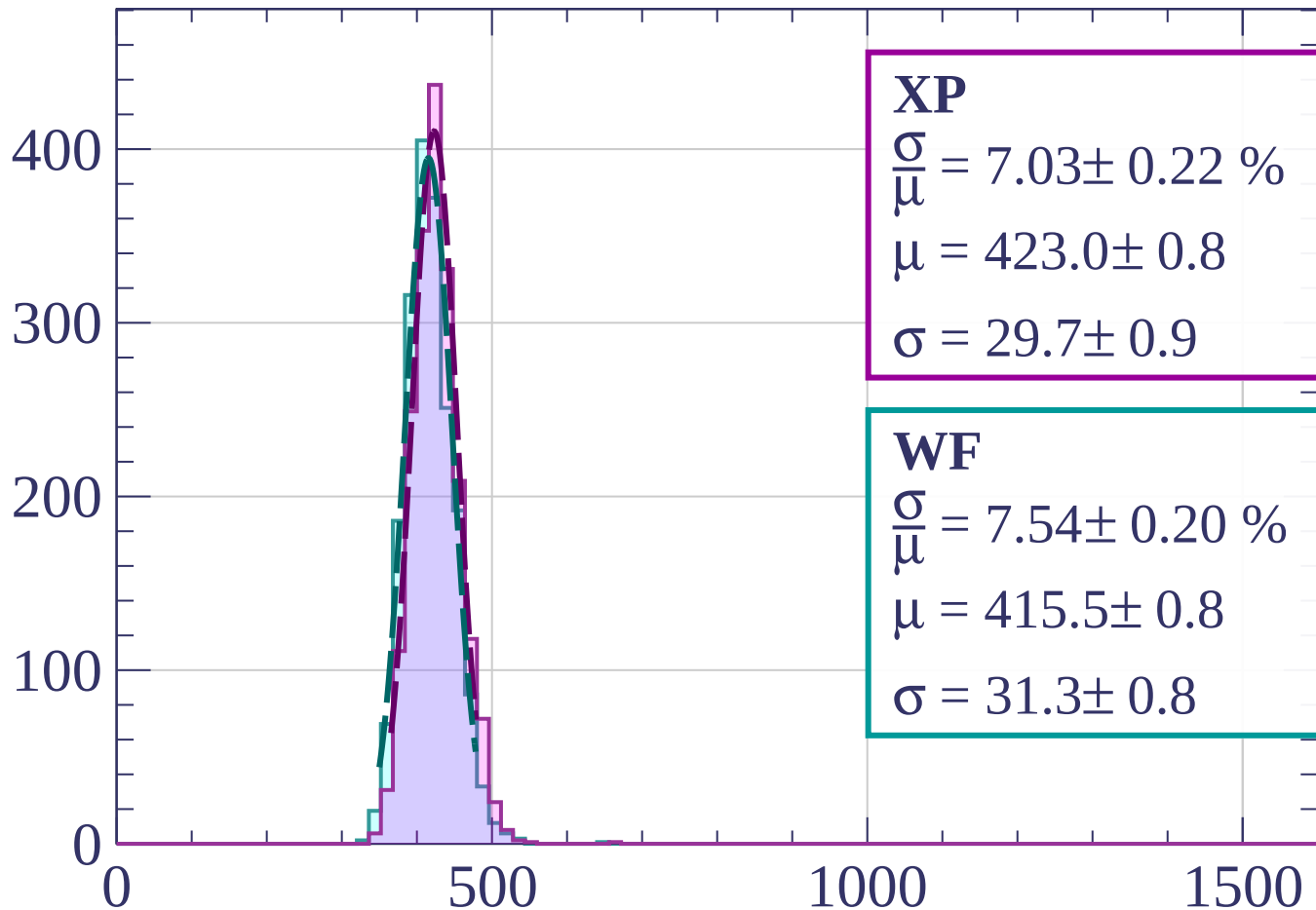
$$\mu = 412.4 \pm 0.8$$

$$\sigma = 31.4 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.22 \%$$

$$\mu = 423.0 \pm 0.8$$

$$\sigma = 29.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.20 \%$$

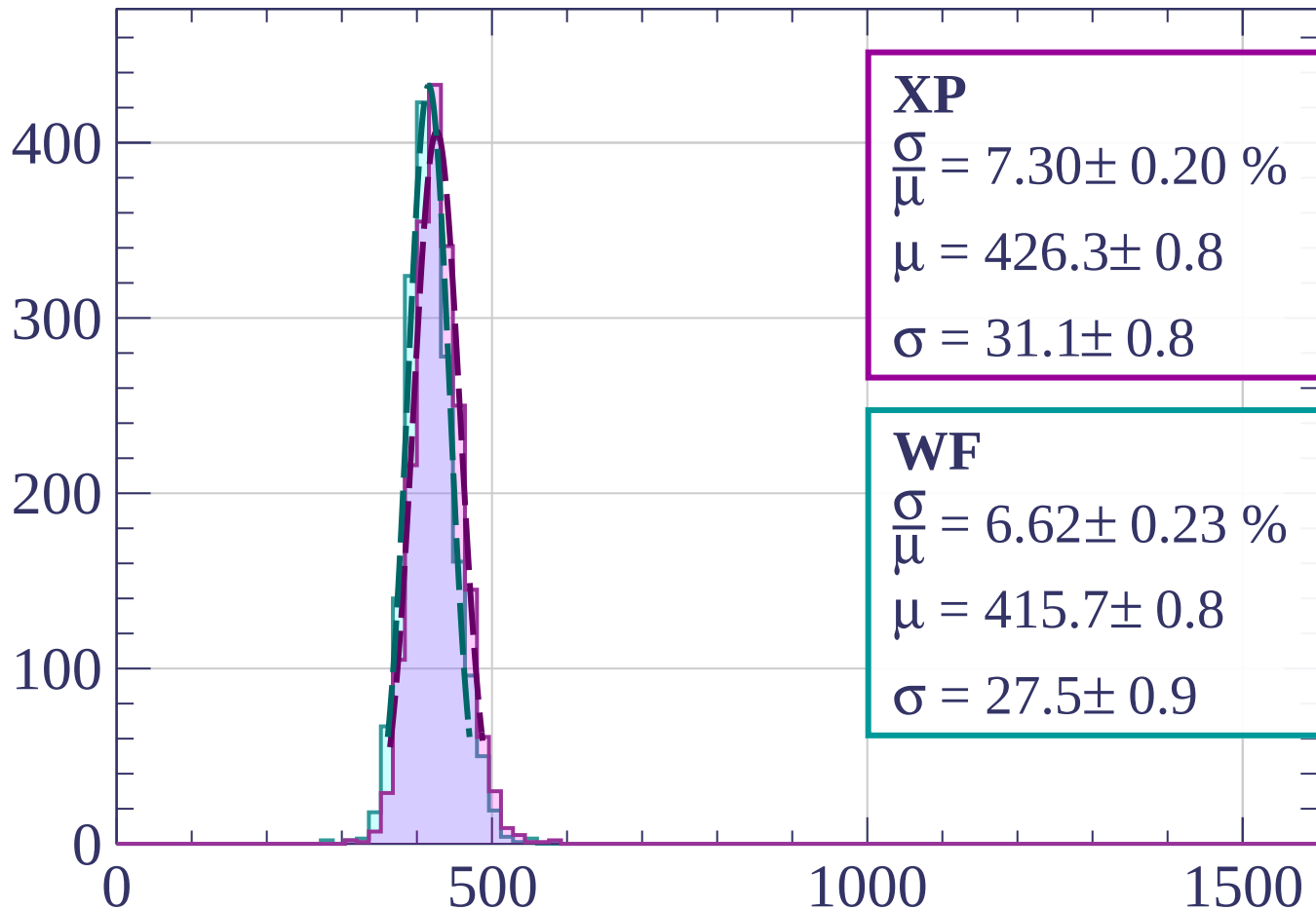
$$\mu = 415.5 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

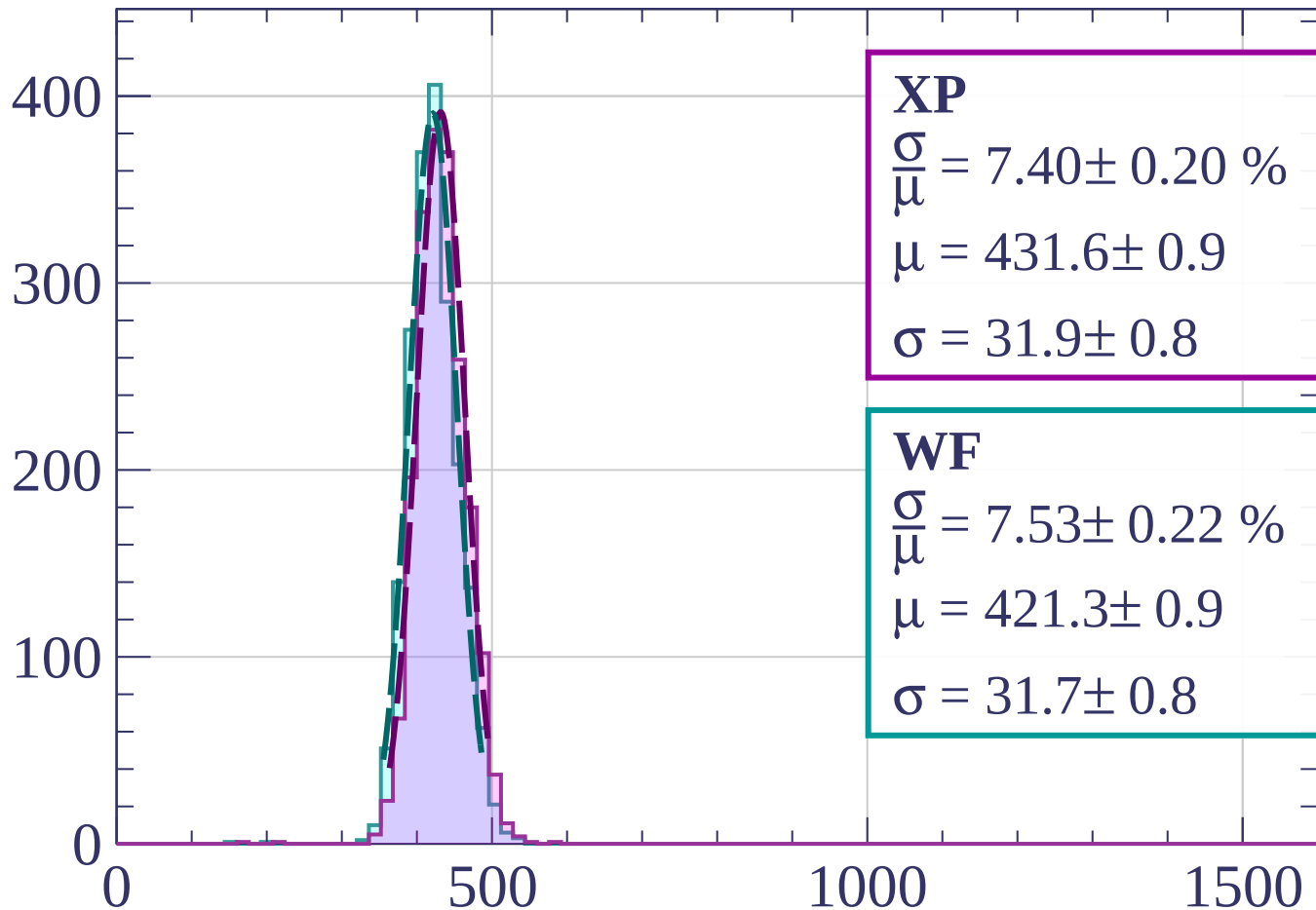
Count



dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.20 \%$$

$$\mu = 431.6 \pm 0.9$$

$$\sigma = 31.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.53 \pm 0.22 \%$$

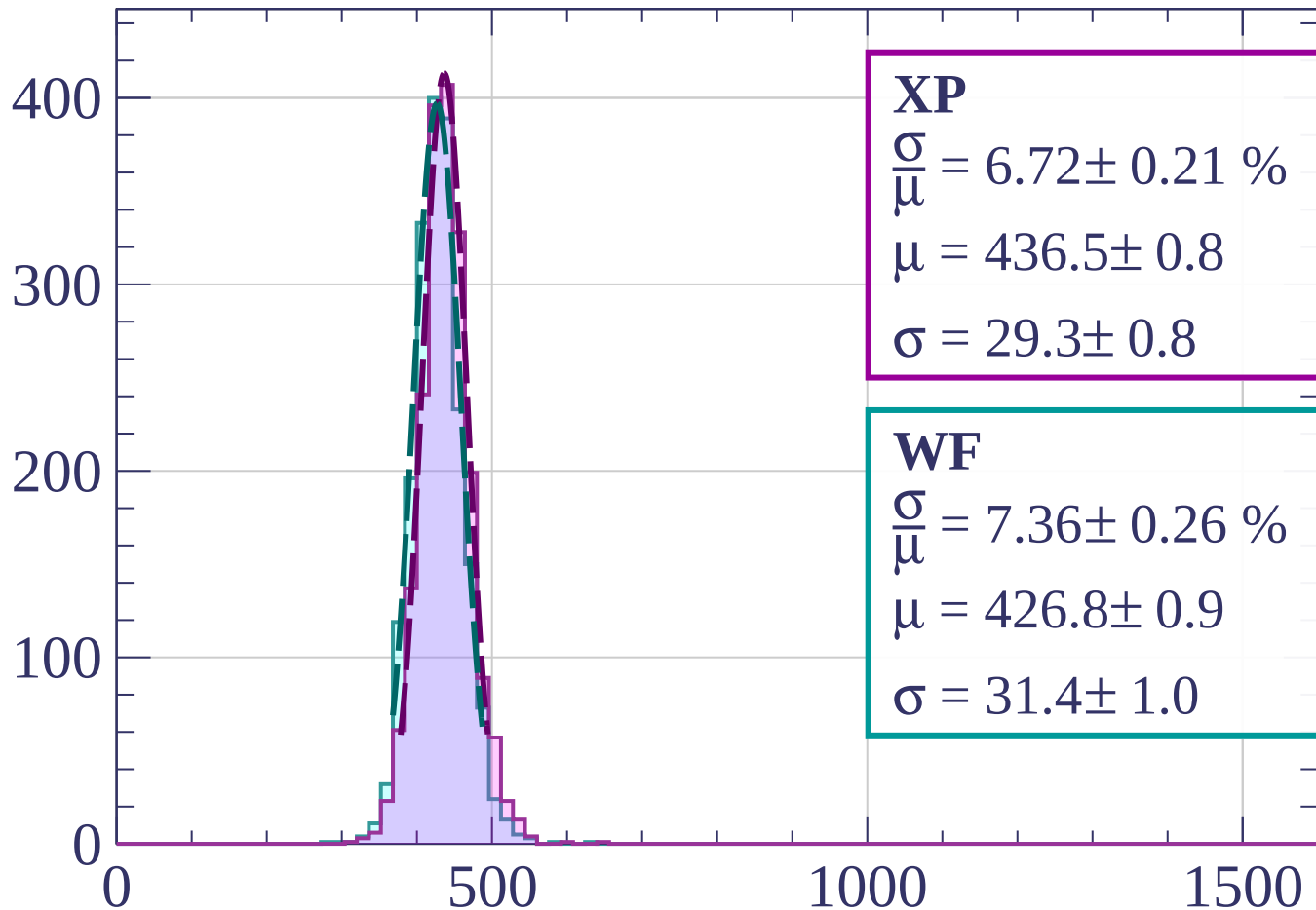
$$\mu = 421.3 \pm 0.9$$

$$\sigma = 31.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

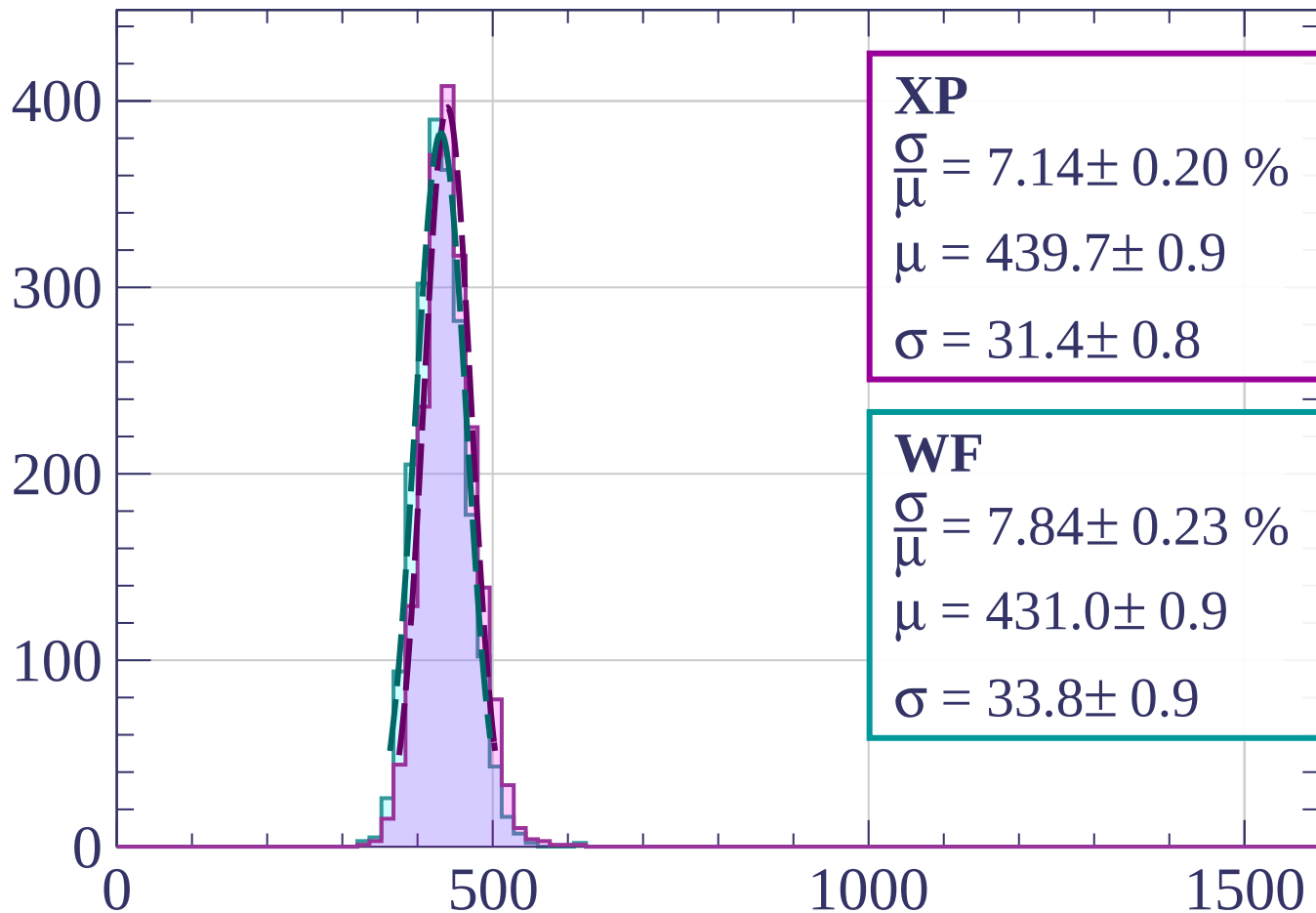
Count



dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.20 \%$$

$$\mu = 439.7 \pm 0.9$$

$$\sigma = 31.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.84 \pm 0.23 \%$$

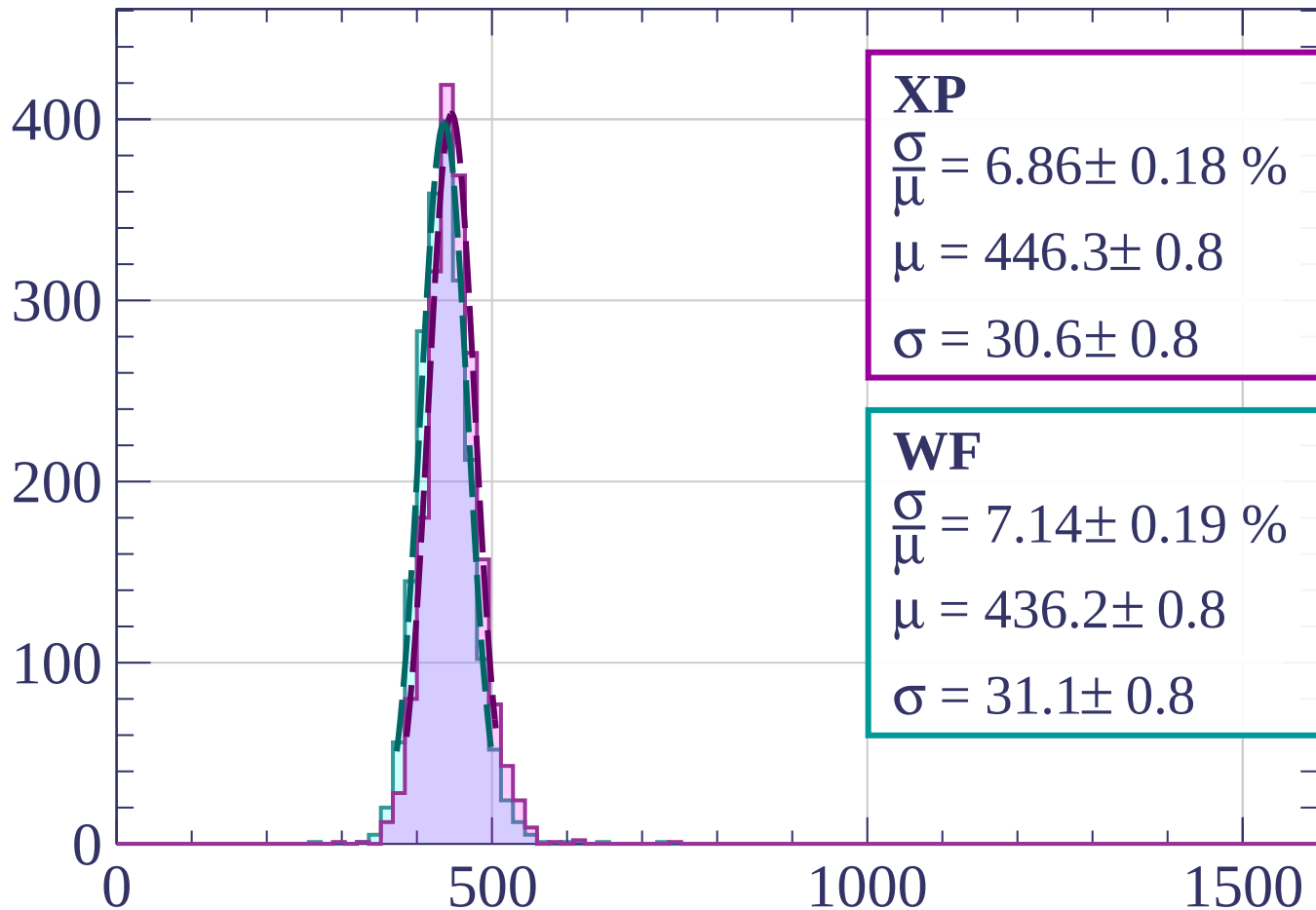
$$\mu = 431.0 \pm 0.9$$

$$\sigma = 33.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $800 < p < 880$

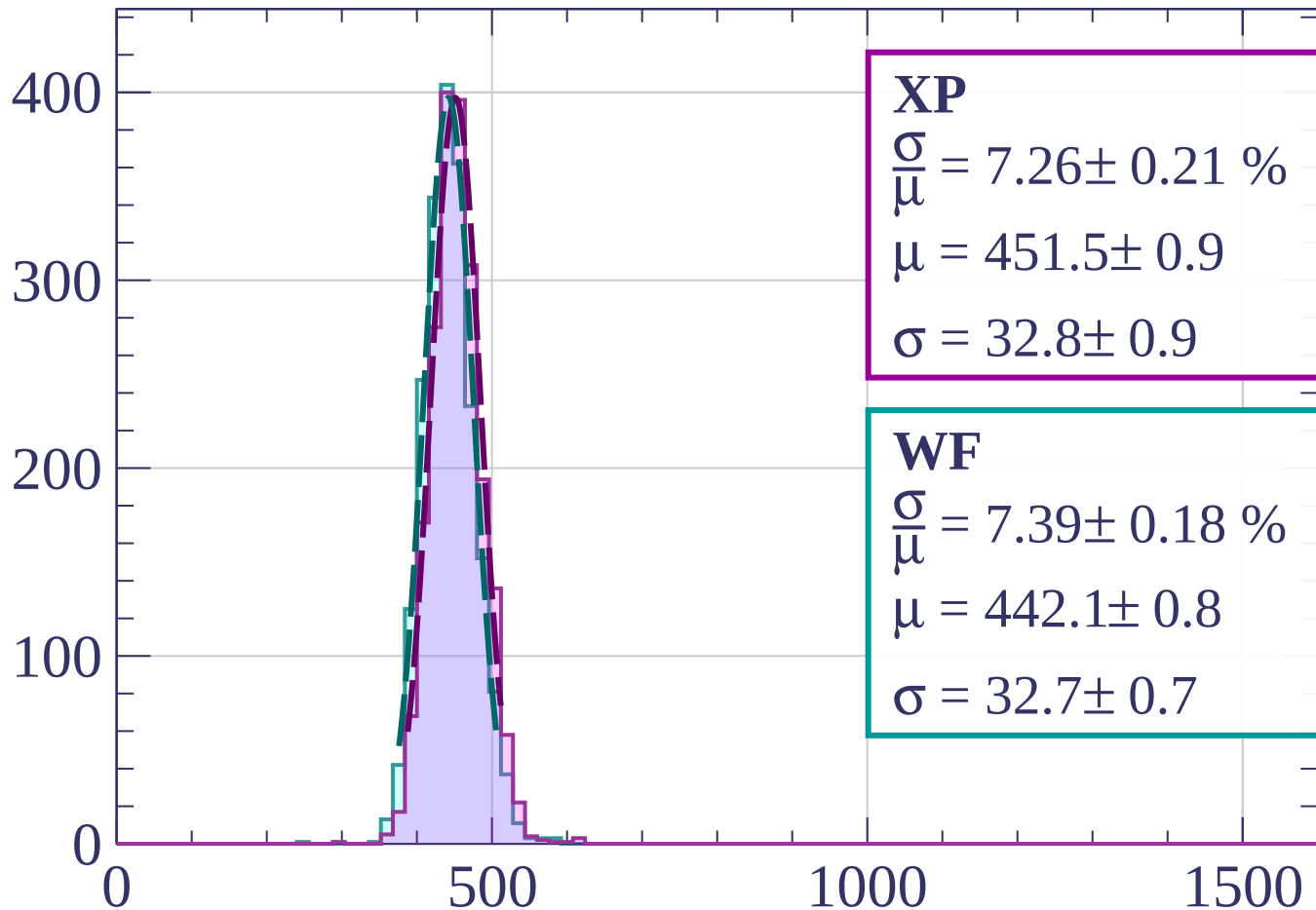
Count



dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.21 \%$$

$$\mu = 451.5 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.18 \%$$

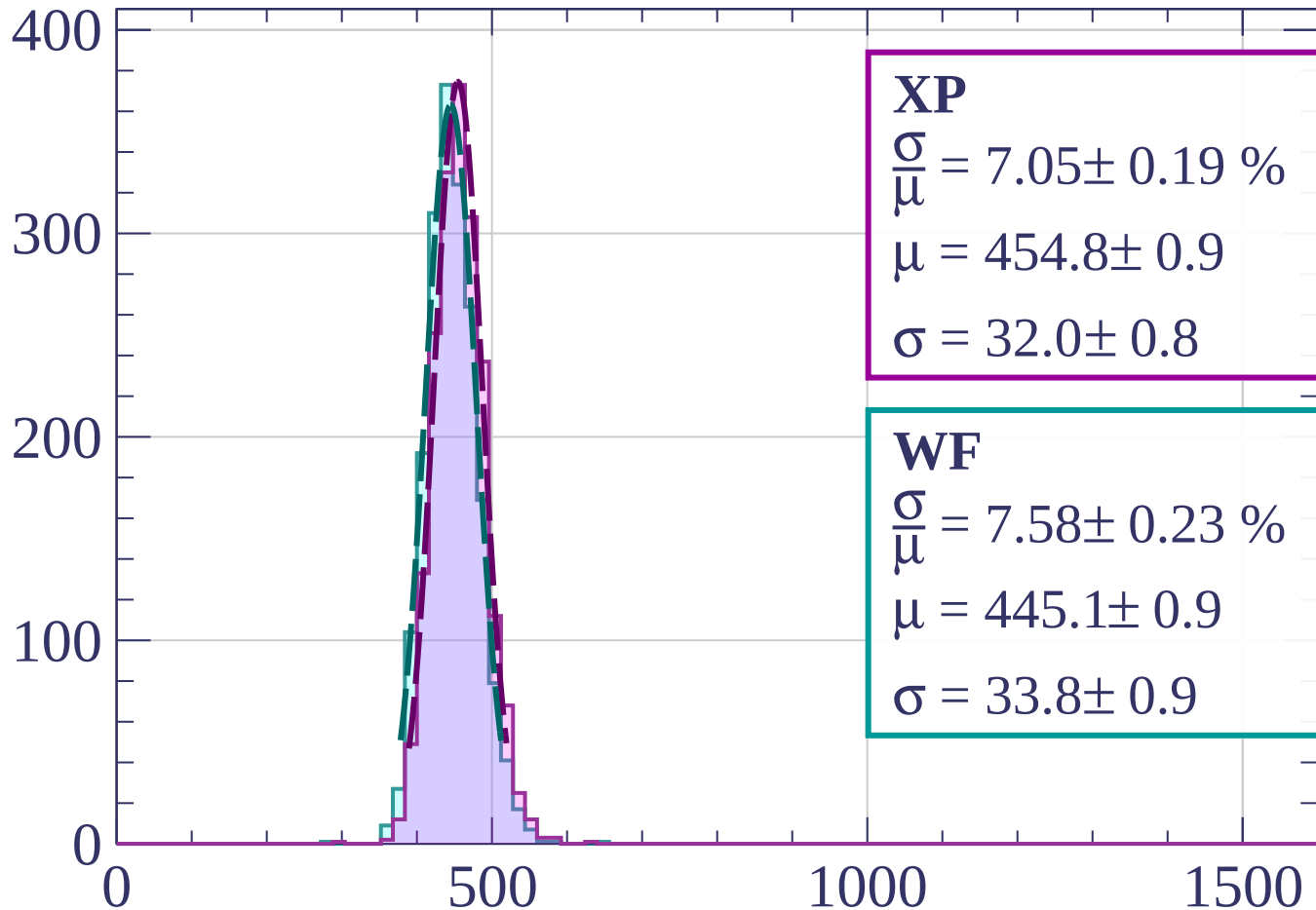
$$\mu = 442.1 \pm 0.8$$

$$\sigma = 32.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 7.05 \pm 0.19 \%$$

$$\mu = 454.8 \pm 0.9$$

$$\sigma = 32.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.58 \pm 0.23 \%$$

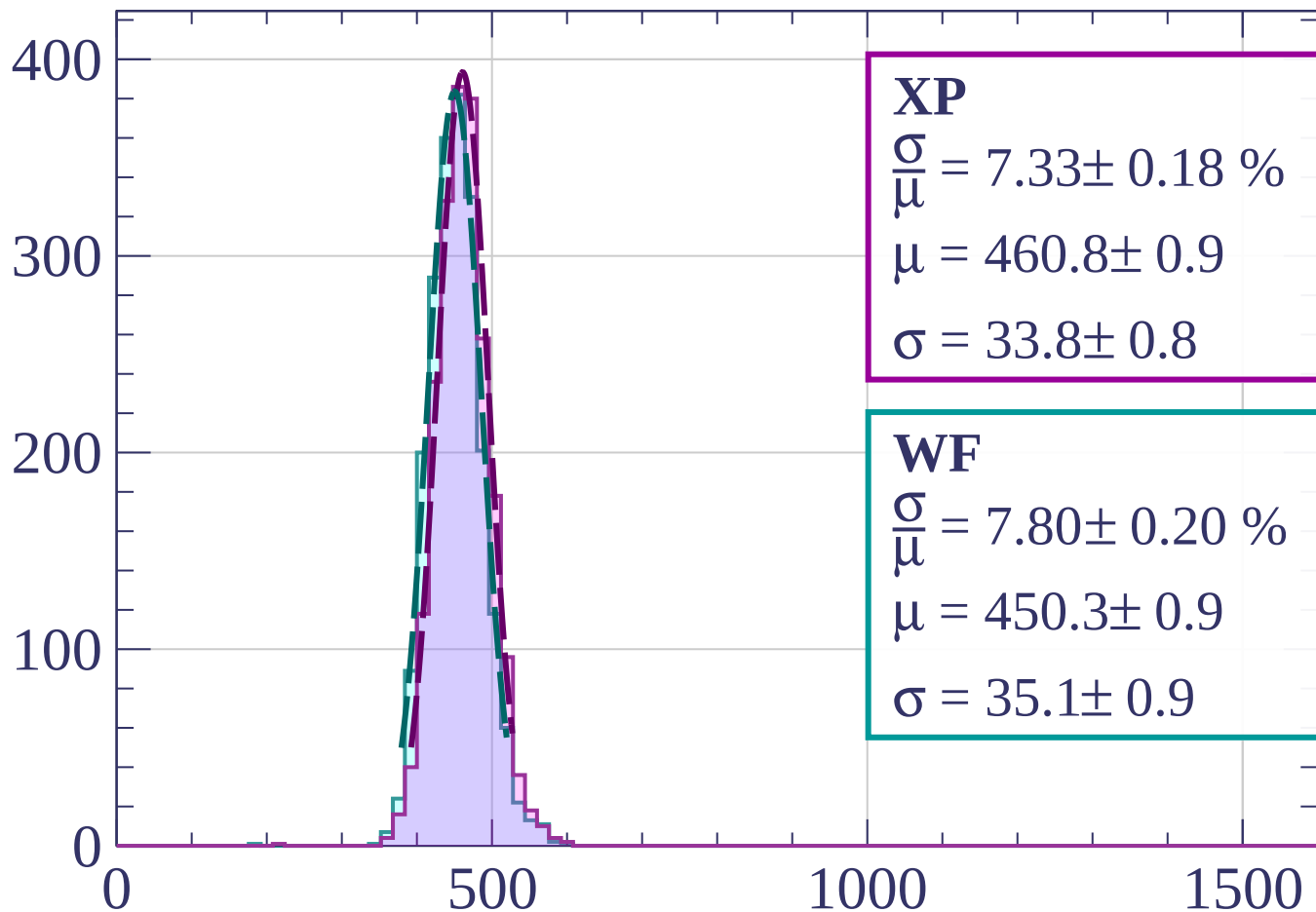
$$\mu = 445.1 \pm 0.9$$

$$\sigma = 33.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.18 \%$$

$$\mu = 460.8 \pm 0.9$$

$$\sigma = 33.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.20 \%$$

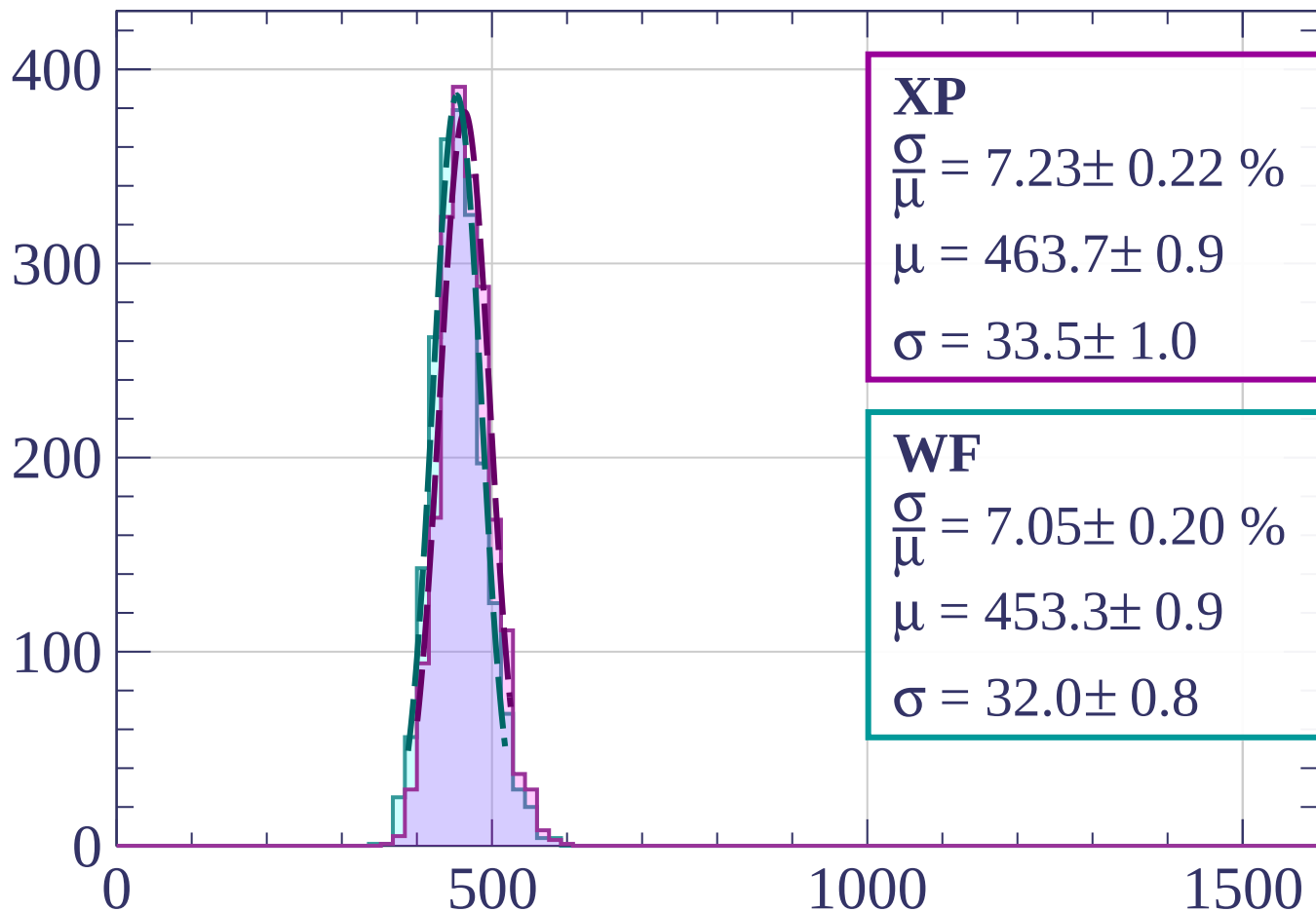
$$\mu = 450.3 \pm 0.9$$

$$\sigma = 35.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

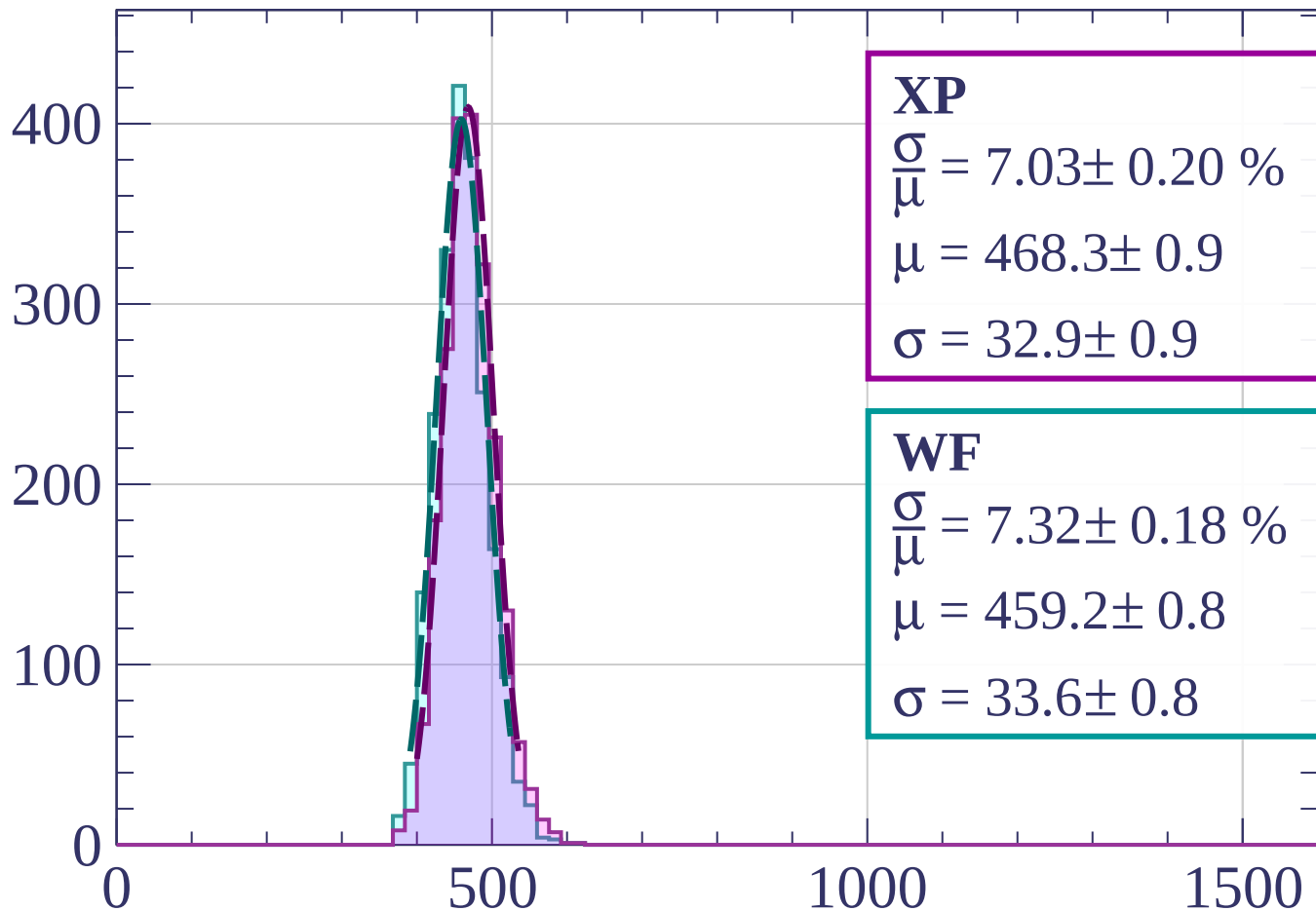
Count



dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.20 \%$$

$$\mu = 468.3 \pm 0.9$$

$$\sigma = 32.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.32 \pm 0.18 \%$$

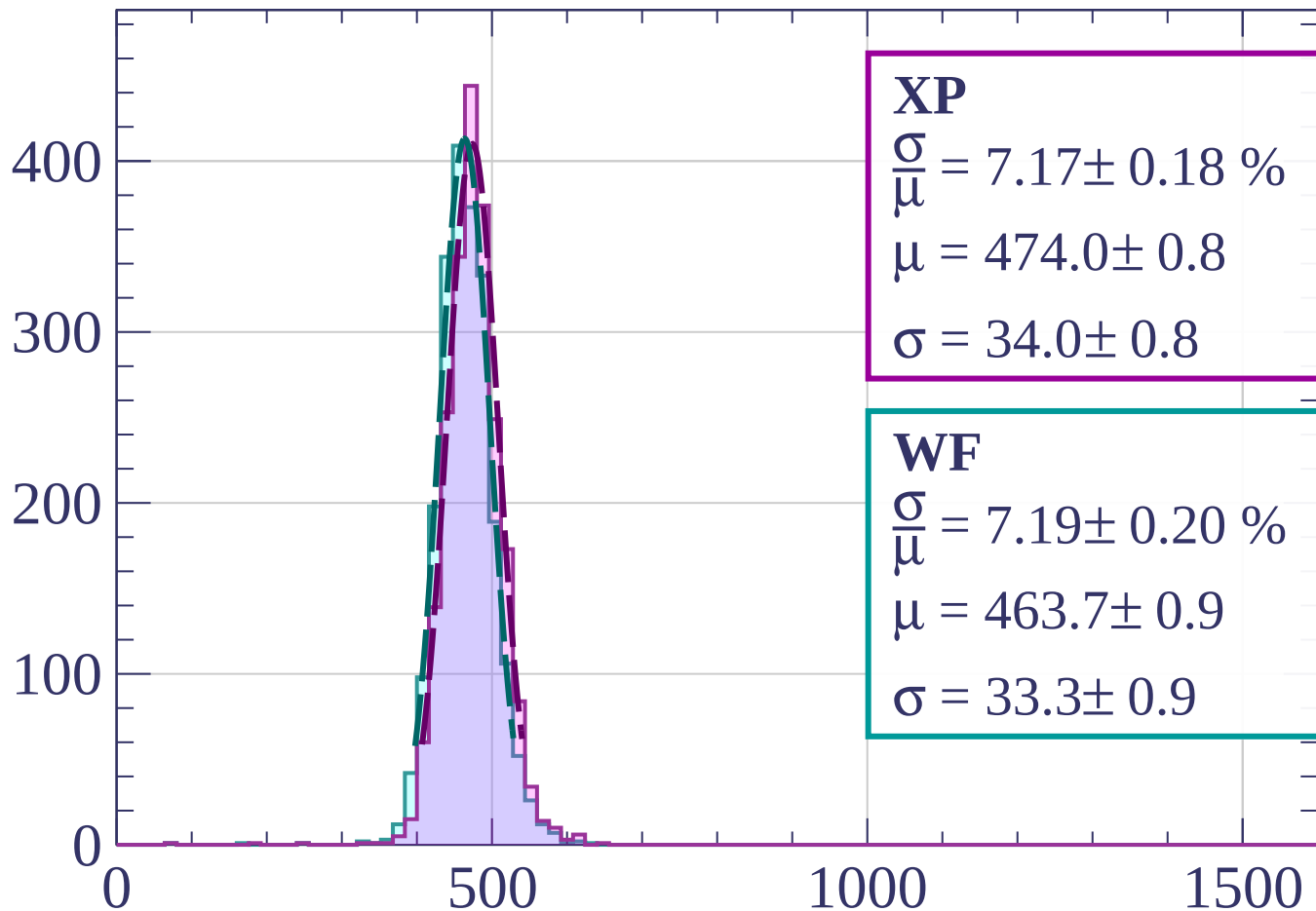
$$\mu = 459.2 \pm 0.8$$

$$\sigma = 33.6 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.17 \pm 0.18 \%$$

$$\mu = 474.0 \pm 0.8$$

$$\sigma = 34.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.19 \pm 0.20 \%$$

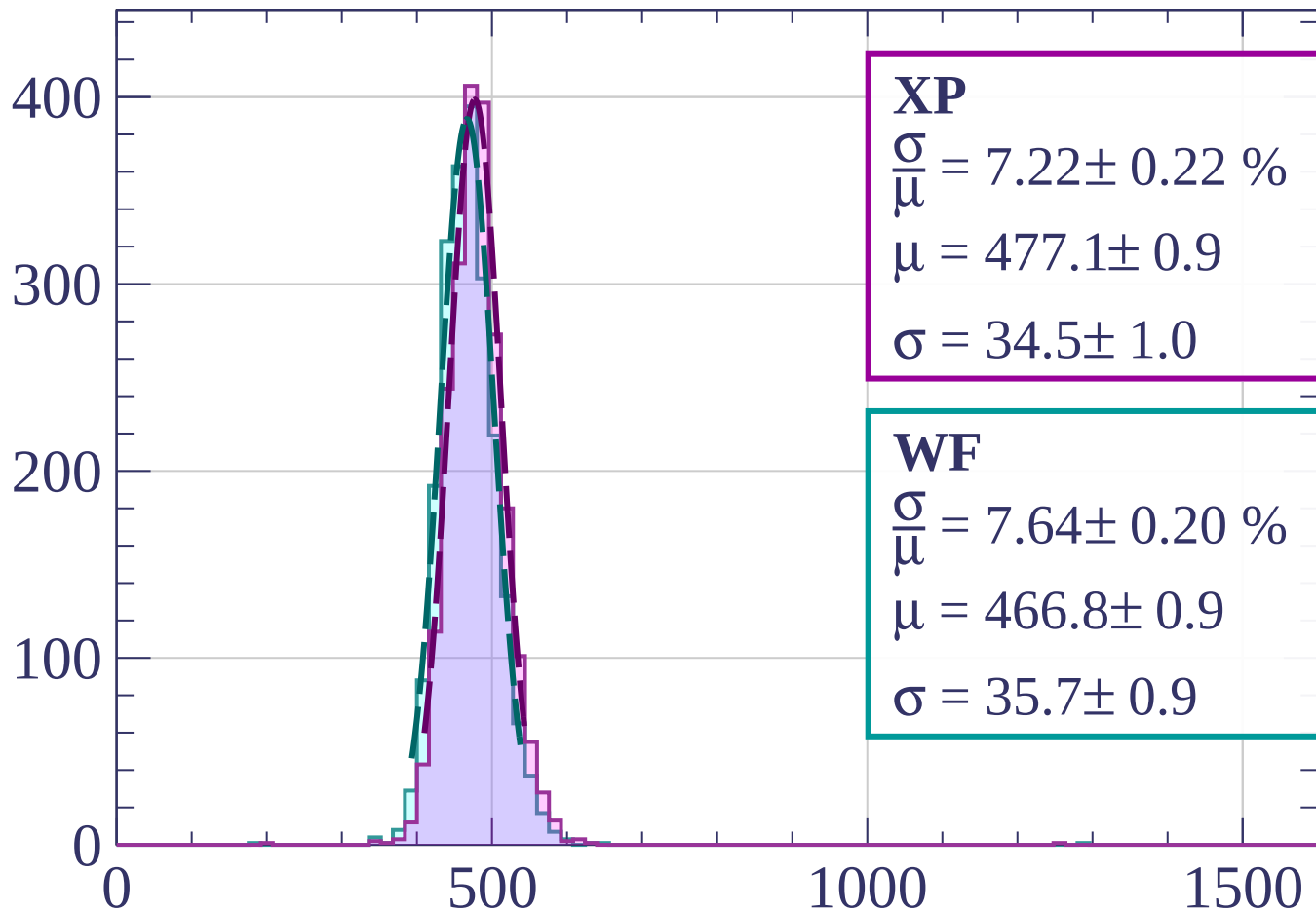
$$\mu = 463.7 \pm 0.9$$

$$\sigma = 33.3 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.22 \pm 0.22 \%$$

$$\mu = 477.1 \pm 0.9$$

$$\sigma = 34.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.64 \pm 0.20 \%$$

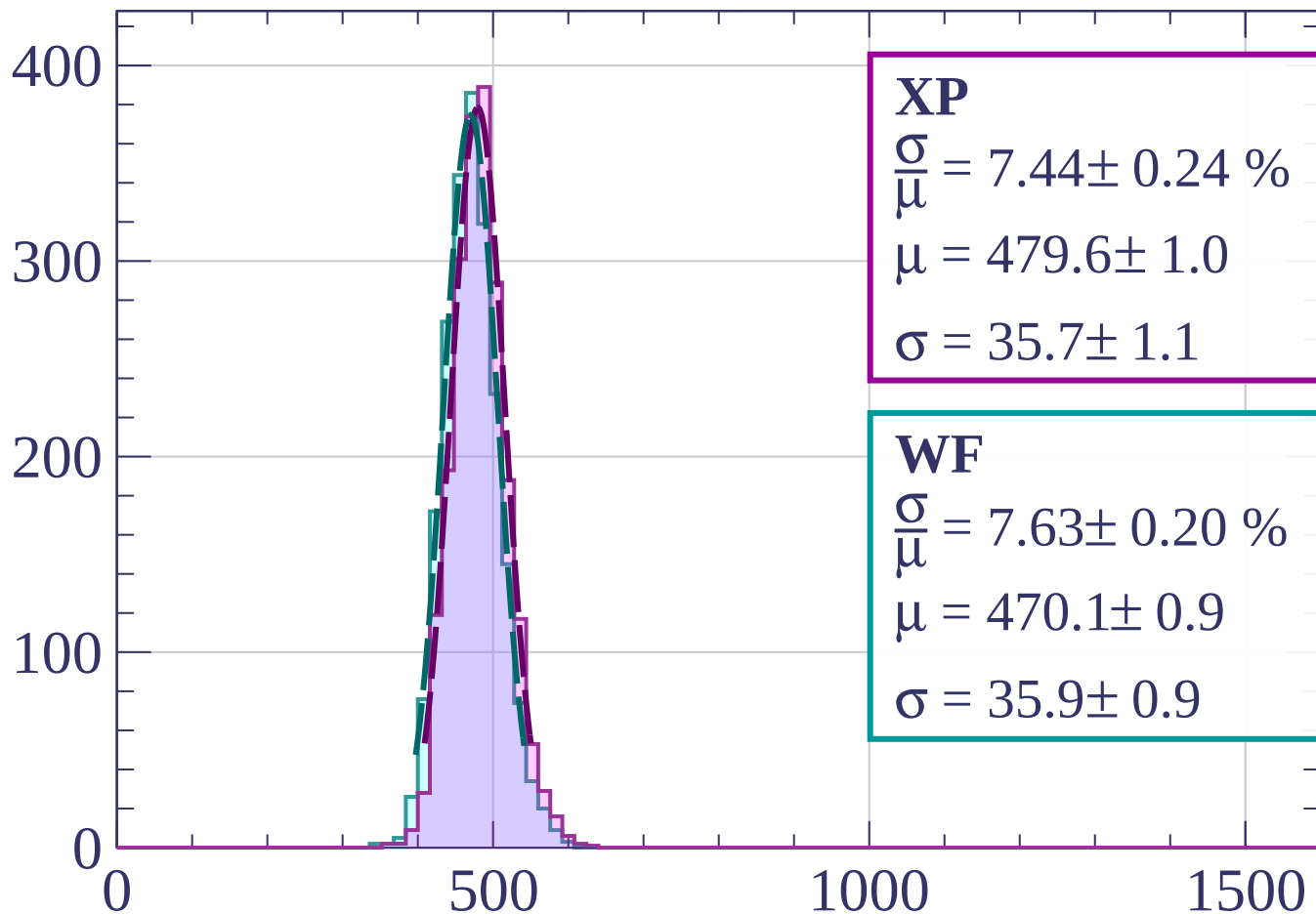
$$\mu = 466.8 \pm 0.9$$

$$\sigma = 35.7 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.24 \%$$

$$\mu = 479.6 \pm 1.0$$

$$\sigma = 35.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.20 \%$$

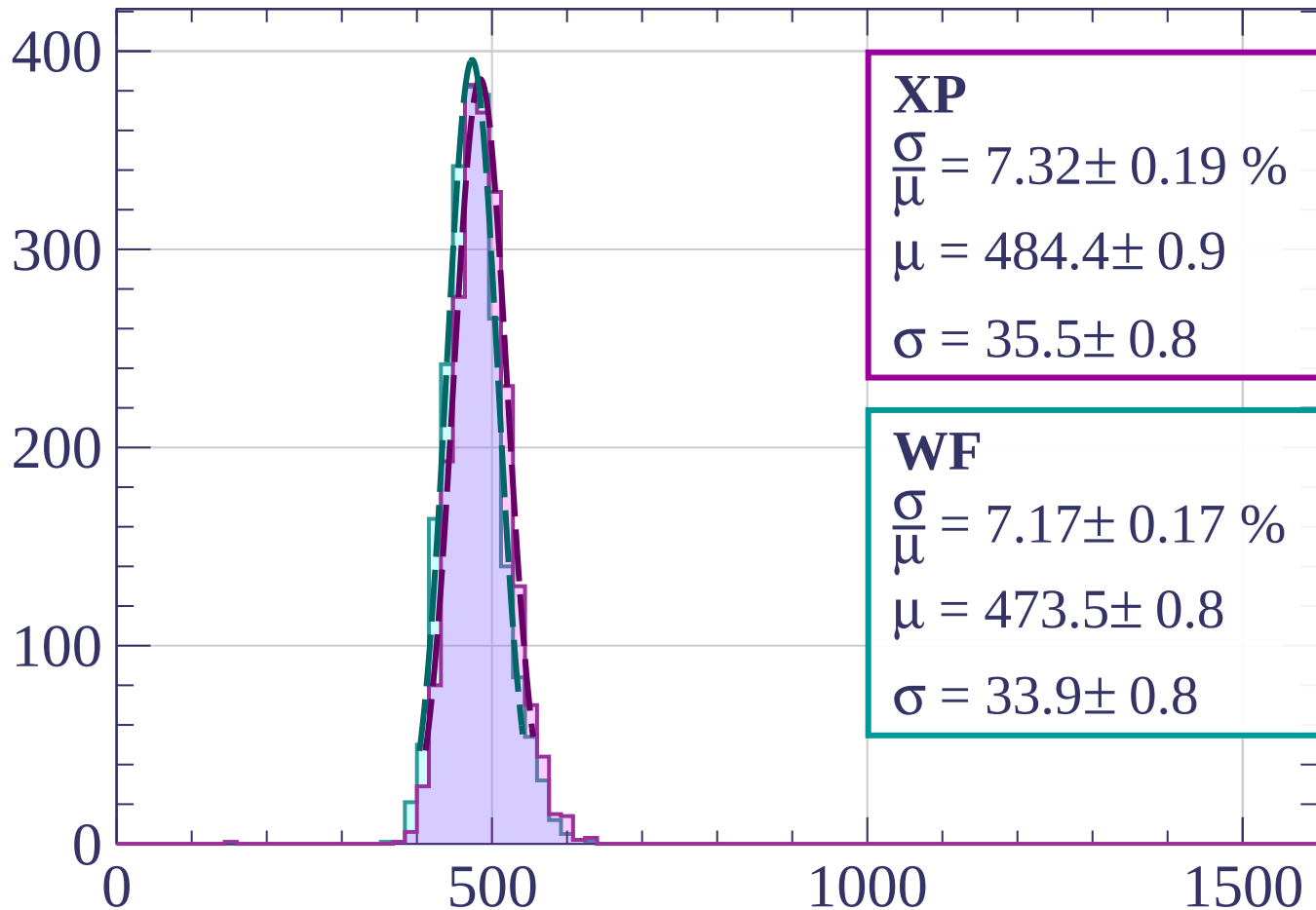
$$\mu = 470.1 \pm 0.9$$

$$\sigma = 35.9 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.19 \%$$

$$\mu = 484.4 \pm 0.9$$

$$\sigma = 35.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.17 \pm 0.17 \%$$

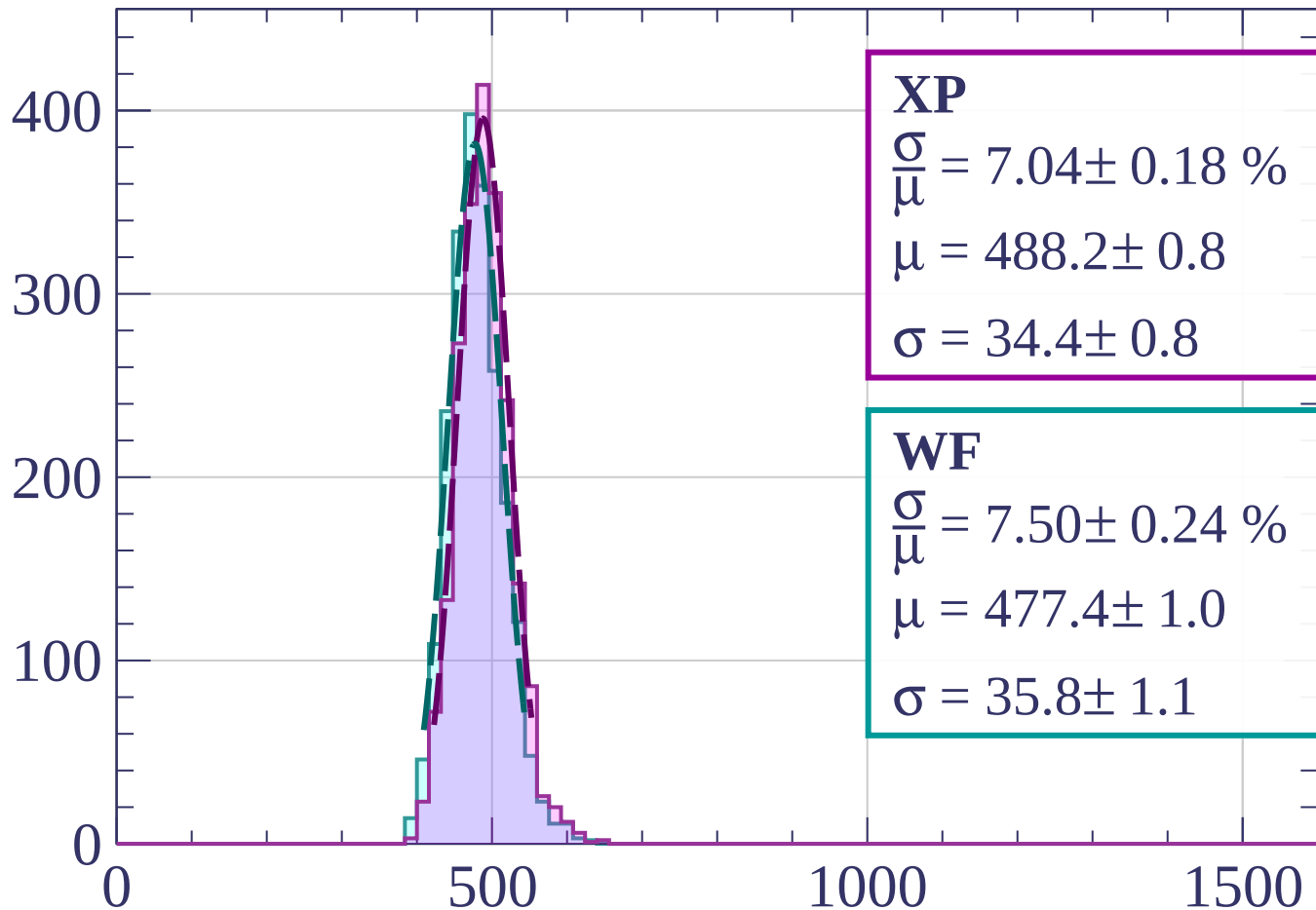
$$\mu = 473.5 \pm 0.8$$

$$\sigma = 33.9 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.18 \%$$

$$\mu = 488.2 \pm 0.8$$

$$\sigma = 34.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.50 \pm 0.24 \%$$

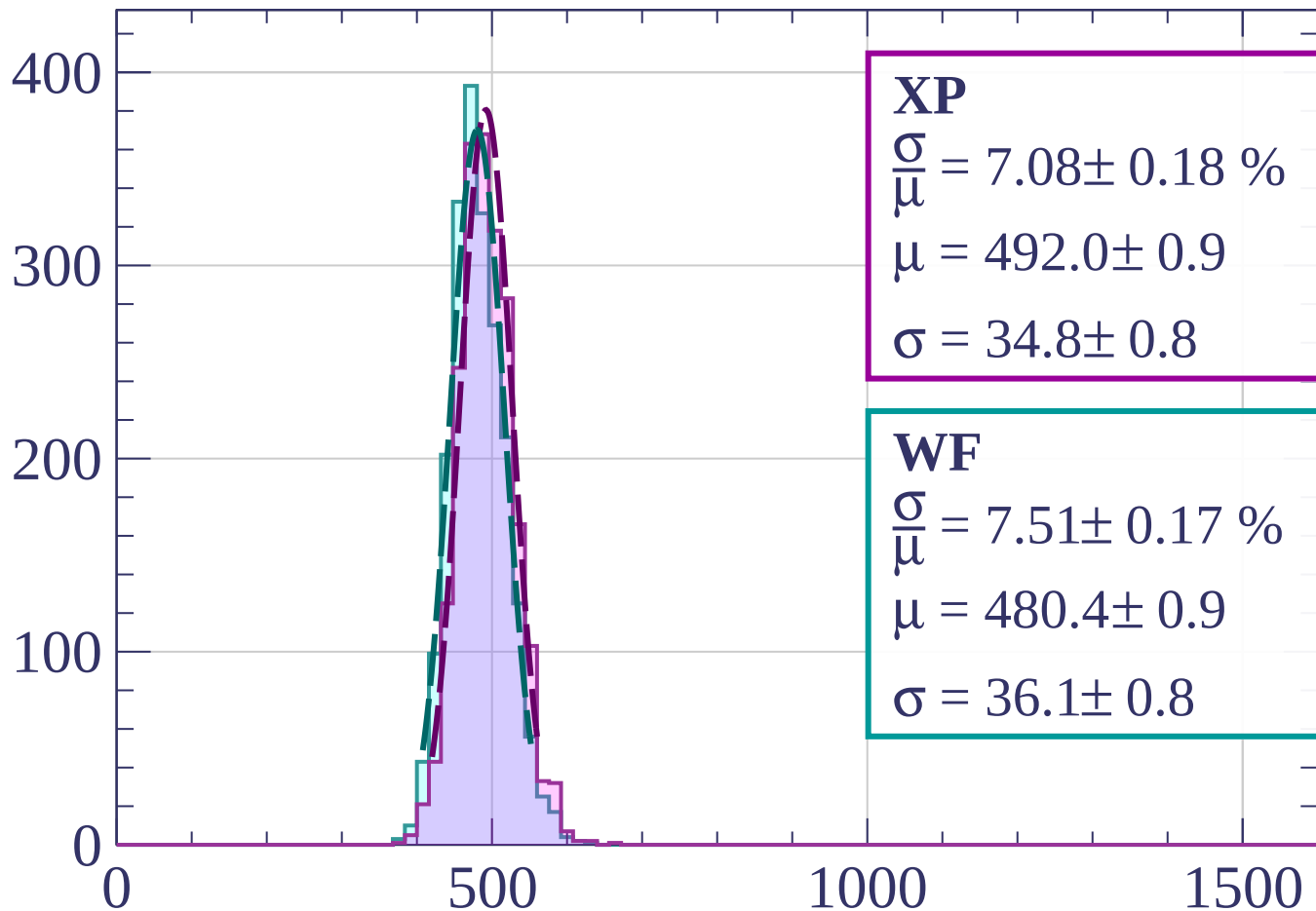
$$\mu = 477.4 \pm 1.0$$

$$\sigma = 35.8 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

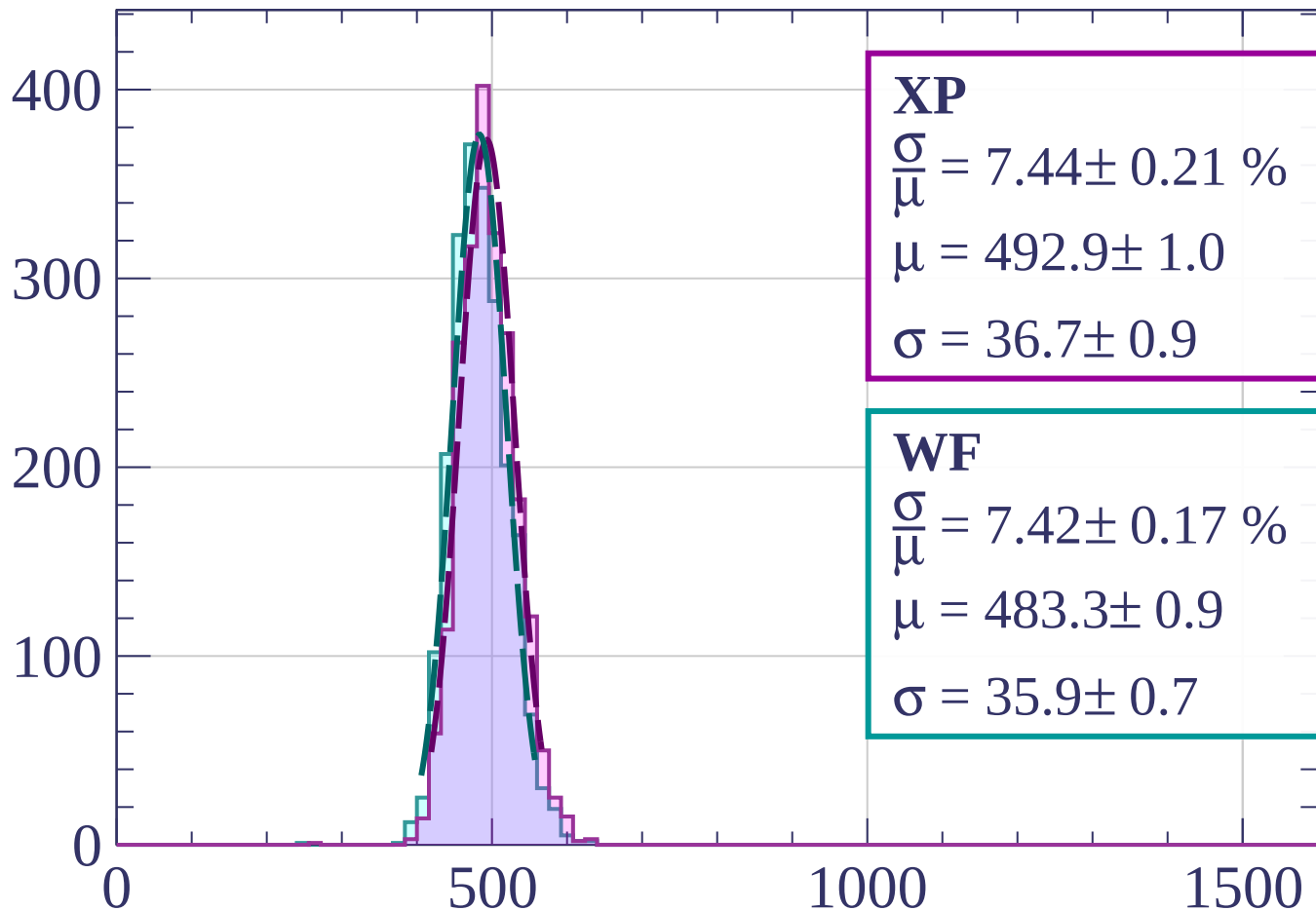
Count



dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.21 \%$$

$$\mu = 492.9 \pm 1.0$$

$$\sigma = 36.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.42 \pm 0.17 \%$$

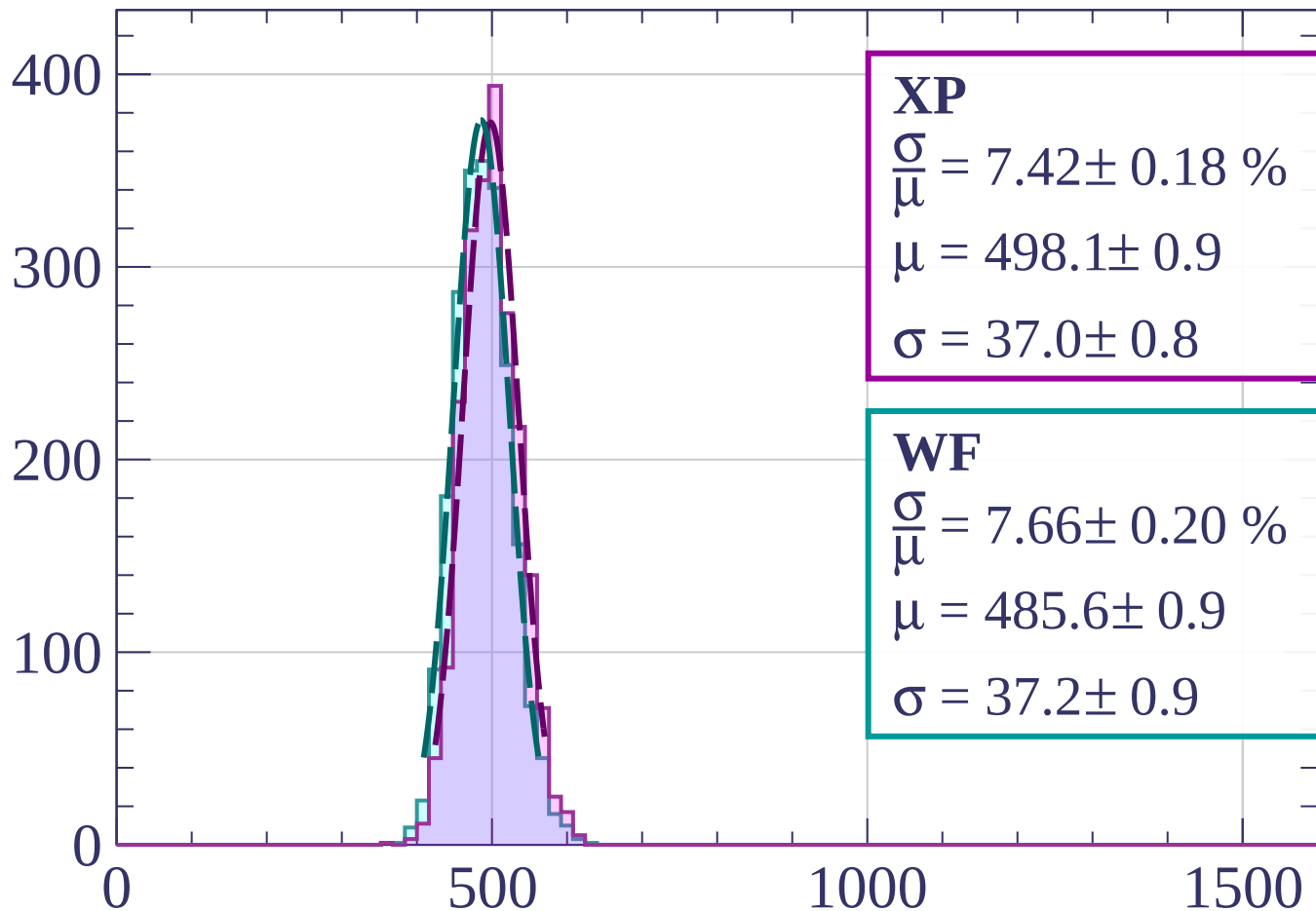
$$\mu = 483.3 \pm 0.9$$

$$\sigma = 35.9 \pm 0.7$$

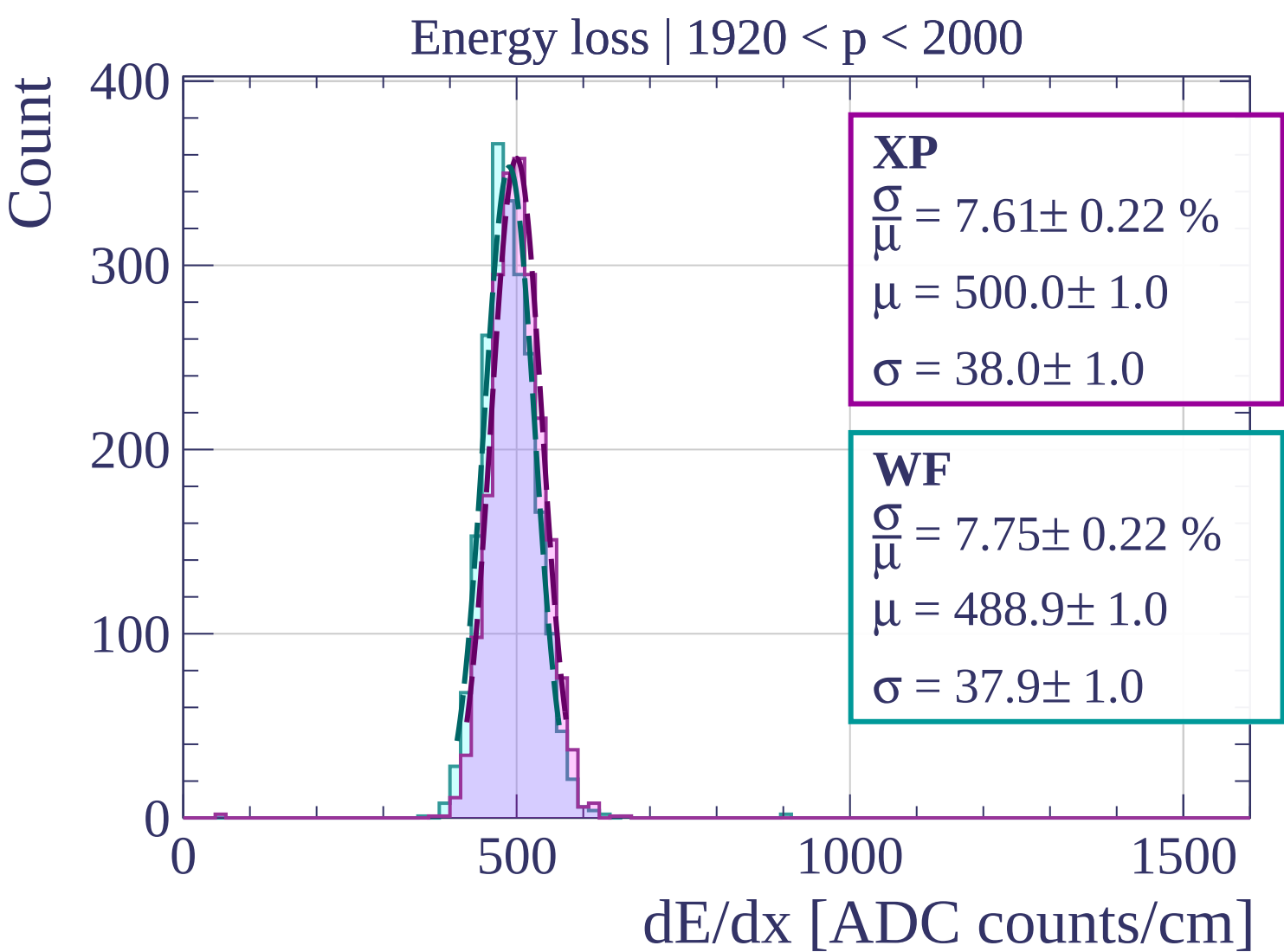
dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count

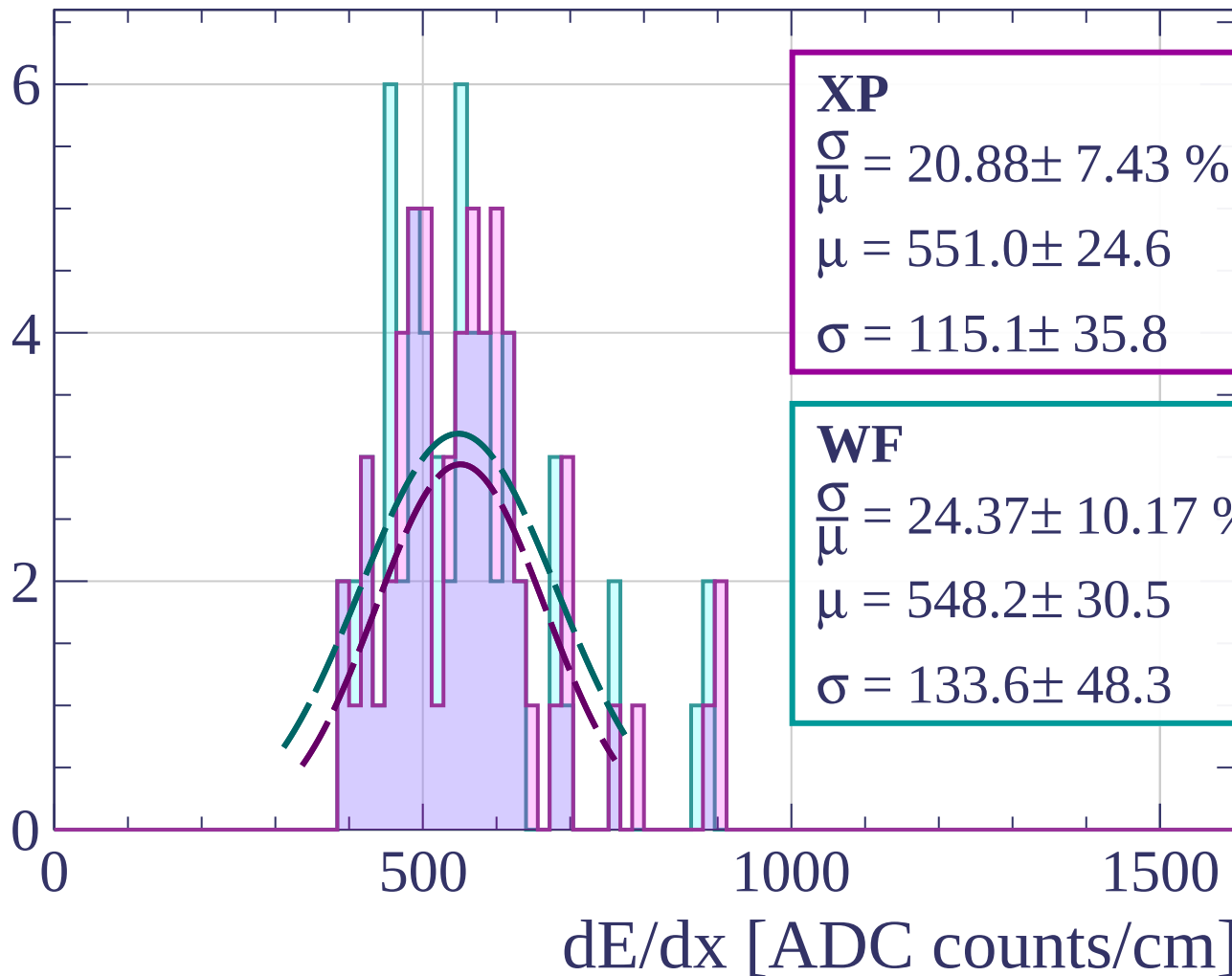


dE/dx [ADC counts/cm]



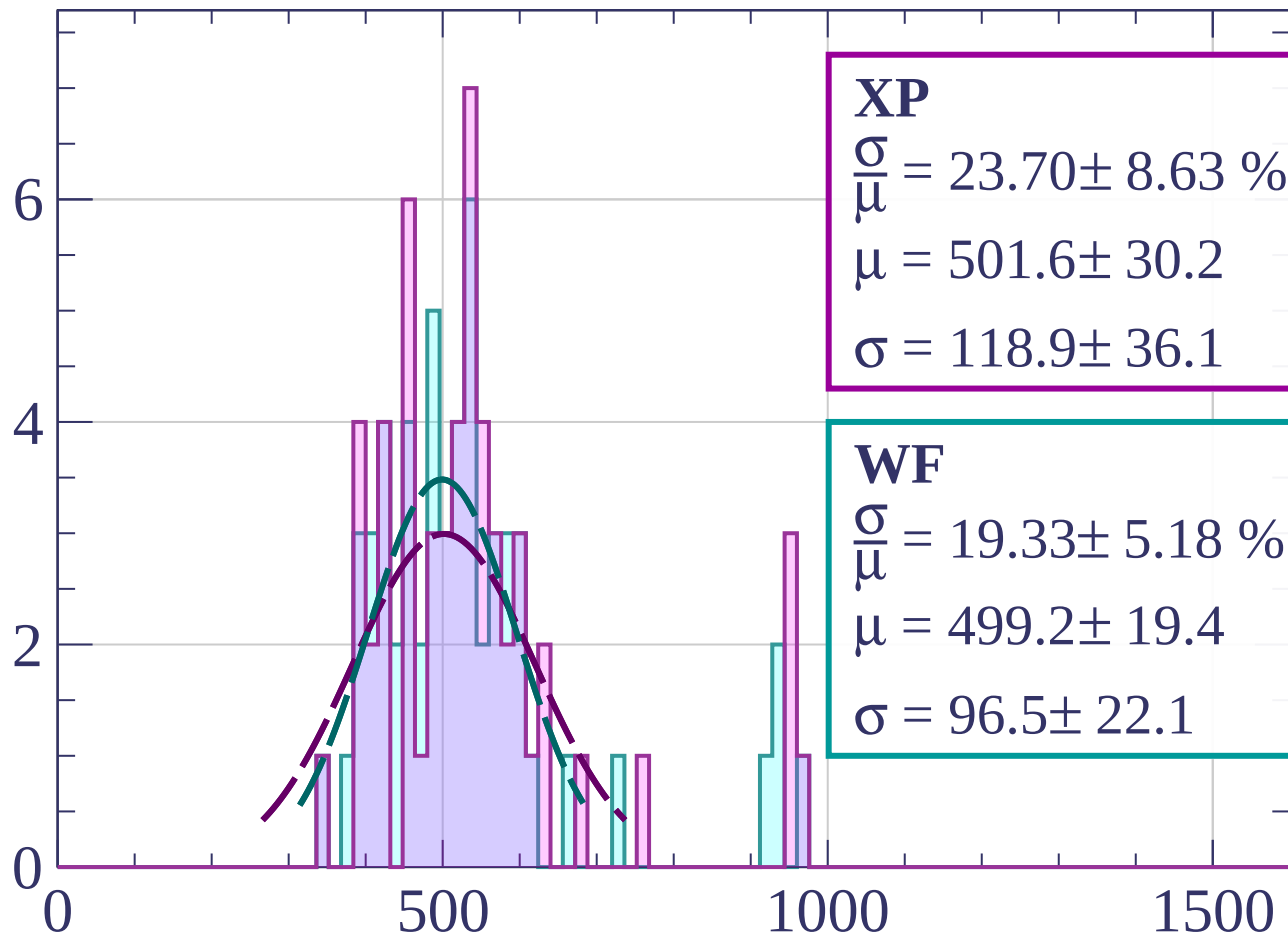
Count

Energy loss | $0 < \phi < 3$



Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 23.70 \pm 8.63 \%$$

$$\mu = 501.6 \pm 30.2$$

$$\sigma = 118.9 \pm 36.1$$

WF

$$\frac{\sigma}{\mu} = 19.33 \pm 5.18 \%$$

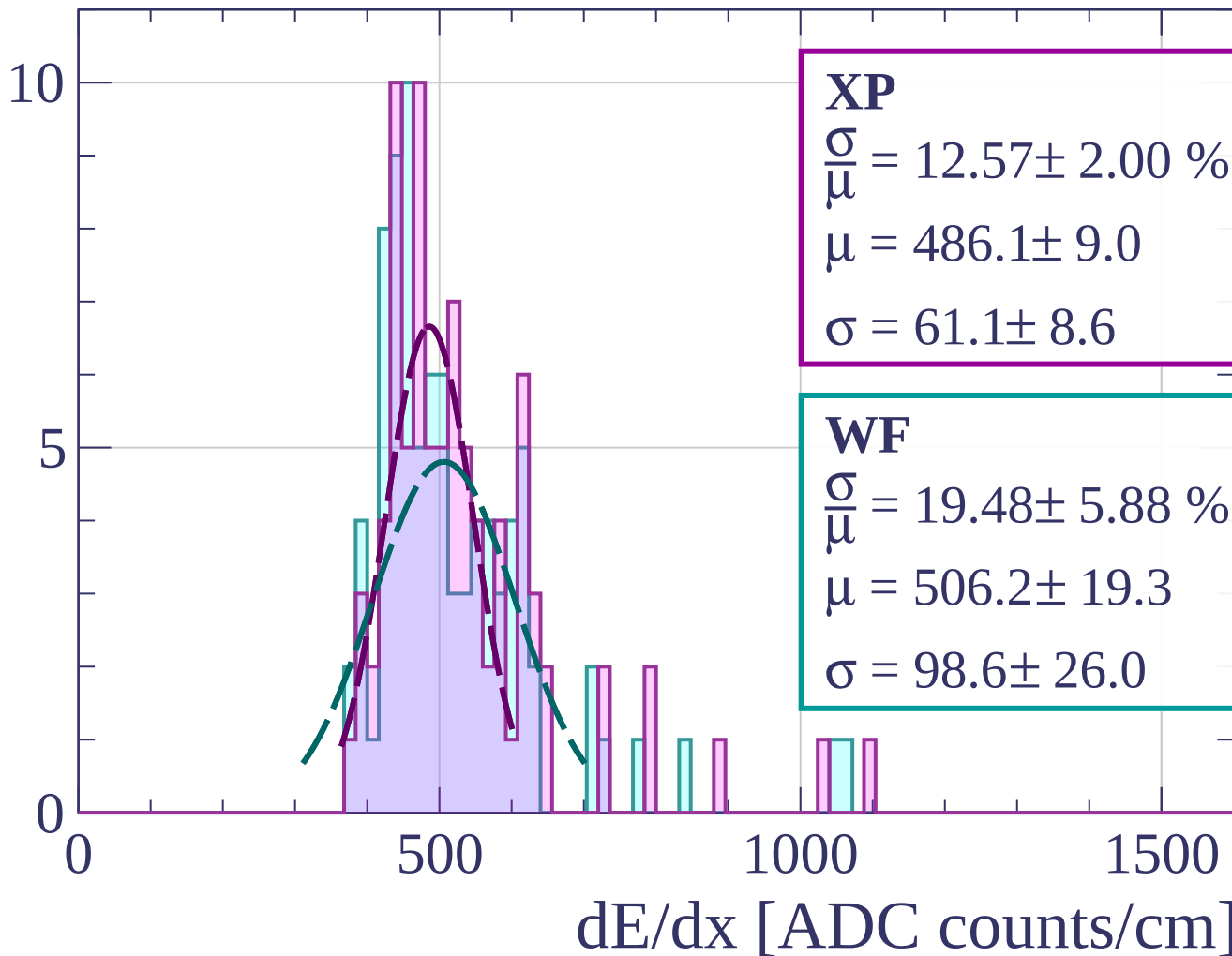
$$\mu = 499.2 \pm 19.4$$

$$\sigma = 96.5 \pm 22.1$$

dE/dx [ADC counts/cm]

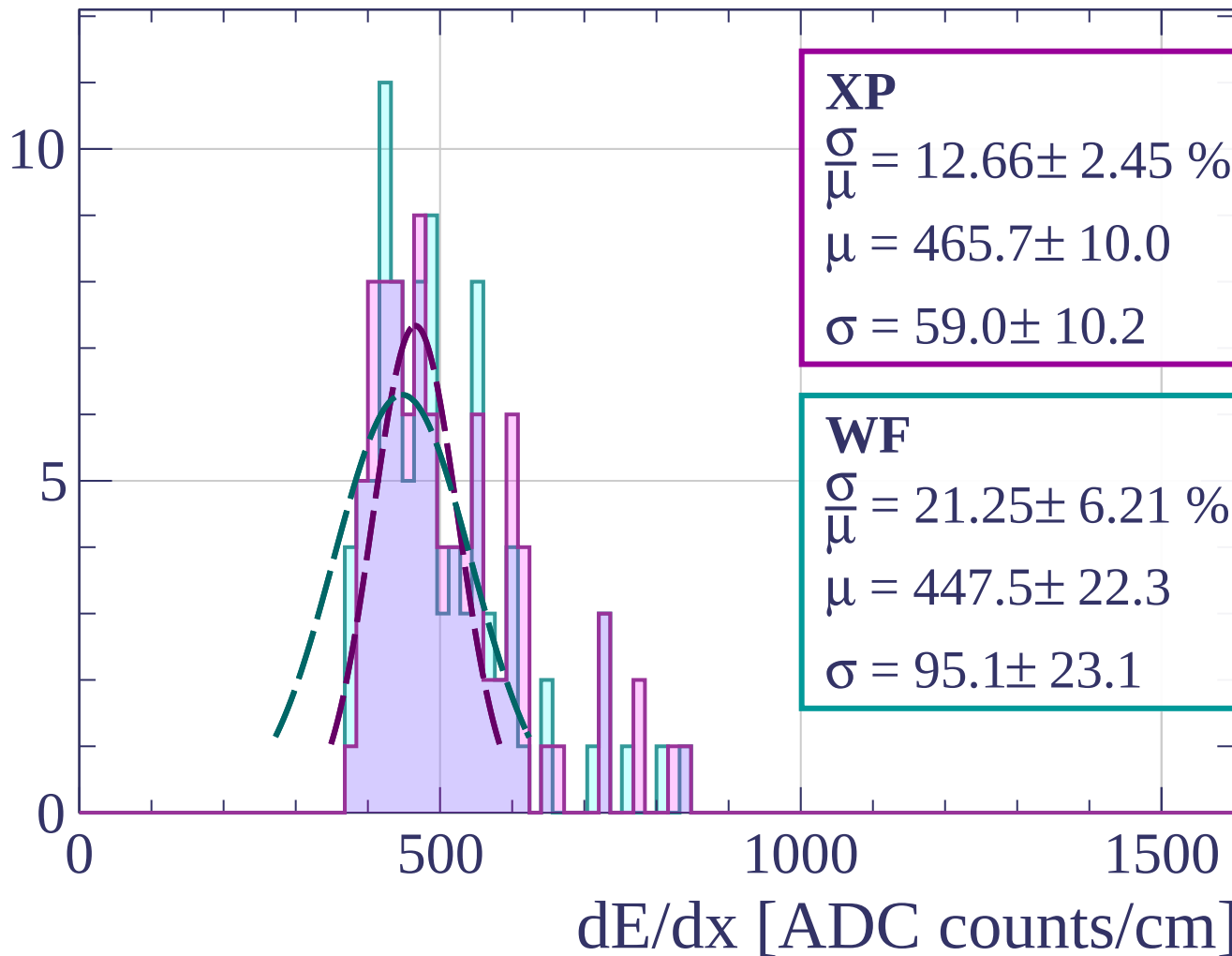
Count

Energy loss | $6 < \phi < 9$



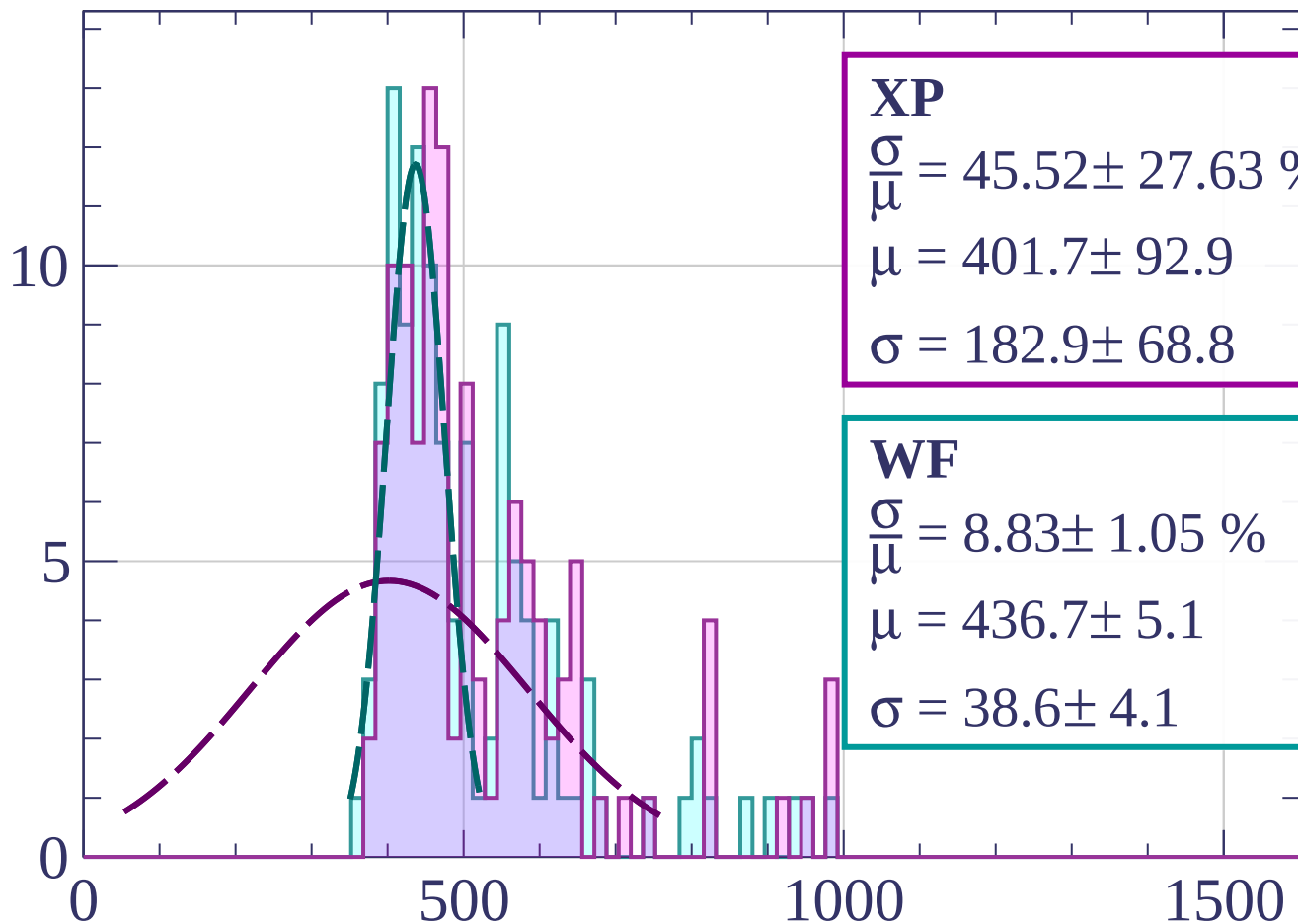
Energy loss | $9 < \phi < 12$

Count



Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 45.52 \pm 27.63 \%$$

$$\mu = 401.7 \pm 92.9$$

$$\sigma = 182.9 \pm 68.8$$

WF

$$\frac{\sigma}{\mu} = 8.83 \pm 1.05 \%$$

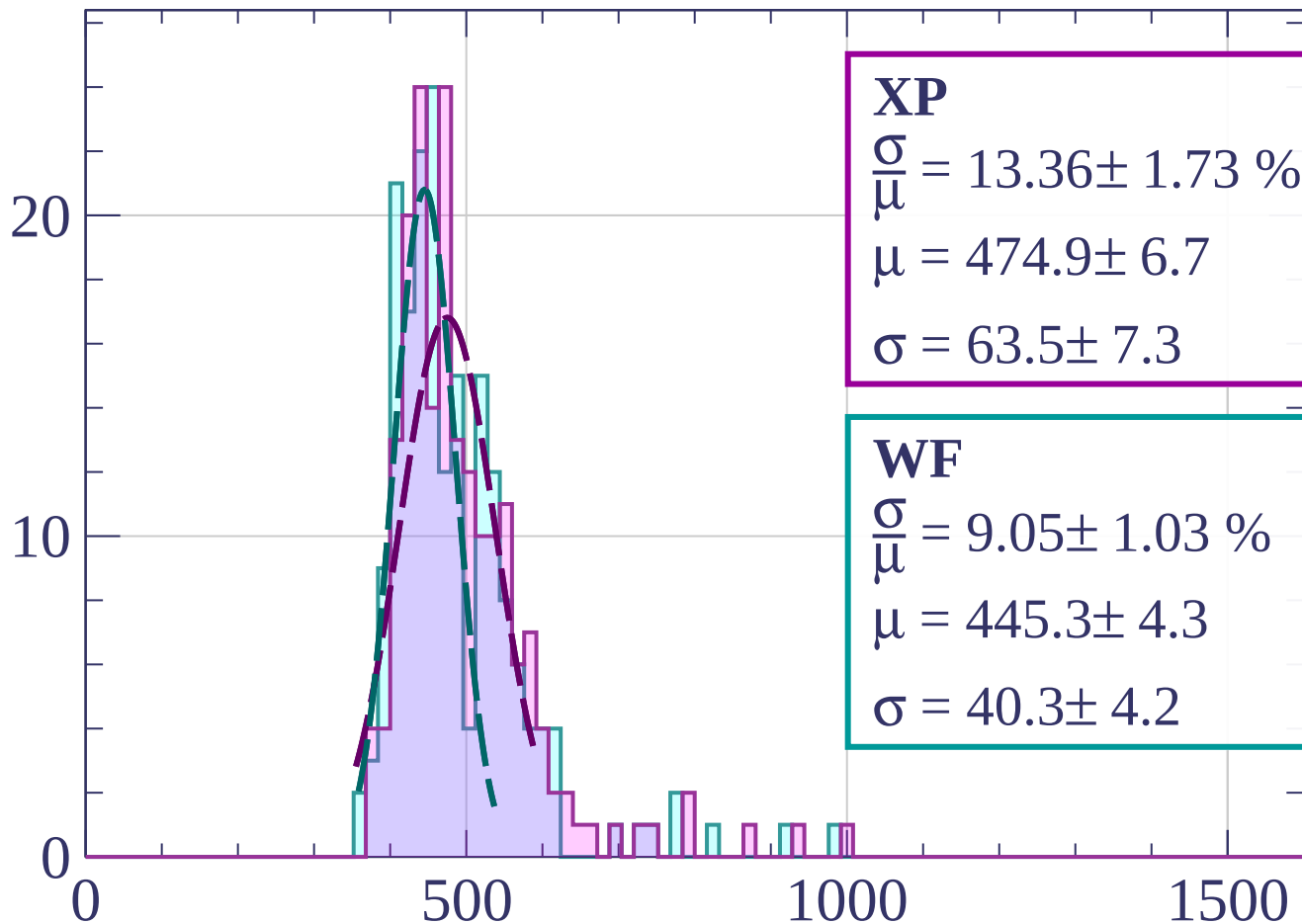
$$\mu = 436.7 \pm 5.1$$

$$\sigma = 38.6 \pm 4.1$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

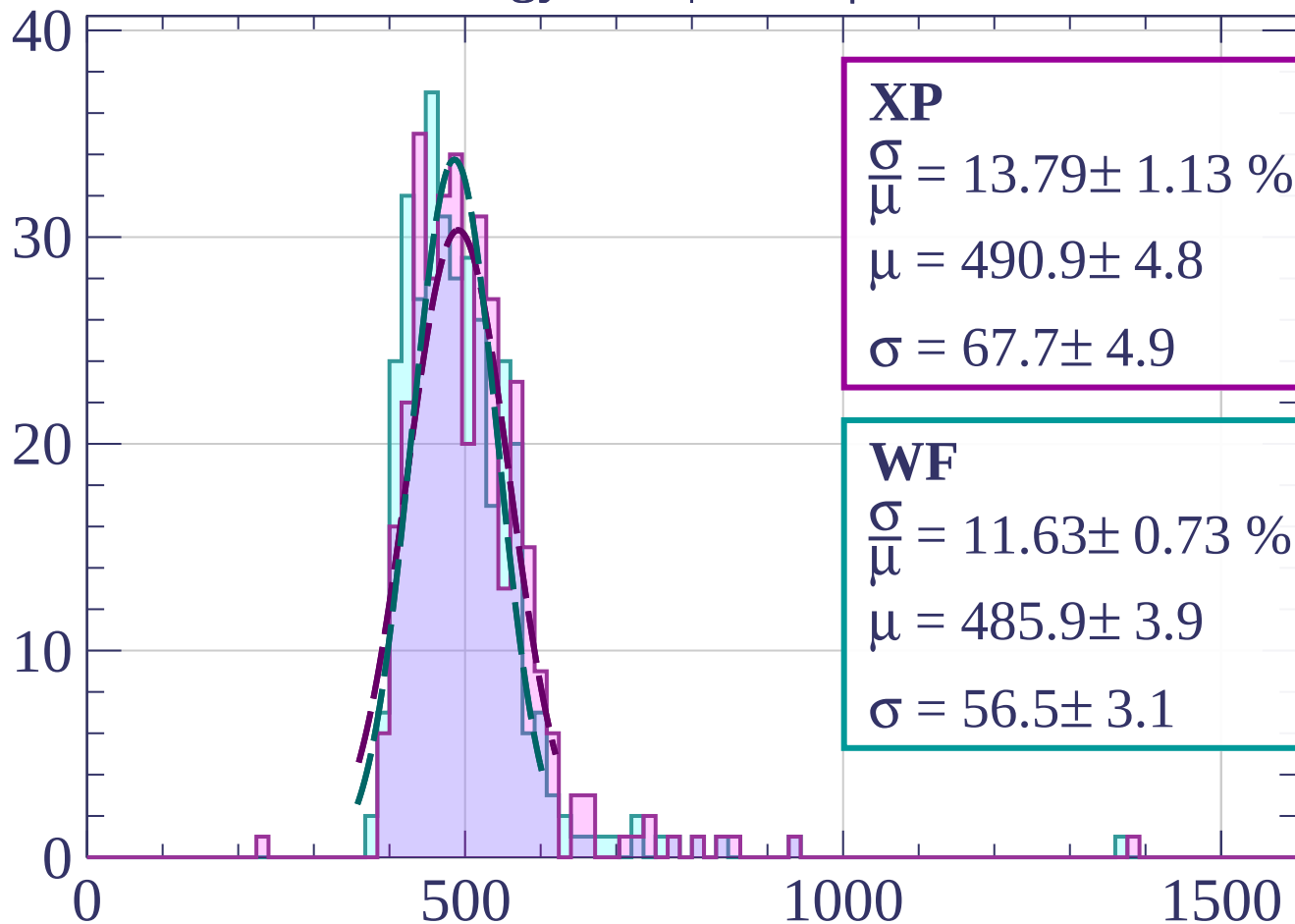
Count



dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

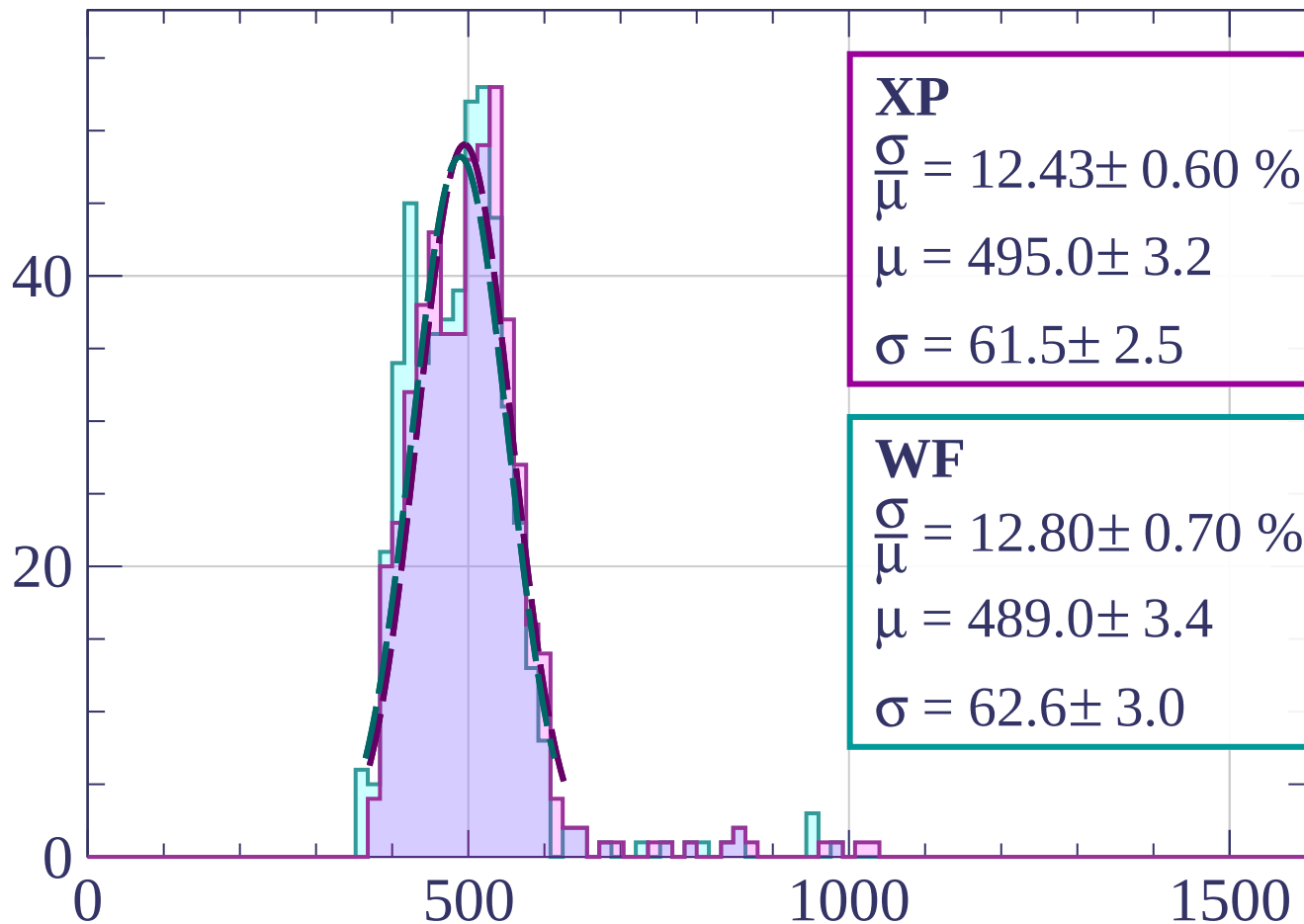
Count



dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 12.43 \pm 0.60 \%$$

$$\mu = 495.0 \pm 3.2$$

$$\sigma = 61.5 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 12.80 \pm 0.70 \%$$

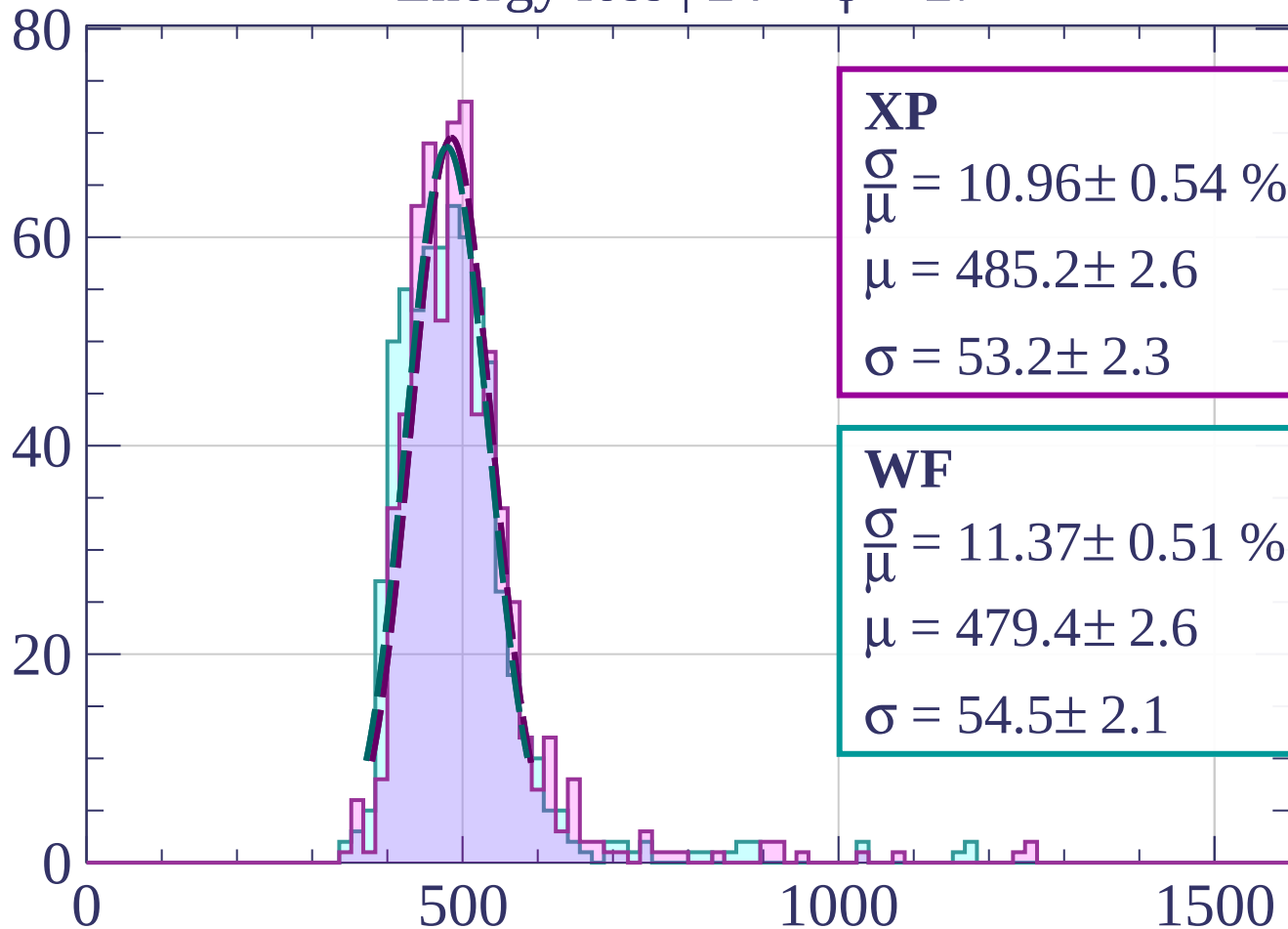
$$\mu = 489.0 \pm 3.4$$

$$\sigma = 62.6 \pm 3.0$$

dE/dx [ADC counts/cm]

Count

Energy loss | $24 < \phi < 27$



XP

$$\frac{\sigma}{\mu} = 10.96 \pm 0.54 \%$$

$$\mu = 485.2 \pm 2.6$$

$$\sigma = 53.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 11.37 \pm 0.51 \%$$

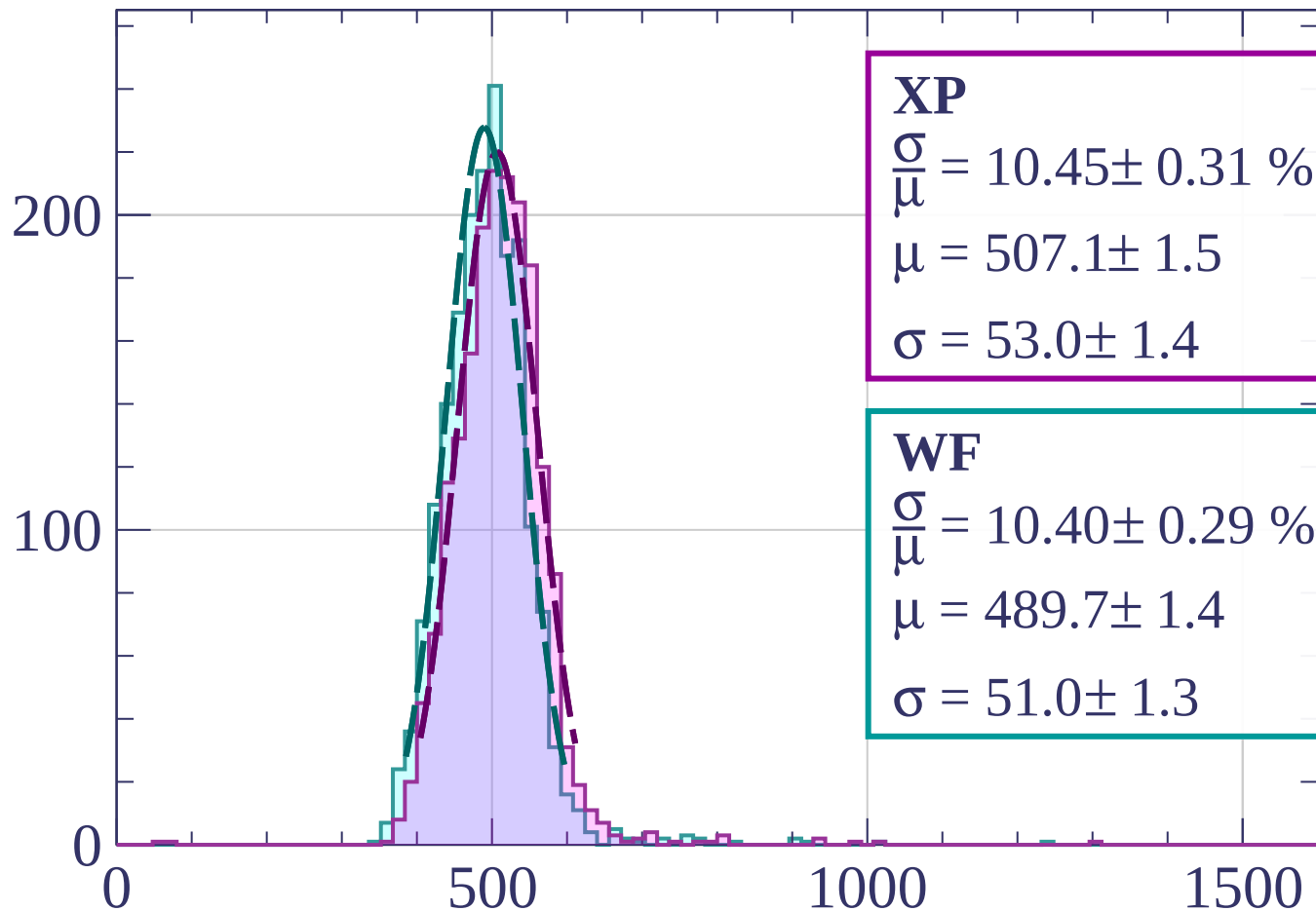
$$\mu = 479.4 \pm 2.6$$

$$\sigma = 54.5 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 10.45 \pm 0.31 \%$$

$$\mu = 507.1 \pm 1.5$$

$$\sigma = 53.0 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.40 \pm 0.29 \%$$

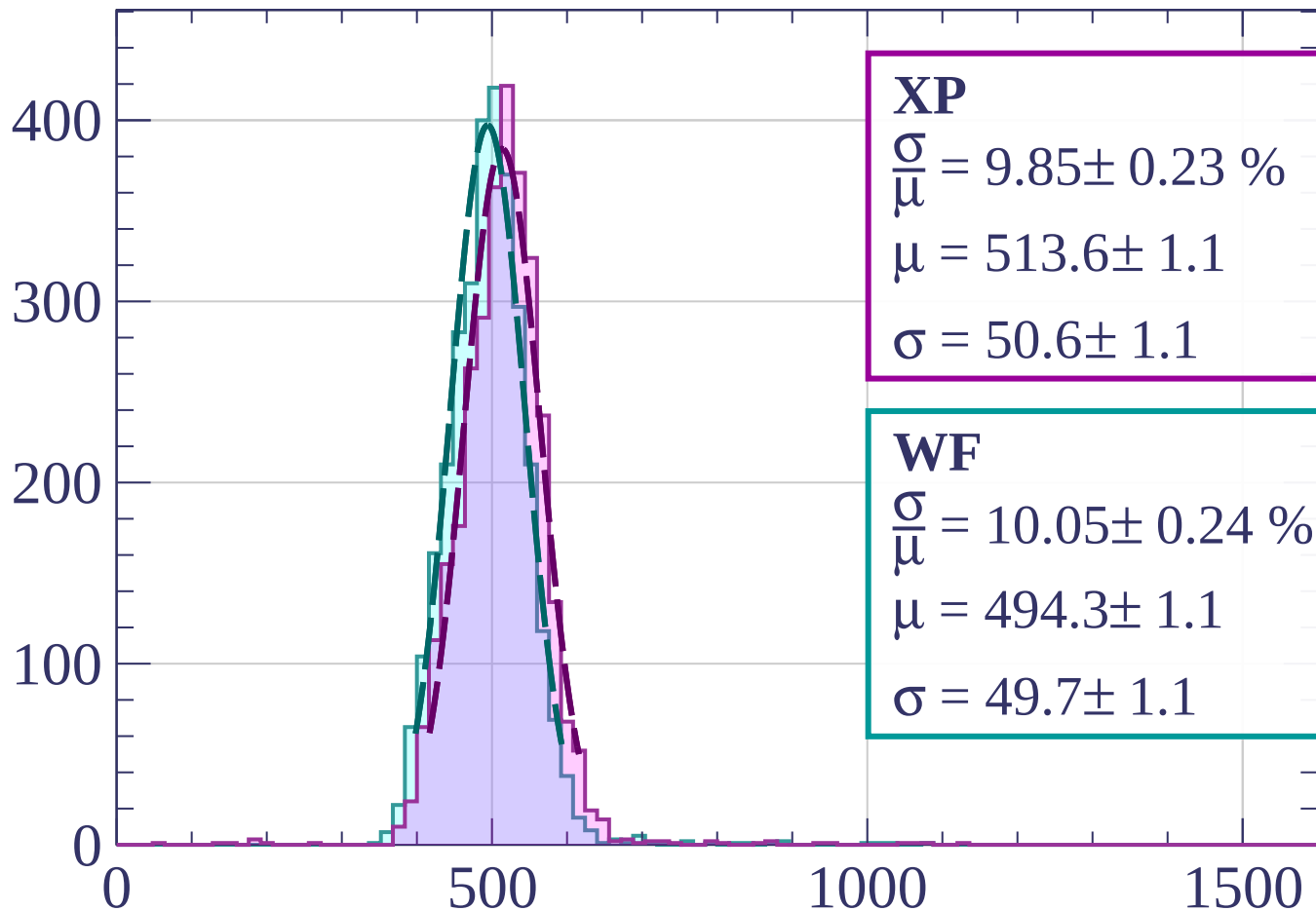
$$\mu = 489.7 \pm 1.4$$

$$\sigma = 51.0 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.23 \%$$

$$\mu = 513.6 \pm 1.1$$

$$\sigma = 50.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.24 \%$$

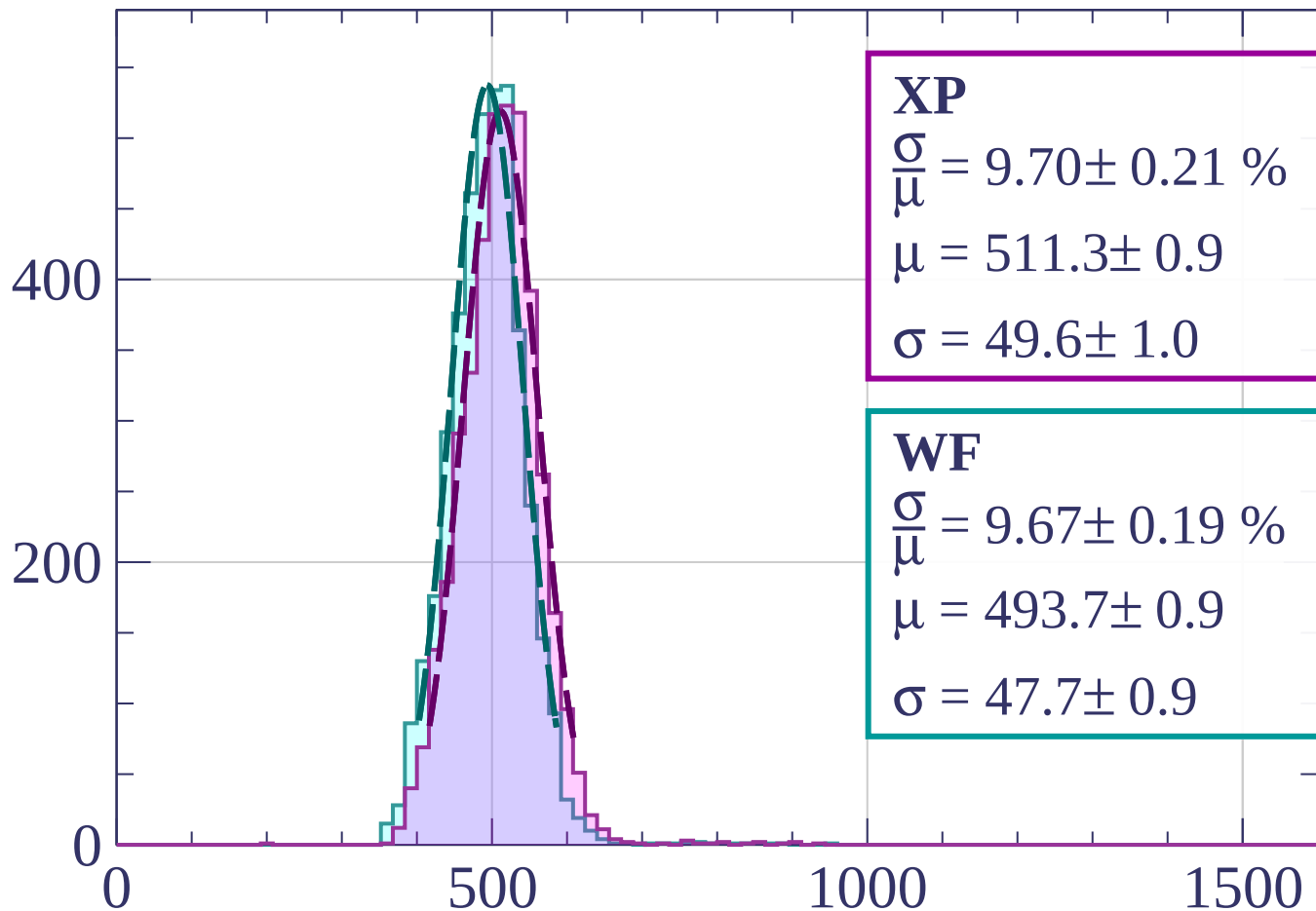
$$\mu = 494.3 \pm 1.1$$

$$\sigma = 49.7 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 9.70 \pm 0.21 \%$$

$$\mu = 511.3 \pm 0.9$$

$$\sigma = 49.6 \pm 1.0$$

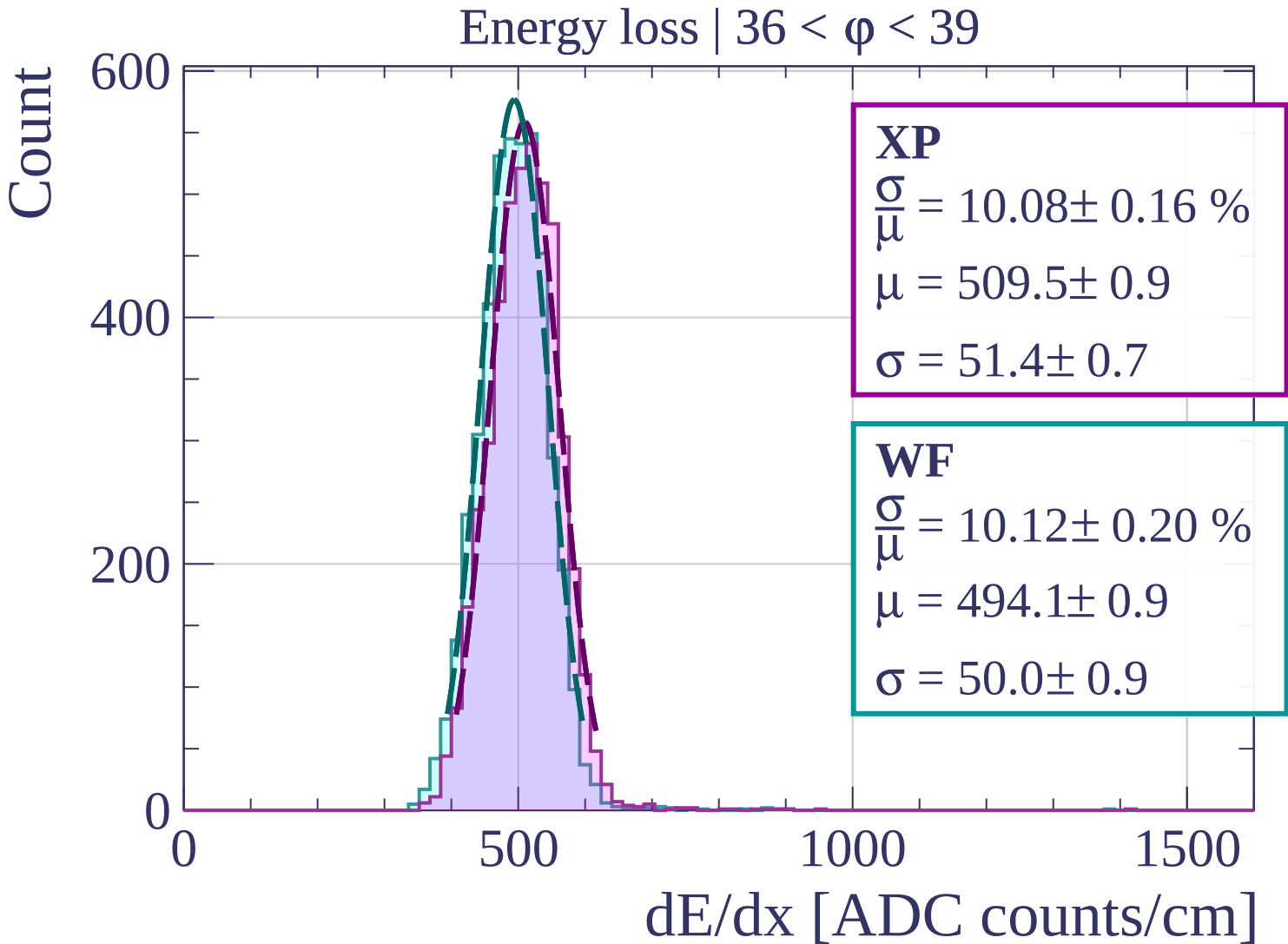
WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.19 \%$$

$$\mu = 493.7 \pm 0.9$$

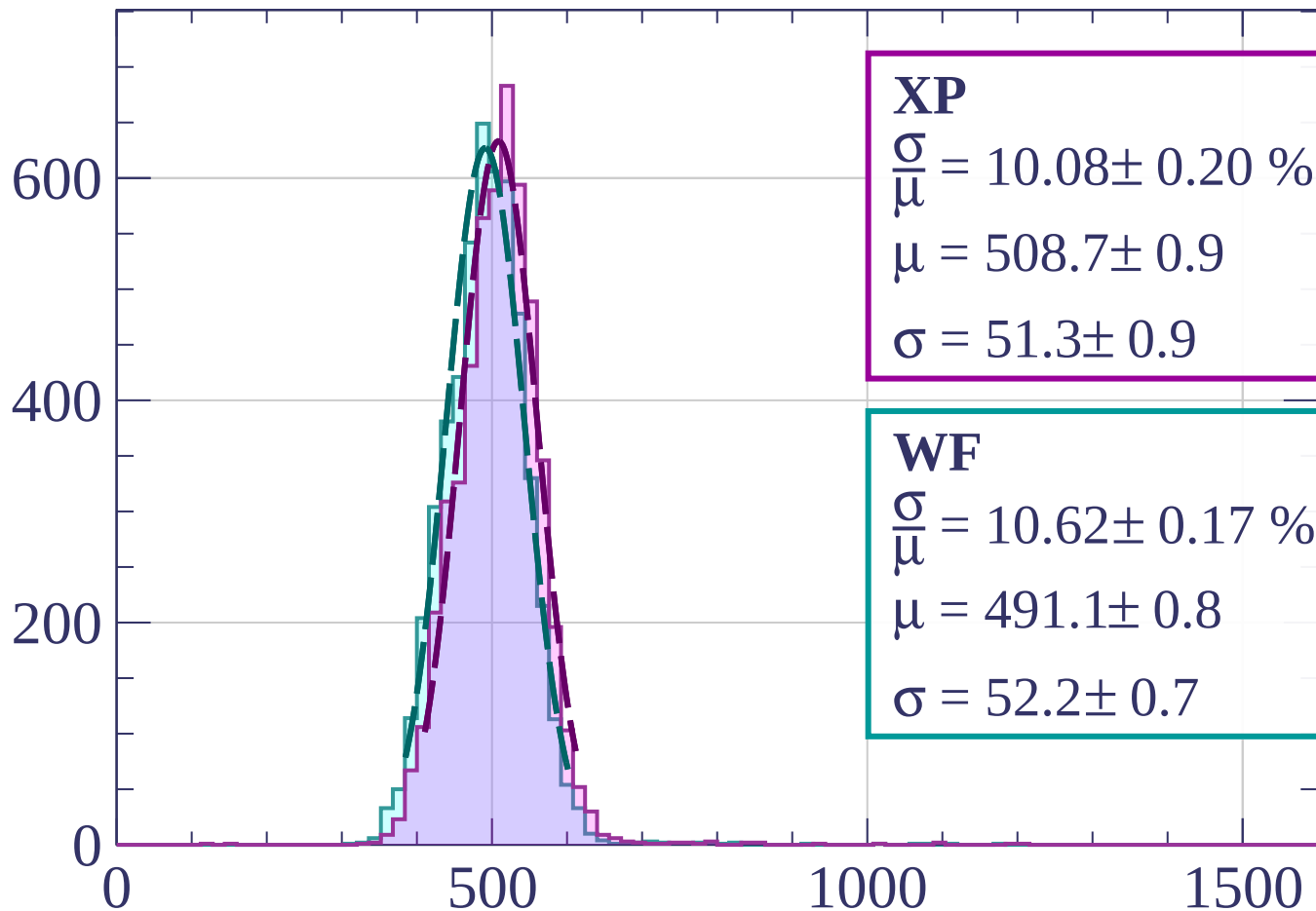
$$\sigma = 47.7 \pm 0.9$$

dE/dx [ADC counts/cm]



Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 10.08 \pm 0.20 \%$$

$$\mu = 508.7 \pm 0.9$$

$$\sigma = 51.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.62 \pm 0.17 \%$$

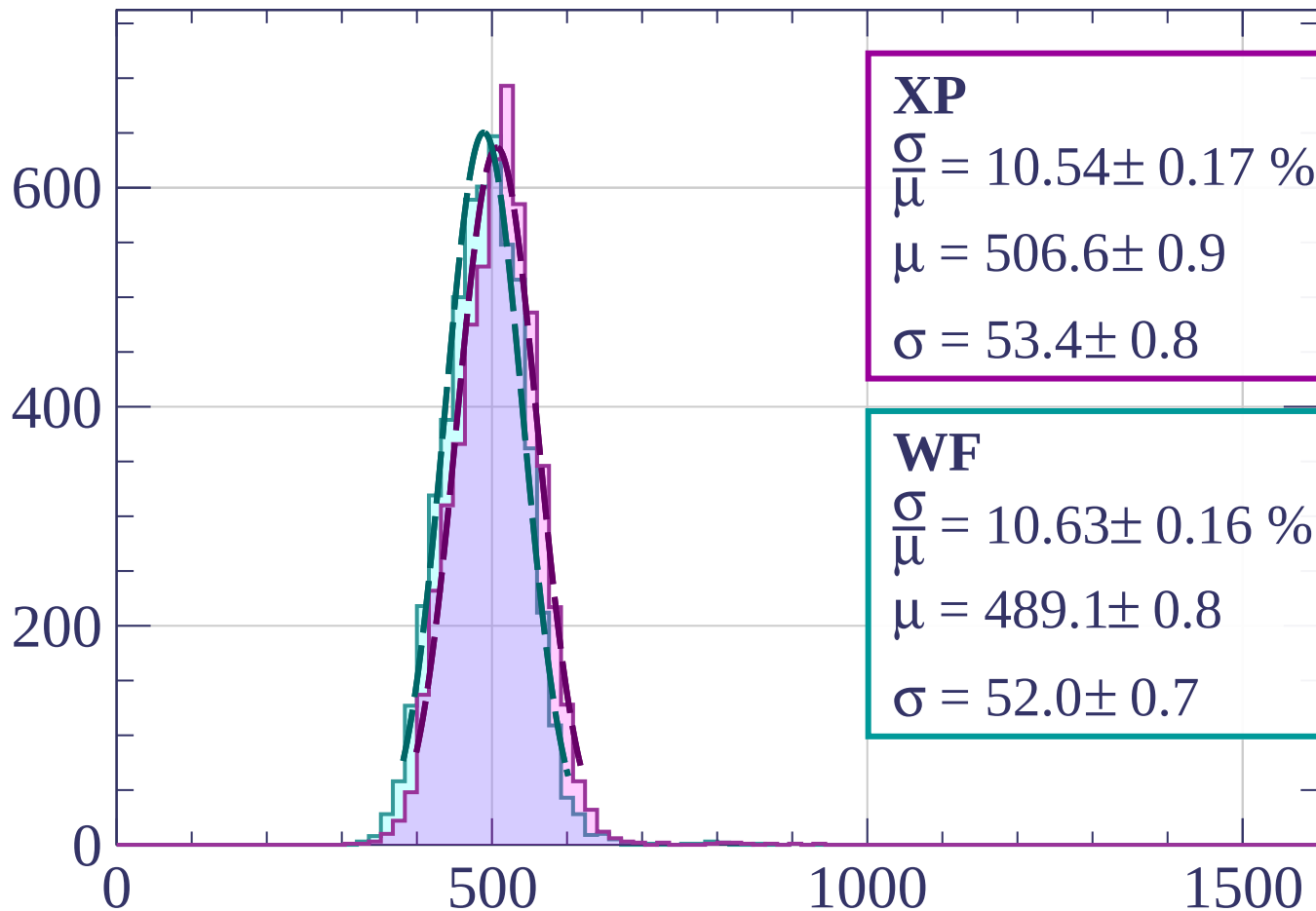
$$\mu = 491.1 \pm 0.8$$

$$\sigma = 52.2 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.17 \%$$

$$\mu = 506.6 \pm 0.9$$

$$\sigma = 53.4 \pm 0.8$$

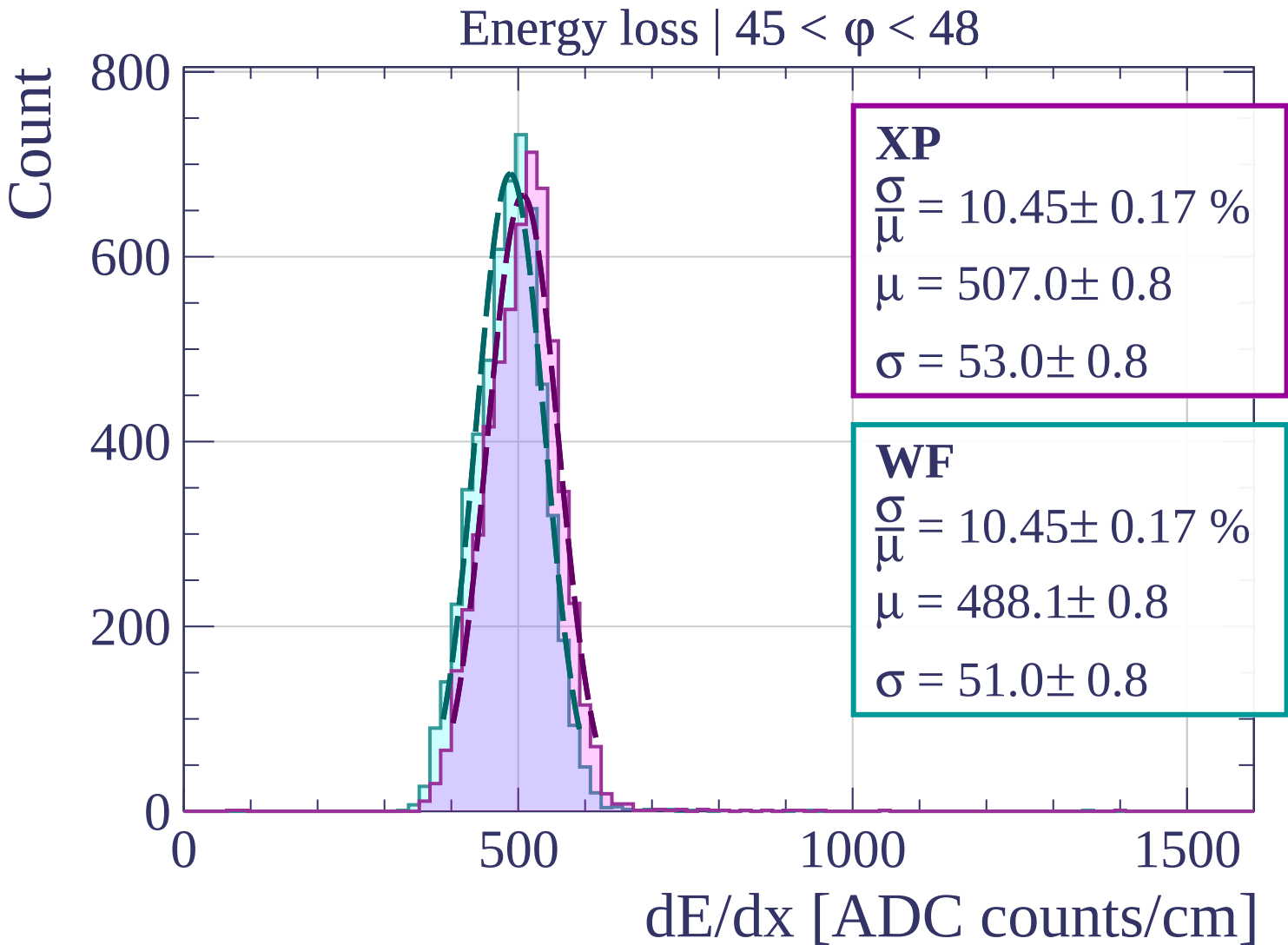
WF

$$\frac{\sigma}{\mu} = 10.63 \pm 0.16 \%$$

$$\mu = 489.1 \pm 0.8$$

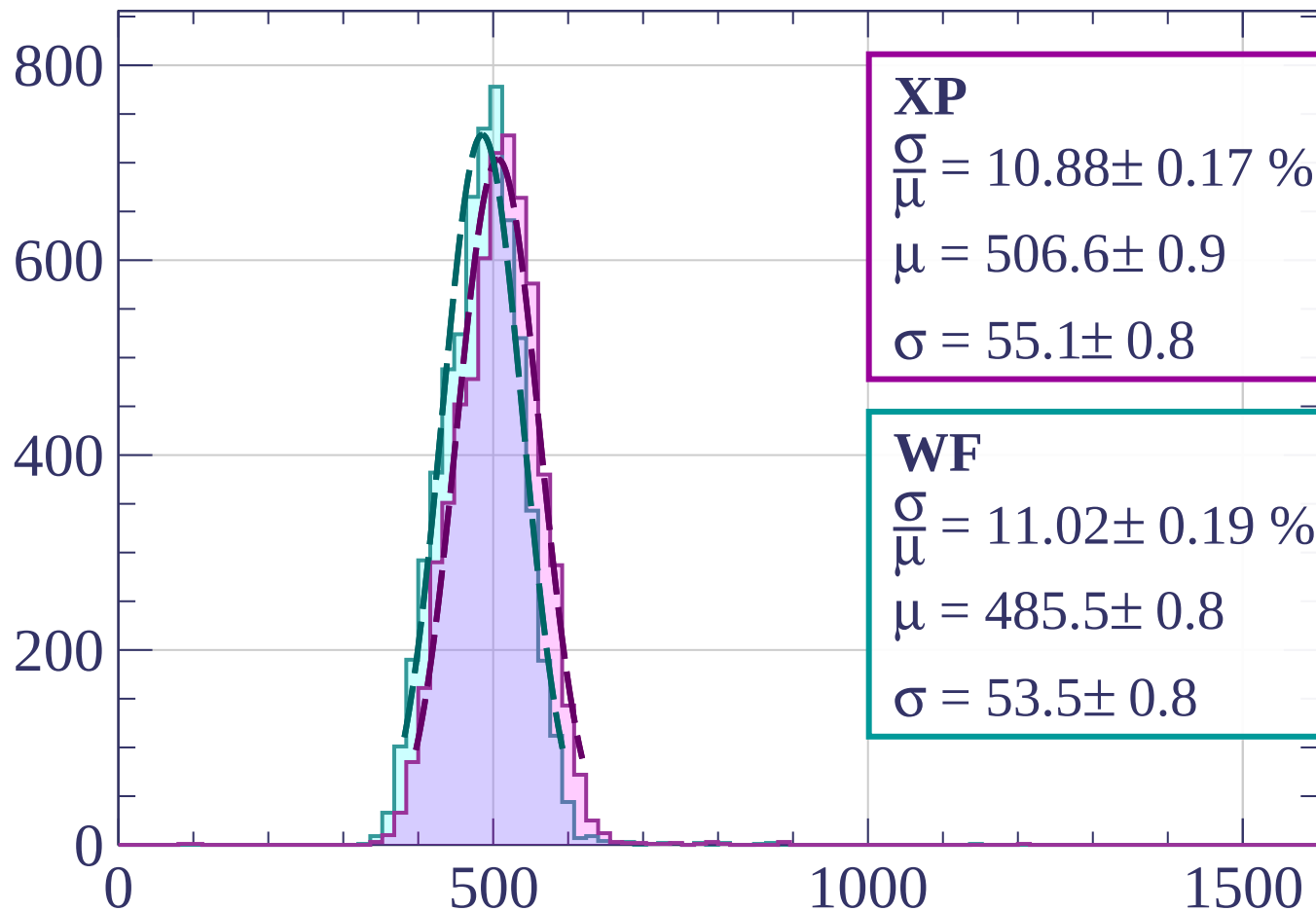
$$\sigma = 52.0 \pm 0.7$$

dE/dx [ADC counts/cm]



Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 10.88 \pm 0.17 \%$$

$$\mu = 506.6 \pm 0.9$$

$$\sigma = 55.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.19 \%$$

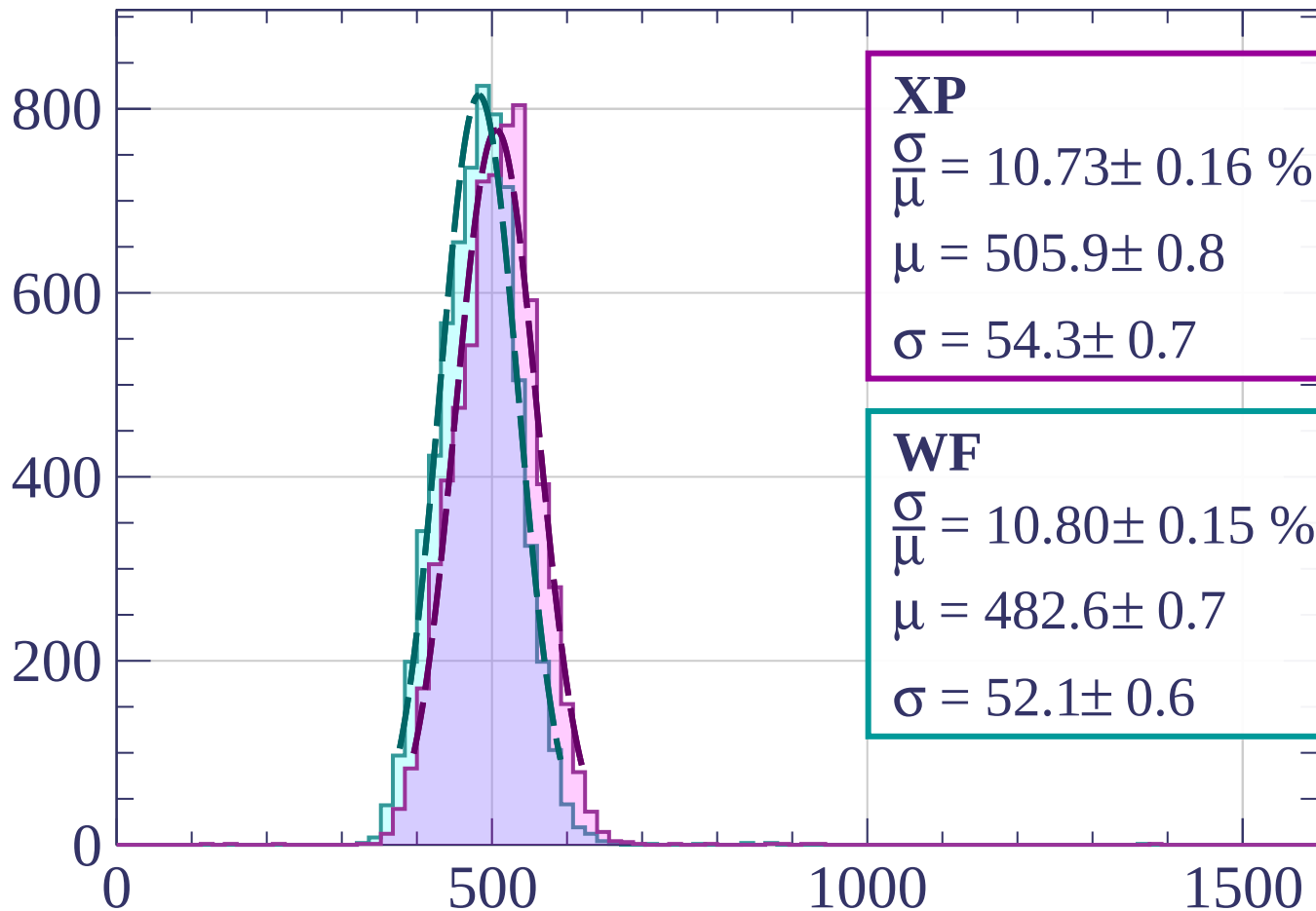
$$\mu = 485.5 \pm 0.8$$

$$\sigma = 53.5 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 10.73 \pm 0.16 \%$$

$$\mu = 505.9 \pm 0.8$$

$$\sigma = 54.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.80 \pm 0.15 \%$$

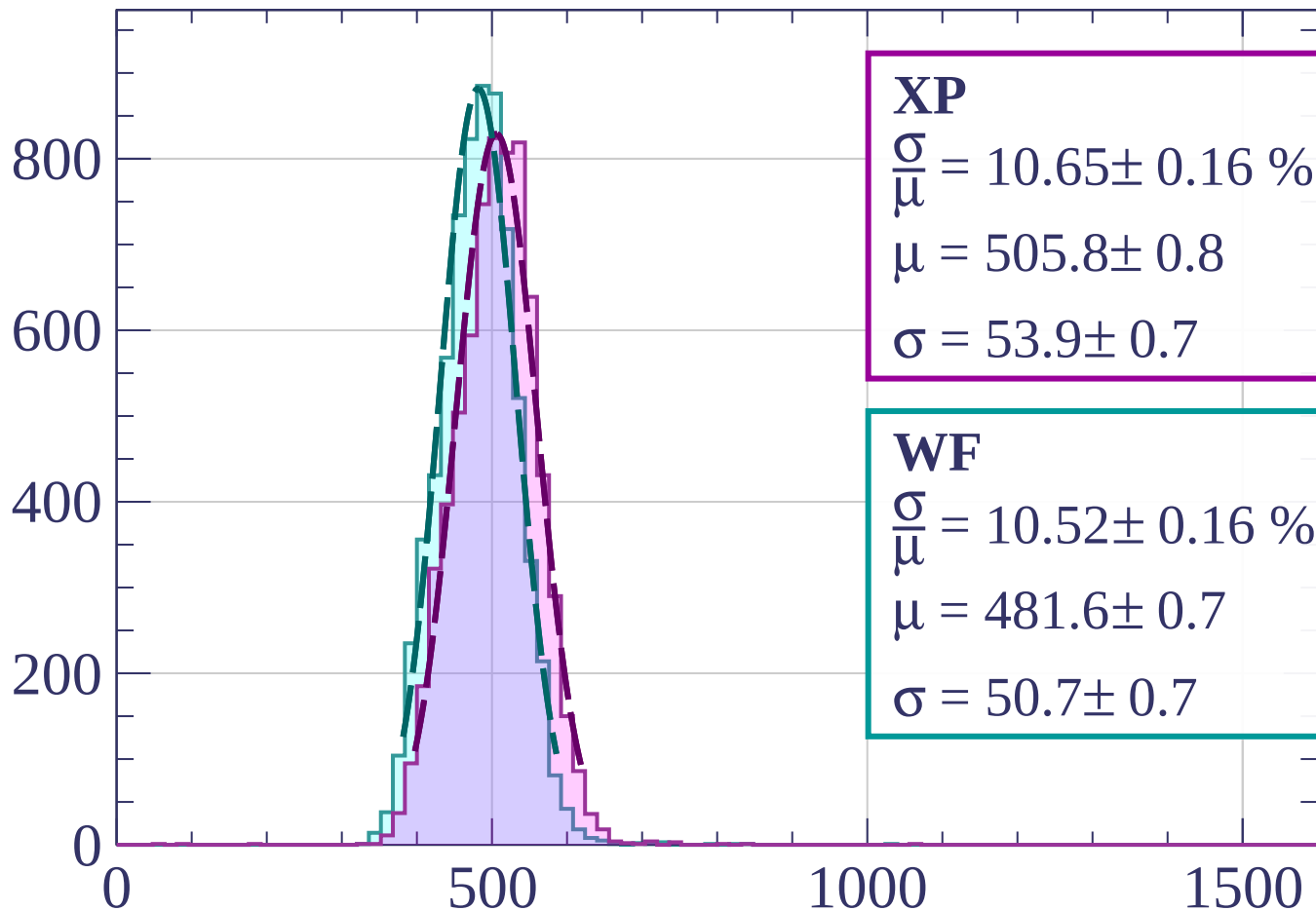
$$\mu = 482.6 \pm 0.7$$

$$\sigma = 52.1 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 10.65 \pm 0.16 \%$$

$$\mu = 505.8 \pm 0.8$$

$$\sigma = 53.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.52 \pm 0.16 \%$$

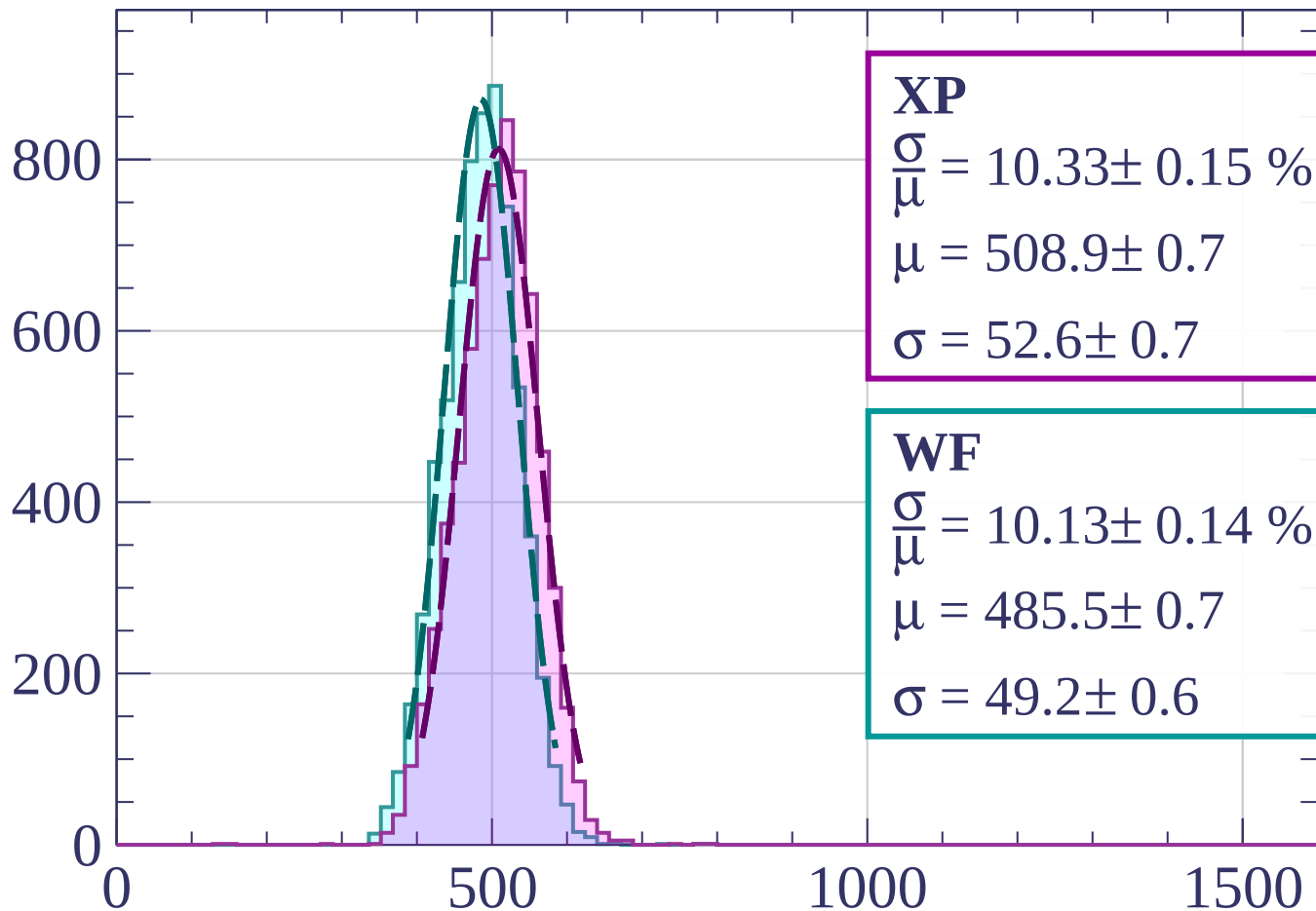
$$\mu = 481.6 \pm 0.7$$

$$\sigma = 50.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.15 \%$$

$$\mu = 508.9 \pm 0.7$$

$$\sigma = 52.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.13 \pm 0.14 \%$$

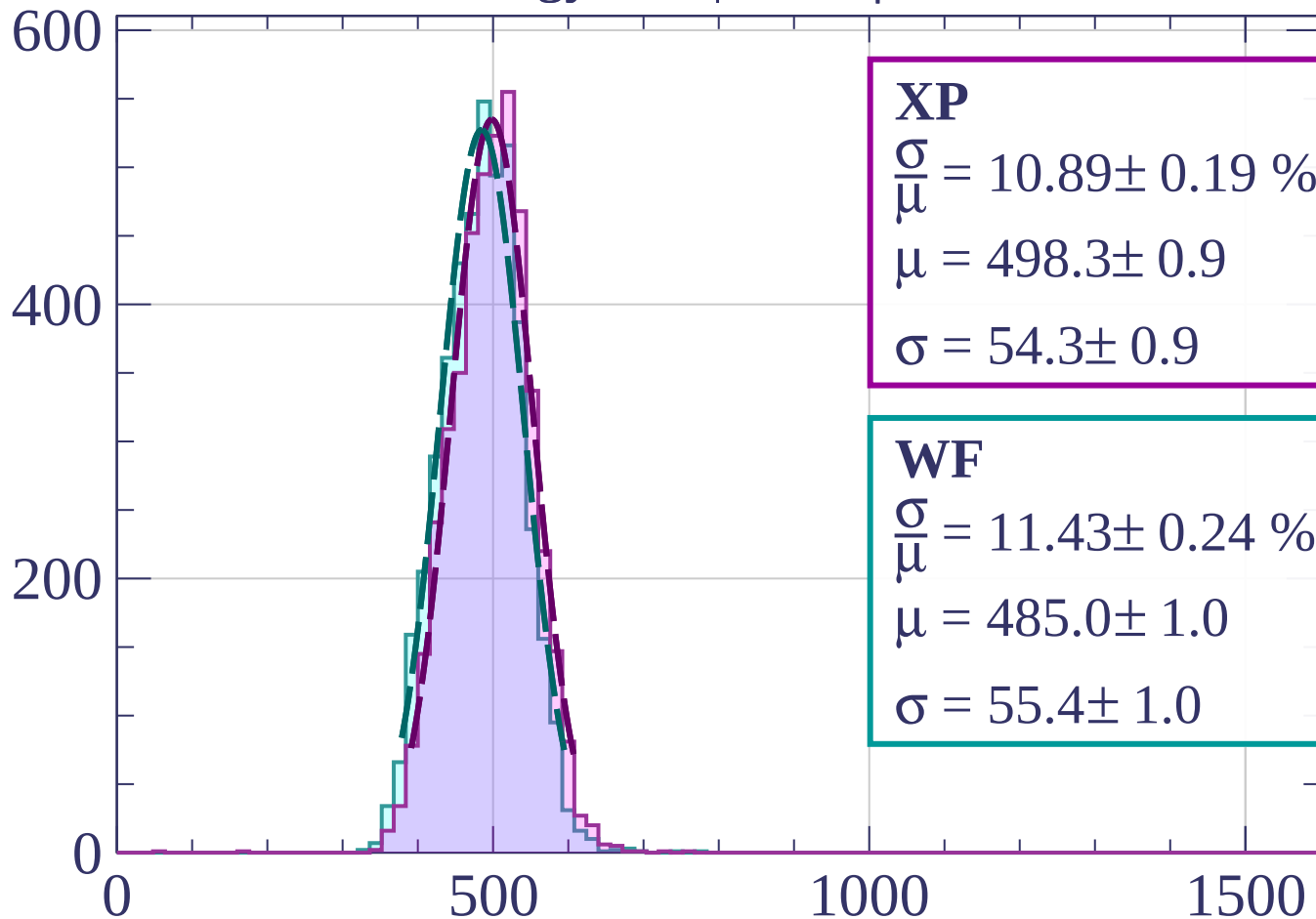
$$\mu = 485.5 \pm 0.7$$

$$\sigma = 49.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 10.89 \pm 0.19 \%$$

$$\mu = 498.3 \pm 0.9$$

$$\sigma = 54.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.43 \pm 0.24 \%$$

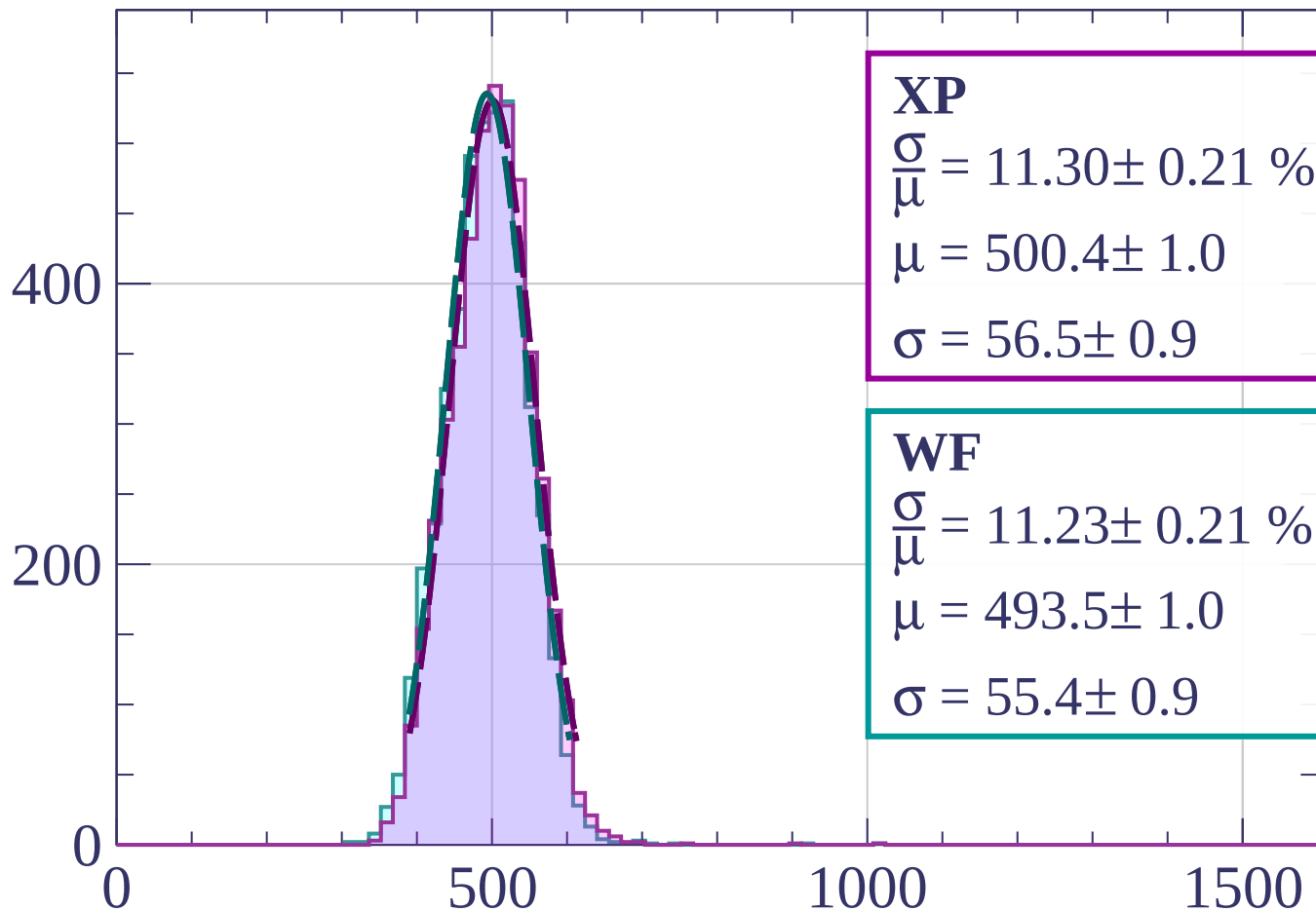
$$\mu = 485.0 \pm 1.0$$

$$\sigma = 55.4 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 11.30 \pm 0.21 \%$$

$$\mu = 500.4 \pm 1.0$$

$$\sigma = 56.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.23 \pm 0.21 \%$$

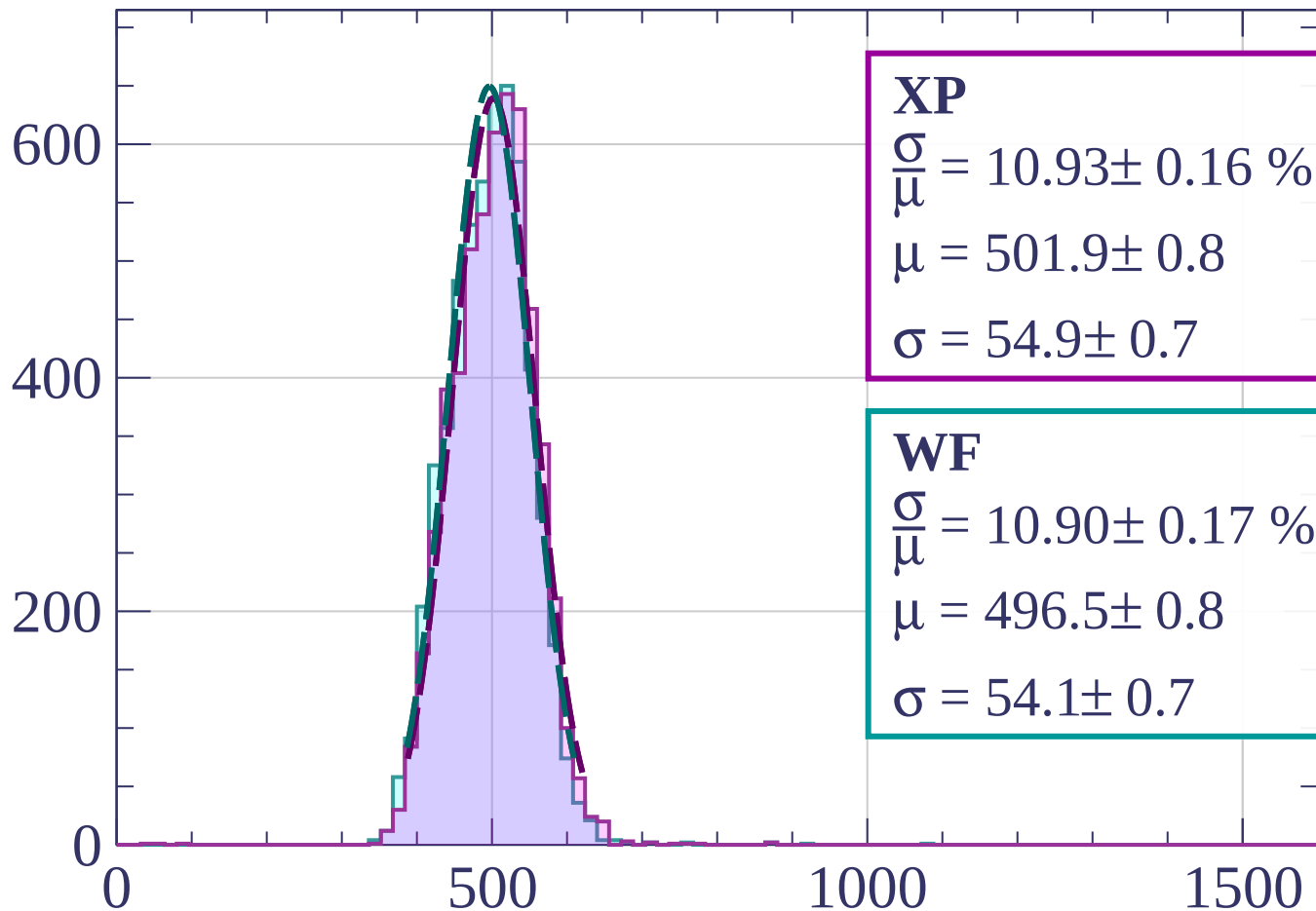
$$\mu = 493.5 \pm 1.0$$

$$\sigma = 55.4 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 10.93 \pm 0.16 \%$$

$$\mu = 501.9 \pm 0.8$$

$$\sigma = 54.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.90 \pm 0.17 \%$$

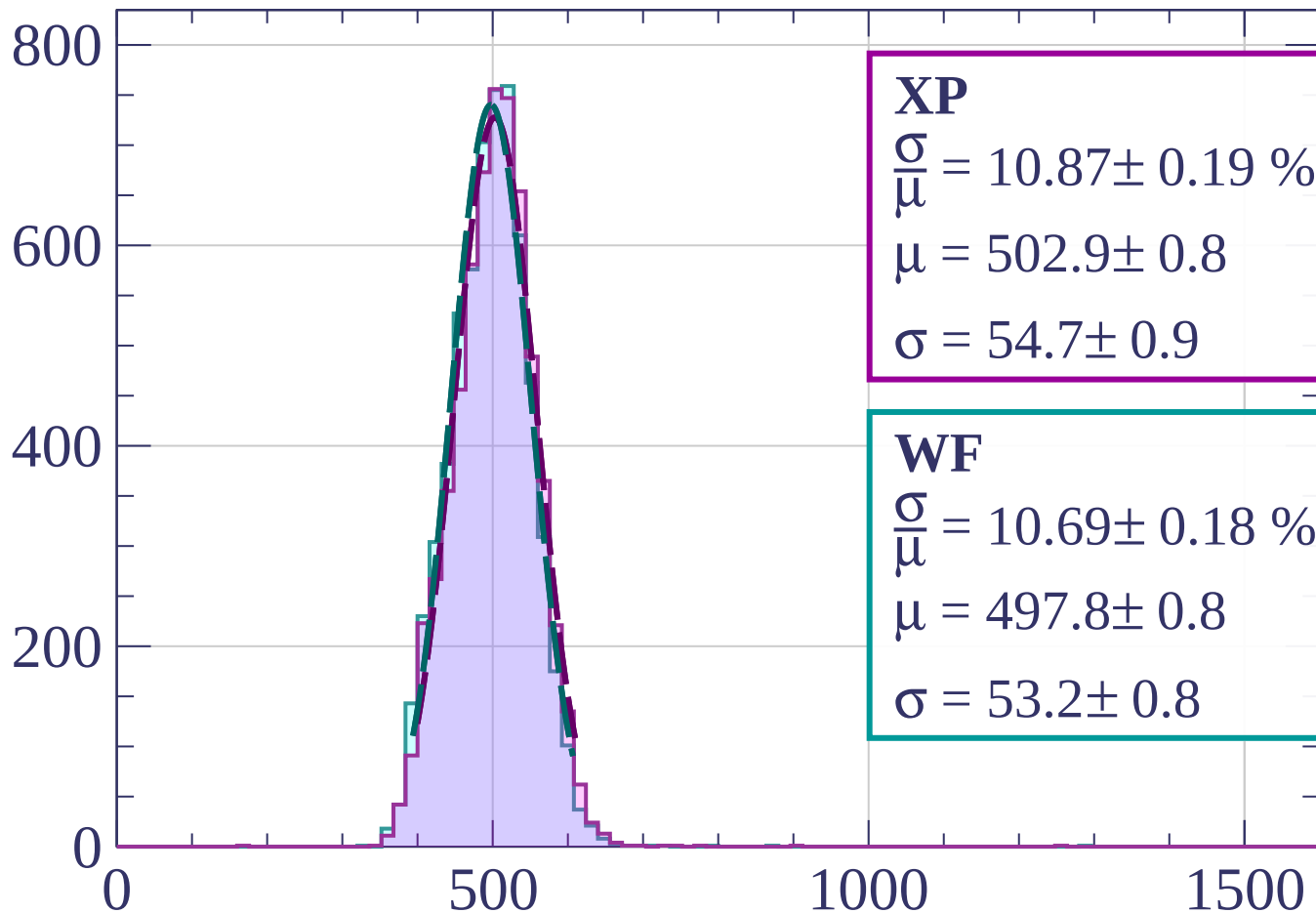
$$\mu = 496.5 \pm 0.8$$

$$\sigma = 54.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 10.87 \pm 0.19 \%$$

$$\mu = 502.9 \pm 0.8$$

$$\sigma = 54.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.69 \pm 0.18 \%$$

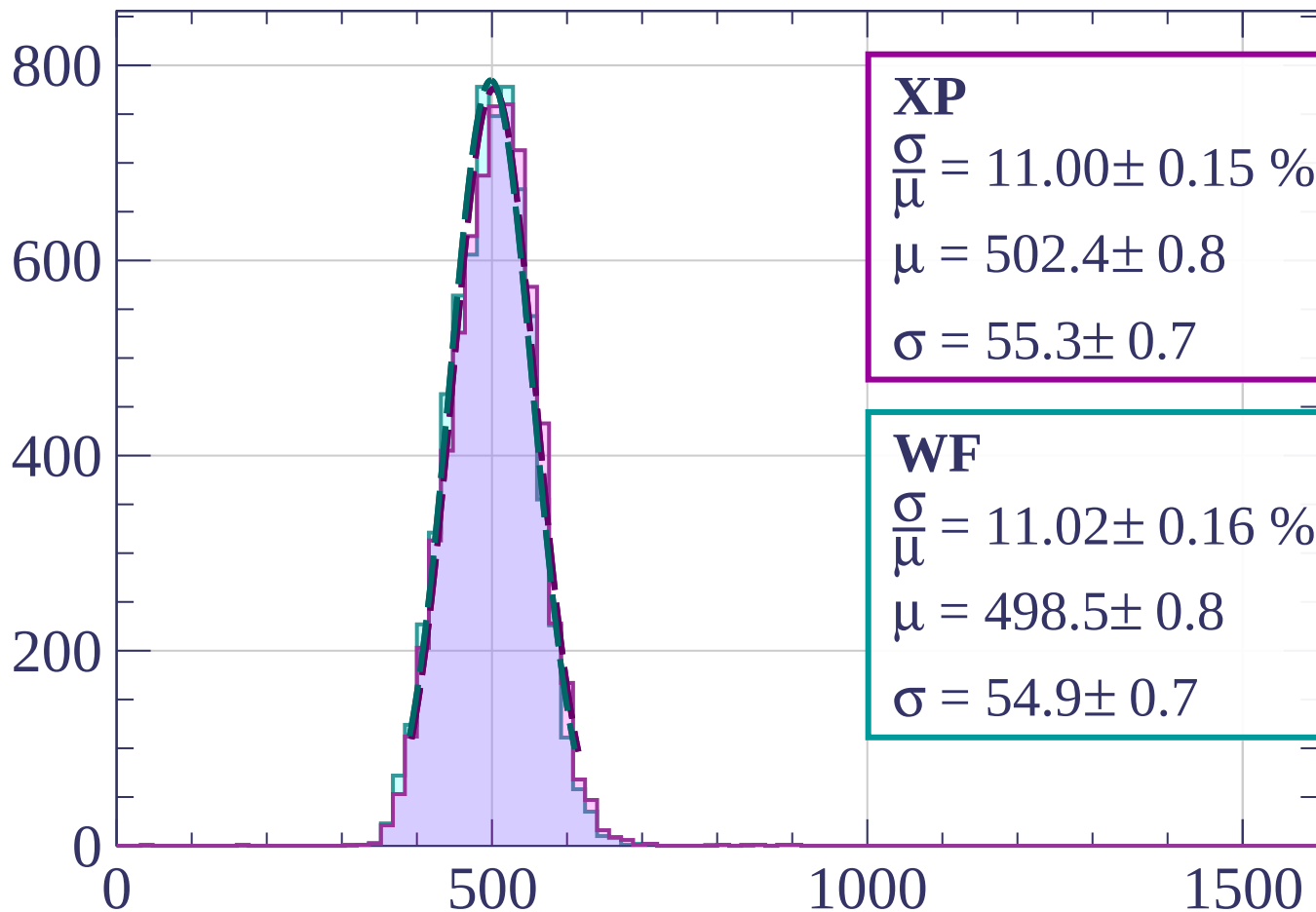
$$\mu = 497.8 \pm 0.8$$

$$\sigma = 53.2 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 11.00 \pm 0.15 \%$$

$$\mu = 502.4 \pm 0.8$$

$$\sigma = 55.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.16 \%$$

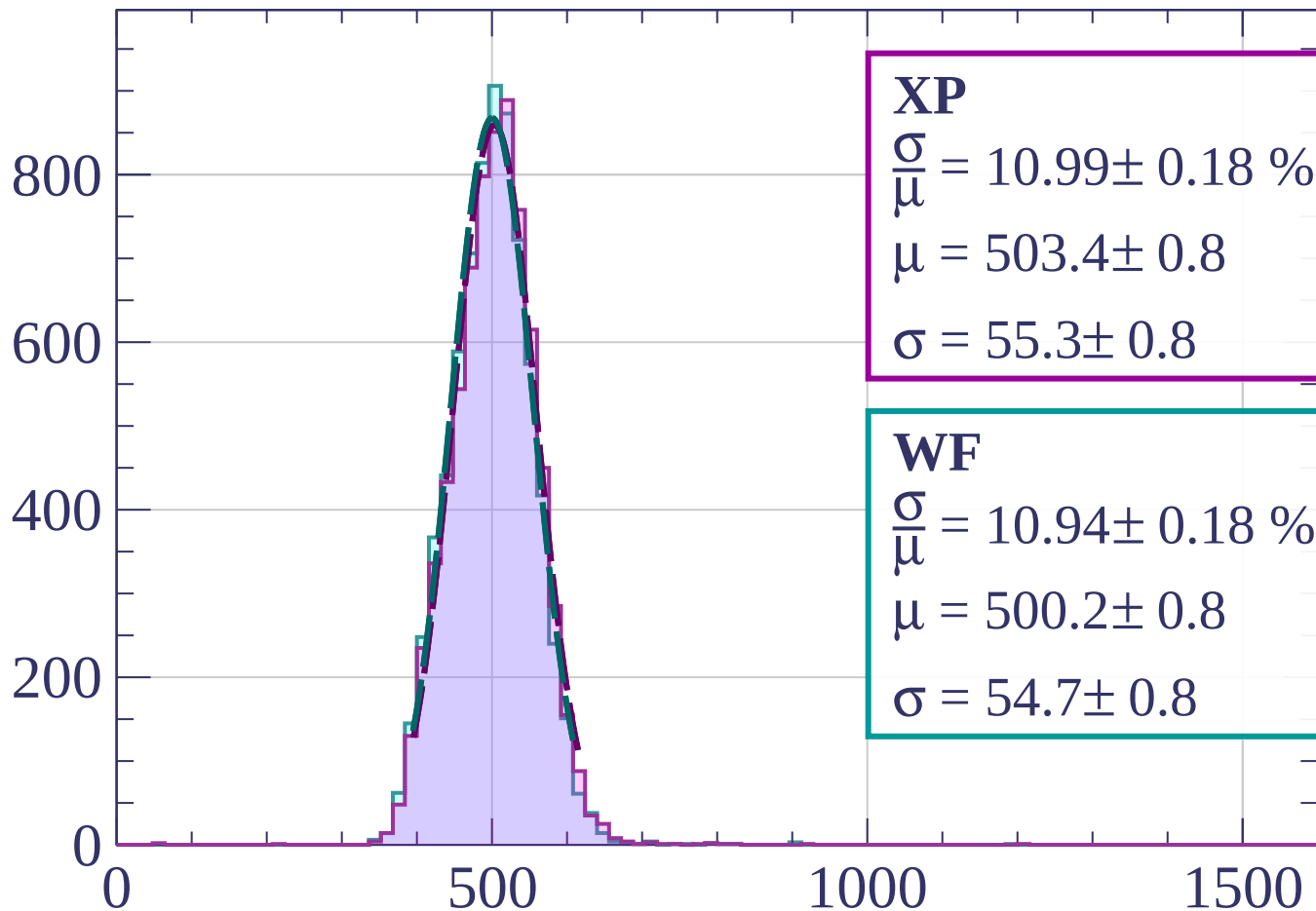
$$\mu = 498.5 \pm 0.8$$

$$\sigma = 54.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 10.99 \pm 0.18 \%$$

$$\mu = 503.4 \pm 0.8$$

$$\sigma = 55.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.94 \pm 0.18 \%$$

$$\mu = 500.2 \pm 0.8$$

$$\sigma = 54.7 \pm 0.8$$

dE/dx [ADC counts/cm]

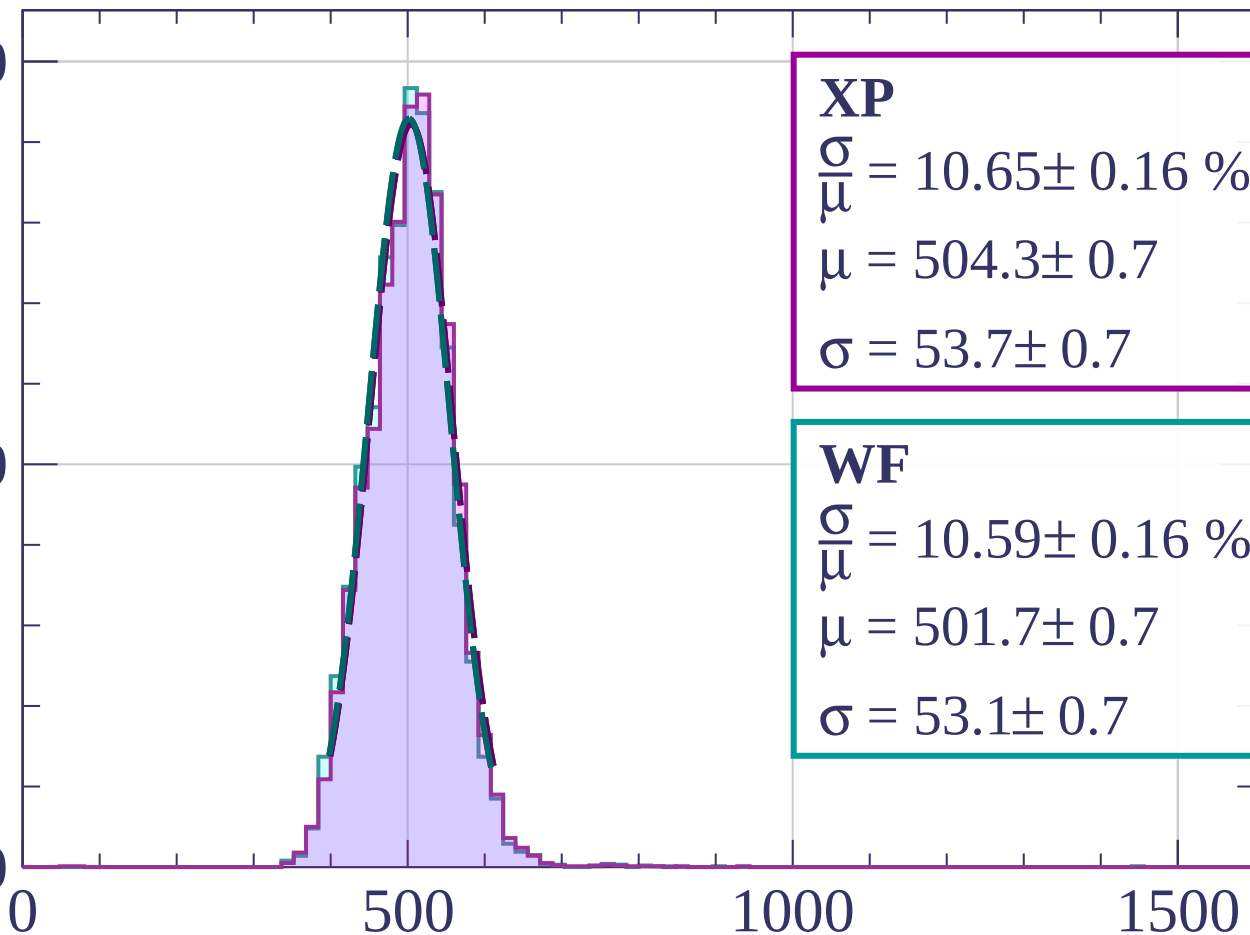
Energy loss | $78 < \phi < 81$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 10.65 \pm 0.16 \%$$

$$\mu = 504.3 \pm 0.7$$

$$\sigma = 53.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.16 \%$$

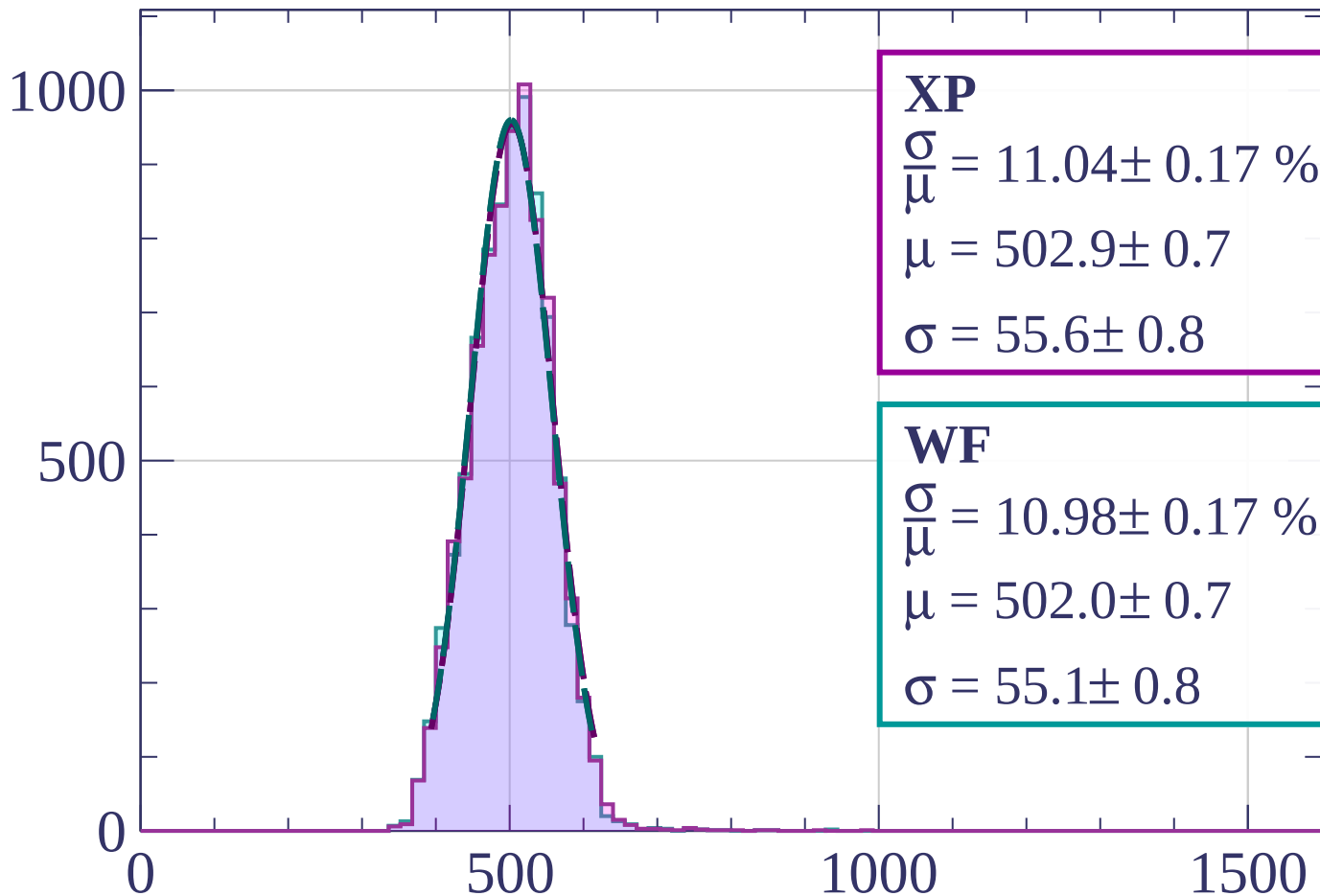
$$\mu = 501.7 \pm 0.7$$

$$\sigma = 53.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \phi < 84$

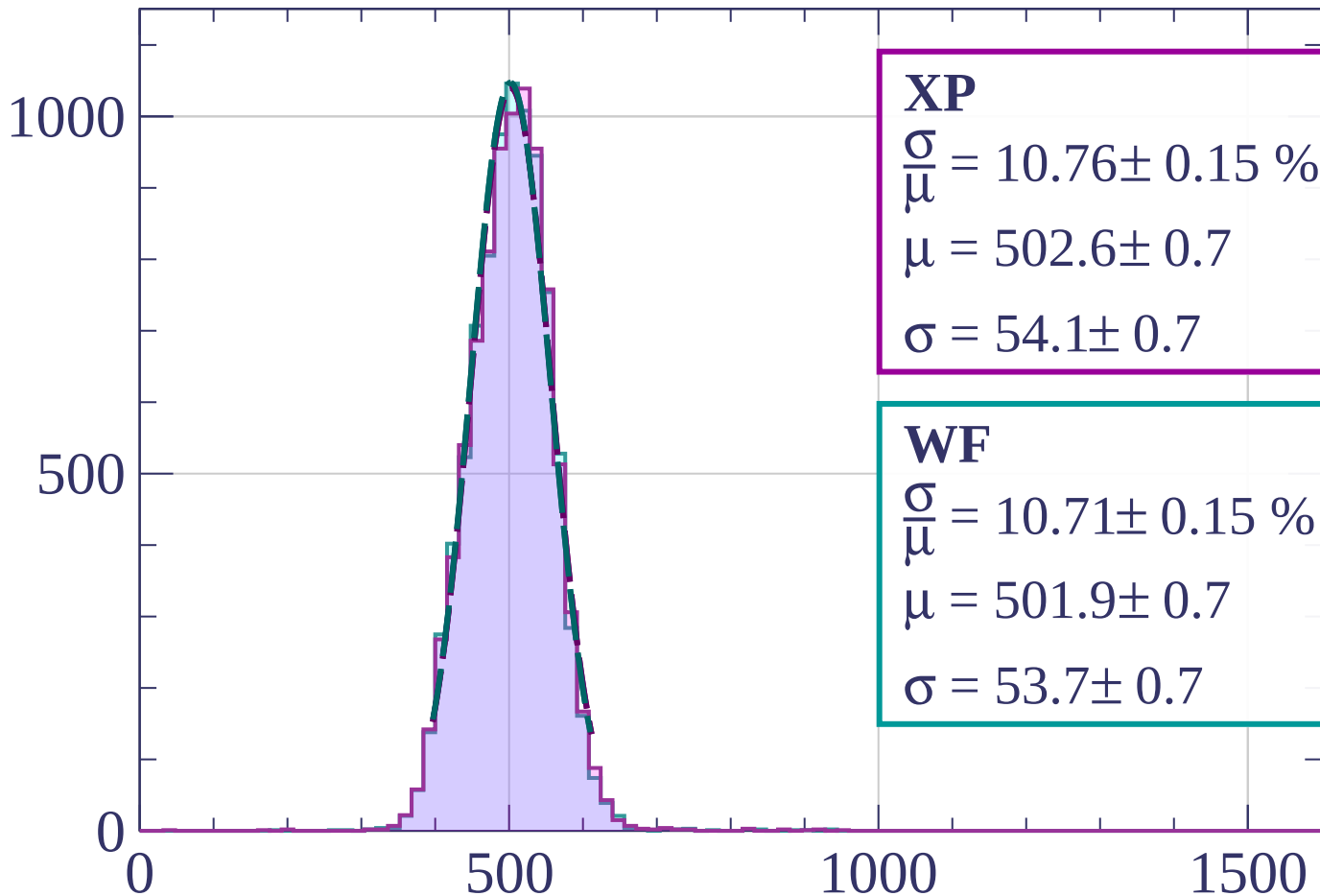
Count



dE/dx [ADC counts/cm]

Energy loss | $84 < \phi < 87$

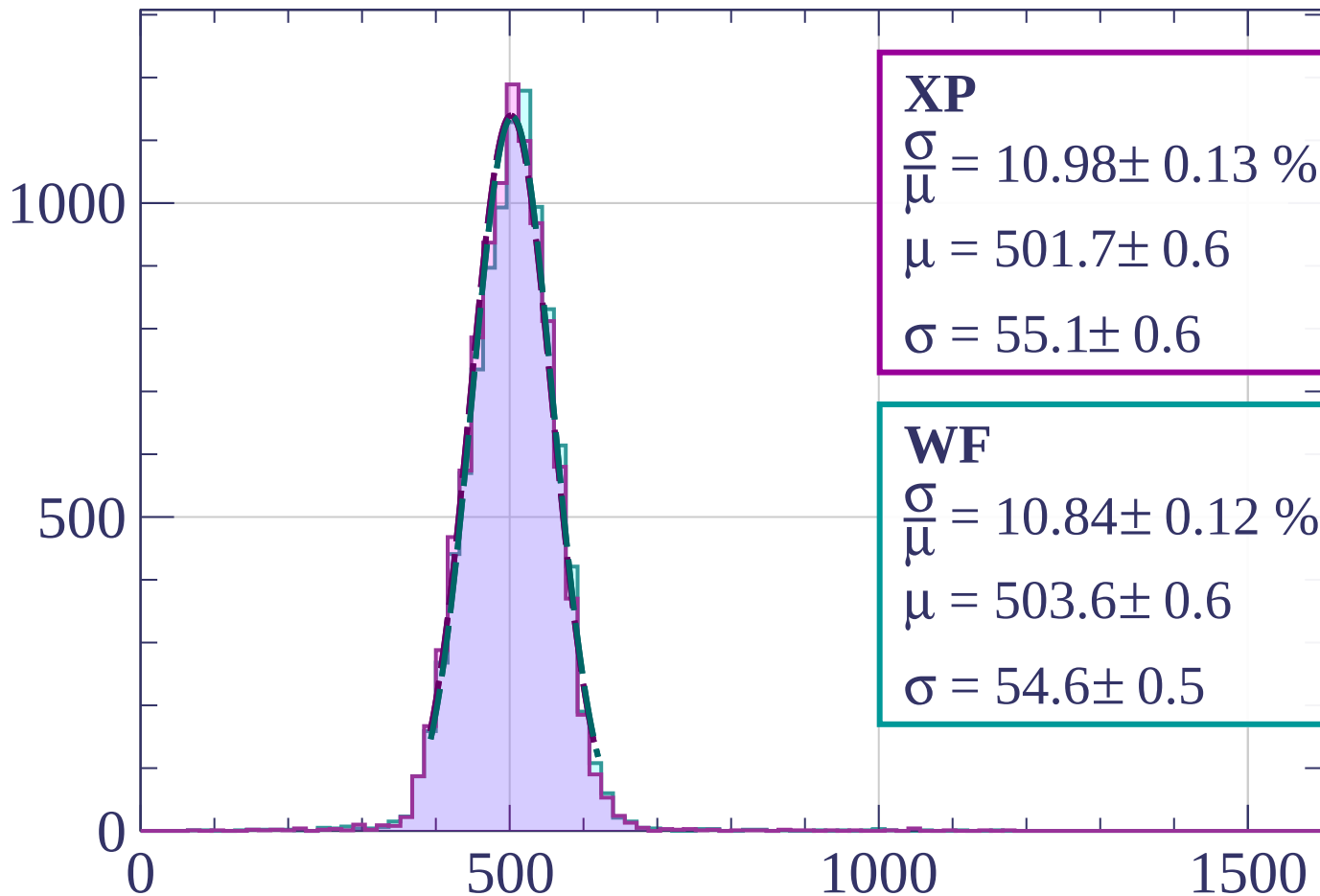
Count



dE/dx [ADC counts/cm]

Energy loss | $87 < \phi < 90$

Count



XP

$$\frac{\sigma}{\mu} = 10.98 \pm 0.13 \%$$

$$\mu = 501.7 \pm 0.6$$

$$\sigma = 55.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.84 \pm 0.12 \%$$

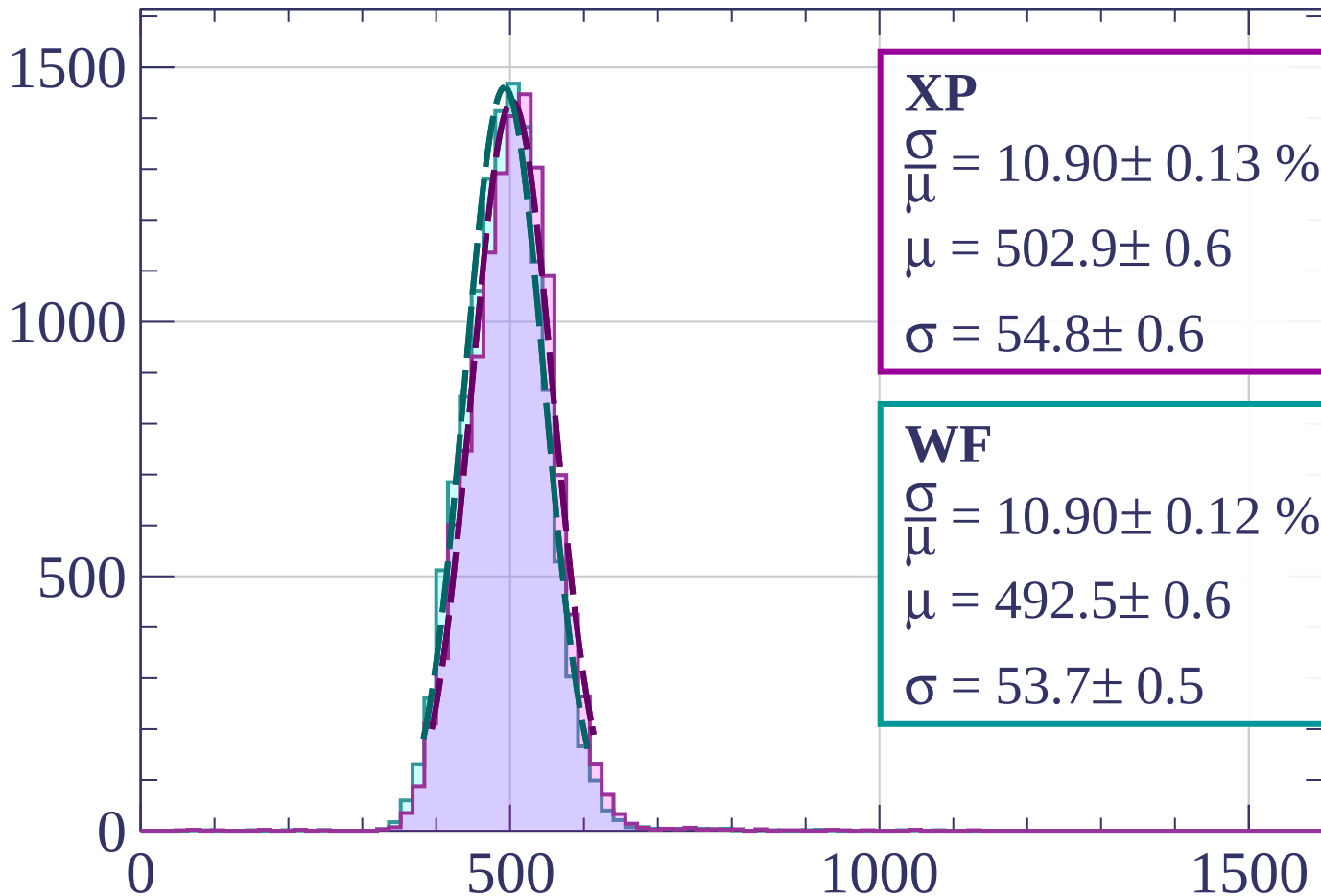
$$\mu = 503.6 \pm 0.6$$

$$\sigma = 54.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 10.90 \pm 0.13 \%$$

$$\mu = 502.9 \pm 0.6$$

$$\sigma = 54.8 \pm 0.6$$

WF

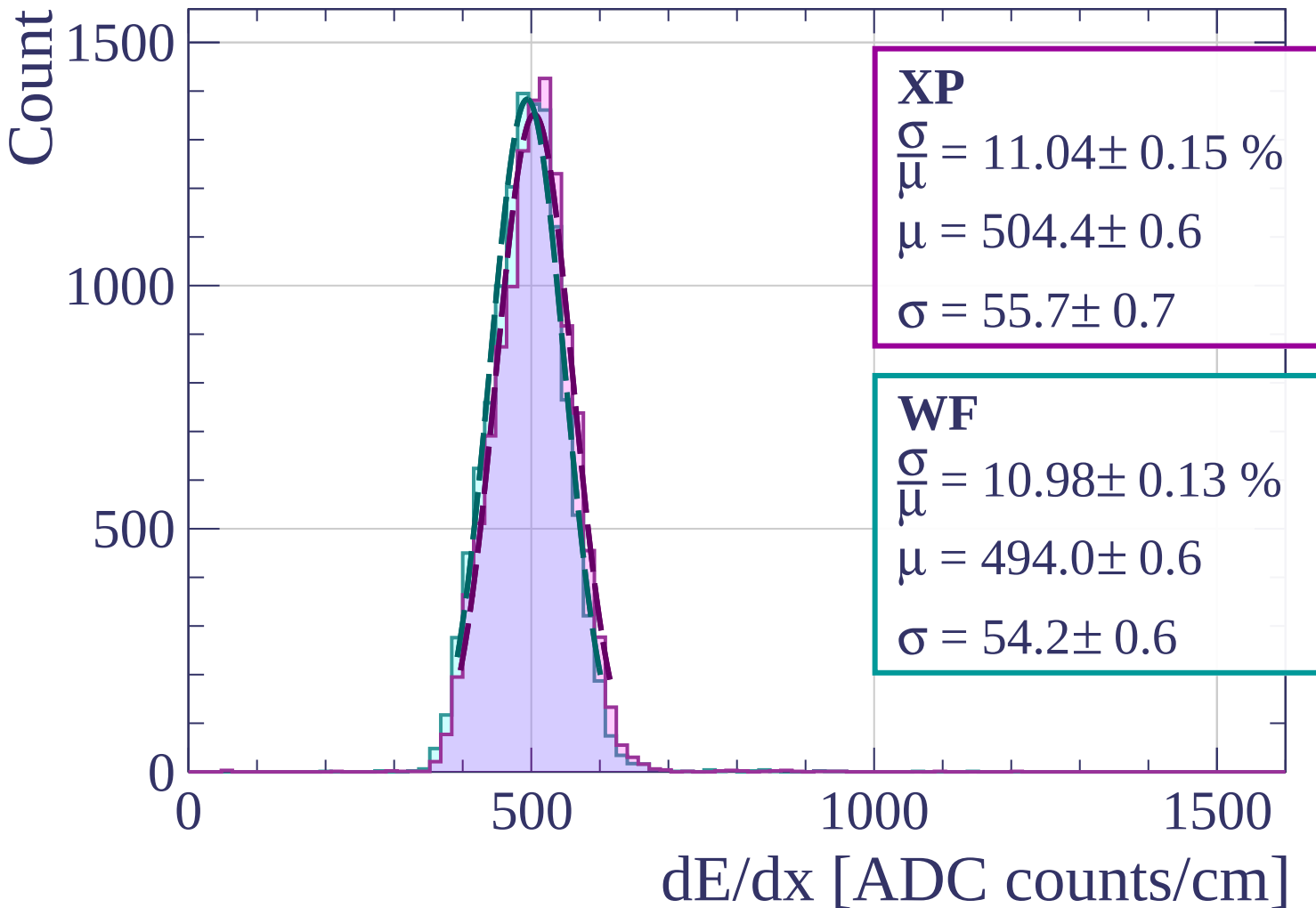
$$\frac{\sigma}{\mu} = 10.90 \pm 0.12 \%$$

$$\mu = 492.5 \pm 0.6$$

$$\sigma = 53.7 \pm 0.5$$

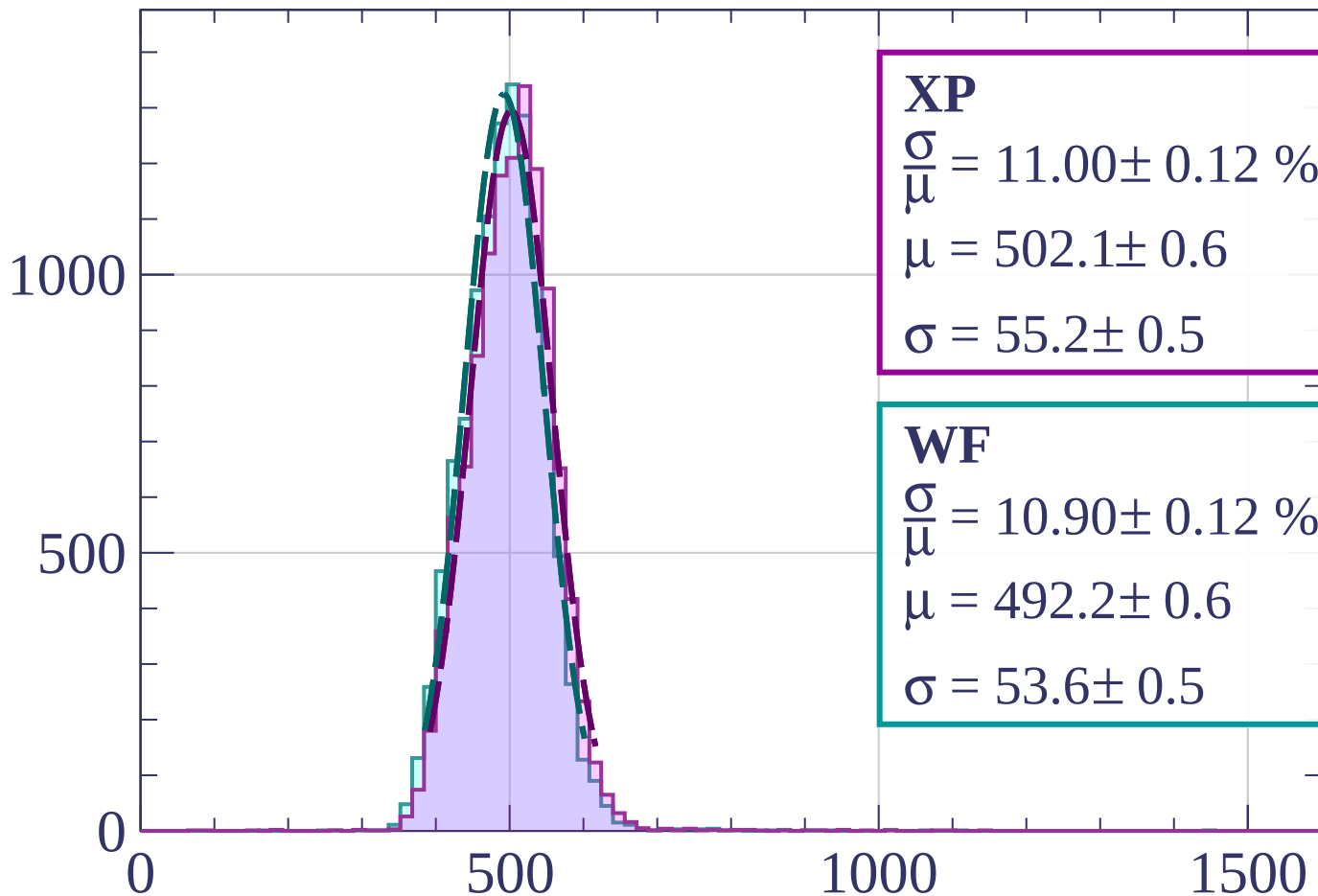
dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 11.00 \pm 0.12 \%$$

$$\mu = 502.1 \pm 0.6$$

$$\sigma = 55.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.90 \pm 0.12 \%$$

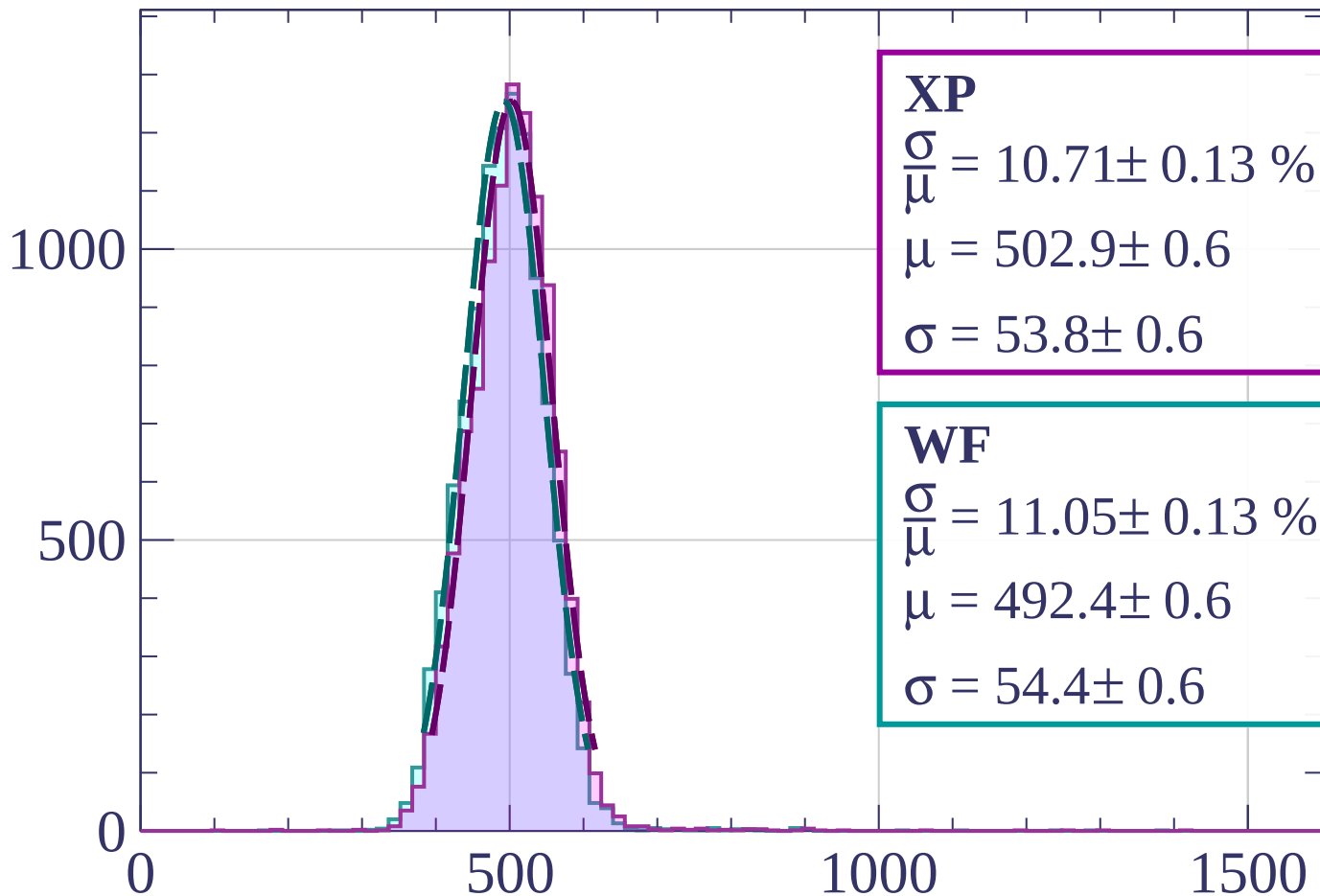
$$\mu = 492.2 \pm 0.6$$

$$\sigma = 53.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.13 \%$$

$$\mu = 502.9 \pm 0.6$$

$$\sigma = 53.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 11.05 \pm 0.13 \%$$

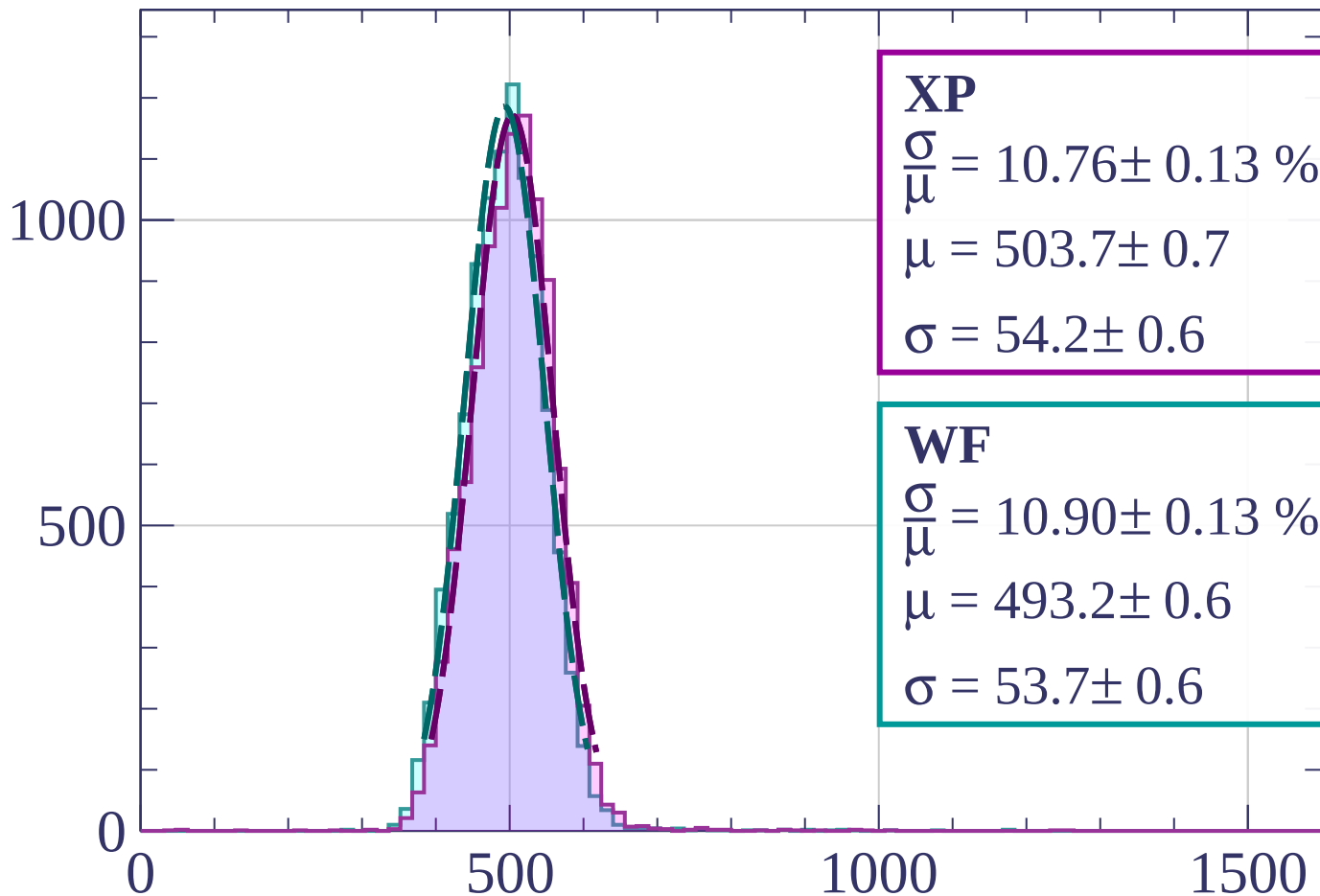
$$\mu = 492.4 \pm 0.6$$

$$\sigma = 54.4 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 10.76 \pm 0.13 \%$$

$$\mu = 503.7 \pm 0.7$$

$$\sigma = 54.2 \pm 0.6$$

WF

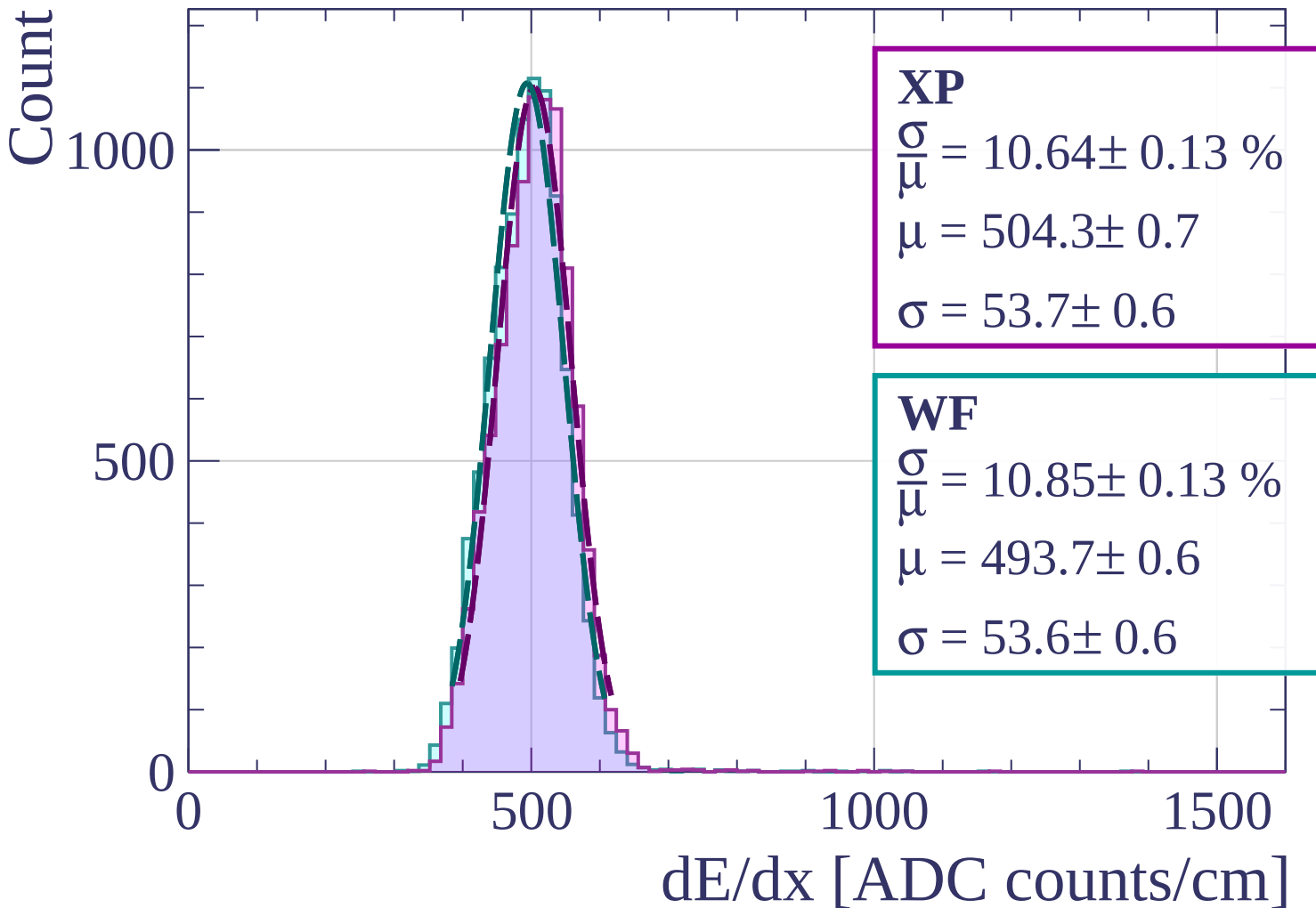
$$\frac{\sigma}{\mu} = 10.90 \pm 0.13 \%$$

$$\mu = 493.2 \pm 0.6$$

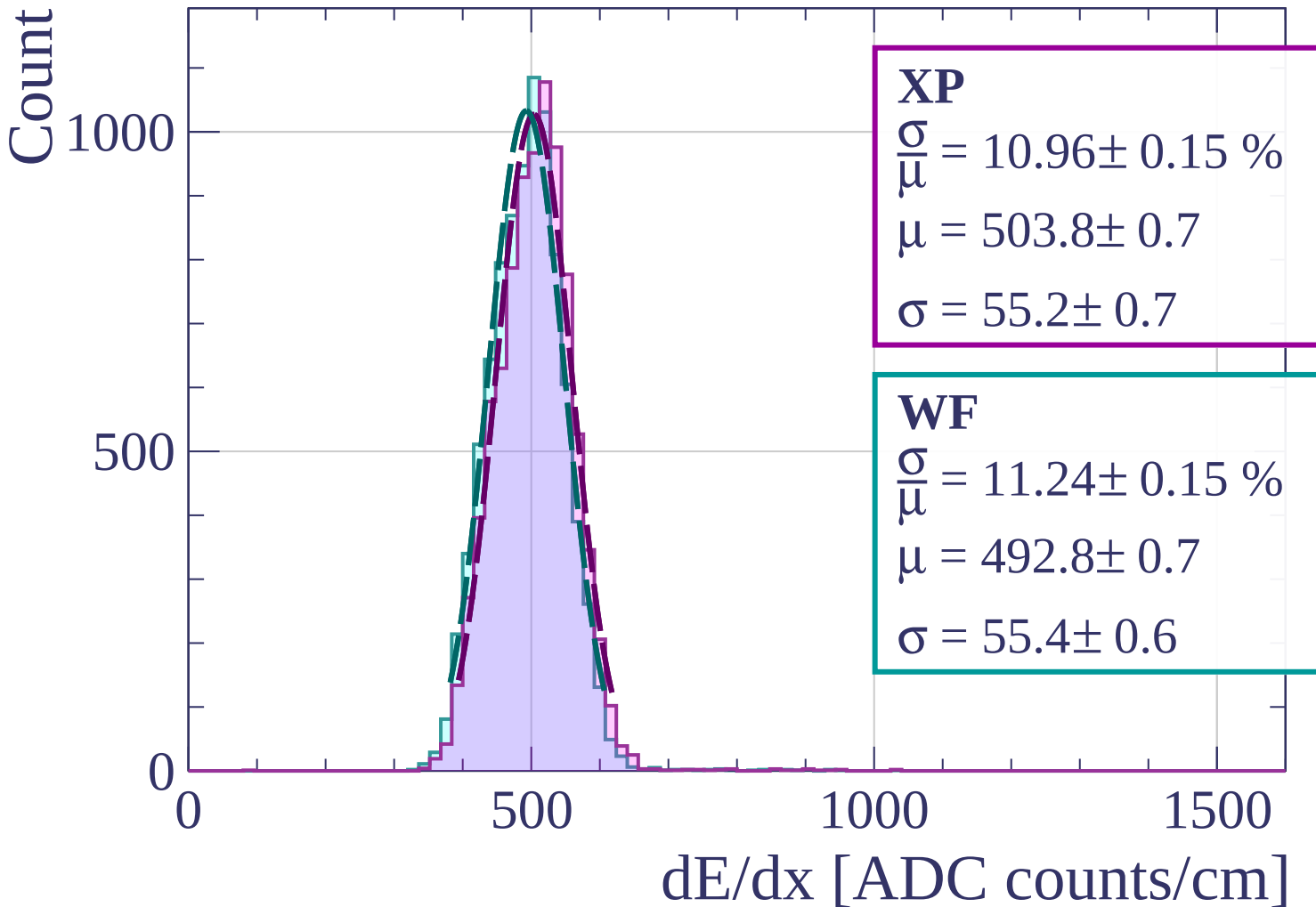
$$\sigma = 53.7 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$



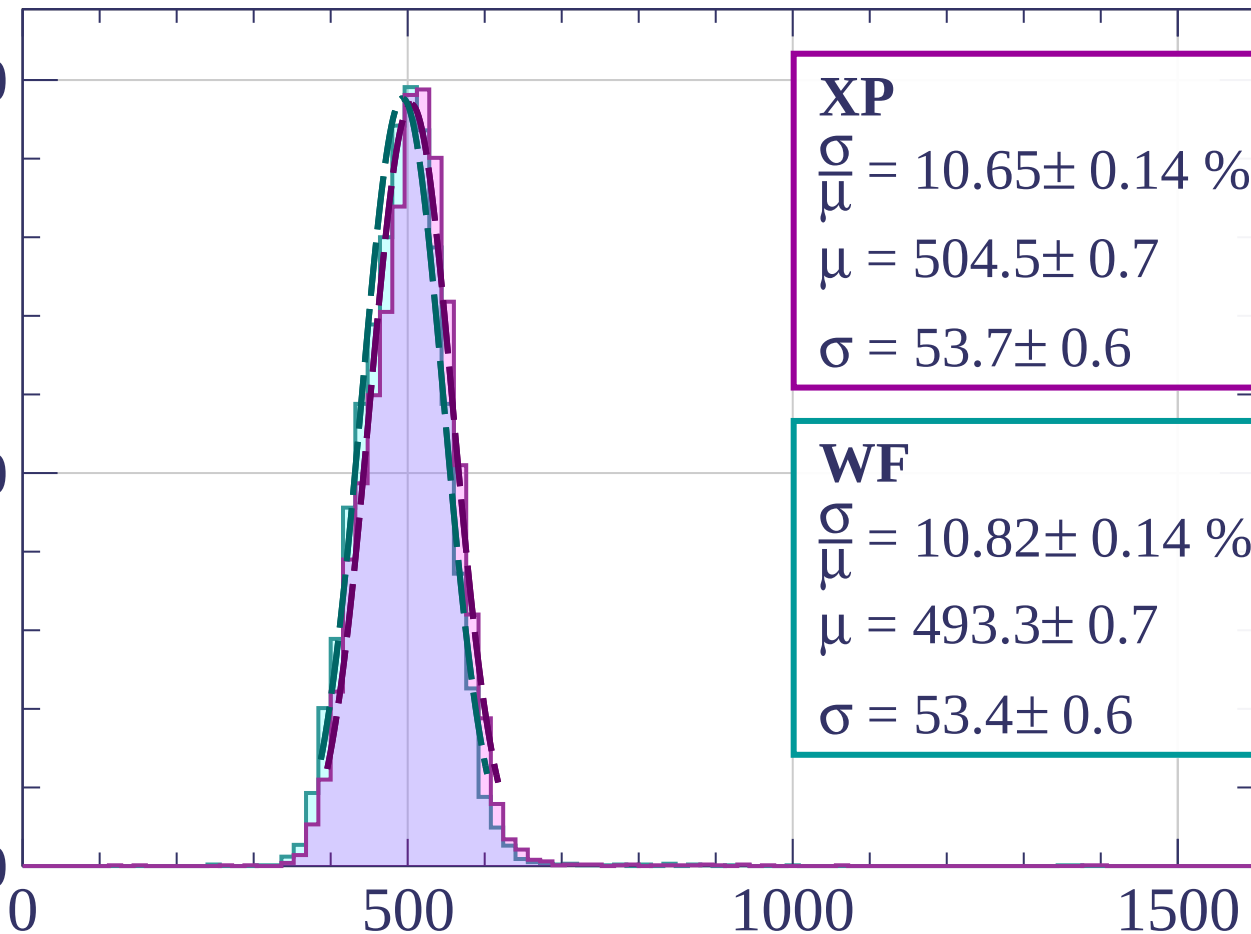
Energy loss | $21 < \theta < 24$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 10.65 \pm 0.14 \%$$

$$\mu = 504.5 \pm 0.7$$

$$\sigma = 53.7 \pm 0.6$$

WF

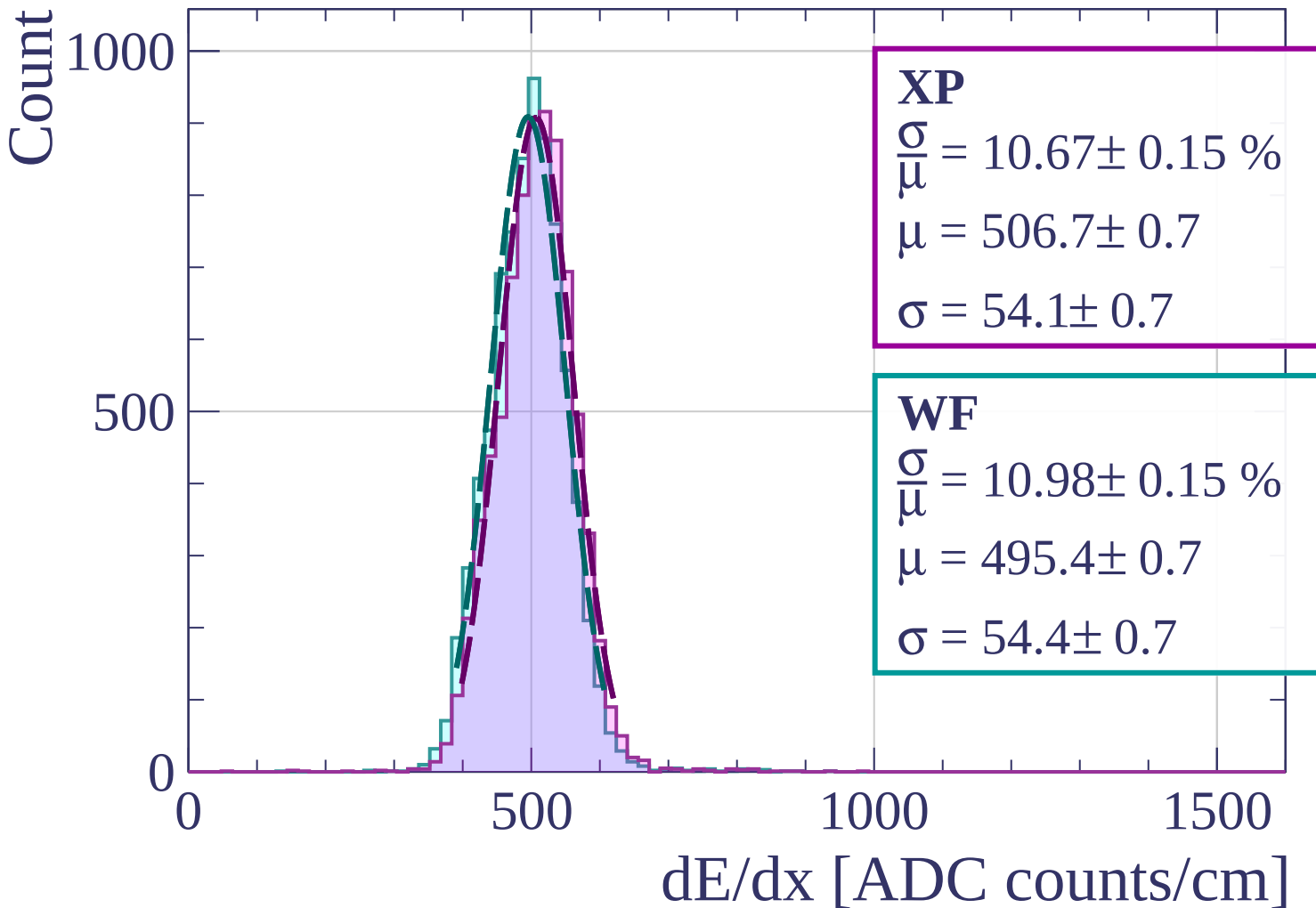
$$\frac{\sigma}{\mu} = 10.82 \pm 0.14 \%$$

$$\mu = 493.3 \pm 0.7$$

$$\sigma = 53.4 \pm 0.6$$

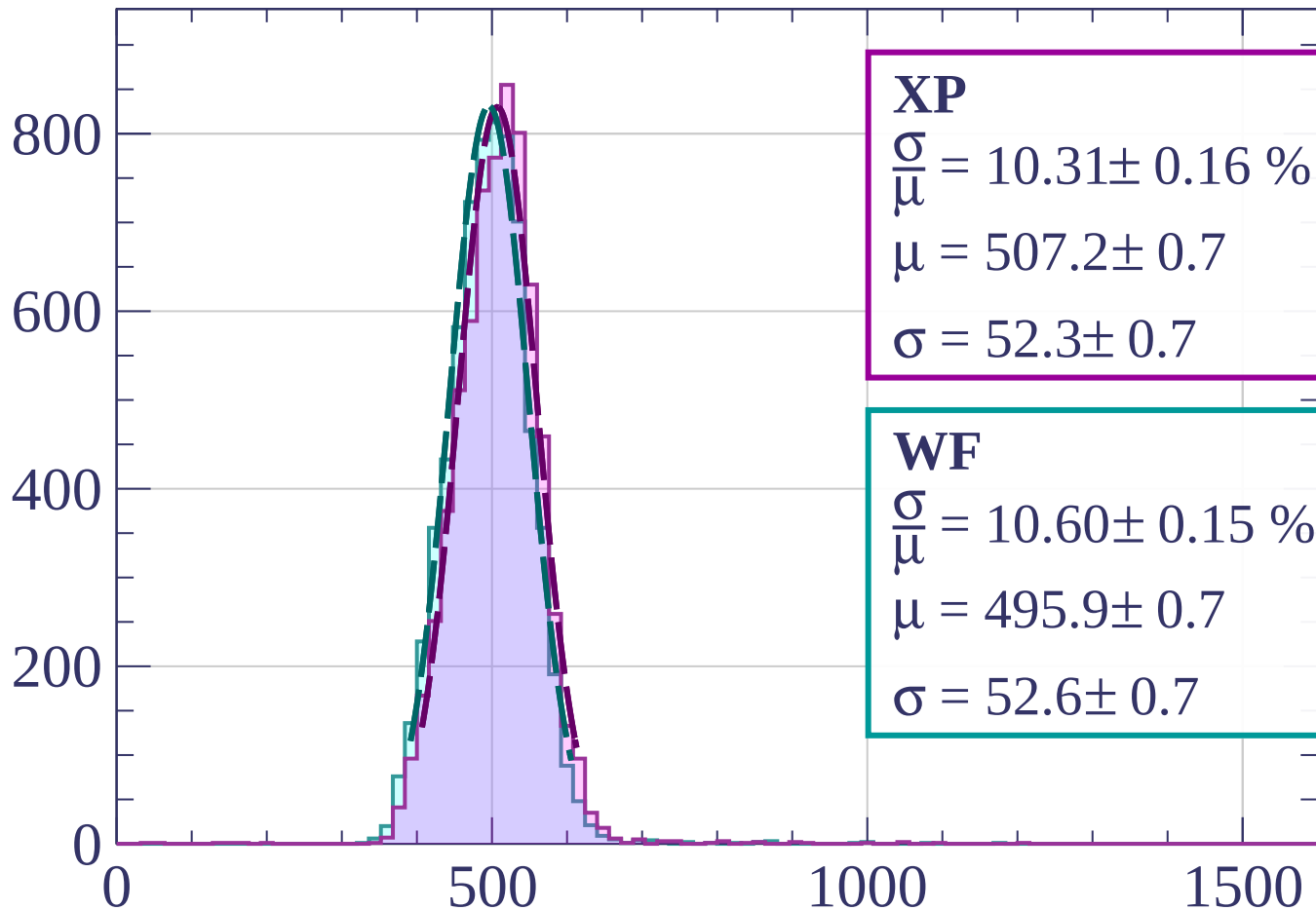
dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$



Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.16 \%$$

$$\mu = 507.2 \pm 0.7$$

$$\sigma = 52.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.60 \pm 0.15 \%$$

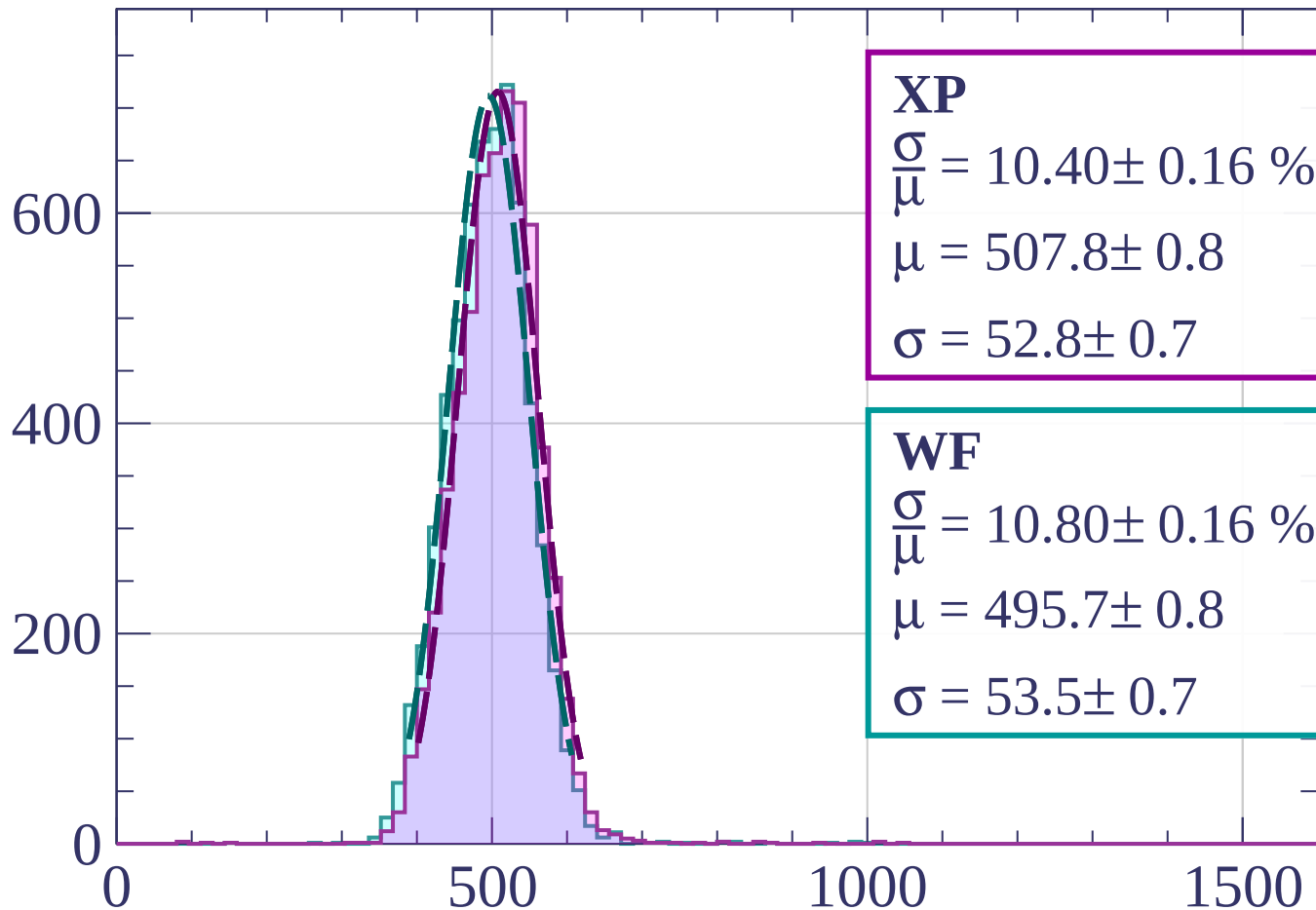
$$\mu = 495.9 \pm 0.7$$

$$\sigma = 52.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.16 \%$$

$$\mu = 507.8 \pm 0.8$$

$$\sigma = 52.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.80 \pm 0.16 \%$$

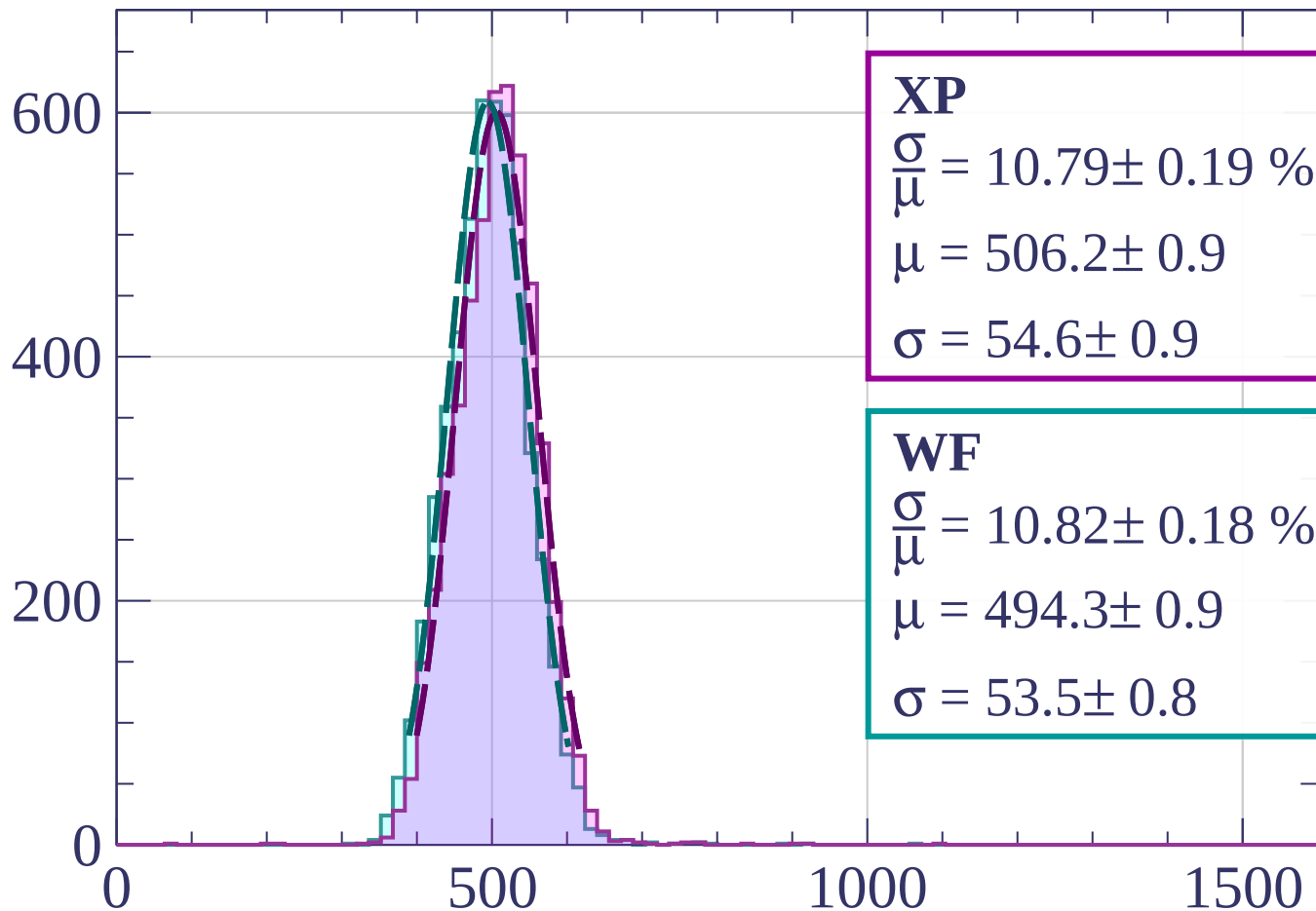
$$\mu = 495.7 \pm 0.8$$

$$\sigma = 53.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 10.79 \pm 0.19 \%$$

$$\mu = 506.2 \pm 0.9$$

$$\sigma = 54.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.82 \pm 0.18 \%$$

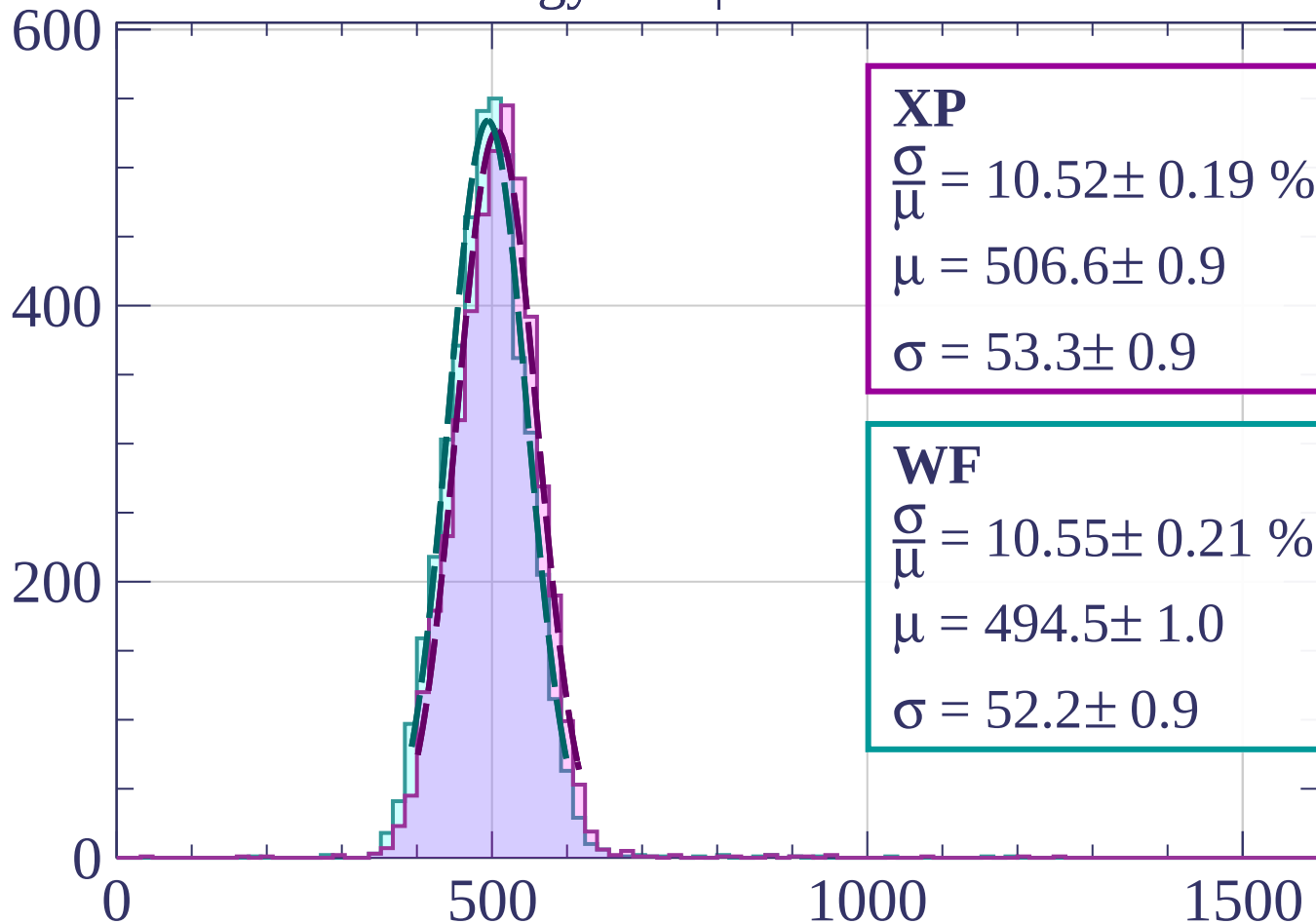
$$\mu = 494.3 \pm 0.9$$

$$\sigma = 53.5 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 10.52 \pm 0.19 \%$$

$$\mu = 506.6 \pm 0.9$$

$$\sigma = 53.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.55 \pm 0.21 \%$$

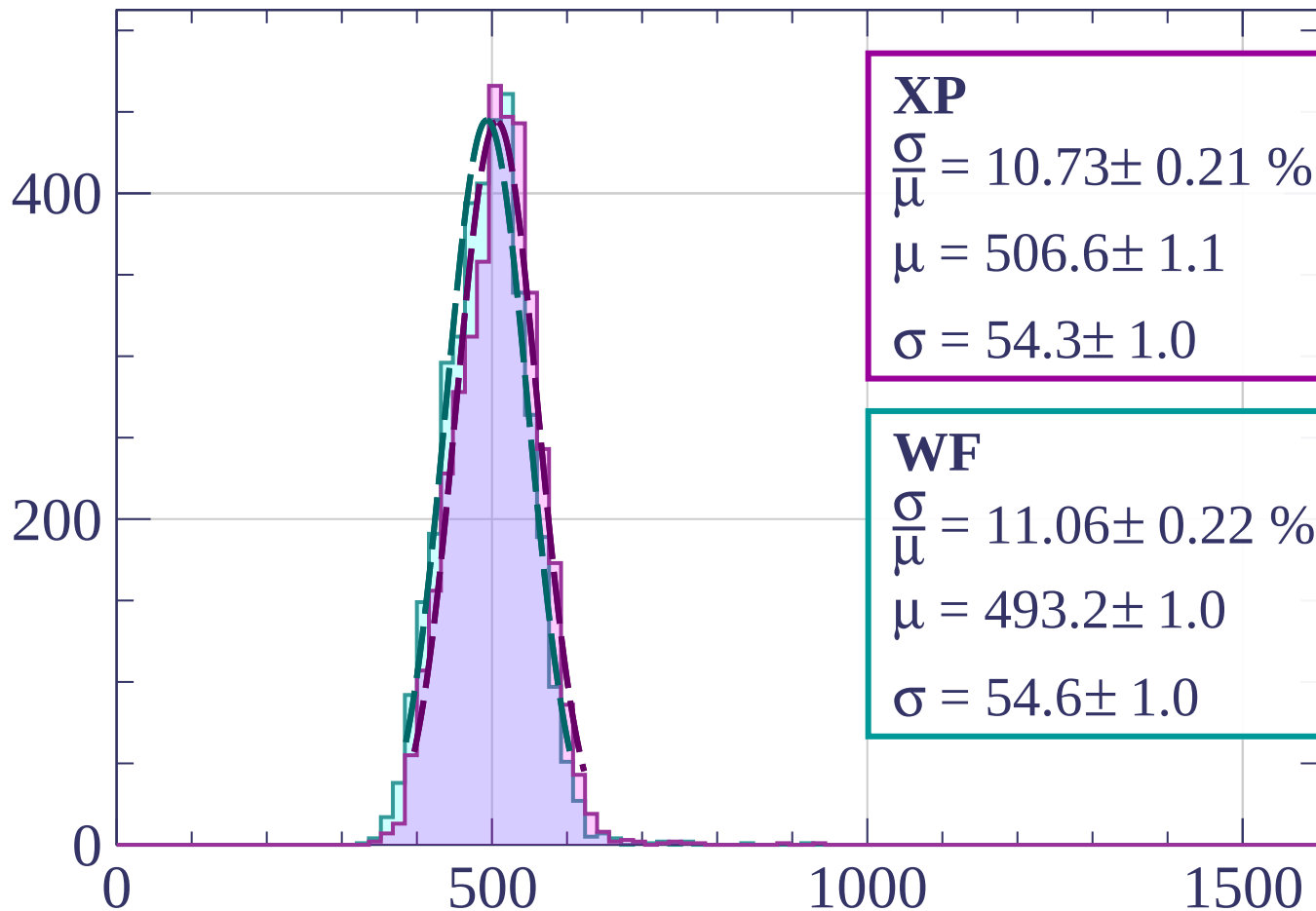
$$\mu = 494.5 \pm 1.0$$

$$\sigma = 52.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

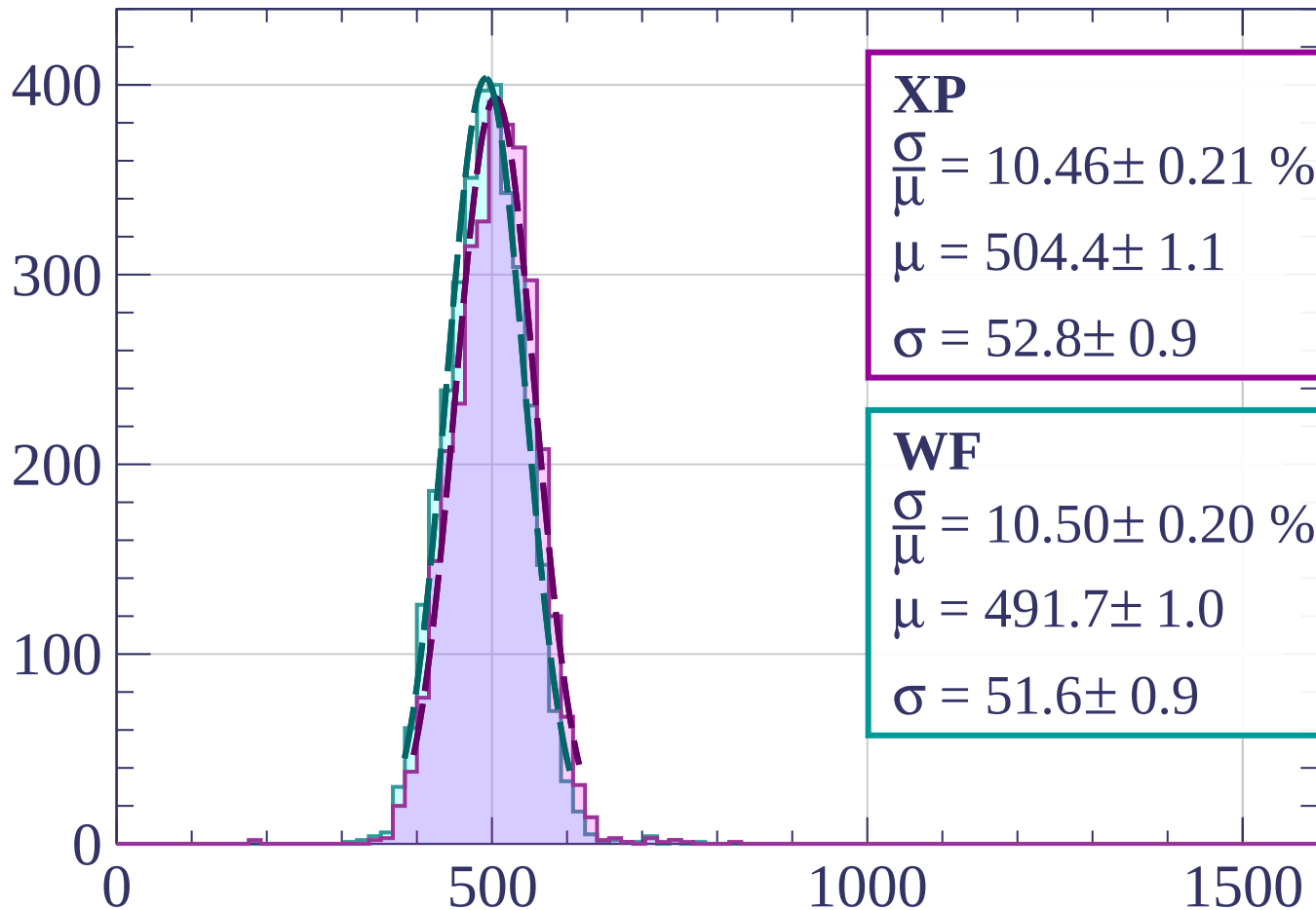
Count



dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 10.46 \pm 0.21 \%$$

$$\mu = 504.4 \pm 1.1$$

$$\sigma = 52.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.50 \pm 0.20 \%$$

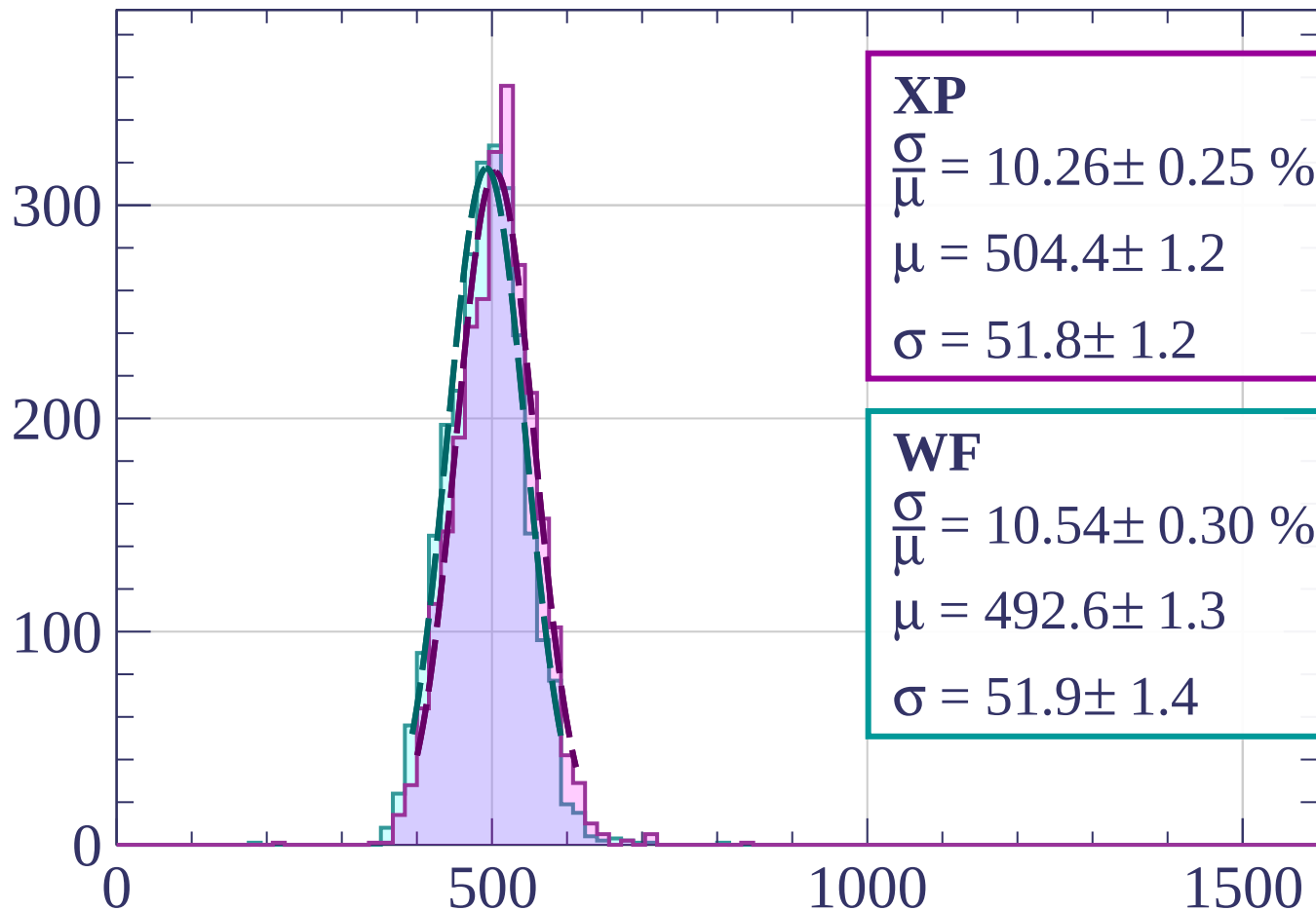
$$\mu = 491.7 \pm 1.0$$

$$\sigma = 51.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 10.26 \pm 0.25 \%$$

$$\mu = 504.4 \pm 1.2$$

$$\sigma = 51.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.30 \%$$

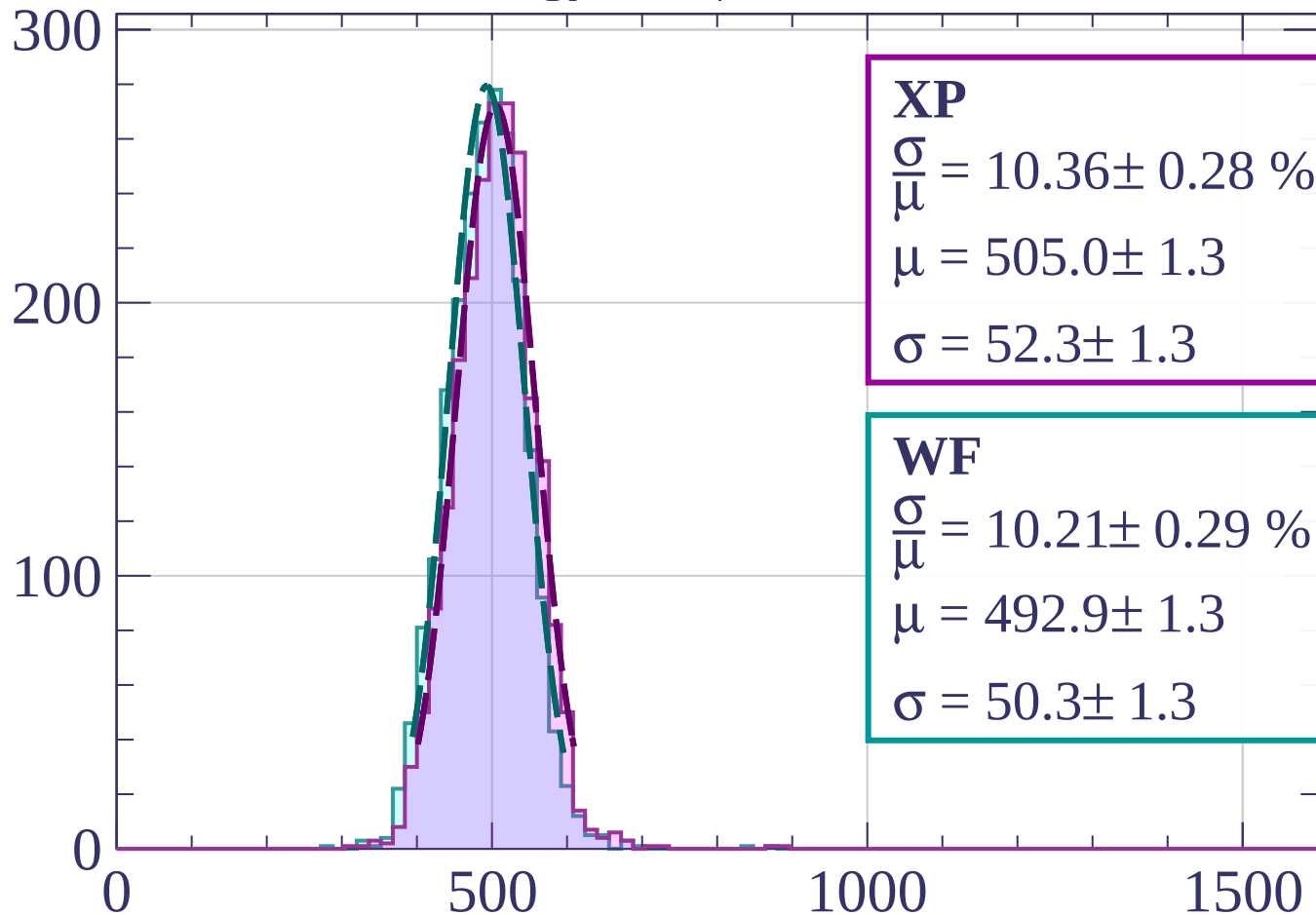
$$\mu = 492.6 \pm 1.3$$

$$\sigma = 51.9 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.28 \%$$

$$\mu = 505.0 \pm 1.3$$

$$\sigma = 52.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 0.29 \%$$

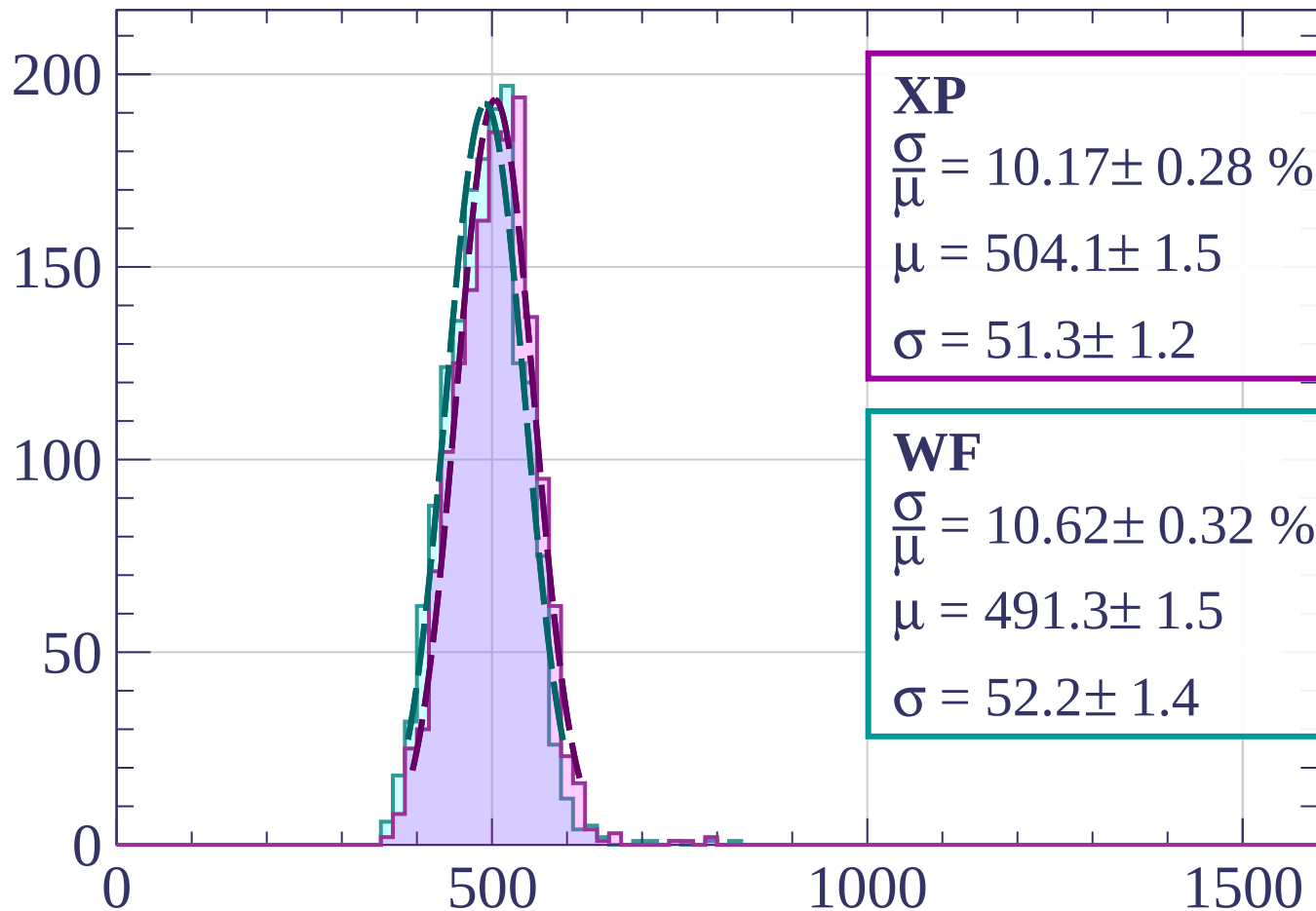
$$\mu = 492.9 \pm 1.3$$

$$\sigma = 50.3 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 10.17 \pm 0.28 \%$$

$$\mu = 504.1 \pm 1.5$$

$$\sigma = 51.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.62 \pm 0.32 \%$$

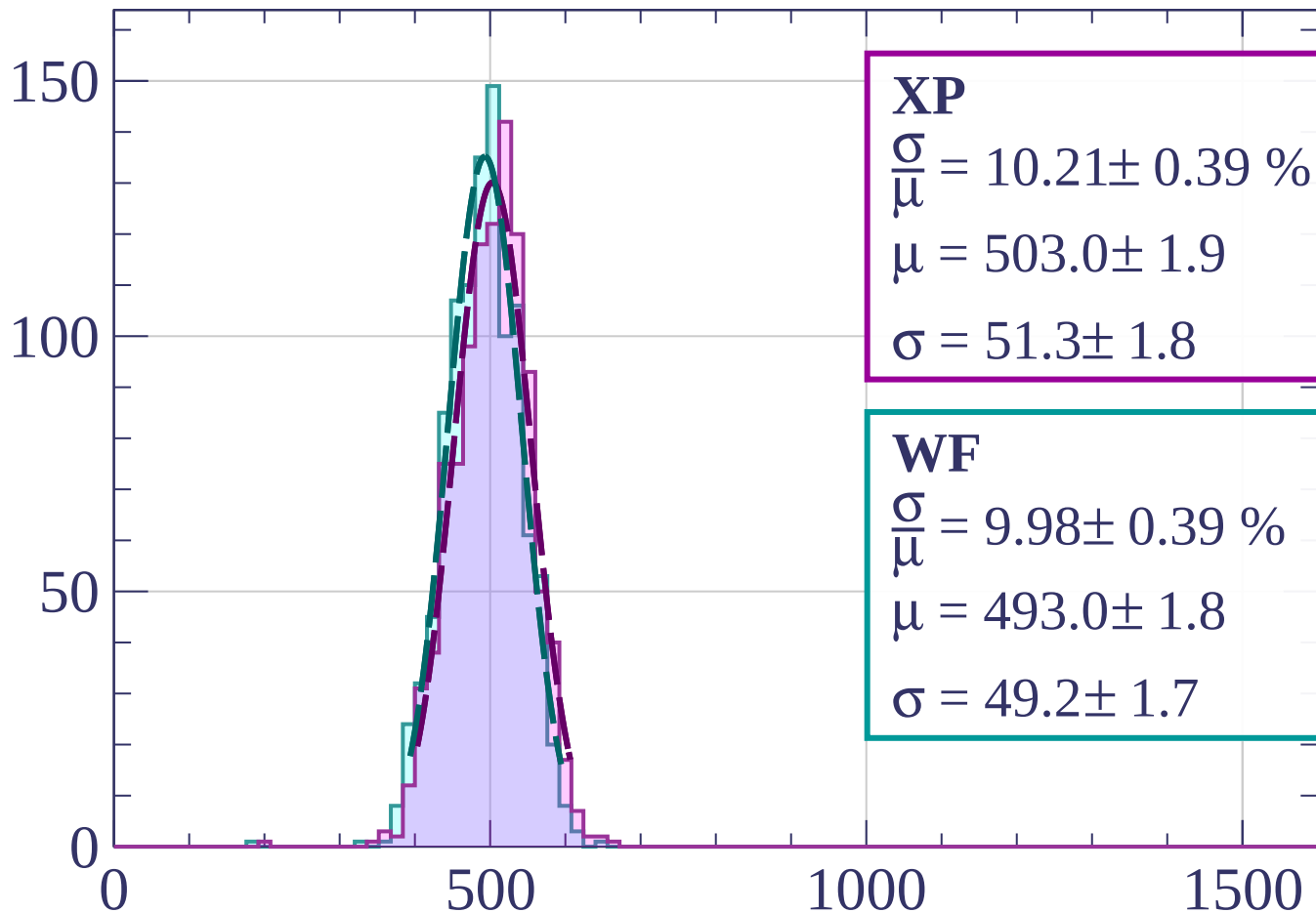
$$\mu = 491.3 \pm 1.5$$

$$\sigma = 52.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.39 \%$$

$$\mu = 503.0 \pm 1.9$$

$$\sigma = 51.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 9.98 \pm 0.39 \%$$

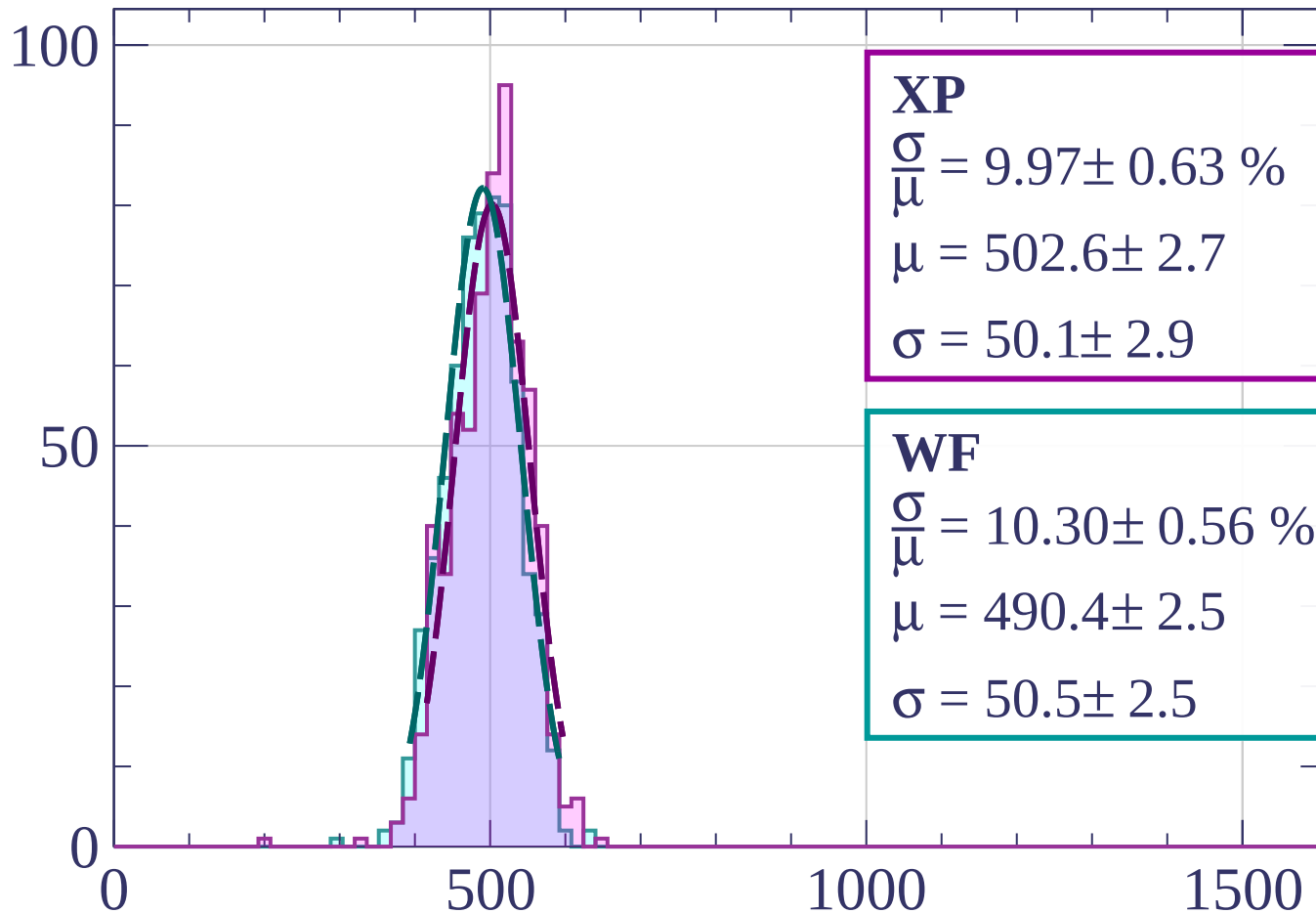
$$\mu = 493.0 \pm 1.8$$

$$\sigma = 49.2 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.63 \%$$

$$\mu = 502.6 \pm 2.7$$

$$\sigma = 50.1 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 10.30 \pm 0.56 \%$$

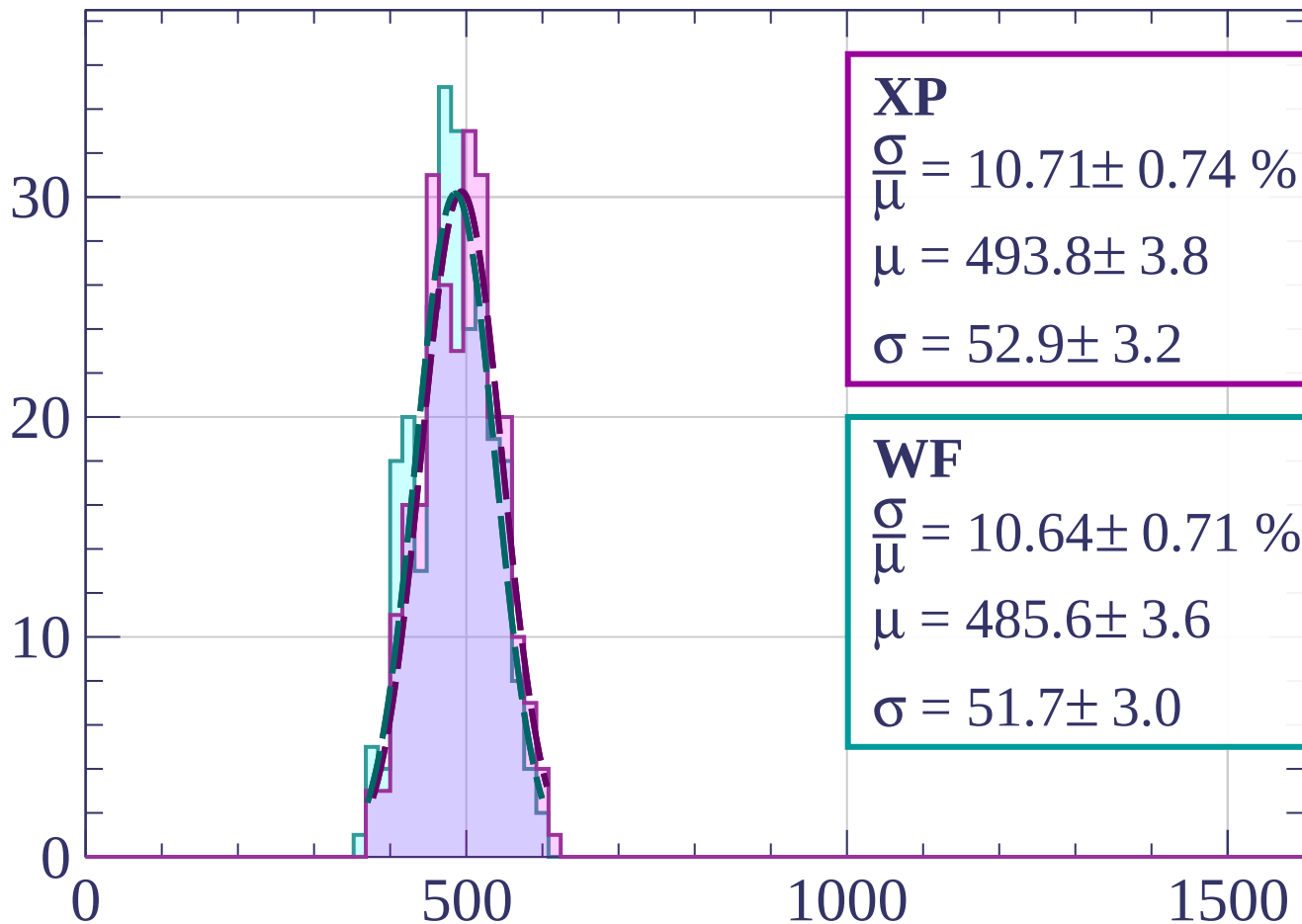
$$\mu = 490.4 \pm 2.5$$

$$\sigma = 50.5 \pm 2.5$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 10.71 \pm 0.74 \%$$

$$\mu = 493.8 \pm 3.8$$

$$\sigma = 52.9 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 10.64 \pm 0.71 \%$$

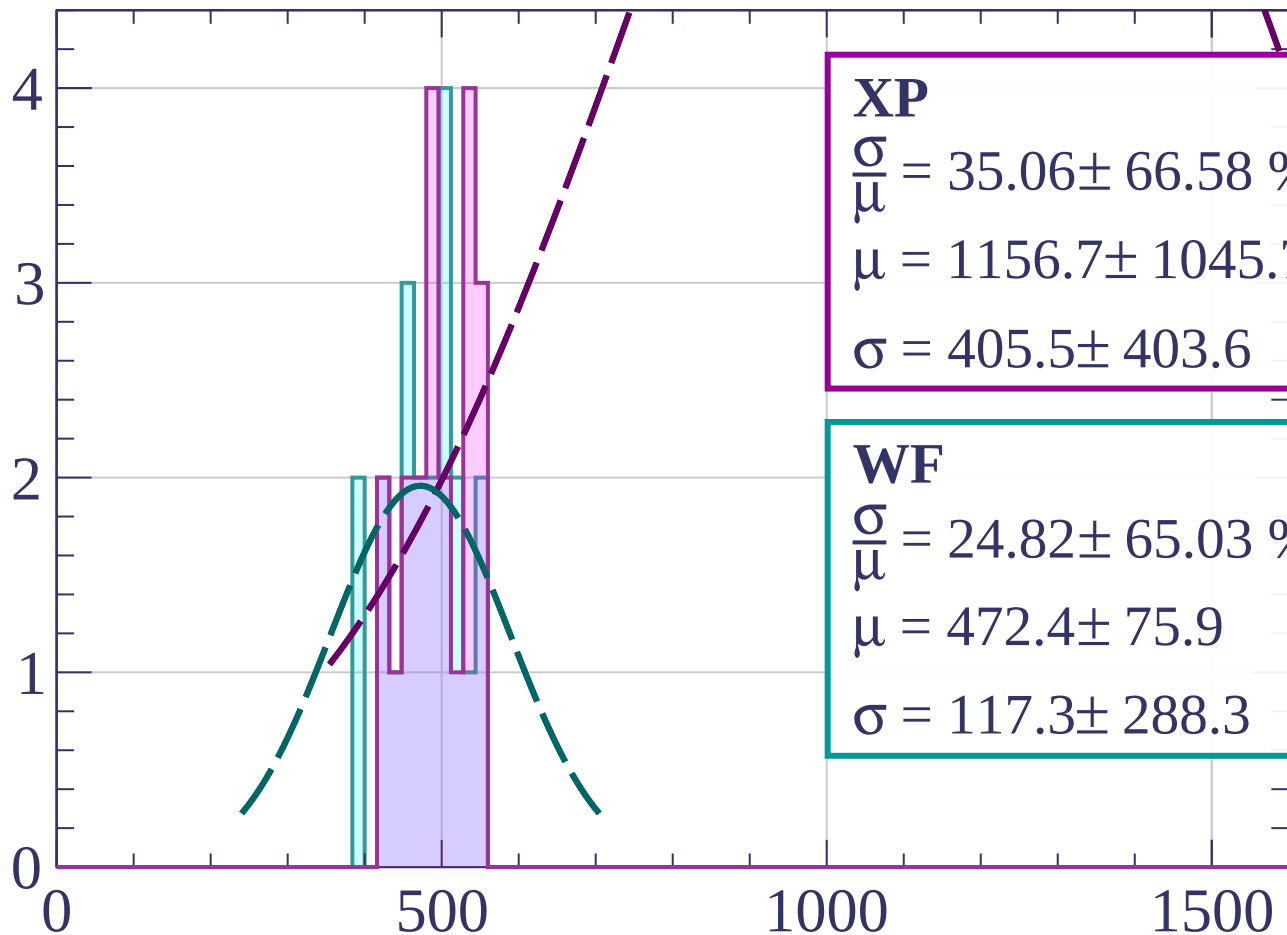
$$\mu = 485.6 \pm 3.6$$

$$\sigma = 51.7 \pm 3.0$$

dE/dx [ADC counts/cm]

Count

Energy loss | $63 < \theta < 66$



XP

$$\frac{\sigma}{\mu} = 35.06 \pm 66.58 \%$$

$$\mu = 1156.7 \pm 1045.7$$

$$\sigma = 405.5 \pm 403.6$$

WF

$$\frac{\sigma}{\mu} = 24.82 \pm 65.03 \%$$

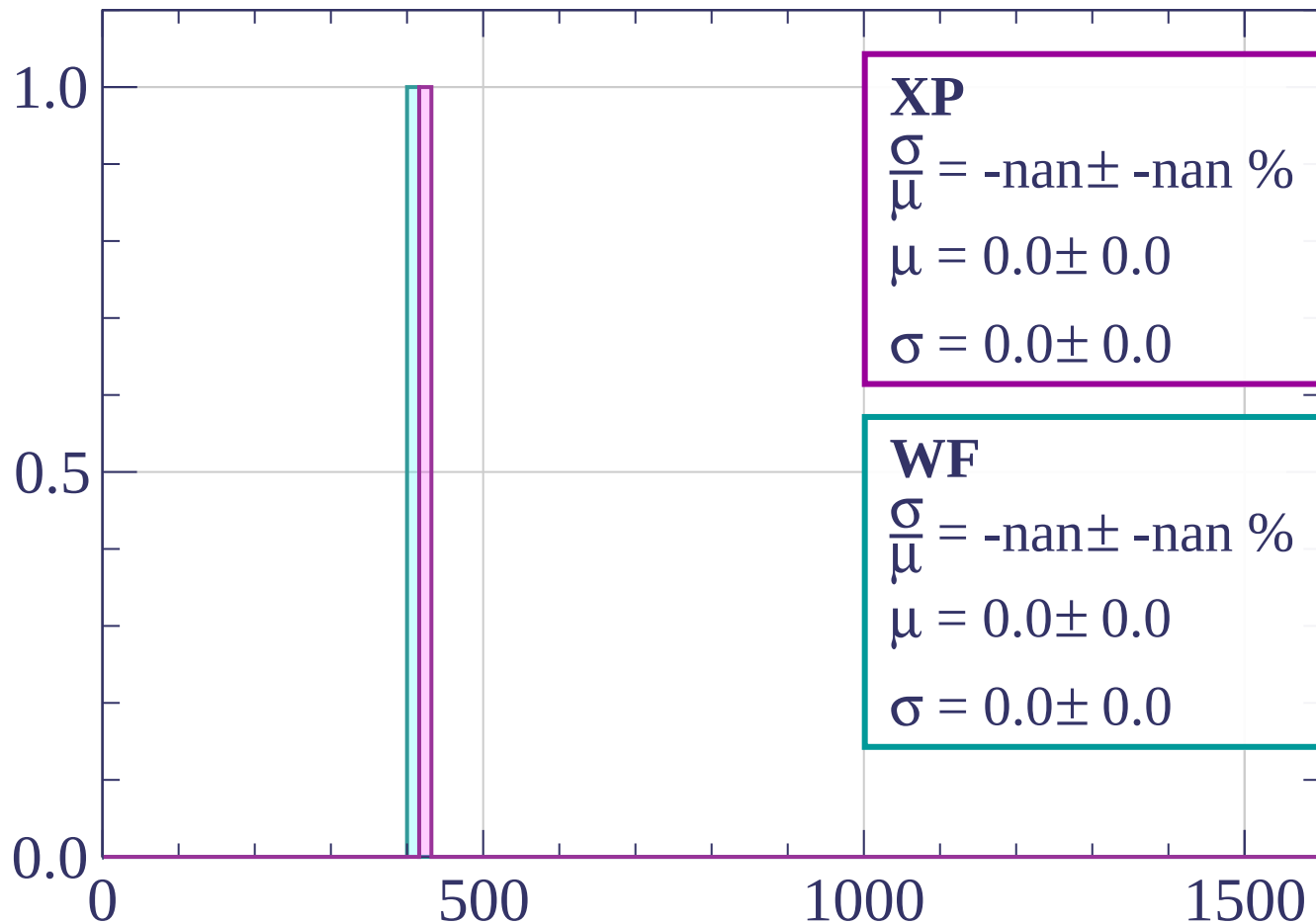
$$\mu = 472.4 \pm 75.9$$

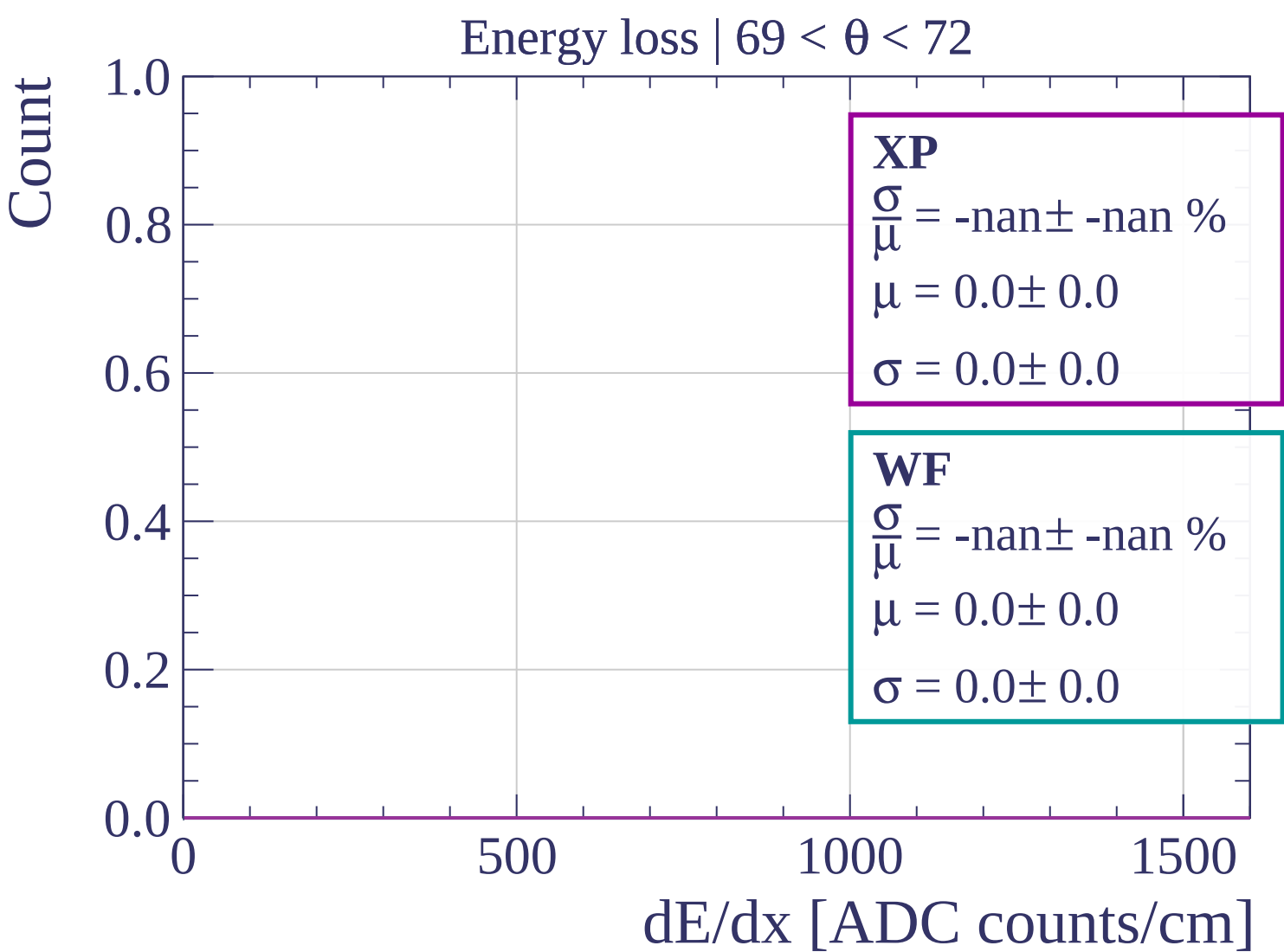
$$\sigma = 117.3 \pm 288.3$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \theta < 69$

Count





Count

Energy loss | $72 < \theta < 75$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

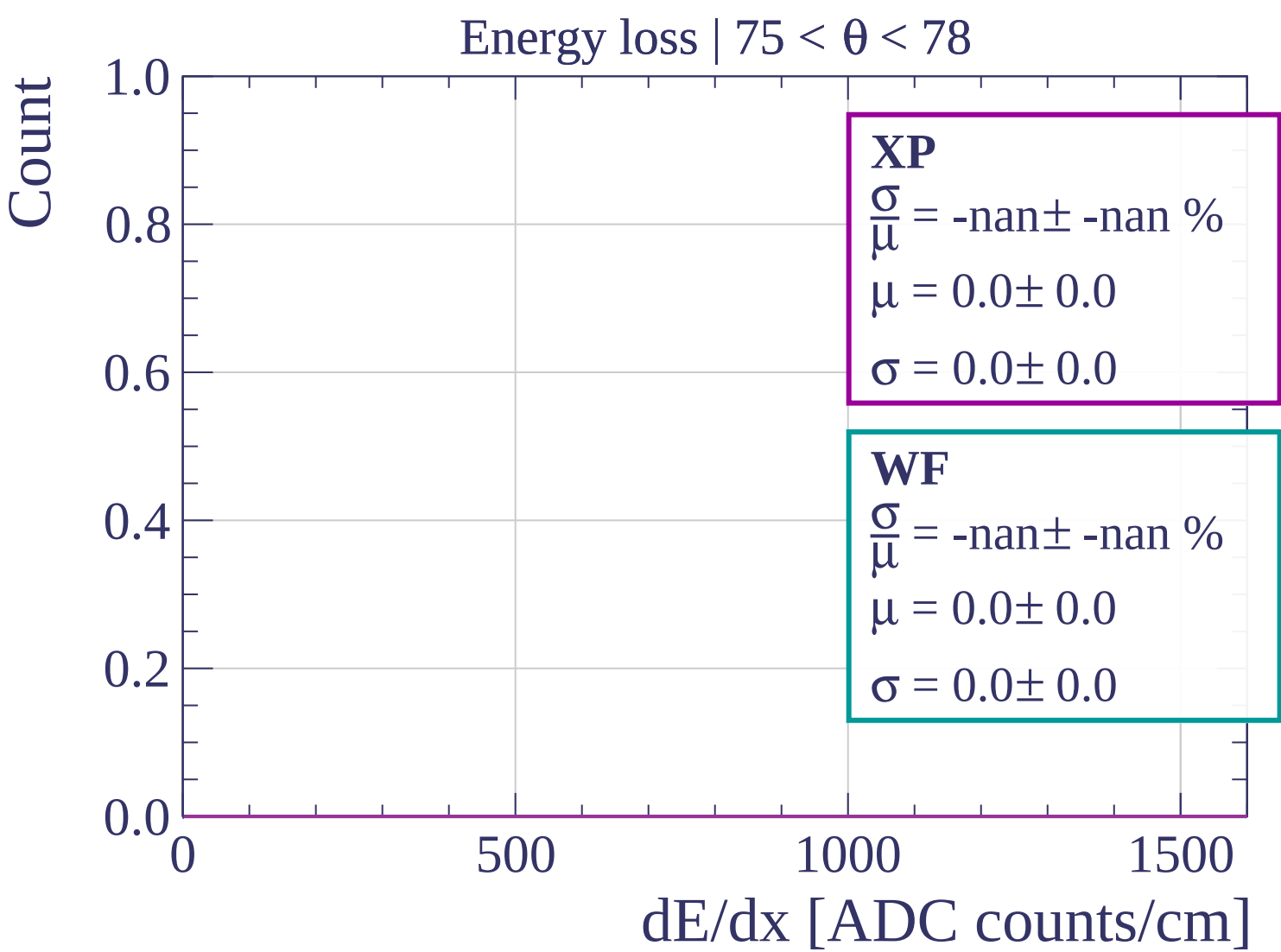
0

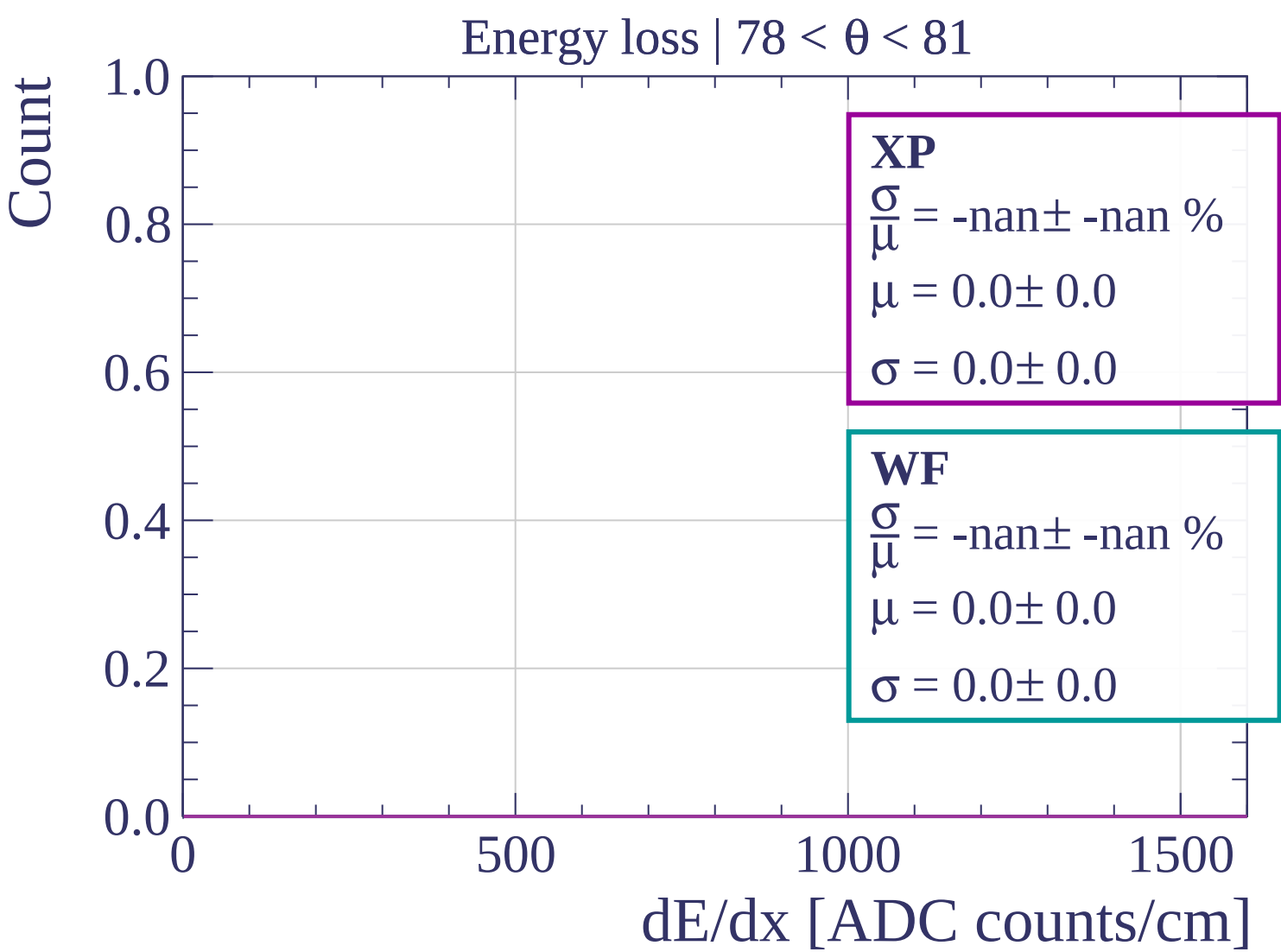
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $81 < \theta < 84$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

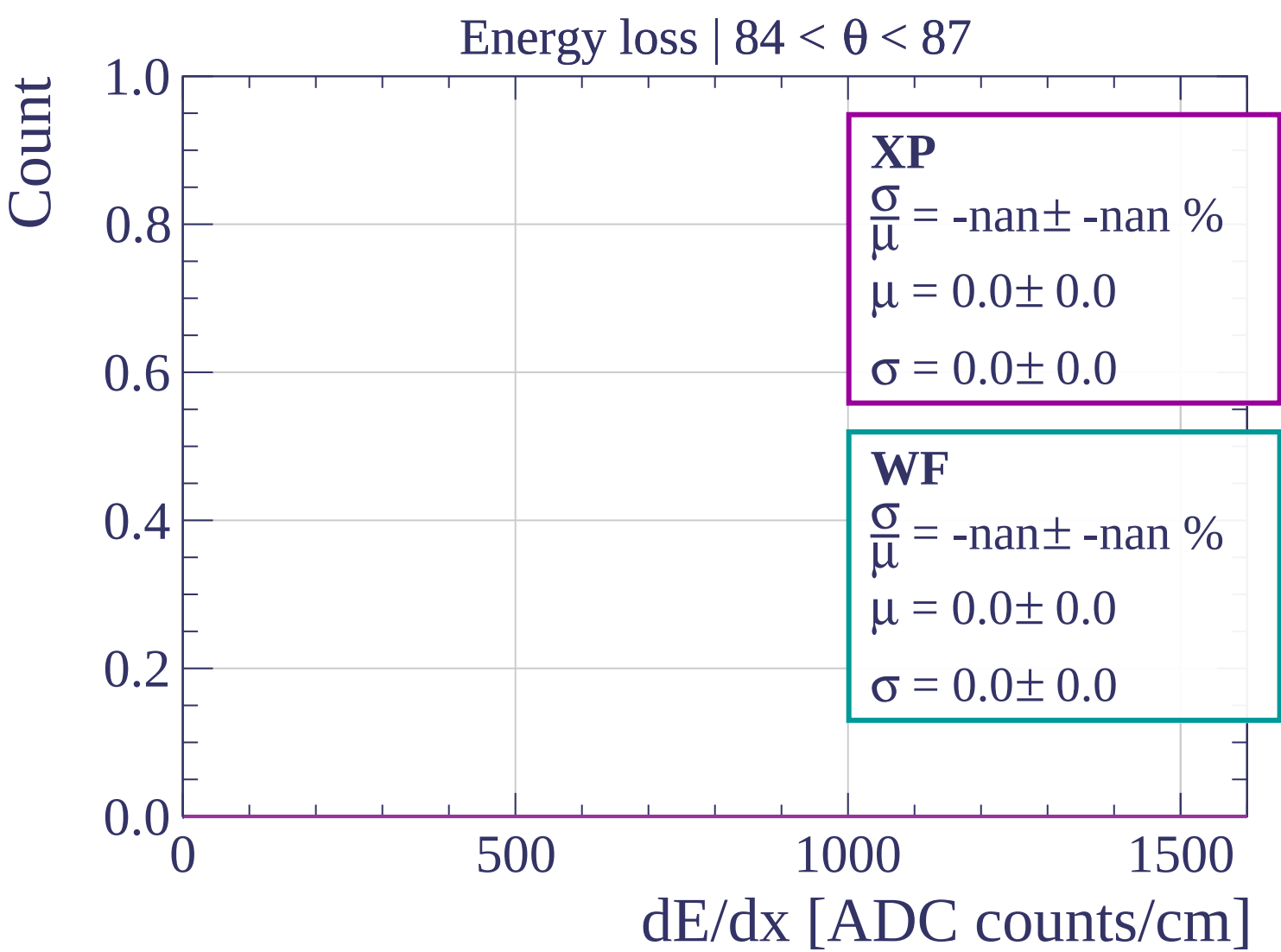
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $87 < \theta < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

